Project Technical Report for the IREC
Team 005 Project Technical Report for the 2026 IREC

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This paper presents *Wedgetail*, NU Rocketry's most innovative entry into the International Rocketry Endeavour Cup for 2026 to compete in the 10000 feet solid COTS category. The rocket is a four-inch platform constructed with SRAD body tubes, and fins, with COTS Wildman couplers, and Von Karman nosecone. The rocket flies on an L1940X Aerotech motor to an expected apogee of 10350 feet under 2 ms^{-1} wind, with a minimum static stability of 2.82 calibre throughout flight. The system utilises a dual separation, dual deployment recovery architype flying on a 24-inch Rocketman drogue parachute, and a 50-inch Recon Recovery main parachute deployed at 1312 feet bringing the rocket down at 28ms^{-1} and 9ms^{-1} respectively. The payload is a technical demonstration of a precision thermoelectric control system designed for the stabilisation of temperature of vital medicine to remote locations. The rocket will also feature an airbrake system, with the mechanical control disabled, for data collection and characterisation. The rocket successfully test flew on an L1420 motor at a High-Powered Launch event to an altitude of 11050 feet, suffering zero damage while collecting data for both payload and airbrake, however the airbrake system failed to deploy leading to the decision to disable it at IREC 2026.

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Nomenclature

C_p = Centre of Pressure
C_G = Centre of Gravity
C_D = Coefficient of Drag
FEA = Finite Element Analysis

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Executive Summary

NU Rocketry's entry into the 2026 International Rocketry Endeavour Cup, Wedgetail, is the newest development from the previous international competition, the Spaceport America Cup 2024. Wedgetail is a four-inch platform flying on a L1490X, to a predicted apogee of 10347 feet under 2 ms^{-1} of average wind, for competition in the 10000 feet solid COTS category.

The rocket's airframe is constructed with SRAD body tubes made of tri-axial fiberglass using a wet-layup process, with a COTS Wildman Von Karmen nosecone and COTS Wildman coupler sections. These tubes have been tested in laboratory compression tests with all tubes withstanding expected loads with large factors of safety. The rocket has 3 clipped-delta carbon fibre fins, surface mounted with 20 mm epoxy filets made and a single layer of 200 gsm carbon fibre as a tip to tip, giving a factor of safety of 1.878 for fin flutter.

The rocket uses a dual-separation dual deployment recovery scheme on a 24-inch Rocketman drogue parachute, and a 50-inch Recon Recovery main parachute deployed at 1312 feet bringing the rocket down at 28ms^{-1} and 9ms^{-1} respectively. The deployment events are triggered by the two flight computers, a Blue Raven and a Telemetrum, ensuring redundancy, with a Raven 4 and a Fluctus as backup flight computers should one of the main computers break. The drogue and main are triggered by 3.0g main black powder charges and a 3.5g backup charge, with the size determined through calculation with an overcompensation added due to the lack of opportunities to experimentally test black powder charge sizes.

Wedgetail features an experimental technological demonstration of a precision temperature control system utilising a Peltier module and PID controller to maintain a set temperature ± 1 C. The payload is constructed of mainly 3.4 mm G10 plates with other components 3D-printed using Acrylonitrile Butadiene Styrene (ABS) for ease of production and prototyping. The system features a machined copper cube with holes for storing temperature sensitive medication, selected for its favourable thermal attributes.

An experimental airbrake system is also flown onboard, utilising a three flap mechanism to actively modify the drag profile of the rocket while ensuring the system imposes no moments on the rocket to avoid attitude changes. The system utilises an Extended Kalman Filter taking in data from a barometer, inertial measurement unit (IMU), magnetometer, and a GPS, which is fed into a model predictive control (MPC) controller to calculate the level of extension required for the flaps.

The rocket has a successful test flight on a L1420 to 11084 feet, with a very stable flight, successful recovery, and data collection in both the payload and the airbrake, however the airbrake system failed to deploy leading to the decision to disable the mechanical element of the airbrake system for IREC 2026 due to the uncertainties around the effect the airbrake could have on the rocket's flight.

Introduction

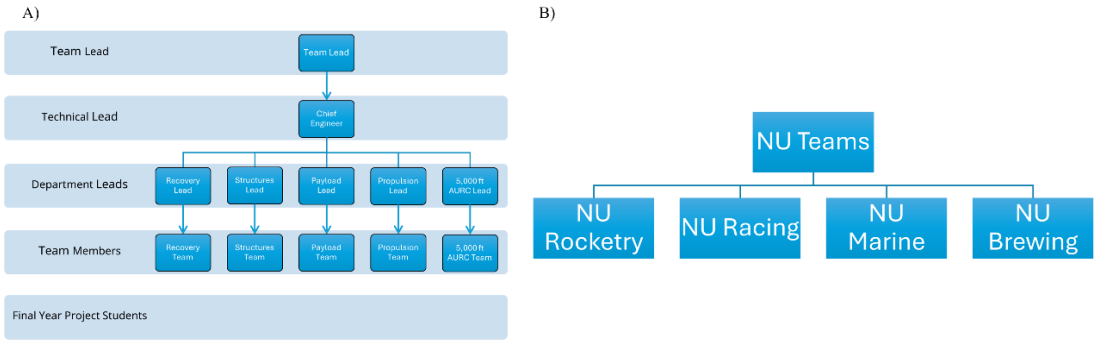

Figure 1 A) NU Rocketry Hierarchy B) NU Teams Hierarchy

For 2026, NU Rocketry is participating in the 10k Solid COTS category flying on a 4-inch platform with two major design goals as well as one major competition goal. Similarly to our previous entry into the 2024 Spaceport America Cup, the rocket was designed to fly on an L2 motor with the added goal of remaining under a 10kg dry mass allowing for a smaller L2 motor, the L1940, to be flown whilst still hitting 10000 feet. The rocket was also to be capable of carrying an experimental airbrake system previously unused in competition for NU Rocketry. The payload will be an experimental technological demonstration of precision thermoelectric control of a volume for the delivery of temperature sensitive cargo. The other major goal was the improvement of our overall point total by at least 1 point over our previous entry at Spaceport America Cup 2024, particularly in the report writing elements of the scoring rubric through more in-depth analysis of all levels of our system, and improvement in our flight performance score.

NU Rocketry exists as one branch of the overall NU Teams structure as seen in Figure 1 B) which consists of four main teams; NU Rocketry, NU Racing, NU Marine, and NU Brewing all of which operate under the University of Newcastle, Australia. NU Teams is overseen by our director Dr Dylan Cuskelly and aided by our technical staff member Mr Malcolm Sidney.

NU Rocketry consists of a tiered leadership structure seen in Figure 1 A) with the Team Lead Mr Nicholas George and Chief Engineer Mr Tate Derrick overseeing the team through a joint leadership approach. The role of the Team Leader is to ensure the team can meet set goals by removing obstacles hindering their progress. The removal of these obstacles allows the technical team to thrive. The role also specifically includes managing stakeholders, organizing logistics, direct management of the final year project team, and future planning. The Chief Engineer's role is to facilitate the technical aspect of the team and ensure the rocket at all times stays compliant with the rules and DTEG. Under the higher leadership level are the department leads, Mr. Nicholas Cummings, Mr. Rylie Heller, and Mr. Brendan Walker who manage the junior members in each of their respective departments, the propulsions lead Mr. Jonah Sessions leading the development of a hybrid motor system, and the 5000ft AURC Team Lead Miss. Jessica Deniss. These departments are then filled with junior and senior engineering students from multiple engineering disciplines predominantly Aerospace, Mechanical, and Mechatronics who complete tasks designated by their respective department lead.

The "Team 005 Project Technical Report to the 2026 IREC" was prepared by members from all levels of hierarchy with the main body prepared by the juniors of the corresponding department to the section, which was reviewed and added to by their Department Lead and the Chief Engineer. As the analysis and overview of the system took a large amount of time compared to other sections, the other sections, such as the Introduction, Mission Concept of Operations, etc. were prepared by the Chief Engineer and the Team Lead to allow other members more time to thoroughly analyse the system.

NU Teams offers multiple avenues for gaining course credit as part of working on larger projects. Students may enrol in either ENGG3200 or ENGG4200 as electives as part of their course requirements while some continue in NU Teams and NU Rocketry to complete their Final Year Project, known at the University of Newcastle as an FYP, usually as a large technical project relevant to the team they are participating in or as part of their leadership

responsibilities allowing for course credits to be given for the work undertaken, however many students exist in the team as an extracurricular activity.

The team utilizes a critical path management system to maintain a successful development path towards the production of the rocket for the International Rocketry Endeavour Cup 2026. The strategy has allowed the team to maintain clear vision on the key components of the system throughout the development, ensuring meaningful progress is being observed by all departments undertaking work upon the rocket. This has allowed for the completion of a full system test of the payload system two weeks before our last international competition platform, as well as the complete development of our avionics stack and recovery system two months before our first test launch. The added time has also ensured multiple launch dry runs to be undertaken resulting in the rocket being launched on the first half of the day of the test launch window.

Figure 2 NU Teams Sponsors

NU Teams currently has 13 total industry partners who not only provide funding for each project, but also graduate and undergraduate job opportunities as part of the NU Teams industry partner program. The 13 industry partners of NU teams currently include Varley, one of Australia's most advanced engineering and manufacturing companies; 3ME a major clean energy innovator in the mining sector; Boeing, a leading, global aerospace company ; 4 Tel; a developer of innovative transport efficiency solutions; SAFEgroup Automation, a leading control systems engineering service; Robotics Systems, a tech hardware development company; Kongsberg, a major defense development company; Sandvik, an underground mining support products developer; Ausgrid, Australia's major power supplier, Ampcontrol, an energy solutions company; ResTech, an accredited research provider; Banlaw, a flow logistics company; and RPC Technologies, a global leader in glass reinforced composite solutions. These industry partners are vital external stakeholders to NU Teams and as a result, NU Rocketry.

NU Teams also has a multitude of sponsors that provide either funding, discounted services, or both to all the teams within NU Teams. Some notable sponsors include Murrays Consulting Solutions, which has helped NU Rocketry develop and improve upon its systems engineering methods and documentation such as detailed test and evaluation plans, and Newcastle Hi-Tensile Bolt (NHTB), a local manufacturer of industry equipment used throughout the rocket.

System Architecture Overview

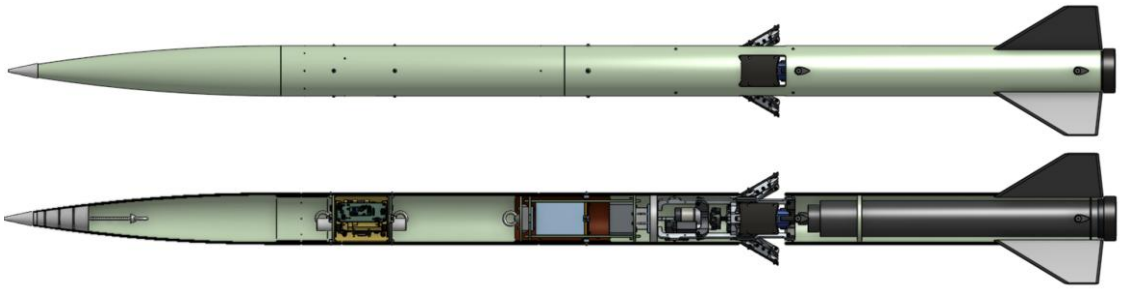

Figure 3 - Rocket Assembly and a cross-section revealing the location of each component and subsystem

The following sections describe both the design and justification of the rocket subsystems in closer detail. The subsystems are broken into the following sections; Structures, Avionics, Recovery Hardware, Payload, Propulsion and Airbrake.

Structural Overview.

Overview

The airframe consists of the forward and aft tubes are constructed using a wet layup method on a 4 inch aluminium mandrel which was used as the team has experience with this method. The mandrel was prepped using a range of sandpaper grits starting with a low grit such as 80 grit and ending on a high grit like 1200 grit, after which 4 layers of partial paste wax are used to polish the surface. This ensures that internal diameter of the body tubes is very smooth and circular making the removal of the cured tube easier and helping the interaction of the sliding coupler joints. The mandrels are prepped this way as it outputs more constant results then using PVA or mylar, even though it requires more effort overall. The tubes have a desired resin to fibre ratio of 1:1.5 which is achieved using heat shrink tape which squeezes out excess resin and helps to remove any voids. This resin ratio results in stronger tubes and lighter tubes as less force is distributed through the weaker resin matrix, allowing the fibres to take more force.

After the tube has cured at ambient temperature for a week, it is then post cured in an industrial oven starting the temperature at 40°C for 30 minutes which is then slowly ramped up to 52°C, held for 64 hrs, and then slowly left to cool back to ambient temperature. The slow ramping of the temperatures is done to prevent embrittlement whilst the post curing process is done to achieve improved mechanical and thermal properties like glass transition temperature.

Nose cone Assembly

1.1. Nose cone

The nosecone used on Wedgetail is 4-inch diameter 5:1 Von Karman nosecone (COTS) with a length of 584 mm. Originally an SRAD nosecone was planned to be made as they are more lightweight and provide more design flexibility however due to team constraints, namely time and manpower, a COTS nosecone was selected. The extra weight of the nosecone proved to be a benefit in the end as an additional 400 g was needed in the nosecone to ensure that the static stability of the rocket was greater than 1.5 even in a worst-case scenario, like the airbrake fully deploying. A Von Karman shape was selected mainly as it has the most favourable specifications at the speeds predicted through OpenRocket simulation, but also as the team has previously used a Von Karman shape. A COTS coupler is epoxied into the nosecone to act as a shoulder since the nosecone separates from the forward tube for main parachute ejection.

1.2. Nose cone Anchor

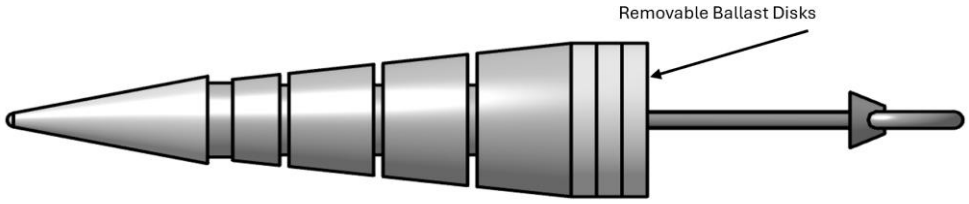

Figure 4 - Nosecone anchor view – Component inserted at the tip of the rocket, detailing the removable Aluminium discs which can be added or removed for mass tuning.

An SRAD anchor is used to mount the COTS nosecone tip with an added threaded rod and eye bolt to act as a mounting point for the main parachute shock cord. This anchor was made from 6061 Aluminium because it is inexpensive and ductile making manufacturing quicker and easier. This anchor was developed to allow for the additional of multiple aluminium disc that can be fixed onto the anchor allowing for apogee tuning in the case of the airbrake being nonfunctional on the launch day. Figure 4 demonstrates the radial slots cut into the anchor, this was done to help provide a second degree of safety, in case the epoxy doesn't bond well with the aluminium, the epoxy will flow into the slots and harden in place mechanically locking the anchor.

Forward Body Tube:

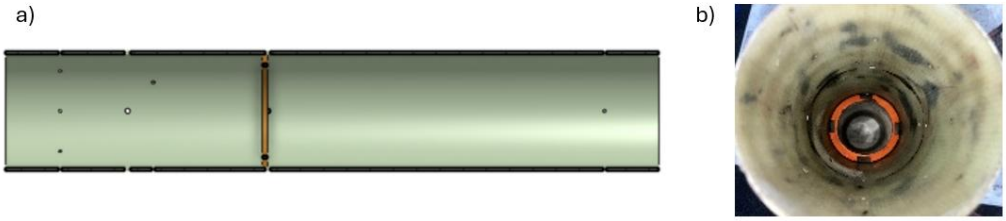

Figure 5 - a) Forward tube cross section - b) AV alignment ring in tube

The forward houses the avionics stack, drogue parachute and interfaces with two sliding couplers with an overall length of 600 mm. The tube is made from two wraps of 595 gsm stitched triaxial e-fiberglass with the main set of unidirectional fibres running longitudinally for improved strength to weight ratio. Previous designs saw 8 layers of lighter 165 gsm 2x2 twill used, however this results in more resin thus making the tubes heavier. It was important to make the tubes as light as possible to ensure that the rocket would have a reasonable amount of overshoot allowing the airbrake enough extra height to utilise its control authority.

The forward tube contains as 3D printed alignment ring demonstrated in Figure 5, which improves ease of use when integrating the avionics. This alignment ring was added as integrating the avionics stack in previous design iterations was a difficult process as aligning all the mounting holes perfectly takes up time on launch day. The tube also contains 3 screw switch holes to allow the avionics to be armed on the pad which also act as static pressure ports. Eight 4-40 nylon shear pins are used to retain the nose cone whilst four shear pins retain the aft tube. Both sets for shear pins have a FOS 1.96, which means one shear pin is redundant in each set. The shear pins were used instead of vent holes as the tubes were determined to not be able to separate due to the pressure difference. This was done to help remove any vibrations from the sliding joints as the shear pins act as fasteners, keeping the joints steady and the removal of large vent holes in the airframe helps with aerodynamic efficiencies by removing the potential drag points.

Aft Body Tube:

The AFT tube contains the aft coupler, payload, airbrake and the motor and is made from the same triaxial fibreglass as the forward tube, with extra reinforcement around the airbrake section. This reinforcement consists of one additional layer of fibreglass with four layers of 200 gsm 2x2 twill carbon fibre which is stepped incrementally to create a smoother transition around the bulged region to help reduce drag and any stress concentrations

It was decided to add additional layers of composites rather than a metal reinforcement system partially due to limitations in manufacturing capability within the required timelines, but it was also decided that the composite system could be lighter than a sufficiently strong aluminium system. Carbon fibre was selected to provide additional reinforcement since this reinforcement is local only around the flap holes and so GPS signals can still be received by the airbrake. Carbon fibre is preferred for this application as buckling is a possible failure mode and the additional stiffness of carbon fibre makes improves strength considerably. The additional layer of triaxial fibreglass was used instead of carbon because this additional reinforcement goes from just below the airbrake flaps to the top of the tube to strengthen the airbrake flap holes and also the upper section of the tube that has several mounting holes that see high stresses from the mass of payload when landing of hard ground.

The flap holes were made by a several step process, while CNC manufacturing is possible, the bulk of the entire aft body tube in combination with composite materials that have health concerns and compressed timelines resulted in using a hand-held router with 3D printed jigs. This process was developed using rapid prototyping and trial and error on scrap body tube sections. The process started with drilling the mounting holes for the airbrake into the Aft body tube, then using 3D printed jig to guide a flush cut composite router bit that was being sprayed with water to make the rough cut of the flap holes. A second jig is used with files to file the flap holes to the final size.

To ensure the safety of flight compression tests were conducted of replica sections that is detailed in the testing section of the airbrake section.

Couplers

Figure 6 - AFT coupler with extra layers of epoxy reduce play in the joint when fully assembled. Note how the fixed side does not have extra epoxy.

A modified COTS coupler was chosen for all sliding joints as currently the team is unable to make tubes with dimensionally accurate outer diameters, thus the effort required to sand down an SRAD tube to size is not warranted. Instead, extra layers of epoxy were added to COTS coupler until correct fitment was achieved as seen in Figure 6 which was used as it is a more time efficient method which allows team members to focus their time on other aspects of the rocket development. The couplers are cut to length such that all sliding and fixed joints are 1 calibre in length. The aft coupler is fixed to the aft tube via four M5 button head fasteners and the payload is in the aft coupler as it allows for the airbrake to be placed behind the C_G such that the airbrakes effect on the C_P doesn't dangerously lower the stability calibre during deployment while also allowing the payload to be easily removed without affecting the rest of the assembly process

Bulkhead

1.3. Avionics Bulkheads

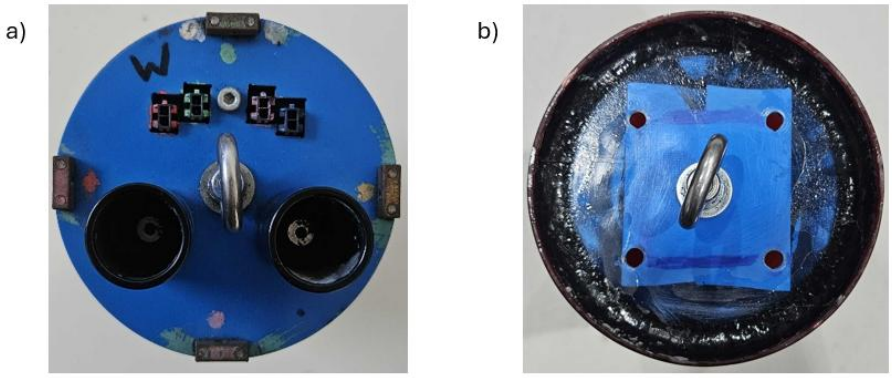

Figure 7 - Bulkheads. Note density of parts allowing optimisation of space and weigh and angled eye bolt on coupler bulkhead.

There are 3 bulkheads used throughout the rocket, two avionics bulkheads and one aft bulkhead. All bulkheads consist of one plate of 3.8 mm G10 plate which was chosen for its high yield strength, low weight and ease of manufacturing using a water jet cutter.

The avionics stack is situated between the avionics bulkheads. These bulkheads serve as the main mounting points for the drogue and main parachutes respectively, along with housing the charge wells for the deployment system. Four steel threaded blocks are attached to the perimeter of the bulkhead, where four M4 fasteners mount the bulkhead to the forward tube. Steel blocks were used instead of aluminium to prevent any cross threading that may occur as it is expected for these to be installed multiple times through its use life. The aft bulkhead is also made from G10 and is permanently fixed to the coupler using an epoxy fillet via the standard procedure. This bulkhead serves as the mounting for the payload

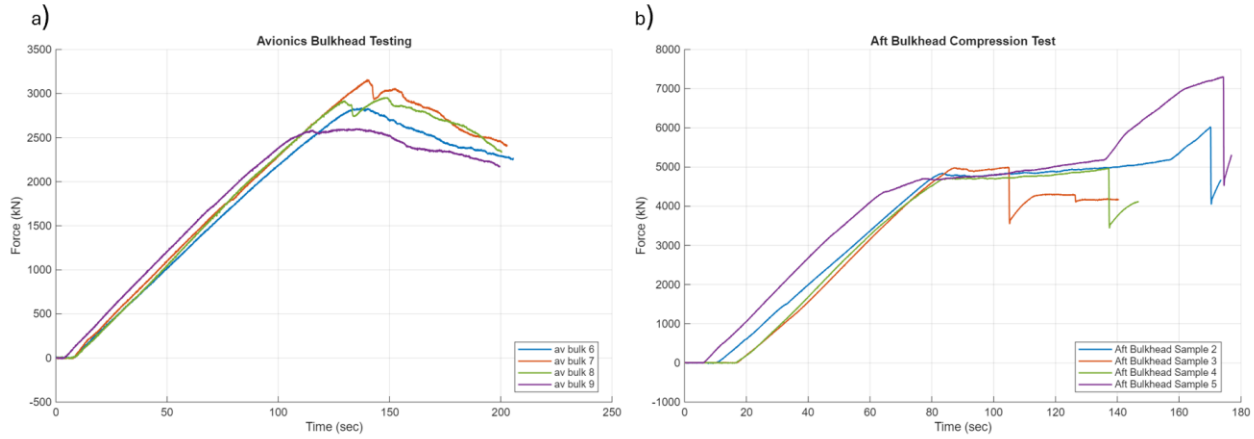

Figure 8 Compression testing data for both avionics and aft bulkheads. Note the lowest yield forces are 2.6kN and 4.7kN for avionics and aft bulkhead respectively

Previous iterations saw the use of a single solid aluminium bulkhead in the 2024 Spaceport America cup however this added too much weight for a 4-inch platform. To ensure required strength is still met, the single layer for 3.8mm G10 was tested seen in Figure 8 which shows that the weakest sample for the avionics and aft bulkheads yielded at 2.6kN and 4.7kN respectively. The maximum expected shock load is 1.6kN for main parachute deployment thus giving a FOS 1.7 for the avionics bulkheads and FOS 2.9 for the aft bulkhead. Even though the bulkheads yield at lower loads, the G10 plates are above the factor of safety margin required in the DTEG. The G10 bulkheads weigh 70 g per bulkhead compared with the 150 g aluminium bulkheads, saving 80 g, whilst simultaneously decreasing the

bulkheads footprint enabling a more compact design. This also greatly decreases the manufacturing complexity as only the one bulkhead plate and the 4 steel blocks are required.

Fins

1. Fin Overview

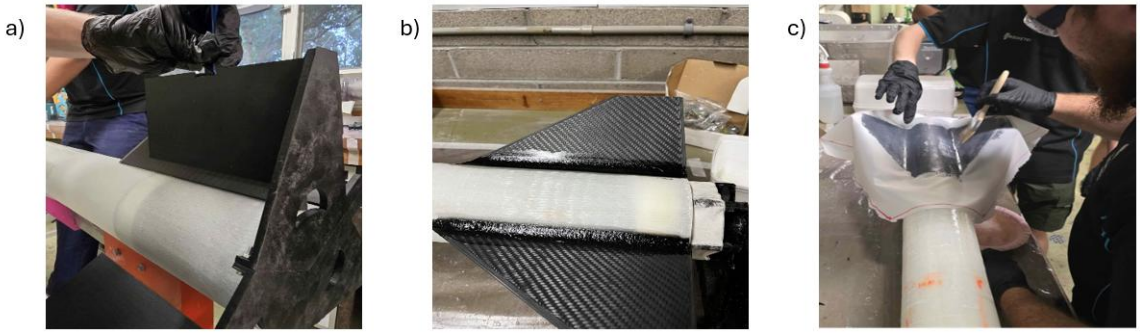

Figure 9 - a) Fig Alignment Jig - b) 20 mm fillets on fins - c) Applying peel ply after tip to tip

Three clipped delta fins made from 5 mm thick carbon fibre plate are attached to the airframe using 20 mm radii fillets reinforced in place with a single layer of 200 gsm carbon fibre as a tip to tip. The clipped delta shape was specifically chosen to help prevent any damage upon landing as the trailing edge is swept forward such that the motor tube hits the ground first. Carbon fibre was selected as it is lightweight and stiff, thus improving stability calibre as less weight is in the aft section which pushes the C_G forward. Three fins were used instead of a four-fin configuration as three fins are lighter and produce less drag overall allowing for a smaller motor to be used. To further reduce surface, drag one layer of peel ply is added during the tip-to-tip layup which soaks up extra resin and is peeled off when cured resulting in a unform surface finish. To help with the lower stability, compared to four fins, the height of each fin is set as 104 mm (one body calibre) above the DTEG requirement. A 30° bevel was machined onto the leading and trailing edges, helping to reduce drag and assist with smooth airflow and therefore increase the predicted apogee.

1.4. Fin Flutter

The NACA TN 4197 Martin method [1] is used to determine fin flutter velocities as it offers a simple way for the team to calculate fin flutter with the resources available. The maximum velocity of velocity of 310 ms^{-1} (1017.06 fts^{-1} occurs at a height of 1144 ft. Assuming the out of plane shear modulus for carbon fibre-epoxy composite is 400,000 psi [2] and a carbon fibre tip-to-tip effectively doubles this shear modulus. At a launch site ambient temperature of 45°C (113°F), results in a flutter velocity of 582.25 ms^{-1} (1910.27 fts^{-1}), this gives a FOS 2.13.

Symbol	*Description*	*Value*	*Units*
c_r	*Root chord length*	*9.055*	*inch*
c_t	*Tip chord length*	*2.756*	*inch*
t	*Fin thickness*	*0.197*	*inch*
b	*Fin semi-span*	*4.094*	*inch*
h	*Altitude (max velocity, AGL)*	*4000*	*ft*
G	*Shear modulus*	*400000*	*Psi*

$$S = \frac{1}{2}(c_r - c_t)b \quad \text{Eq 1}$$

$$AR = \frac{b^2}{S} \quad \text{Eq 2}$$

$$\lambda = \frac{c_t}{c_r} \quad \text{Eq 3}$$

$$T = 59 - 0.00356h \quad \text{Eq 4}$$

$$P = \frac{2116}{114} \times \left(\frac{T + 459.7}{518.6}\right)^{5.256} \quad \text{Eq 5}$$

$$a = \sqrt{1.4 \times 1716.59 \times (T + 460)} \quad \text{Eq 6}$$

$$V_f = a\sqrt{\frac{G}{\frac{1.337AR^3(\lambda + 1)P}{2(AR + 2)\left(\frac{t}{c_r}\right)^3}}} \quad \text{Eq 7}$$

Motor Tube Mounting

1.5. Aft Alignment ring:

The Aft alignment ring is a single 3D-printed 9mm ABS ring bonded via epoxy to a fiberglass inner ring, which was hand sanded down to a 5mm thickness to achieve the required fit. The inner ring makes direct contact with the motor case to provide precise radial alignment for the L-1940x-PS motor case, with fiberglass selected for its superior temperature resistance, ensuring compatibility with the thermal demands of the motor case during operation.

1.6. Motor Tube:

The 3-inch filament-wound fiberglass motor-tube (COTS) has 3.048mm (0.12 inch) wall thickness and was purchased from Wildman Rocketry as the team doesn't have the capabilities to manufacture tubes with the dimensional accuracy required for motor tubes. The motor-tube was chosen for its service temperature 123ºC (255Fº) [3] making it suitable to tolerate the motor case temperature. A failure in the motor mount system could result in catastrophic vehicle loss, so to mitigate this risk, the team conducted extensive structural testing as shown in Figure 11 confirming that a single-ring configuration provides sufficient structural integrity.

1.7. Centring Ring:

A single centring ring was used to centre the motor tube to the aft section which was waterjet was cut from a custom 1.5mm (0.05in) carbon fibre plate. The plate was prepared on a flat non-porous melamine board waxed evenly with a partial paste, creating a smooth waxed surface for 6 layers of 200gsm carbon fibre sheets to be stacked on top of each other and wet laid up (Refer to appendix H layup instructions). In 2024 Spaceport America Cup, the rocket utilised a 4-centering ring motor configuration which research found that the lowest strength configuration presented a 6.75 safety factor to the motors thrust, and a 3.9 safety factor to an extreme landing of 160G's. Results suggested that the centring ring strength has minimal effect on the performance of the assembly therefore, a single ring motor configuration was implemented, as this can reduce the rear weight of the rocket by around 400g, allowing higher rocket flights and increasing the stability of the rocket while still remaining structural integrity.

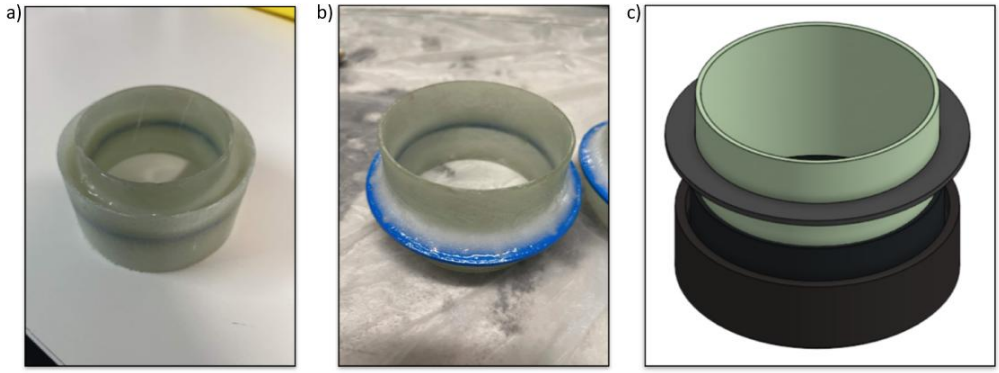

Figure 10 - a) Centering ring around motor tube - b) Section of motor tube assembly - c) Full assembly of motor tube

1.8. Motor Retainer:

The 75-mm flanged motor retainer assembly (COTS) was purchased from Wildman Rocketry and was epoxied onto the exposed end of the motor-tube of the aft section (refer to Appendix H layup instructions) The threaded

interface of the retainer was masked with flash tape, ensuring no epoxy ingresses into the threads. This maintained proper engagement and allowed for reliable installation and removal of the motor retention cap.

1.9. Assembly Testing

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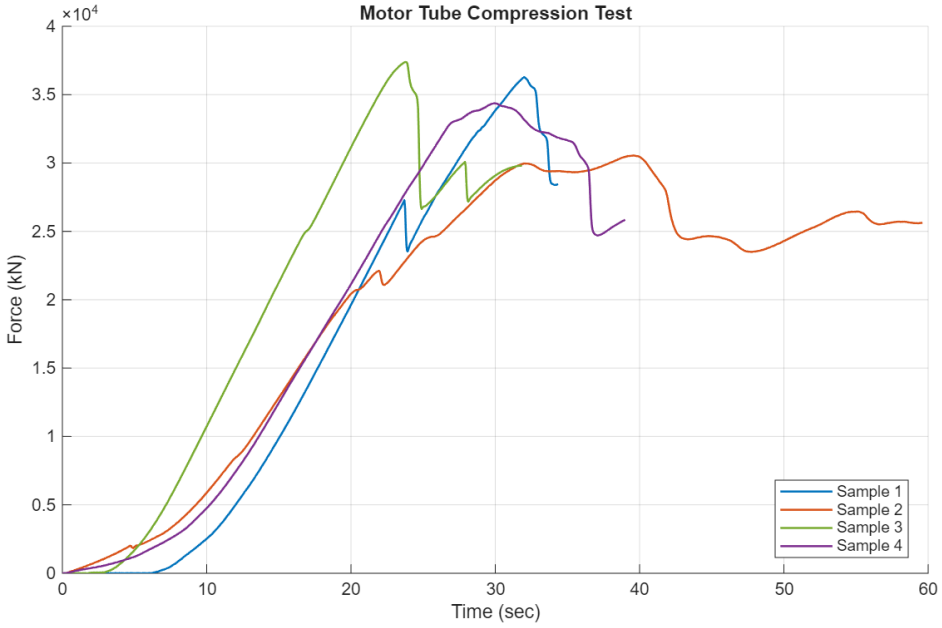


Figure 11 - Graphed results of motor tube compression test

Previous interactions in the 2024 Spaceport America Cup utilised a 4-centering ring motor configuration, however while more rings correlate to a higher maximum force this leads to heavier weight in aft sections. To save weight a single carbon fibre centring ring configuration was implemented. Figure 5 shows each test sample experiences a roughly linear increase in compressive force relative to time until reaching their respective yield strengths; Sample 3 recorded the highest yield at approximately 35,000 N, Sample 4 exhibited the lowest at 25,000 N, and Samples 1 and 2 yielded at 30,000 N and 27,000 N. The average yield force of 29,250 N corresponds to a FoS of 12.78, withstanding the thrust requirement of 2,289 N produced by the L1940X-PS motor [2]. This demonstrates that the motor tube design comfortably withstands the expected compressive loads under operational conditions, this indicates that lighter motor tube designs with stiff materials can be used without compromising the structural integrity of the assembly based on an assumption of 300g during a hard landing with the full weight of the aft section acting vertically on compacted soil, both fillets have achieved adequate bonding, confirming that the adhesive joints meet the structural demands of the design.

A. Propulsion

Propulsion Overview

Wedgetail utilizes an Aerotech L1940X reaching a predicted apogee 10347 feet in 2ms^{-1} average wind. This reaches the target apogee of 10000 feet while also leaving overhead room to account for wind changes, and simulation limitations ensuring the team will score as high as possible.

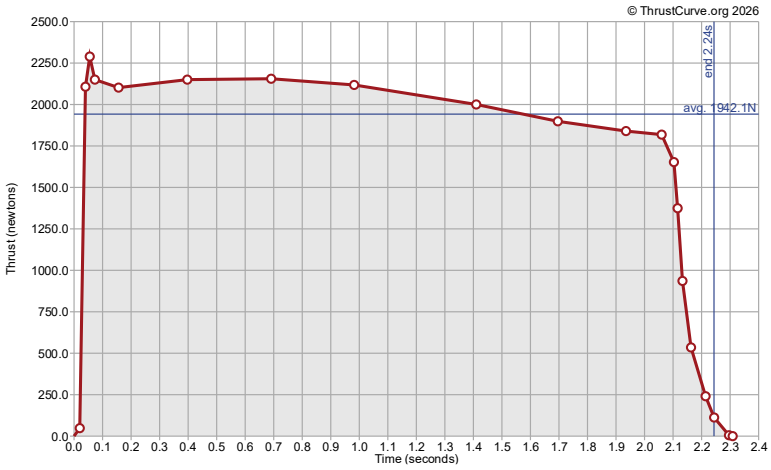


Figure 12 L1940 Thrust Curve

Propulsion Philosophy

The choice in motor was nearly entirely driven by motor availabilities in Australia, as L3 motors are difficult to purchase and no-members hold a current L3 certification. Initially, the rocket was designed around an Aerotech L1420 as NU Rocketry owns multiple of them and test flew the rocket using this motor, however, when purchasing motors for IREC, the team encountered availability issues and had to find a replacement motor which was originally selected to be a Cesaroni L2375 as the team own a four grain case to use this motor. This motor worked extremely well when the airbrake was still being deployed, but the decision to disable the airbrake meant the rocket was severely overshooting the 10000 feet target. This led to the switch to the L1490 as it flew to a predicted apogee of 10347 feet under 2ms^{-1} average wind, which left a large amount of overhead to account for wind changes, and for simulation inaccuracies.

Avionics

1. Flight computers

Wedgetail's avionics bay is constructed to house three COTS avionics system. Two flight computers, a Blue Raven and a Telemetrum v4.0, are used to record flight data and control deployment of energetic charges, and a TeleGPS v4.0. These computers were selected for a combination of reasons including team experience with the Blue Raven, frequency assignment and optimising the avionics stack for space and mass while not sacrificing redundancy on either GPS or flight computer systems.

The Blue Raven is equipped with four pyro-technic channels. The channels are Apogee, Main, 3rd and 4th. Apogee is the primary channel for the deployment of the 24-inch drogue parachute with the 3rd being its backup channel. The Main and 4th channel are used to deploy the 50-inch main parachute, with the 4th channel being the backup. The Blue Raven was selected due to the team's prior experience with the system and its ability to control both primary and backup chargers for both main and drogue.

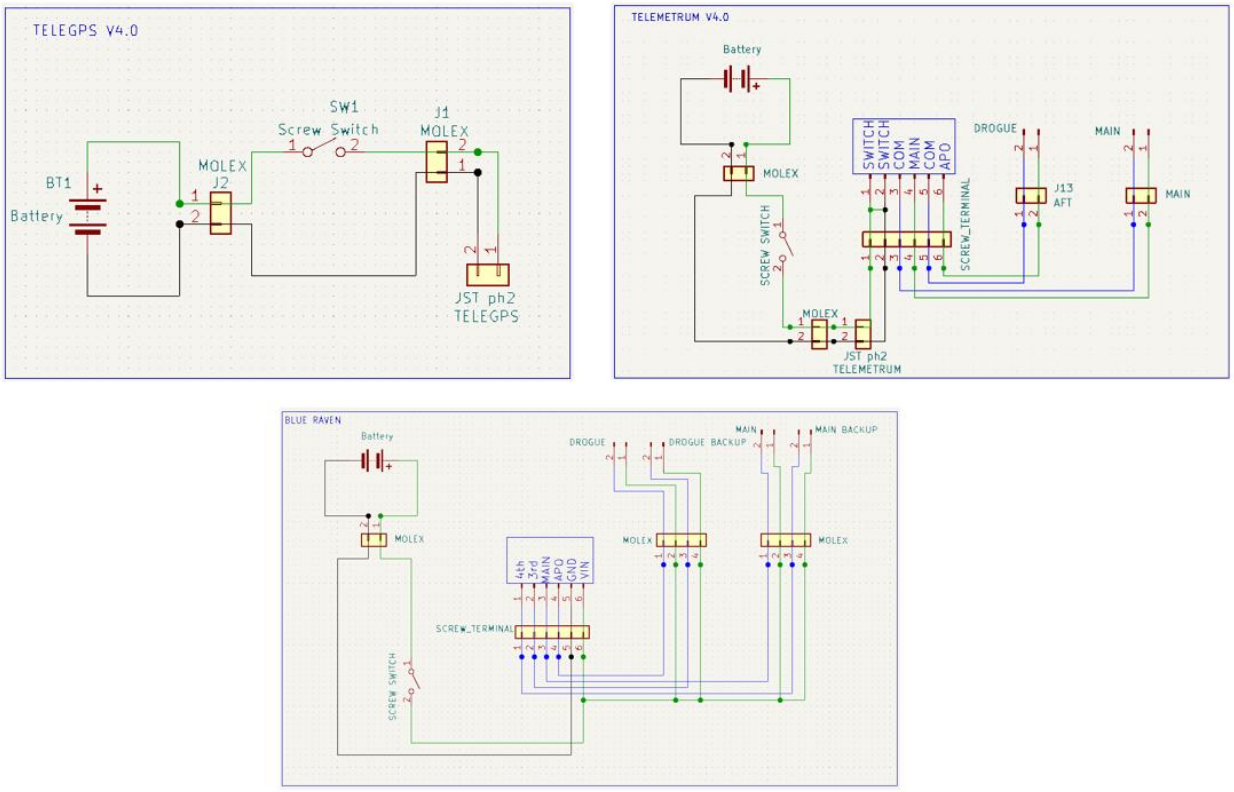


Figure 13 - Flight computer wiring diagrams

The Telemetrum has two pyro-technic channels, Main and Apogee. Apogee is to deploy the drogue parachute, and Main channel is for the deployment of the main parachute. The Telemetrum allowed the team to simplify the avionics design by integrating both flight computer and GPS functionality and operating on the 70 cm ham band simplified the frequency assignments. While the team didn't have experience with this computer prior to the test flight the redundancy of the Blue Raven combined with the well tested COTS nature of this device allowed the team to make these optimisations.

The TeleGPS was selected as the redundant GPS as it also operates on 70 cm easing the frequency assignment issues. Currently one member of the team has an Australian Standard Amateur Radio license with three others currently working through this process, Australia has a reciprocal agreement with the USA allowing operators to act under the restrictions of the General license without any other requirements apart from appending W5/ to the callsign.

A Fluctus and Raven 4 were also purchased as backups for the Blue Raven and Telemetrum. Raven 4 was selected as backup due to the same pinout and functionality as the Blue Raven and the Fluctus selected for its history within the team while the integrated GPS ensure redundancy for both GPS and flight computers. Like for like backups were not an option due to concerns with every avionics system, having dissimilar systems means if one of these concerns is validated the backup will be available.

Each device is mounted onto separate power distribution boards (PDBs), powered by independent 18650 lithium-ion batteries to achieve complete isolation. Each PDB is equipped with a USB-C connector and a dedicated lithium-ion charging IC for the purpose of having easily accessible battery charging, eliminating the need to disassemble the avionics bay when topping up the cells. An external dedicated charger with storage voltage capabilities is used for primary charging. The power switching method for the PDBs are custom screw switches located on the top half of the boards, these screw switches can be accessed from the outside of Wedgetail and are use the same tool as most of the rockets assembly for switching.

Battery selection

Li-Ion 18650s were selected for all computers as prior testing showed they have more than 24 hours of runtime for most computers. The team conducted testing of the cells purchased to determine the true capacity of the cells and found an average of 2330 mAh with a variation of 100 mAh. This testing was conducted on a battery charger with capacity counting discharging from 4.2 V to 3.4 V, this testing stopped above the minimum cell voltage to prevent damage to the cell and provided a conservative capacity reading.

Testing the flight computers power consumption was done with a benchtop power supply and proved the manufactures estimates correct. From these tests an estimated runtime was determined for each computer. The requirement of 24 hours of runtime was established, and all cells meet this for all computers. A live test was conducted whereby the computers were powered from the cells for approx. 8 hours to validate the predicted runtimes. This test ensures that the cells can be tested to expected power on times while not risking damage to the cells from undercharging as a max runtime test would do.

While using the same cells for all computers is less optimal for weight, size, and introduces unprotected cells where protected cells could potentially be used, it allows for more redundancy, ease of use and removes the risk of failure caused by connecting the incorrect cell to a flight computer and having it trigger protections during a deployment event, browning out the flight computer.

Avionics Stack Design

The avionics stack is enclosed by two fiberglass bulkheads mounted using steel blocks directly to the forward tube with ABS 3d printed mount for the PDBs. On the drogue side bulkhead, a 3D printed battery housing is attached on top of the power distribution board mount with a bolted in lid providing access to the three 18650 lithium-ion cells. Each cell has a nickel strip spot welded on with two 18 AWG wires soldered to them. These wires pass through the lid of the battery mount and have a 1x2 Molex micro-fit connector to plug into a PDB. Each battery is interchangeable between all PDBs removing the risk of connecting the wrong battery. This setup was designed to minimise the mass and size of the avionics while ensuring needed thermal characteristics and easy maintenance.

All wiring used within the avionics structure utilizes crimped Molex micro-fit or JST connectors which are assigned and painted a colour to avoid confusion when assembling. Each harness has a different colour at each end allowing each socket and plug pair to be painted the same. Located in the assembly procedures for Wedgetail is a master list for the connections outlining the purpose of the connection along with its assigned colour however the system is designed to be assembled purely based on colour matching.

Antenna mounts were designed into the battery housing for all antennas potentially installed. While running the antennas radiating element parallel to batteries is not optimal, the avionics stack design does not have a better place for them and the spacing designed into the battery mount has proven to be sufficient for a stable telemetry lock throughout the entire test flight including when recovering the system.

A shroud was implemented to limit the amount of black powder residue from getting in contact with flight computers due to its corrosive properties. This ABS 3d printed part was designed for easy installation and removal and implements a clip style connection onto existing bulkhead bolts again reducing mass and complexity.

A key criterion in designing the avionics stack was ensuring modularity of power distribution boards and flight computers. One factor in achieving this is the standard size of 102 mm by 45 mm. This allows for any power distribution board to be replaced with another and was determined to be the minimum size to mount all our flight computers with clearance for wiring and strain relief.

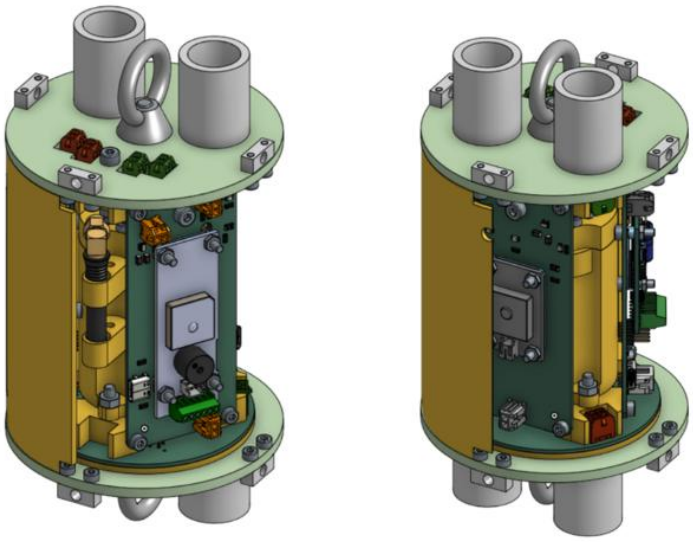


Figure 14 - Finalised avionics stack

Ejection charges and settings

Wedgetail uses 3.0 g primary charges and 3.5 g backups for both the main and drogue side. Initial testing on a mock-up rocket with correct volumes for the sections showed that 2.4 g charges were sufficient. Plans for testing on the complete rocket were disrupted by Australian laws and launch opportunities. We were able to run a single test, and it was decided that testing a larger than previously tested charge of 3.0 g in the nose section would be the lowest risk test as it has the most shear pins and a slightly higher volume, a charge that proves sufficient for this side will thus be sufficient for the aft. With the panic fire mode setup on both flight computers and the recovery system being rated for that the main means that not testing the drogue side is the lowest risk given the limits in place. This charge was safe and deployed the nose approx. 2/3rd of the length of the shock cord. This test along with the previous testing and calculations that show the main and drogue side need very similar sized charges allowed the team to confidently fly wedgetail with 3.0 g and 3.5 g charges for both parachutes.

Settings for deployment are similar on all computers, the drogue is set for each manufacturer's recommended apogee detection, and the main is configured to deploy while descending through 400 m or if the rocket reaches 40 ms^{-1} descent rate. The Raven has its backup drogue charge set with a barometric trigger including Mach lockout and 1 s delay. The backup main charge is configured 1 s after reaching 400 m or 50 ms^{-1} descent rate. These settings proved safe and effective during the test flight with the raven firing the primary charge successfully deploying the drogue and the primary charge from the Telemetrum deploying the main.

Recovery Hardware

1. Parachute selection

Wedgetail utilises a dual deployment recovery system with a separate drogue and main parachute. The main will be a recon recovery 50 inch parachute with a CD of 1.55 and a Rocketman 24 inch 0.97 CD drogue. Our test flight flew on a repaired Rocketman 48 inch parachute, seen in Figure 15 as the team has encountered significant issues with shipping parachutes and our intended flight parachutes have yet to arrive, these will serve as a backup for competition in case the arranged pickup of parachutes encounters an issue. A major factor in the selection of these chutes was the oscillations and instability seen in recovery systems of similar size using high CD, small, light parachutes. Selecting lower CD and larger chutes traded space and weight for stability and thus safety of recovery.

Figure 15 Repaired Main Parachute

These parachutes are protected by and contained in a Nomex deployment bag with the main bag including a pilot chute sewn in, this is then wrapped in a Nomex blanket in a burrito style with the closed end faced towards the ejection charges to further reduce the risk of charring of the chutes. This proved quite effective in test flights showing no signs of ejection gasses meeting the parachutes. The pilot chute integrated with the main bag also proved effective in reducing the opening loads of the main

Load calculations

Parachute shock load calculations were done to establish needed strengths of all recovery load routes. These calculations were done by taking the max expected velocity for deployment in a worst-case scenario and working out the drag force provided by each parachute in this condition. This worst-case velocity for the drogue assumed launching into significantly more than 11 ms^{-1} wind, looking at horizontal velocity for apogee and adding 20 ms^{-1} of vertical velocity to account for flight computers being late to trigger and requiring the backup charge. The worst case for the main was this horizontal velocity with a 50 ms^{-1} vertical velocity as this is the panic fire velocity set in the Blue Raven's configuration. This resulted in a force of 1.6kN for the main and 800 N for the drogue.

Harnesses and hardware

All eye nuts in the recovery system are M6 forged stainless. The nose anchor is an eye nut attached via a threaded rod to an aluminium weight and anchor epoxied into the front of the nosecone, all other eye nuts are bolted with a M6 socket head cap screw and washer through 3.5 mm G10 plate. The main side attaches to eye nuts via M6 jaw-jaw combination swivel and shackle, and to the main via a M6 quick link and the swivel included on the parachutes. The drogue size has M4 quick links and 500 lb rated Wildman swivels.

The shock cord on the main side is 40 ft of 2200 lb rated braided Kevlar from Wildman, attached in the centre to the nosecone, the shorter end is connected to the forward tube and the longer end to the main. The drogue has 40 ft of 900 lb rated braided Kevlar again from Wildman, attached to the forward section in the centre, the booster section on the short side and the drogue on the longer side. This rigging was selected to reduce the chance of tangling under both sets of parachutes and ideally reduce the shock loading on the heaviest component being the booster section. This configuration was flown during the test flight and post recovery inspection revealed no tangling, and only minor damage to the stitching of the centre loop of the drogue side.

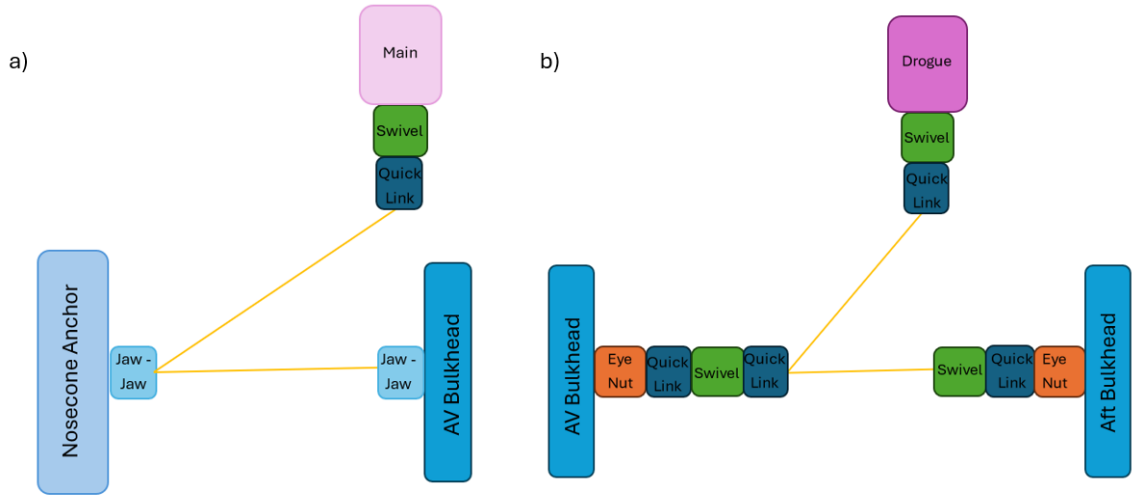


Figure 16 - a) Main recovery harnessing - b) Drogue recovery harnessing

Payload Architecture

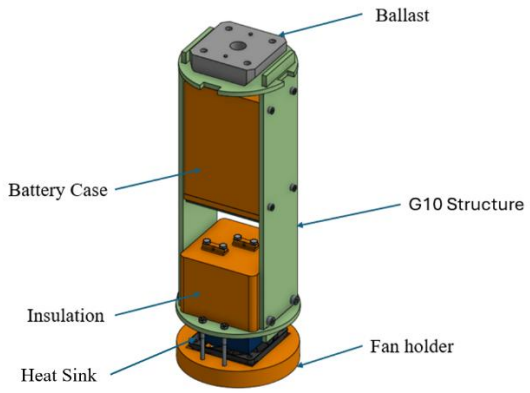


Figure 17 – Finalised Payload

2. System Overview

The payload is a technology demonstration that has been designed to demonstrate proof of concept for small scale thermoelectric control systems inside sounding rockets. This has applications for the rapid delivery of temperature sensitive cargo by means of rocketry to remote locations such as space stations. The demonstration consists of a system which regulates the temperature of a copper cube using the Peltier effect, this is accomplished using a Peltier module connected to a Full H-Bridge Motor Driver allowing both heating and cooling functions as well as power variation. The technology being demonstrated is the ability to control this temperature by means of the above method.

In detail, the payload system operates as follows; prior to integration, a Molex plug is inserted into the payload PCB which allows the batteries to supply power to the system idle electronics. The onboard microcontroller keeps the unnecessary systems in an idle state until it receives an activation signal on the pad. This activation system consists of a constant active high logic state on a GPIO pin which is supplied by a screw switch, turned during the rockets on-

pad arming procedure. During this idle time, the payload will perform two basic functions; reading and logging temperatures at different points within itself as well as measuring and logging battery voltage. Once the screw switch is activated, the microcontroller attempts to regulate the temperature of the copper block. This process uses a PID controller, fed by the output of a thermistor temperature sensor which sits inside the copper block to generate a PWM duty cycle which controls both the direction of heat pumped to/from the copper block and the current delivered (power). To improve the smoothness of the controller and PWM signal, a rolling digital average is applied over 50 samples to remove ADC noise.

Payload Airframe/Hardware

The payload's airframe is constructed of two 234.8mm (9.2in) long, 4mm (0.16in) wide G10 plates connected through several 10mm epoxy resin fillets (according to standard procedures) to upper and lower G10 baseplates. G10 was chosen due to its high yield strength, low weight and ease of manufacturing as well as the teams extensive expertise testing and working with the material. The payload (Figure 17) has a total length of approx. 285mm and has a major diameter of 3.9 Inch (99mm), the payload weighs 2070g and has a removable ballast weight (in the spacer) of 373g. The payload is held (via the spacer) to the aft bulkhead using four M4 socket head cap screw bolts. The goal of the airframe was to have a nominal length less than or equal to an equivalent 3U CubeSat length and be able to survive an estimated deceleration of 100g. The G10 plates were waterjet cut to allow for prototyping capability, for instance the lower G10 plate required precise 2D geometry to make room for the Peltier device. The payload hardware consisted of several student designed components manufactured using 3D printing and machined copper. The battery container consists of two 3D printed parts which have M3 brass heat set inserts inserted into them, these parts are held to the airframe using eight M3 bolts. Two COTS 2s battery packs can be held in this space. Additionally, the lid of the container has mounting for the PCB which consists of 4 M2 heat set inserts, these parts were printed from ABS, this is due to its high impact resistance and high glass transition temperature.

The copper container consists of a cube with several drilled holes which are representative of the space available for a medical product. It also consists of four M4 tapped holes, which restricts its vertical movement along the G10 plates. Previously this block was manufactured out of steel while testing was taking place due to the ready availability and low cost of manufacturing with this material. To ensure compliance with Tripoli safety standards, the decision to switch materials was made and settled on copper despite the significant cost increase due to the following material properties. Copper possesses high thermal density in comparison to other available materials (aluminium, copper) and has low thermal resistance. As a ductile metal, copper is explicitly allowed by the DTEG. The container itself is contained in 3D printed insulation which was sliced using an infill pattern which generates air gaps, theoretically increasing its thermal resistance.

Payload Electronics

The PCB is designed for compatibility with the payload structure, fitting neatly into the rectangular center tower. One side contains all daughter boards, plugs and other tall components while the other is populated only with compact components such as SMD (surface mount) resistors and capacitors. This allows the PCB to mount directly to the battery housing lid with minimal spacing.

Figure 18 shows the KiCad model of the latest PCB design, using a mixture of off the shelf daughter boards used for simplicity and custom circuitry designed to support IC chips where daughter boards are less available or where more specific needs are applicable.

The PCB is broken into 3 main sections; Power, Microcontroller and peripherals. The power segment is supplied by dual RadioMaster 7.4V 5000mAh 2s Li-ion Batteries wired in series to increase battery life. (up to 10 000mAh at 7.4V). Each battery is protected by a 3A fuse to prevent damage to batteries or the board from short circuits or battery voltage imbalances. The rest of the power section consists of a 5V regulator for supplying power to the microcontroller and a capacitor bank to reduce ripple and voltage spikes along the 5V rail. The 5V regulator used is a commercial daughter board chosen for simplicity and capacity for supplying up to 500mA current.

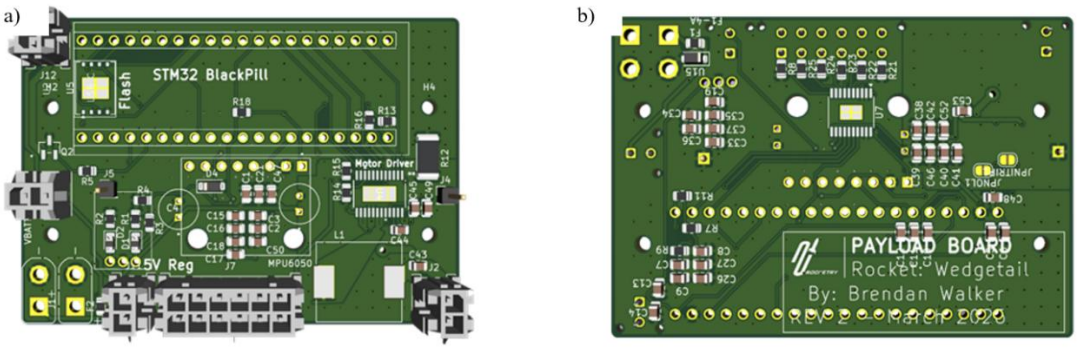


Figure 18 - a) Payload PCB Front - b) Payload PCB Back

The peripherals section contains a number of external devices which serve as interfaces between the microcontroller and the real world. Due to the low resolution of the microcontrollers inbuilt ADC (12-bit) an external 16-bit ADC was selected and used for converting analog signals to a discrete value. The selected ADC communicates with the microcontroller using the SPI protocol and provides high accuracy, low noise voltage measurements of critical systems. This ADC is also equipped with configurable gain, allowing for current measurements of the Peltier module using a small (10 mOhm) resistor and a 32x digital gain stage. Other signals measured is the systems battery voltage, necessary for ensuring batteries do not over discharge and become damage and measurement of the voltage across thermistors. The resistance property of thermistors changes with temperature, using a voltage divider, the ADC and a Steinhart Hart model (for accurately modelling the logarithmic relationship between resistance and temperature), the temperature of the thermistor and therefore its environment can be calculated by the microcontroller.

To record data from the payload, a flash storage IC was selected, this allows the safe storage of data after the system is switched off, without the need for an SD card which is much more susceptible to data loss caused by forces of flight.

An external motor driver IC was selected for controlling the Peltier module based on several factors. Firstly a Full-H bridge driver was required as in addition to controlling the power delivery to the module, control over the direction of the current was required in order to allow the system to both heat and cool. An IC was selected over a conventional breakout board due to the huge height of most commercial motor driver boards which would struggle to fit inside the space allocated for the PCB, since the system operates at 7.4V the large heatsink is not necessary. Additionally, the cost of the IC and supporting circuitry is less expensive than purchasing breakout boards.

An accelerometer breakout board is included in the design with the intention of communicating when flight events (take off, chute deployment, landing) occur and allow temperature data to be categorised by phase of flight.

An external fan is used to regulate temperature of a heat sink attached to the external side of the Peltier module and is switched on and of using a BJT circuit, providing basic control.

The selected Peltier module is a 40x40mm square, rated up to 13V and predicted to provide around 10W of heat pumping when running at 1.5A [4]. This module provides a balance between power consumption and heat pumping at the selected system voltage (7.4V) while fitting neatly above the steal cube and beneath a heatsink.

Payload Risk Assessment

The payload actively performs temperature changes on both its surrounding environment and internal components, this has the potential to create some risk to both people and the rocket body. Additionally, some other standard risks associated with batteries are present. A full risk assessment table is shown in the Appendix (Table 1: Payload Risk Assessment).

Airbrake System

1. System Overview

An airbrake subsystem was developed to enable altitude control during the coast phase of flight by actively increasing aerodynamic drag to minimize apogee overshoot. The primary objective of the system was to safely and reliably target an apogee of 10,000 feet for the 2026 IREC 10K COTS category. The system was ultimately not deployed at IREC, as insufficient flight testing was completed to verify system performance with the confidence required for a competition flight. The airbrake assembly is located in the aft section of the rocket after of the CG, positioned just above the motor and below the payload as seen in Figure 3.

Figure 19 - Airbrake subassembly with flaps mounted and extended

1. PCB overview

The airbrake system is controlled via a dedicated Printed Circuit Board (PCB), as shown in Figure 20 - Airbrake PCB mounted in place on the Airbrake sub-assembly. Its primary function is to reliably sense the rocket's state and control the deployment of the airbrake flaps while operating under the dynamic conditions experienced during flight.

The PCB is a custom two-layer design developed in KiCad and manufactured and assembled by JLCPCB. A key feature of this design is the integration of the STM32F411 microcontroller directly onto the board. Embedding the microcontroller reduces overall board size, particularly height.

State estimation is achieved through an assortment of onboard sensors. An IMU provides acceleration and angular rate information, a barometer measures pressure, a MAX-M10S GPS module supplies position and velocity data, and a magnetometer supports axis referencing. Data from these sensors are fused using a Kalman filter, improving the accuracy and robustness of the estimated position, orientation, and drag coefficient used by the airbrake control system.

Additional onboard components include a 512 MB flash memory chip for logging sensor data and filter states, an RGB LED for status indication, and a motor driver for actuator control. The system is powered by one 3s 18650 LI-Ion pack supplying 11.1 V with 2000 mAh.

Figure 20 - Airbrake PCB mounted in place on the Airbrake sub-assembly

Firmware overview

The flight software was designed to prioritise safety, reliability, and deterministic operation, whilst ensuring the logging of comprehensive flight data. The secondary objective is the precise deployment of the airbrake system to achieve target apogee.

The system is implemented using FreeRTOS, which was selected for its scheduling, low memory overhead, and suitability for safety critical embedded applications. FreeRTOS enables semi-concurrent execution of software subsystems through prioritised tasks, ensuring that sensor acquisition, state estimation, control actuation and data logging functions without timing conflicts.

The software architecture consists of tasks organised by priority according to their specific timing needs. The sensor acquisition task polls the IMU, barometer, and magnetometer at fixed rates for each sensor, while receiving data from the GPS when available, maintaining current measurements across all sensors. Sensor data is written to a shared buffer, protected by mutexes, and is utilised by the state estimation task, which executes the extended Kalman filter (EKF) using the latest available sensor data. The resulting state vector is then published to a secondary structure accessed by both the control task, which computes actuator commands based on the estimated state; and the data logging task, which writes to the onboard flash memory. Inter-task communication is handled through FreeRTOS queues and shared memory, decoupling task execution while ensuring safe access to shared data structures. This prioritisation enables the system to maintain safe operation under high computational load while maximising time given to tasks in need.

Fail safe behaviour is embedded throughout the software architecture to ensure compliance with DTEG requirements. The system also stops actuation just before apogee to reduce the risk of causing tangles or otherwise interfering with the safe recovery.

Control System

The airbrake takes all the sensor data created and feeds them asynchronously into an extended Kalman filter for calculating both inertial navigation data and current drag coefficients for airbrake control. This is fed into an MPC controller that estimates the required C_D to achieve the apogee target. This targeted C_D is achieved with nested PIDs controlling the flap deployment with feedback from the EKFs C_D prediction. This system achieved good results in initial testing for both accuracy and disturbance rejection however integration and flight testing was limited and produced no results to confirm this setup.

This architecture was selected to be the most reliable system when considering the lack of validated CFD to provide data for a more deterministic and preplanned control system. Unlike a system that maps planned trajectories for achieving apogee this system dynamically calculates the current trajectory, increasing processing requirements of the embedded system but reducing the concerns with preflight data creation.

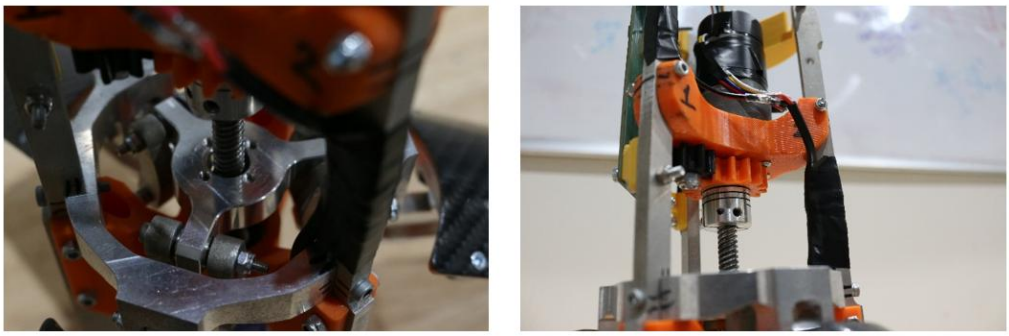


Figure 21 - Airbrake motor and lead screw. Note all flaps attaching to the mechanism identically

Mechanical design

The mechanical design of this system has been iterated on for several years by several team members. The core of the system is a brushed DC motor with encoder that turns a lead screw, this rotation is converted by a lever mechanism into the flaps extending away slightly and rotating up into the airflow. This ensures that the system is always in a safe state as there is no mechanical way to create a moment on the vehicle. During initial design phase multiple architectures were considered including sliding flaps, direct rotational controlled flaps and the selected leadscrew design. While airbrakes that slide out linearly into the airflow are more compact designs, they often bind up when airflow on the flaps create a torque and bind up the sliding mechanism. Direct rotation of the flaps was also considered and rejected due to the risk of uneven deployment of the flaps resulting in a torque and requiring complex safety mechanisms.

The flaps themselves are an ABS 3D print with a single layer of carbon fibre laminated to the ABS and this is mounted to a machined aluminium mount that extends and bolts in all four corners to reduce the likelihood of fin flutter happening to these flaps and creating a significant flight safety issue. This is connected via aluminium linkages to the central arm mount ring.

The main structure consists of 3 legs that are locked together by the central aluminium ring and the ABS 3D printed battery and motor mounts. The legs have 4 threaded holes each for mounting to the aft body tube, the position of the mounting holes shave been selected to help transfer axial loads across the columns of the flap holes as well as resisting bending. The mounting holes also prevent the columns from buckling outwards while the internal structure resists inward buckling.

Due to the team's lack of validated CFD the worst case for stability was assumed, meaning the CP would move entirely to the location of the flaps resulting in a stability of approx. 2.3 cal (after motor burnout).

Testing

1.1. Function Testing

The Airbrake control system requires a large data set of flight variables in order to characterize and efficiently employ the Extended Kalman Filter. This requirement has meant test launches, both at competition scale and on smaller test bed platforms, have become even more valuable for the team than ever. Initially, the Airbrake PCB was launched on a smaller platform to 1000 ft, to characterize and test onboard sensors, then re-launched twice at a high-power launch weekend to 2000 ft then 3000 ft. These tests ultimately resulted in zero data being saved however improvements were made post each flight solving major issues with the codebase. While these launches were not successful in generating data, they were successful in improving the state of the airbrake system.

The remaining high power launch window was then used for a full-scale test of the entire platform including the airbrake system which logged vital flight data, however, the system did not actuate resulting in the inability to identify the drag characteristics and the resulting effect on the rocket's flight path. Overall, the system test was successful as it allowed for a large amount of flight data to be gathered which can be used in the further development of the airbrake system, but the inability to deploy and the lack of available testing windows has hindered the use of the airbrake system.

For IREC 2026, the decision has been made to not deploy the airbrake system due to the unknown effects it could have on the flight of the rocket. This decision comes from a design philosophy, flying an untested system is not

best practice and while the system was designed to be safe in the worse known case the uncertainty of the entire system means there are several unknowns for the real-life performance which may have any number of impacts. This has resulted in multiple changes being required to wedgetail before competition, however these minor modifications were deemed less risky than deploying an untested airbrake.

Development of this system will continue with more test flights being planned as this airbrake lends great flexibility and control to the design process of all future competition rockets.

1.2 Mechanical Testing

As the holes in the airframe required to facilitate to flaps are quite large, an significant effort is required to create confidence in the structure integrity of the compromised section of body tube. This is was achieved by compression tests of an replica section of the airbrake and surrounding body tube.

Physical tests are preferred by NU Rocketry at this stage in the team's development as the skills to do FEA is somewhat limited and to compound this issue FEA of composite materials is difficult to get accurate simulations and this uncertainty results in a low confidence of any FEA results for this application and indeed this view can be applied to most structural components of the rocket as physical testing is greatly preferred over FEA.

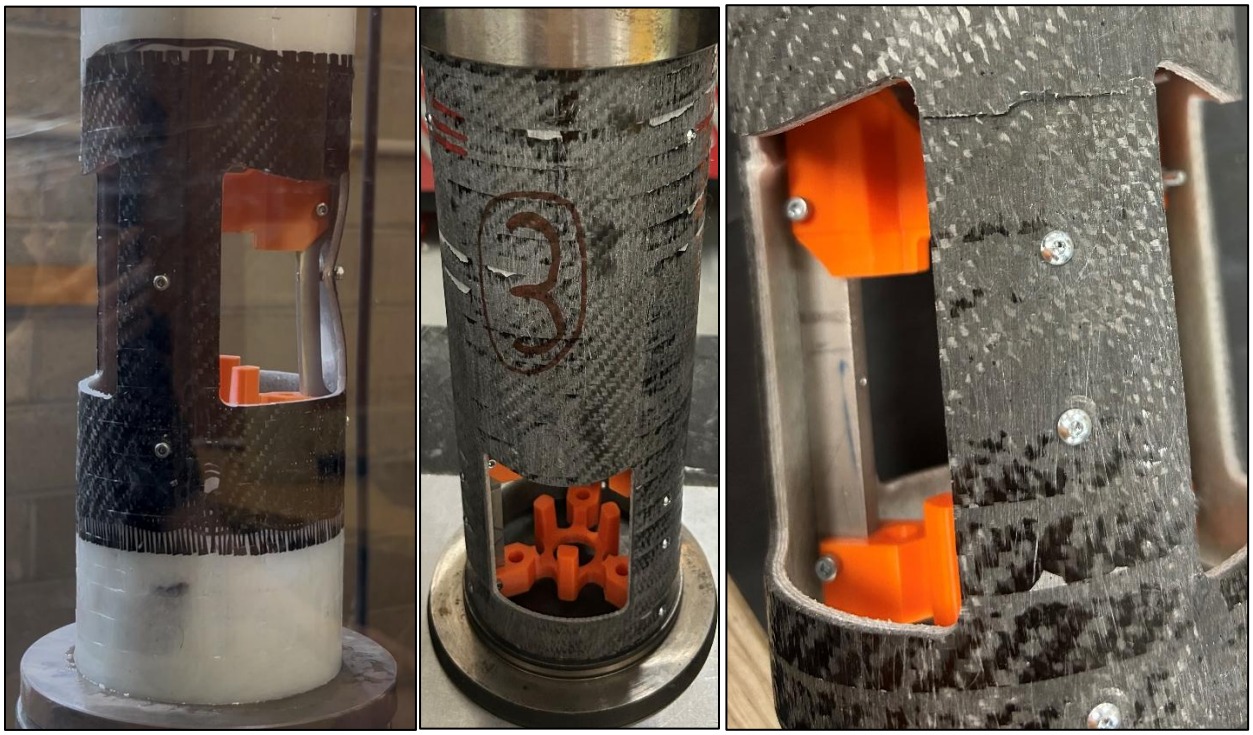


Figure 22. Left; Initial test ~30 kN. Centre; Updated design 3rd test ~53 kN. Right; Updated design 2nd test ~51 kN.

The initial test was conducted and failed in a Euler Buckling inwards at 29.7 kN compressive load, this is significantly lower than the standard body tube compressive strength and so several design changes were made. Firstly, the gap between the structures legs and the body tube in that area was filled in, secondly the bolt in the middle of the column was changed to two bolts that are placed equally apart in the column area, this was done to avoid weakening the buckling stress concentration point in the middle of the column.

To confirm the structural integrity of this design 3 additional tests were done. These tests resulted in compressive failure in all samples typically in 2 of the 3 columns, this can be seen in the Right image of Figure 22. The first sample failed at 52.3 kN, the second sample failed at 50.6 kN, the third sample failed at 52.9 kN, thus giving an average of 51.9 kN with acceptable deviation. This value is above the strength of the standard body tube at 47 kN which provides acceptable confidence of the integrity of this design. Further confidence was gained by the successful test flight of Wedgetail.

Mission Concept of Operations

The following describes the phases of nominal operation across the IREC 2026 flight in the form of a concept of operations (CONOPS). The graphic CONOPS is to be used as a high-level overview of the nominal phases of flight whereas the written CONOPS is to be used as a more detailed description of each phase of nominal flight. Each phase is labelled with a time after liftoff, these times are predicted times based on Open Rocket simulation and may vary slightly across the nominal flight.

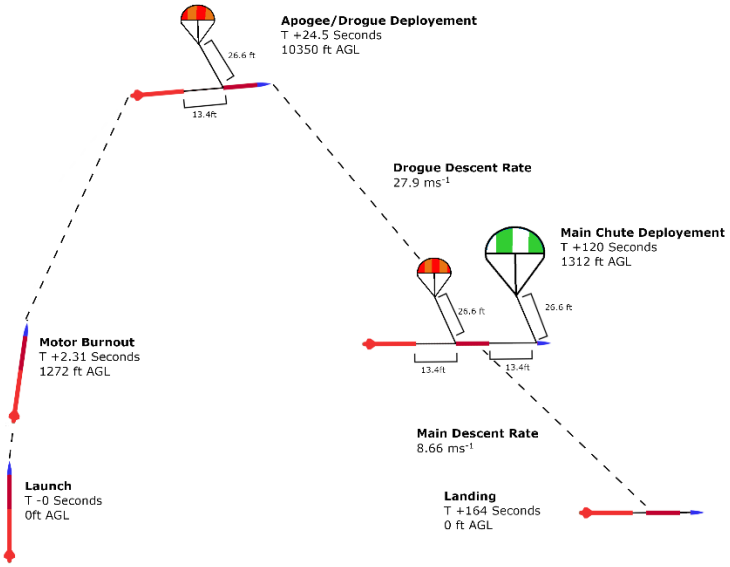


Figure 23 CONOPS Visualisation

B. Pre-Launch

Following a successful flight safety review, payload inspection, and an LCO inspection in accordance with SA Cup IREC Range Standard Operating Procedures 2024 [4] the rocket is loaded onto the launch rail and erected into the upright position. The Tele-GPS and the airbrake are then turned on, allowing time for both to attain a GPS lock. Following the Tele-GPS gaining a lock, confirmation from IREC personnel is gained and all non-essential personnel are evacuated from the pad allowing for the arming of both flight computers. Confirmation is then gained from the Range Manager to install ignitors following which the rocket is in a launch ready state and all team members are to evacuate the pad.

Launch

Proceeding the launch readiness poll and after the final countdown is complete, the FIRE signal is sent across the firing line to the ignitor. Once this FIRE signal has reached the ignitor at L +0 seconds the motor ignition process begins, and the rocket achieves lift off soon after.

Motor Burnout

At T +2.31 seconds, motor burnout is reached and the L1940 motor is no longer producing thrust coasting the rocket towards apogee

Apogee/Drogue Deployment

At T +24.5 seconds, the vertical velocity of the rocket reaches 0 ms^{-1} at which both flight computers trigger the apogee event, igniting the 3.0 g primary drogue charge, separating the four shear pins connecting the aft tube and forward tube together. The nominally red and orange drogue parachute is then released from forward tube. At T +25.5, the apogee backup event is triggered by the Raven, igniting the 3.5g backup drogue charge to ensure deployment. Following full deployment of the drogue parachute, the rocket should be descending at 27.9 ms^{-1}.

C. Main Chute Deployment

At T +120 seconds, following the point the rocket descends past 1312 ft AGL, the main deployment event is triggered by the flight computers, igniting the 3.0g primary main charge, separating the eight shear pins connecting the forward tube and the nose cone together. The nominally green and white drogue parachute is then released from the forward tube. At T +121 seconds, the main backup event is triggered by the Raven, igniting the 3.5g backup main charge to ensure deployment. Following full deployment of the main parachute, the rocket should be descending at 8.66 ms^{-1}

Landing

At T +164 seconds, the rocket touches down onto the ground at 8.66 ms^{-1} at which point, the recovery process begins following the SA Cup IREC Range Standard Operating Procedures 2024 [4].

Conclusions and Lessons Learned

Technical Learnings

Overall Learnings

As NU Rocketry is a relatively young team, many large technical changes have been undertaken in comparison to the previous instalment with many shortcomings identified. The main concern was the lack of testing being undertaken on key components which has been rapidly improved upon in this year's competition. However, the team significantly struggled due to the lack of prototyping that could be employed caused by both unfavorable timelines, and the human want to fully develop systems. The team similarly struggled more with lack of time as key tests were not able to be performed or performed extremely behind schedule, lowering the quality of solutions being produced and leading to significant design changes.

Structures Learnings

The structures department saw a large shift in production methodology and material selection due to an increased need to reduce the overall weight through a multitude of design changes, the largest of which being a change of fiberglass material to tri-axial weave. This change lowered our tube weight significantly, however the added testing required, slowed down competition progress but the overall gain was well worth the negatives. Another change was the introduction of carbon fins for the same goal of reduction of the rocket with the added benefit of stability. This change proved extremely beneficial to the rocket design itself but the added machining time did affect other areas of the rocket and as such more consideration will be included for machining time required for individual parts.

Payload Learnings

Payload undertook a large overhaul, primarily in the department itself as previous payloads were more considered afterthoughts, with an emphasis on a clear design cycle and rigorous prototyping. This mentality shift allowed for a fully developed payload to be deployed and used in a test platform, which was unable to be achieved in previous years. The payload department also started the reduction of breakout boards and daughterboards towards embedding systems onto PCBs directly, a significant advancement in our PCB development process. The change came with a number of learning difficulties, particularly how NU Rocketry approves and checks PCBs as few members had experience with boards produced in this manner. This change overall reduced the size of all the PCBs produced by the payload department, simplifying the mechanical design process and training new members in PCB Design.

Recovery Learnings

The recovery department undertook changes mainly in relation to the deployment of Wedgetail's parachutes, and our GPS frequency transmissions. The main parachute was changed to utilize a deployment bag with a sewn in pilot chute which vastly lowered our opening loads while not significantly increasing complexity which will be used in all competition platforms moving forward. The department also drastically reduced the overall weight of the AV-stack through the removal of threaded rod or aluminum stacks, allowing for the rocket to fly on a small L2 motor while still getting to 10000 feet.

Airbrake Learnings

The airbrake department saw the largest number of complications regarding the implementation of the system itself. The mechanical system itself went through a manner of revisions primarily to reduce overall weight, while also maintaining strength in the sections of the tube where the airbrake is housed. The software required much more progress than previously predicted and had a much higher level of complexity which, when combined with the timeline

constraints in place, had a negative effect on not only the airbrake department, but other departments that interfaced with it. The learnings from this project will be employed in future in the next revision of the airbrake, allowing for more in-depth, realistic goals and timelines for the department itself.

Managerial Learnings

The upper levels of leadership encountered manner issues that the team had previously put off in the development of the IREC rocket. One of the most major issues was the upskilling of juniors, as many seniors from previous years had left the team leading to a large loss of knowledge which the team had to rapidly re-learn. This issue has become the forefront of all leadership positions resulting in the formation of the 5000 feet Australian University Rocketry Cup (AURC) department, meant to teach juniors the key skills to use on the larger competition platform. It is also being addressed through the focus on juniors completing higher level tasks than in previous years, removing the dependence on senior members, and the knowledge gap between members, through tasks such as juniors developing CAD designs, body tubes, PCBs or other skills which require experience and time to learn.

Another key issue was the extremely tight timelines caused by the design cycle the team had employed previously whereby a new rocket was built every year, which has been ratified through the introduction of a realistic two year design cycle allowing for more development time, and the reduction of stress to all NU Rocketry members in future.

In the introduction of the two-year design cycle, the team has now put a much larger focus on utilizing every available launch window more efficiently through the testing of smaller test-bed platforms to test competition prototypes. It has also emphasized the need for clearer design goals and milestones to ensure the team remains moving towards a common goal and maintaining beneficial progress.

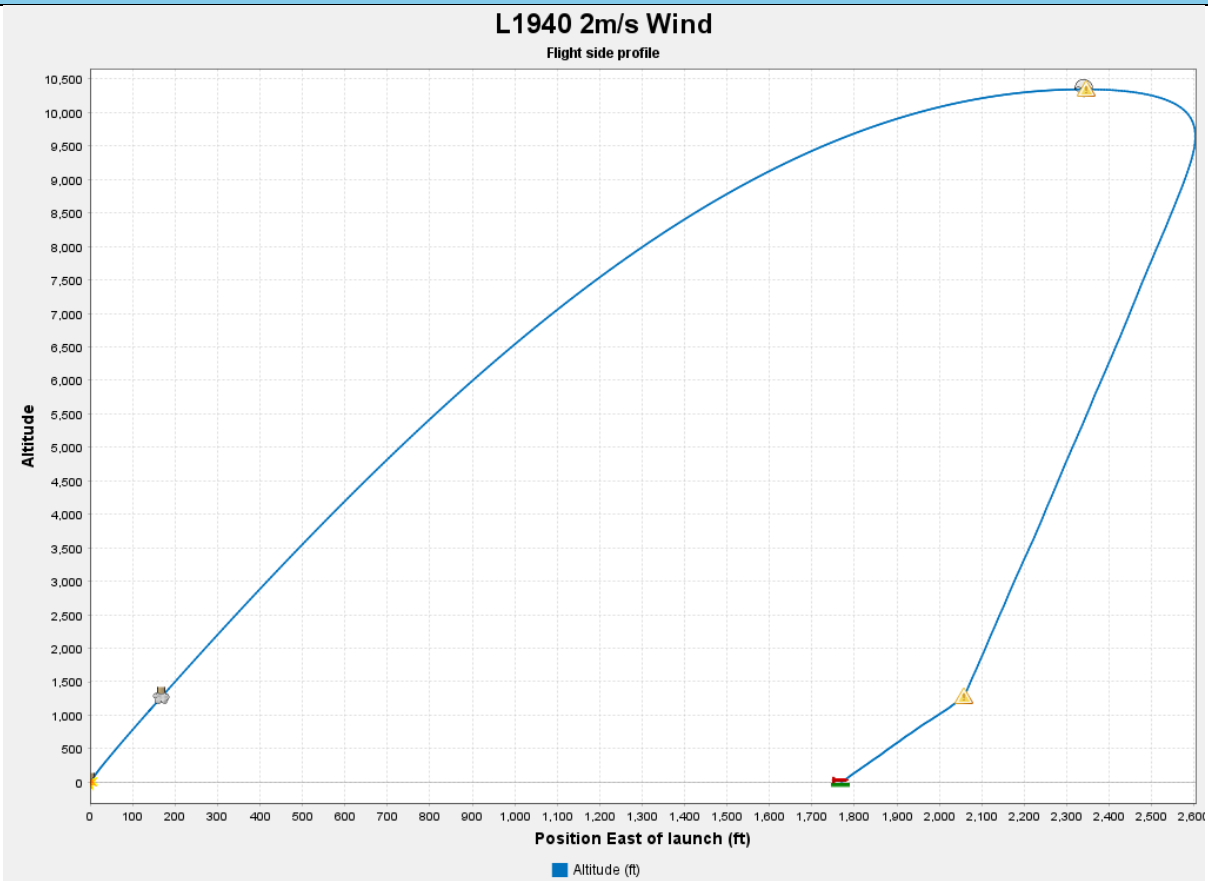
Appendix A- System Weights, Measures, And Performance Data

All simulated information provided by Openrocket with an average wind speed of 2 ms^{-1}

Rocket Information		Propulsion	
Number of Stages	1	Motor Type	Solid
Length	2354 mm	Cots?	Yes
Airframe Diameter	4 inch	Manufacturer	Aerotech
Number of Fins	3	Letter Classification	L
Fin Semi-span	104 mm	Average Thrust	1,939.3 N
Fin Tip Chord	70 mm	Total Impulse	4,329.2 Ns
Fin Thickness	6 mm	Motor Burn Time	2.2 s
Weight on Launch Rail	13351 g	Propellant Weight	1,825 g
Payload Weight	2070 g	Empty Motor case/structure weight	1333 g
Lift-Off Weight	13351 g		
Centre of Pressure	1770 mm		
Centre of Gravity	1420 mm		
Flight Data		**Recovery Information**	
Liftoff Thrust to Weight	15	Altimeters used	Blue Raven Telemetrum TeleGPS
Rail Departure Velocity	40.4 ms^{-1}	COTS?	Yes
Minimum Static Margin	2.82	Redundant	Yes
Maximum Acceleration	162 ms^{-2}	Drogue Primary Charge	3.0 g
Maximum Velocity	310 ms^{-1}	Backup Charge	3.5 g
Fin-Flutter Velocity	582.25 ms^{-1}	Drogue Deployment Altitude	Apogee
Target Apogee	10000 ft	Drogue Descent Rate	28 ms^{-1}
Predicted Apogee	10347 ft	Main Primary Charge	3.0 g
		Backup Charge	3.5 g
		Main Deployment Altitude	400 m

	Main Descent Rate	9 ms^{-1}
	Parachute Shock Loads	Main 1.6 KN Drogue 800 N
	Mechanical Links	M6 Eye nuts M6 Jaw Jaws M6 Quick link M4 Quick link 500lb Swivel

Flight Profile Graph



Appendix B- Project Test Report

Project Test Report	
Test Name: Tube Compression Test	**Date: 08/04/2026**

Test Description:

The objective of this test was to prove flight worthiness of the rocket before it has to fly at the Internation Rocketry Endeavour Cup 2026. This test will be conducted using an L1420 motor and has an estimated OpenRocket Apogee of 10500 feet.

Test Result Notes:
The test was completed at 1:17pm on the 12/04/2026. The completion of this test resulted in a successful flight of the rocket and recovery.
The rocket severely overshot the expected apogee due to some unforeseen issues in the Openrocket simulation reaching an apogee of 11087 feet.

Certified: T. Derrick

Checked: R. Heller

Completion Criteria	
Electronics Armed	SUCCESSFUL
Ignition	SUCCESSFUL
Nominal Boost Phase	SUCCESSFUL
Drogue Deployment	SUCCESSFUL
Main Deployment	SUCCESSFUL
Recovery	SUCCESSFUL
Damage	NO DAMAGE
Payload Operational	SUCCESSFUL
Airbrake deployment	FAIL

Test Media:

DESIGN.
BUILD.
TEST.

Project Test Report	ROCKETRY
Test Name: Tube Compression Test	**Date: 08/04/2026**

Test Description: The objective of this test was to evaluate the axial compression strength of the rocket tube. This was done to ensure structural integrity throughout the flight

Test Result Notes:
The test demonstrated a yield strength just above 47kN before multiple cracks formed laterally.

Certified: Rylie Heller

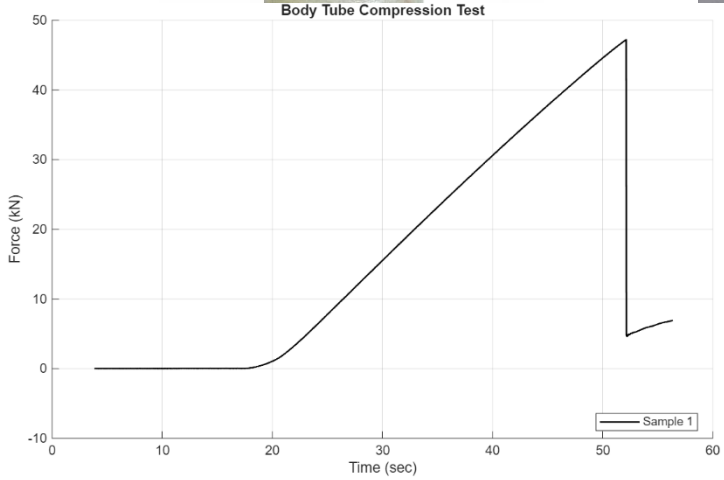
Checked: Tate Derrick

Completion Criteria	
Tube withstands 30 kN	SUCCESSFUL
	SUCCESSFUL
	SUCCESSFUL

Test Media:

a)

b)



Project Test Report	ROCKETRY

Test Name: AV bulkhead Test	**Date: 13/04/2026**

Test Description: This test is to be conducted to test the total strength and point of frailer for the avionics bulkhead under the condition subject to it during flight. The purpose is to find the force at which the component critical fails.

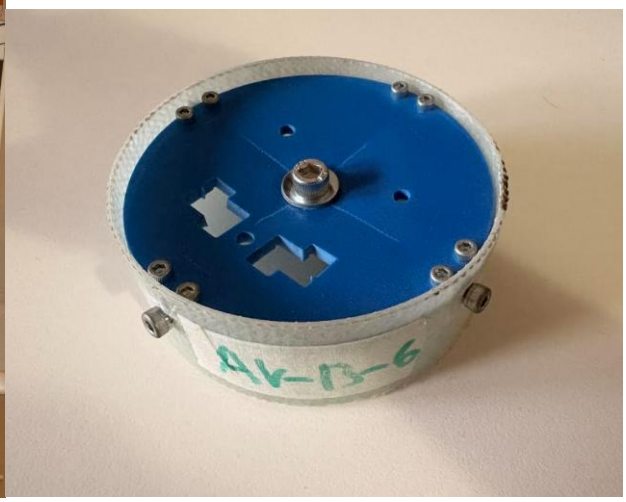
Test Result Notes: this test was completed at 10:02am on 13/04/2026. This tested the structural integrity of the AV bulkhead showing the strength to be at least 2.6kN

Certified: Rylie Heller

Checked: Tate Derrick

Completion Criteria	
M4 bolt withstanding 1.6kN	SUCCESSFUL
Steel blocks withstanding 1.6kN	SUCCESSFUL
G10 Bulkhead withstanding 1.6kN	SUCCESSFUL

Test Media:



Shimadzu universal testing machine with the AV bulkhead and testing configuration

AV bulkhead failing at 2.6kN

Project Test Report	
Test Name: Aft Bulkhead Compression Test	**Date: 08/04/2026**

Test Description: The objective of this test was to evaluate the compressive strength of the aft bulkhead assembly, validating the bulkhead and coupler design while ensuring the bulkhead can withstand expected shock loads.

Test Result Notes:
Each of the four test samples demonstrated a yield strength of roughly 4.8kN, before cracking of the G10 sheet. Each sample then experienced delamination of the upper fillet from the coupler wall, occurring between 5 and 7.3kN across the samples.

Certified: Rylie Heller

Checked: Tate Derrick

Completion Criteria	
No bearing failure of tube at 1.6kN	SUCCESSFUL
No delamination of coupler	SUCCESSFUL
Fillet delamination above 1.6kN newtons	SUCCESSFUL

Test Media:

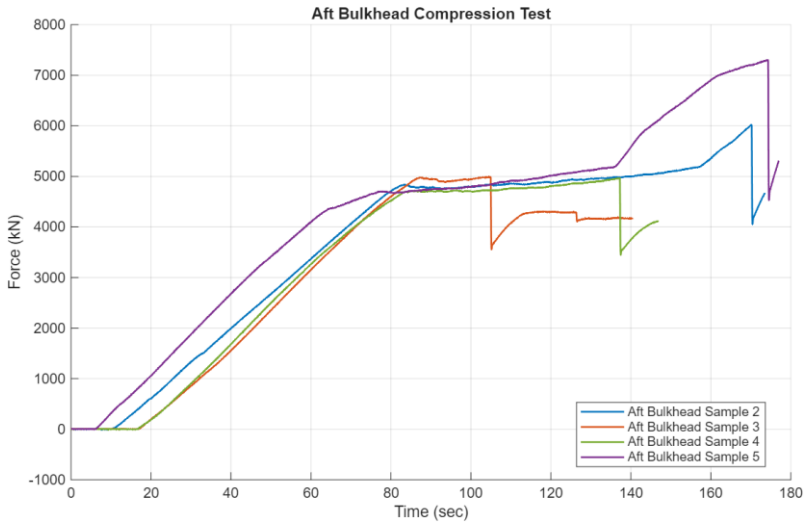


Figure 24- Graphed data from compression testing of aft bulkhead

a) b) c)

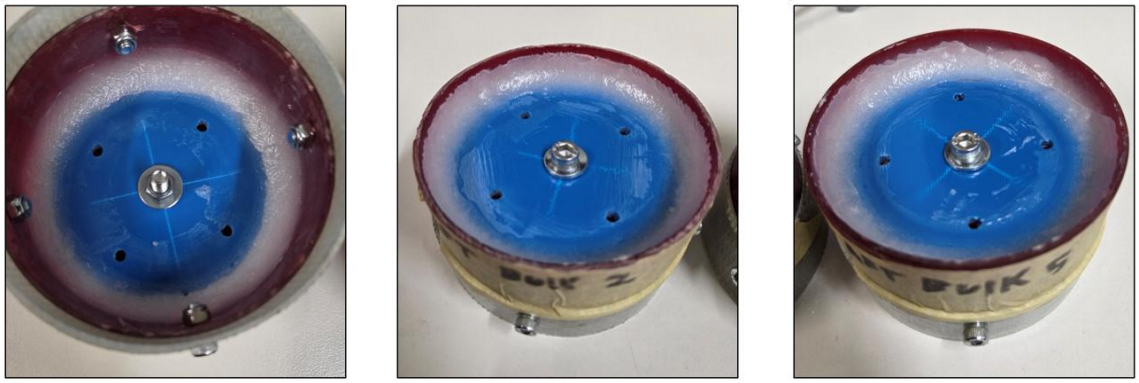


Figure 25 - Test results a) Cracking of G10 plate b) Delamination of fillet c) Delamination and cracking of fillet

Project Test Report	ROCKETRY
Test Name: Motor Tube Tensile Testing	**Date: 08/04/2026**

Test Description: The objective of this test was to evaluate the axial compressive strength of motor tube samples under pressure loading conditions determining the yield force. Ensuring that structures maintain integrity against the minimum required threshold of 2289N.

Test Result Notes:
The test was successfully completed on the 8th April at 3:00pm the completion of this test demonstrated the structural integrity of the motor-tube beyond the design goal exhibiting no damage at 2289N ultimately failing at 40kN.

Certified: Rylie Heller

Checked: Tate Derrick

Completion Criteria	
Fiberglass tube + motor tubes withstand 30kN	SUCCESSFUL
All samples withstand safety factor	SUCCESSFUL
No delamination of epoxy fillets	SUCCESSFUL

Test Media:

a) b) c)

A) Shimadzu universal testing machine with the motor-tube sample 1 and testing configuration, B) delamination of epoxy, C) cracks in motor tube

Project Test Report	ROCKETRY
Test Name: Airbrake Compression Testing	**Date: 20/04/2026**

Test Description: The objective of this test was to evaluate the axial compressive strength of the airbrake section, ensuring that its stronger in compression than compared to the standard triaxial tube.

Test Result Notes:	**Completion Criteria**	
The test was successfully completed with the following the three samples failing at the following forces 52.3 kN, 50.6 kN and 52.9 kN. Therefore exceeding the compressive strength of the body tubes.	Airbrake section failed past 47.0 kN	SUCCESSFUL
Certified: Rylie Heller		
Checked: Tate Derrick		

Test Media:

Project Test Report	ROCKETRY

Test Name: Payload Bench Test	Date: 28/3/2026

Test Description: In an open laboratory environment, validate the following objectives:

1. The controller can modulate temperature of steal block to 22 ± 0.5ºC
2. Microcontroller successfully records readable data to flash chip
3. Confirm the battery management system allows the payload to run safely on battery power

Test Result Notes: Payload was sat in a lab, set to target 22 degrees C and connected only to battery power. The system failed to write data to the flash chip however other objectives were validated using a live output which team members watched as the test took place.

Certified: B. Walker

Checked: T. Derrick

Completion Criteria	
The controller can modulate temperature of steal block to 22 ± 0.5ºC	SUCCESSFUL
Microcontroller successfully records readable data to flash chip	FAIL
System performed other objectives using managed battery power	SUCCESSFUL

Test Media:

Project Test Report	ROCKETRY

Test Name: Payload Flight Test	**Date: 12/04/2026**

Test Description: Full flight test of the payload with three main goals:

1. Confirm the payload safely holds up against forces of flight
2. Attempt to hold the steel cube at 15ºC ± 0.5 for the duration of flight and recovery
3. Collect data on the operation of the temperature control system

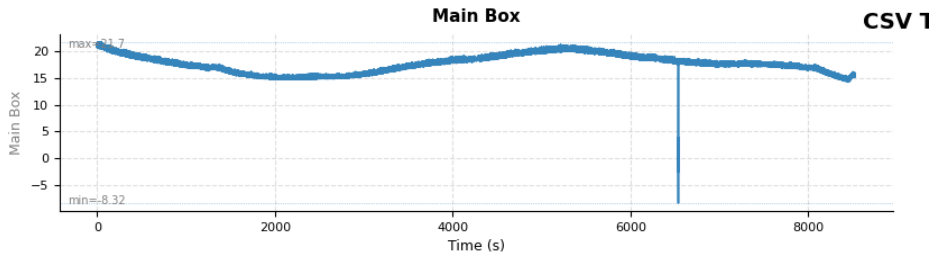
Test Result Notes: Payload flew and survived with no damage to any components or the rocket body. Data was recorded the entire flight and recovery. Reviewing the data showed that the cube did reach 15 degrees and was held there for a short time before an excess heat buildup on the heatsink pushed more heat into the cube than the controller could remove.
Data below shows that as the heat sink temperature increased, the temperature measured inside the box also increased.

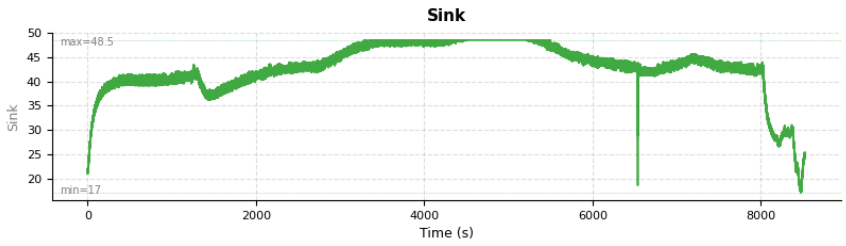
Certified: B. Walker

Checked: T. Derrick

Completion Criteria	
Safe flight with no damage to payload components or rocket	SUCCESSFUL
Hold the steel cube at 15ºC ± 0.5 for the duration of flight and recovery	FAILED
Collect data on the operation of the temperature control system	SUCCESSFUL

Test Media:





Project Test Report	ROCKETRY

Test Name: Heating Test	**Date: 26/04/2026**

Test Description: The payload was placed in the rocket tube and given the target of 40ºC. The goals for this test were:

1. Confirm the controller also functions correctly in heating mode
2. Demonstrate ability to target and maintain a higher temperature point in preparation for IREC
3. Microcontroller successfully records readable data to flash chip

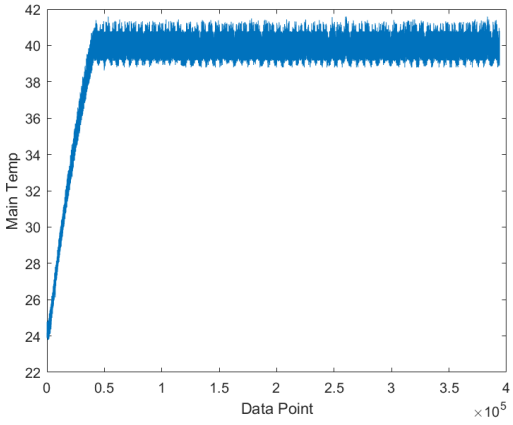
Test Result Notes: Test was completed in a room temperature lab and run for 2-3 hours. Data below showed that the system managed to target and maintain 40 degrees C consistently in this environment. There was a significant amount of ADC noise present, using a digital filter such as a rolling average is recommended to reduce this effect.

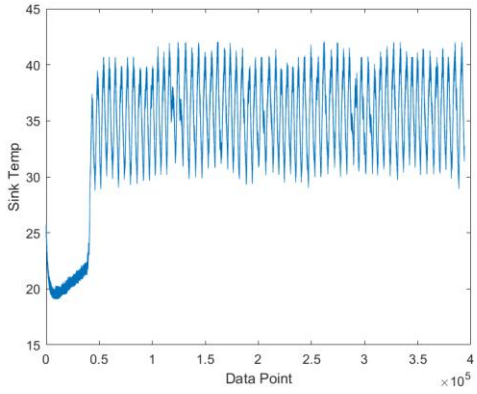
Certified: B. Walker

Checked: T. Derrick

Completion Criteria	
Controller functions correctly in heating mode	SUCCESS
Ability to successfully target and maintain a higher temperature target	SUCCESS
Microcontroller successfully records readable data to flash chip	SUCCESS

Test Media:





Appendix C- Hazard Analyses

Table 1: Payload Risk Assessment

Risk	**Mitigation**	**Likelihood (After Mitigation)**	**Potential for Harm**
Burns from touching hot components.	Avoid handling heated components (heat sink, Peltier module) immediately after recovery. Wait for at least 10 minutes after system shutoff before attempting to handle.	Low	Low – minor skin burns
Damage/weakening of rocket body from generated heat	If rocket body temperatures rise above 60ºC (140ºF), deactivate the Peltier module until temperatures drop back to safe levels for the epoxy used in rocket construction (<57ºC (134ºF)	Low	High
Battery Fire	All batteries used in the payload system are connected to the PCB with a 3A rated fuse to protect batteries from rapid discharge.	Low	High – high temps from battery fire can damage rocket structure

Activity / Task / Location: High Powered Rocket Launch / International Rocket Engineering Competition 2026 / Midlands, Texas, USA	**Reviewed / Approved By:** Malcolm Sidney **Signature and Date:**
Hazard Analysis Developed by: Austin Gogarty / Nicholas George	**Date:** 24/10/25

Risk Matrix

Likelihood

N.B. For more details regarding use of this matrix / definitions refer to final page of this document

Consequence	Rare	Unlikely	Possible	Likely	Almost Certain
Severe *Eg. Potential Fatality or Injury or Illness with permanent disability*	MEDIUM	MEDIUM	HIGH	EXTREME	EXTREME
Major *Eg. Potential Lost Time Injury (but non-permanent disability)*	LOW	MEDIUM	MEDIUM	HIGH	EXTREME
Moderate *Eg. Potential Medical Treatment injury or illness (but no lost time)*	LOW	LOW	MEDIUM	MEDIUM	HIGH
Minor *Eg. Potential First Aid injury*	LOW	LOW	LOW	MEDIUM	MEDIUM
Minimal *Eg. Hazard or near miss requiring reporting and follow up action*	LOW	LOW	LOW	LOW	LOW

Actions required based on Hazard Analysis

Extreme	An “extreme” risk requires immediate assessment and senior staff consideration is required; a detailed mitigation plan must be developed, and consideration should be given to ceasing the activity unless the risk can be reduced to a level of high or less; regular monitoring and reported on to the relevant management/steering committee; Target resolution should be within 1 month.
High	A “high” risk may also require immediate assessment and senior staff consideration; a mitigation plan must be developed; regular monitoring and reported on to the relevant management/steering committee. Target resolution (ideally reduction to medium or low level of risk) should be within 3 months.
Medium	A mitigation plan must be developed; existing controls need to be reviewed. Target resolution (ideally reduction to low level of risk) should be within 1 year.
Low	Risk is tolerable; manage by well established, routine processes/procedures and be mindful of changes to nature of risks.

Hazardous Material	Storage	Handling	Transportation	Risk Rating based on Risk Matrix	Control Methods	Residual Risk Rating based on Risk Matrix
Fuel Grains	Leave fuel grains in original packaging until required to be assembled. Store in explosives storage cabinet, free from any flammables or ignition sources	Handle with care. Keep stored until required to assemble into motor casing and do not come in contact with any ignition sources or flammables during use.	Transport in original packaging in the portable explosives magazine. Ensuring it is free of any flammables or ignition sources.	**HIGH**	Only properly trained and certified individuals shall have access to handle fuel grains during the assembly of the motor. Use nitrile gloves whilst handling.	**MEDIUM**
Ignitors	Store in cool, dry, metal magazine, free from any ignition sources, flammables, and explosives	Handle with care, Keep stored until required and do not come in contact with any ignition sources during use	Transport in cool, dry metal magazine, free from any ignition sources, flammables, and explosives	**MEDIUM**	keep wire ends twisted until ready to be armed. Transport separate to explosives and only once required	**LOW**
Black Powder	Store in explosive storage cabinet, free of any flammables or ignition sources	Wear hearing protection, safety glasses and shield along with long drill cotton and steal capped boots.	Transport in portable explosives magazine. Ensuring it is free of any flammables or ignition sources. Only transport quantities required, decanted into 200g vials	**HIGH**	Only properly trained and certified individuals shall have access to handle black powder. The use of a safety glove box to contain black powder in case of spills	**MEDIUM**

Li-ion Battery Cells	Store in Li-ion transport bags, at storage voltages, ensuring they are free of any sharp objects or flammables	Handle with care, ensure to avoid sharp objects and high temperatures. Do not handle unless required	Transport in Li-ion transport bags as per manufactures requirement. Only transport the quantities required.	**MEDIUM**	Inspect batteries before and after usage, check the environment surrounding is free of any sharp objects and only handle as a last step when required to power systems.	**LOW**
Assembled Motor	Store in explosive storage cabinet, free of any flammables or ignition sources	Handle with care, Keep stored until required and do not come in contact with any ignition sources during use.	Transport in portable explosives magazine. Ensuring it is free of any flammables or ignition sources. If required to transport for flight, insert motor into rocket immediately after removal from the explosives magazine and ensure, ignition source is transported separate to the rocket	**HIGH**	Only properly trained and certified individuals shall have access to handle the assembled motor. If integrated into the rocket for transport, ensure the rocket is never pointed toward any personnel and the ignitor is transported separately to the rocket	**MEDIUM**

Summary of Requirements based on Hazard Analysis		Review Period / Date
Personal Protective Equipment	Safety Glasses, Steel Capped Boots, Long Drill Cotton Clothing, Face Sheild, Hearing Protection, Nitrile Gloves	01/2027
Other Equipment and Equipment Protection	Safety Glove Box, Explosives Cabinet, Explosives Transport Magazine, Li-ion Storage Bags	01/2027
Training Requirements	NSW Security Clearance, NSW Pyrotechnics License	01/2027
Procedures, SOPs etc	IREC 2025 Range SOP, University of Newcastle Black Powder Decanting, Ground Testing and Launch Standard Operating Procedure	01/2027
Relevant Legislation etc.	AS 2187.1-1998 Explosives-Storage, transport and use Part 1: Storage WHS Act 2011 (NSW) & Regulations / Codes of Practice Australian code for the transport of explosives by road and rail – 3rd edition Eye Protection *AS/NZS 1337* Safety footwear *AS/NZS 2210.2* CASA UNMANNED AIRCRAFT AND ROCKETS: ROCKETS *Tripoli Rocketry Association Safety Code* IREC 2025 Range Standard Operating Procedures	01/2027

Questions to ask in order to determine the hazards relating to the task:

A Could people be injured or made sick by things such as:

- Noise
- Light
- Radiation
- Toxicity
- Infection
- High or low temperatures
- Electricity
- Moving or falling things (or people)
- Flammable or explosive materials
- Things under tension or pressure (compressed gas or liquid; springs)
- Any other energy sources or stresses
- Biohazardous material
- Laser

B Can workplace practices cause injury or sickness?

- Are there heavy or awkward lifting jobs?
- Can people work in a comfortable posture?
- If the work is repetitive, can people take breaks?
- Are people properly trained?
- Do people follow correct work practices?
- Are there adequate facilities for the work being performed?
- Are universal safety precautions for biohazards followed?
- Is there poor housekeeping? Look out for clutter
- Torn or slippery flooring
- Sharp objects sticking out
- Obstacles

C Imagine that a child was to enter your work area?

- What would you warn them to be extra careful of?
- What would do to reduce the harm to them?

D What could go wrong?

- What if equipment is misused?
- What might people do that they shouldn't
- How could someone be killed?
- How could people be injured?
- What may make people ill?
- Are there any special emergency procedures required?

E Are procedures or organisational systems missing or not being followed?

- Standard Operating Procedures?
- Risk Assessments?
- Induction or training?
- Management of change?
- Safety Inspections?
- Hazard reporting?
- Contractor Management?

F What kinds of injuries could possibly occur?

- Broken bones
- Eye damage
- Hearing problems
- Strains or sprains
- Cuts or abrasions
- Bruises
- Burns
- Lung problems including inhalation injury/ infection
- Skin contact
- Poisoning
- Needle-stick injury
- Psychological illness or injury

THE UNIVERSITY OF NEWCASTLE AUSTRALIA

ROCKETRY

How to Assess Risk

Step 1 – Consider the Consequences

What are the potential consequences of an incident occurring?
Consider what could reasonably happen as well as what may actually happen.

Look at the descriptions and choose the most suitable Consequence.

Consequence	
Serious	Potential Fatality or Injury or Illness with permanent disability
Major	Potential Lost Time Injury requiring time off work (but non-permanent disability)
Moderate	Potential medical treatment Injury or Illness but no lost time
Minor	Potential First Aid Injury
Minimal	No injury but hazard exists or near miss occurred requiring reporting and follow up action

Step 2 – Consider the Likelihood

What is the likelihood of the consequence identified in step 1 happening?

Consider this with the current controls in place.

Look at the descriptions and choose the most suitable Likelihood.

Likelihood	
Almost Certain	The event could be expected to occur in most circumstances: "This is a common problem here".
Likely	The event has a reasonable chance of occurring in usual conditions: "It has happened here before".
Possible	The event might occur occasionally, has occurred sometime: "Has infrequently happened here before".
Unlikely	The event has a small chance of occurring. "It has not happened here but has occurred elsewhere".
Rare	Very unlikely to occur. "It would be extremely rare for it to occur here".

Step 3 – Calculate the Risk Rating

A. Take Step 1 rating and select the correct column.

B. Take Step 2 Rating and select the correct line.

C. The calculated risk rating is where the two ratings cross

CONSEQUENCE	LIKELIHOOD: Rare	Unlikely	Possibly	Likely	Almost Certain
Serious	MEDIUM	MEDIUM	HIGH	EXTREME	EXTREME
Major	LOW	MEDIUM	MEDIUM	HIGH	EXTREME
Moderate	LOW	LOW	MEDIUM	MEDIUM	HIGH
Minor	LOW	LOW	LOW	MEDIUM	MEDIUM
Minimal	LOW	LOW	LOW	LOW	LOW

Controlling the Risk: Risk control is a method of managing the risk with the primary emphasis on controlling the hazards at source. For a risk that is assessed as "extreme" or "high", steps should be taken immediately to minimize risk of injury. The method of ensuring that risks are controlled effectively is by using the "hierarchy of controls". The Hierarchy of Controls are:

Control Type	Example
Eliminate	Removing the hazard, eg taking a hazardous piece of equipment out of service.
Substitute	Replacing a hazardous substance or process with a less hazardous one, eg substituting a hazardous substance with a non-hazardous substance.
Engineering	Redesign a process or piece of equipment to make it less hazardous, Isolating the hazard from the person at risk, eg using a guard or barrier, or containing the hazard in an enclosure.
Administrative	Adopting safe work practices or providing appropriate training, instruction or information.
Personal Protective Equipment (PPE)	The use of personal protective equipment could include using gloves, glasses, earmuffs, aprons, safety footwear, dust masks. **NOTE: This is a last resort control and should be used in conjunction with higher level controls.**

Appendix D- Risk Assessment

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Activity / Task / Location: High Powered Rocket Launch/ International Rocket Engineering Competition 2026 / Midlands, Texas, USA	**Reviewed / Approved By:** Malcolm Sidney **Signature and Date:**
Risk Assessment Developed by: Austin Gogarty / Nicholas George	**Date:** 24/10/2025

Risk Matrix

Likelihood

N.B. For more details regarding use of this matrix / definitions refer to final page of this document

Consequence	Rare	Unlikely	Possible	Likely	Almost Certain
Severe *Eg. Potential Fatality or Injury or Illness with permanent disability*	MEDIUM	MEDIUM	HIGH	EXTREME	EXTREME
Major *Eg. Potential Lost Time Injury (but non-permanent disability)*	LOW	MEDIUM	MEDIUM	HIGH	EXTREME
Moderate *Eg. Potential Medical Treatment injury or illness (but no lost time)*	LOW	LOW	MEDIUM	MEDIUM	HIGH
Minor *Eg. Potential First Aid injury*	LOW	LOW	LOW	MEDIUM	MEDIUM
Minimal *Eg. Hazard or near miss requiring reporting and follow up action*	LOW	LOW	LOW	LOW	LOW

Actions required based on Risk Assessment

Extreme	An “extreme” risk requires immediate assessment and senior staff consideration is required; a detailed mitigation plan must be developed, and consideration should be given to ceasing the activity unless the risk can be reduced to a level of high or less; regular monitoring and reported on to the relevant management/steering committee; Target resolution should be within 1 month.
High	A “high” risk may also require immediate assessment and senior staff consideration; a mitigation plan must be developed; regular monitoring and reported on to the relevant management/steering committee. Target resolution (ideally reduction to medium or low level of risk) should be within 3 months.
Medium	A mitigation plan must be developed; existing controls need to be reviewed. Target resolution (ideally reduction to low level of risk) should be within 1 year.
Low	Risk is tolerable; manage by well established, routine processes/procedures and be mindful of changes to nature of risks.

Hazard Identification and initial Risk Rating				Control measures and Residual Risk Rating		Remaining Hazards
What are the steps of the activity / items of equipment?	What are the possible causes?	What are the potential hazards?	Risk Rating based on Risk Matrix	What control methods or measures will be used to reduce the likelihood and/or the consequence of an illness or injury from those hazards?	Residual Risk Rating based on Risk Matrix	What hazard remains?
Freight						
	Shipping Rocket	Fire / Explosion	HIGH	Ensure all components are well cleaned of explosive residue before freight, no hazardous goods to be shipped with logistics company. Explosive to be purchased at destination not shipped. Batteries to be transported inside certified Li-ion transport bags in carry-on luggage.	LOW	Fire / Explosion
		Theft, seizure of goods	HIGH	Shipping with reputable logistics company. Informing the logistics company of the exact contents of the freight Not shipping any hazardous goods Only shipping goods registered in ATA Carnet	LOW	Theft, seizure of goods
	Manual Handling	Slips, trips and falls	MEDIUM	Clear work areas of debris	LOW	Slips, trips and falls

				Eyes on path, Leader to manage the loading and unloading. Leather steel toed safety boots must be worn		
		Falling objects	MEDIUM	Clear work areas of debris Eyes on path, Leader to manage the loading and unloading. Leather steel toed safety boots must be worn	LOW	Falling objects
		strains and sprains	MEDIUM	Two person lifts for items over 15KG Use mechanical aids where possible	LOW	Strains and sprains
International Travel						
	Flights	Misplacing personnel	HIGH	Ensure all personnel are aware of the itinerary. Ensure all personnel are contactable. Ensure all personnel have required documentation. Ensure all personnel are accounted for at various key stages of travel	LOW	Misplacing personnel
		Misplacing Goods	MEDIUM	Ensure all goods are correctly labelled including all contact information. Ensure all goods are measured and weighed before arrival at the airport.	LOW	Misplacing Goods
		Transportation of Batteries resulting in Fire / Explosion	HIGH	Limit quantities of batteries transported to only what is required and below the airline allowance (100 Wh).	LOW	Transportation of Batteries resulting in Fire / Explosion

				Batteries to be transported inside certified Li-ion transport bags in carry-on luggage. Batteries must charged/discharged to a suitable voltage for transport.		
Event Transport						
	Preparing gear for loading, Loading/Unloading gear into vehicles	Slips, trips and falls	MEDIUM	Clear work areas of debris Eyes on path, Leader to manage the loading and unloading. Leather steel toed safety boots must be worn	LOW	Slips, trips and falls
		Falling objects	MEDIUM	Clear work areas of debris Eyes on path, Leader to manage the loading and unloading. Leather steel toed safety boots must be worn	LOW	Falling objects
		strains and sprains	MEDIUM	Two person lifts for items over 15KG Use mechanical aids where possible	LOW	strains and sprains
		Struck by moving vehicles, entangled, crushed	HIGH	Avoid areas with moving vehicles, organise transportation vehicle where possible. Park transportation vehicles as close as practical to operating area.	LOW	Struck by moving vehicles, entangled, crushed
	Loading/Unloading explosives magazine	Fire and explosion	HIGH	Storing black powder and rocket propellant in container outlined in AS2187.1-1998. A minimum of one person must be holding a current pyrotechnicians	LOW	fire/explosion

				licence for model rocket propellant. A member of the professional staff shall be designated as the person in charge of the magazine and is the only person permitted to move the magazine or access its contents. Following storage and transportation requirements outline in AS2187.1-1998. Leather steel toed safety boots required		
		Slips, trips and falls	MEDIUM	Clear work areas of debris Two person lifts for items over 15 KG, Eyes on path, Leader to manage the loading and unloading. Leather steel toed safety boots must be worn	LOW	Slips, trips and falls
		Falling objects	MEDIUM	Clear work areas of debris Eyes on path, Leader to manage the loading and unloading. Leather steel toed safety boots must be worn	LOW	Falling objects
		strains and sprains	MEDIUM	Two person lifts for items over 15KG Use mechanical aids where possible	LOW	strains and sprains
		Struck by moving vehicles, entangled, crushed	HIGH	Avoid areas with moving vehicles, organise transportation vehicle where possible. Park transportation vehicles as close as practical to operating area.	LOW	Struck by moving vehicles, entangled, crushed

		Theft of explosives	HIGH	The nominated magazine custodian shall follow the procedures outlined in the management of explosives and rocket motors plan. Including magazine security and restricting access to authorised personnel.	LOW	Theft of explosives
	Travelling in vehicles to/from launch site/accommodation/flights	Fatigue, road and traffic hazards, fire / explosion	EXTREME	Vehicles suited to the load of the rocket and terrain of the trip will be used. Vehicles fitted with safety features such as ABS, TC, airbags and inertia reel seat belts will be used. All drivers hold a current Australian license. Drivers will be well rested. Drivers will drive to the conditions. Following transportation procedures outlined in "Australian code for the transport of explosives by road and rail – 3rd edition". Refer to the 'Transportation of Explosives' section of the Universities Management of Explosives and Rocket Motors Plan, ensure the NEQ of explosives maintains a Category 1 (Low Risk) at all times. Drivers must not drive longer then 2hrs without a rest stop. Plan fuel stops ahead of time, organising to have the vehicles refuelled prior to half fuel remaining.	LOW	Fatigue, road and traffic hazards, fire / explosion

				fire extinguisher must be carried in all vehicles carrying explosives		
	Storage of explosives magazine while traveling	fire/explosion, theft	HIGH	While at the launch site the magazine shall be under constant supervision by a correctly trained and certified UON staff or student when not secured inside a vehicle. The magazine shall not be stored in any accommodation at any time. Refer to the 'Transportation of Explosives' section of the Universities Management of Explosives and Rocket Motors Plan.	LOW	Fire/explosion, theft
Rocket Assembly						
	Loading/Unloading rocket propellant and black powder	strains and sprains, fire/explosion	EXTREME	Storing black powder and rocket propellant in containers outlined in AS2187.1-2018. Two person lifts for items over 15KG. A member of the professional staff shall be designated as the person in charge of the magazine and is the only person permitted to move the magazine or access its contents. Store the dangerous materials with processes outline in AS2187.1-2018, the person in charge of the explosives magazine shall follow the storage procedures outline in AS2187.1-1998 Clause 4.6.5.2. Ensure updates are made to	LOW	falling objects, strains and sprains, hit by moving vehicles, fire/explosion

				Chemwatch for any changes in explosives or quantities within the magazine. Leather steel toed safety boots		
	Preparation of black powder charges	fire/explosion	EXTREME	Follow Newcastle University, Black Powder Testing SOP 1.3 Follow Newcastle University, Black Powder Testing RA 1.2 Safety glasses to be worn at all times. Hearing protection and safety visor to be worn when black powder outside of glovebox.	MEDIUM	Fire / Explosion
	Assembling rocket	Slips, trips and falls, falling objects, strains and sprains, cuts/abrasions	HIGH	Clear work area before assembly Ensure tools are being used for only their intended purposes. Keep hand away from pinch points and sharp objects. Steel toed safety boosts required.	LOW	Slips, trips and falls, falling objects, strains and sprains
Rocket Pre-Flight						
	Prepare rockets for launch	Slips, trips and falls, falling objects, strains and sprains, hit by moving vehicles, fire/explosion	EXTREME	Clear paths of debris Motor ignitor shall not be connected to the launch control system until all flight electronics are active. Launch control system must have arming switch with a removable key or interlock. Eyes on path, the leader shall inspect all rockets prior to launch.	LOW	falling objects, strains and sprains, hit by moving vehicles, fire/explosion

				safety glasses for all personnel, hearing protection and mask to be worn at all times by personnel in contact with the rocket.		
	Pre-mature detonation of recovery charge	Fire/Explosion, falling objects	HIGH	A member of the professional staff shall be designated as the person in charge of the magazine and is the only person permitted to move the magazine or access its contents. Check condition of black powder during removal, do not attempt to launch black powder deemed to not be in usable state. Never store motor ignitors within the explosive's magazine Safety glasses when explosives are outside of the magazine	LOW	Fire/Explosion, falling objects
	Pre-mature detonation of motor	Fire/Explosion, falling objects	HIGH	A member of the professional staff shall be designated as the person in charge of the magazine and is the only person permitted to move the magazine or access its contents. Check condition of motors during removal, do not attempt to launch damaged or out of date motors. Have damaged or out of date motors disposed of. Never store motor ignitors within the explosive's magazine	LOW	Fire/Explosion, falling objects

				Safety glasses when explosives are outside of the magazine		
	Awaiting other rocket launches	Falling objects, fire/explosion	EXTREME	Engineering: Rockets must launch from a stable platform. Administrative: While and after installing an igniter the rocket must remain pointed in a safe direction. No rockets shall launch when the sustained surface winds exceed 32kph. Activity shall cease whenever a thunderstorm has been detected within 16km of the launch site. Follow directions given by leader. Personnel must be located at a safe distance outlined in Tripoli safety code safe distance table (60m for I540 635Ns motors).	MEDIUM	Falling objects, fire/explosion
		dehydration, sun burns, heat stroke	EXTREME	sunscreen, hats, sunglasses required. an adequate supply of water is required to be accessible at all times. frequently reapply sunscreen. take frequent breaks from direct sun exposure.	MEDIUM	dehydration, sun burns, heat stroke
Rocket Launch						
	Launching rocket	Falling objects, fire/explosion	HIGH	Rockets must launch from a stable platform, with a barrier installed underneath the rockets to prevent property damage.	LOW	Falling objects, fire/explosion

				While and after installing an igniter the rocket must remain pointed in a safe direction. No rockets shall launch when the sustained surface winds exceed 32kph. Activity shall cease whenever a thunderstorm has been detected within 16km of the launch site. Follow directions given by leader. The minimum size of the launch area must meet the value outline in CASA UNMANNED AIRCRAFT AND ROCKETS: ROCKETS clause 7.2.1. Personnel must be located at a safe distance outlined in Tripoli safety code safe distance table. Public access areas to the launch site will be patrolled by rocketry personnel to prevent members of the public entering the operating area (750m diameter circle around the launch site).		
	Rocket deviates from the nominal flight path	Falling objects, fire/explosion	EXTREME	While and after installing an igniter the rocket must remain pointed in a safe direction. No rockets shall launch when the sustained surface winds exceed 32kph. Activity shall cease whenever a thunderstorm has been detected within 16km of the launch site. Follow directions given by leader.	MEDIUM	Falling objects, fire/explosion

				The minimum size of the launch area must meet the value outline in CASA UNMANNED AIRCRAFT AND ROCKETS: ROCKETS clause 7.2.1 (f) (750m diameter for I540 635Ns motors). Personnel must be located at a safe distance outlined in Tripoli safety code safe distance table (60m for I540 635Ns motors). Public access areas to the launch site will be patrolled by rocketry personnel to prevent members of the public entering the operating area (750m diameter circle around the launch site).		
	Recovery system fails to deploy	Falling objects, fire/explosion	EXTREME	The minimum size of the launch area must meet the value outline in CASA UNMANNED AIRCRAFT AND ROCKETS: ROCKETS clause 7.2.1 (f) (750m diameter for I540 635Ns motors). Personnel must be located at a safe distance outlined in Tripoli safety code safe distance table (60m for I540 635Ns motors). Public access areas to the launch site will be patrolled by rocketry personnel to prevent members of the public entering the operating area (750m diameter circle around the launch site).	MEDIUM	Falling objects, fire/explosion

	Recovery system deploys but parachutes do not release	Falling objects, fire/explosion	EXTREME	The minimum size of the launch area must meet the value outline in CASA UNMANNED AIRCRAFT AND ROCKETS: ROCKETS clause 7.2.1 (f). Personnel must be located at a safe distance outlined in Tripoli safety code safe distance table. Public access areas to the launch site will be patrolled by rocketry personnel to prevent members of the public entering the operating area (750m diameter circle around the launch site).	MEDIUM	Falling objects, fire/explosion
Upset Conditions						
	Rocket motor does not ignite, but does when approached on the pad	Falling objects, fire/explosion	HIGH	Check condition of motors during removal, do not attempt to launch damaged or out of date motors. Have damaged or out of date motors disposed of. Never store motor ignitors within the explosives magazine	LOW	Falling objects, fire/explosion
	Severe weather conditions	Slips, trips and falls, falling objects, strains and sprains, cuts/abrasions, fire/explosion	HIGH	No rockets shall launch when the sustained surface winds exceed 32kph. Activity shall cease whenever a thunderstorm has been detected within 16km of the launch site. Follow directions given by leader.	LOW	Slips, trips and falls, falling objects, strains and sprains, cuts/abrasions, fire/explosion
	Evacuation	Slips, trips and falls, falling objects, strains and sprains,	HIGH	Evacuate in an orderly manner to the nearest evacuation point.	LOW	Slips, trips and falls, falling objects, strains and sprains,

		cuts/abrasions, fire/explosion		leave straight away, attempt where possible to dis-arm any armed energetics. Check that all personnel are accounted for at the evacuation point Follow the direction of the university technical staff member		cuts/abrasions, fire/explosion
	Fire	Slips, trips and falls, falling objects, strains and sprains, cuts/abrasions, fire/explosion	HIGH	Evacuate in an orderly manner to the nearest evacuation point. leave straight away, attempt where possible to dis-arm any armed energetics. Check that all personnel are accounted for at the evacuation point Follow the direction of the university technical staff member	LOW	Slips, trips and falls, falling objects, strains and sprains, cuts/abrasions, fire/explosion
Rocket Recovery						
	Recovering Rocket	dehydration, sun burns, heat stroke	EXTREME	Sunscreen, hats, sunglasses required. an adequate supply of water is required to be accessible at all times. frequently reapply sunscreen. take frequent breaks from direct sun exposure. Ensure you are adherent to the Spaceport America Cup 2024 Standard Range Operating Procedures at all times	MEDIUM	dehydration, sun burns, heat stroke
		Slips, trips and falls	HIGH	Clear paths of debris	LOW	Slips, trips and falls

				Steel toed safety boots must be worn at all times Ensure you are adherent to the Spaceport America Cup 2024 Standard Range Operating Procedures at all times.		
		Sprains and Strains	HIGH	Two person lifts for items over 15KG Use mechanical aids where possible	LOW	Sprains and Strains
		Bites and stings	EXTREME	Steel toed safety boots must be worn at all times. Long length clothing must be worn during the retrieval of a rocket. Ensure you are adherent to the Spaceport America Cup 2024 Standard Range Operating Procedures at all times	MEDIUM	Bites and stings
		Becoming disoriented	HIGH	Ensure you carry a radio and /or operational device at all times Never retrieve a rocket alone or venture from the retrieval group. Ensure you are adherent to the Spaceport America Cup 2024 Standard Range Operating Procedures at all times	LOW	Becoming disoriented

Summary of Requirements based on Risk Assessment		Review Period / Date
Personal Protective Equipment	Safety glasses, leather safety boots, hats, sunglasses, face shield, hearing protection, long clothing	01/2027
Other Equipment and Equipment Protection	Safety glove box, fire extinguisher	01/2027
Training Requirements	NSW Pyrotechnicians Licence for Model rocket propellant & associated Security Clearance, NSW Drivers Licence	01/2027
Procedures, SOPs etc	IREC 2025 Range SOP, The University of Newcastle Launch SOP, The University of Newcastle Black Powder SOP	01/2027
Relevant Legislation etc.	AS 2187.1-1998 Explosives-Storage, transport and use Part 1: Storage WHS Act 2011 (NSW) & Regulations / Codes of Practice Australian code for the transport of explosives by road and rail – 3rd edition Eye Protection *AS/NZS 1337* Safety footwear *AS/NZS 2210.2* CASA UNMANNED AIRCRAFT AND ROCKETS: ROCKETS *Tripoli Rocketry Association Safety Code* IREC 2025 Range Standard Operating Procedures	01/2027

Questions to ask in order to determine the hazards relating to the task:

A Could people be injured or made sick by things such as:

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- Radiation
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- Electricity
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- Any other energy sources or stresses
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- Laser

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- Are there heavy or awkward lifting jobs?
- Can people work in a comfortable posture?
- If the work is repetitive, can people take breaks?
- Are people properly trained?
- Do people follow correct work practices?
- Are there adequate facilities for the work being performed?
- Are universal safety precautions for biohazards followed?
- Is there poor housekeeping? Look out for clutter
- Torn or slippery flooring
- Sharp objects sticking out
- Obstacles

C Imagine that a child was to enter your work area?

- What would you warn them to be extra careful of?
- What would do to reduce the harm to them?

D What could go wrong?

- What if equipment is misused?
- What might people do that they shouldn't
- How could someone be killed?
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- Needle-stick injury
- Psychological illness or injury

THE UNIVERSITY OF NEWCASTLE AUSTRALIA

ROCKETRY

How to Assess Risk

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Consider what could reasonably happen as well as what may actually happen.

Look at the descriptions and choose the most suitable Consequence.

Step 2 – Consider the Likelihood

What is the likelihood of the consequence identified in step 1 happening?

Consider this with the current controls in place.

Look at the descriptions and choose the most suitable Likelihood.

Step 3 – Calculate the Risk Rating

A. Take Step 1 rating and select the correct column.

B. Take Step 2 Rating and select the correct line.

C. The calculated risk rating is where the two ratings cross

Consequence		Likelihood	
Serious	Potential Fatality or Injury or Illness with permanent disability	Almost Certain	The event could be expected to occur in most circumstances: "This is a common problem here".
Major	Potential Lost Time Injury requiring time off work (but non-permanent disability)	Likely	The event has a reasonable chance of occurring in usual conditions: "It has happened here before".
Moderate	Potential medical treatment Injury or Illness but no lost time	Possible	The event might occur occasionally, has occurred sometime: "Has infrequently happened here before".
Minor	Potential First Aid Injury	Unlikely	The event has a small chance of occurring. "It has not happened here but has occurred elsewhere".
Minimal	No injury but hazard exists or near miss occurred requiring reporting and follow up action	Rare	Very unlikely to occur. "It would be extremely rare for it to occur here".

CONSEQUENCE	LIKELIHOOD				
	Rare	Unlikely	Possibly	Likely	Almost Certain
Serious	MEDIUM	MEDIUM	HIGH	EXTREME	EXTREME
Major	LOW	MEDIUM	MEDIUM	HIGH	EXTREME
Moderate	LOW	LOW	MEDIUM	MEDIUM	HIGH
Minor	LOW	LOW	LOW	MEDIUM	MEDIUM
Minimal	LOW	LOW	LOW	LOW	LOW

Controlling the Risk: Risk control is a method of managing the risk with the primary emphasis on controlling the hazards at source. For a risk that is assessed as "extreme" or "high", steps should be taken immediately to minimize risk of injury. The method of ensuring that risks are controlled effectively is by using the "hierarchy of controls". The Hierarchy of Controls are:

Control Type	Example
Eliminate	Removing the hazard, eg taking a hazardous piece of equipment out of service.
Substitute	Replacing a hazardous substance or process with a less hazardous one, eg substituting a hazardous substance with a non-hazardous substance.
Engineering	Redesign a process or piece of equipment to make it less hazardous, Isolating the hazard from the person at risk, eg using a guard or barrier, or containing the hazard in an enclosure.
Administrative	Adopting safe work practices or providing appropriate training, instruction or information.
Personal Protective Equipment (PPE)	The use of personal protective equipment could include using gloves, glasses, earmuffs, aprons, safety footwear, dust masks. **NOTE: This is a last resort control and should be used in conjunction with higher level controls.**

Appendix E – Checklists

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Purpose	The purpose of this check-sheet is to track and assure that the IREC 2026 rocket is recovered
Applicability	Wedgetail
Notes	"IREC26-CL001 Assembly Checklist", "IREC26-CL002 Preflight Checklist" and "IREC26-CL003 Launch Checklist" must be completed prior to this checklist

Flight Information			
Predicted Apogee (m)	**Max Velocity (m/s)**	**Velocity Off Rail (m/s)**	**Total Weight (kg)**
Max Acceleration (m/s^2)	**CP (From Nose Cone)**	**CG (From Nose Cone)**	**Stability Calibre**

Final Sign Off		
Signing confirms the checker/s are certain all information filled out within this document is correct		
Name/Signature	**Date**	**Time**

Procedure	**Initial and/or Value**
Observation of rocket flight and expected condition **Write one of the following:** expected nominal, expected minor damage or expected excessive damage	
All recovery team members in proper attire according to Range Standard Operating Procedures	
Recovery team has water supply based on Range Standard Operating Procedures **Note:** 1 gallon per recovery operation per person is required	
Flier and recovery team reported to MCC	
Recovery team fill out "recovery personnel Information"	
Team has Communications and Tracking Pack	
Authorization to depart MCC received	
Record "Radio #"	

Record departure time	
Initial radio call made to MCO made verifying Communications and Tracking Pack is Functioning properly. Received confirmation communication and tracking features are functioning nominally. **Note:** Recovery team must return to MCC if features are not functioning nominally	
Note: Every radio call to the MCC shall include the following: **Unique Identification:** The calling team's issued Communication & Tracking Pack's "Radio #" **Walk Direction:** The calling team's current direction of travel using eight-point cardinal coordinates (The MCO will assist the team in orienting themselves prior to their departure from the MCC.) **Health & Status:** Any other comments relevant to the team's progress and capacity to proceed A radio call MUST be made to MCC within **15 minutes** of the last or as requested by MCO. **Note:** Calls should be made within **10 minutes** of the last	
Recovery Team locates rocket and initiates a regular radio call to indicate to the MCC that the rocket has been located	
Warning: Appropriate PPE used (face shield and safety glasses) Launch Lead approaches rocket	
Switch off Raven then Telemetrum. Rocket is **now Safed**	
Switch off Tele GPS	
Switch off Airbrake	
Warning: If unburnt energetics are found within the charge wells, remove unburnt charge, place in separate recovery bag with no electronics for inspection and disassembly back at site Inspect charge wells for unburnt energetics.	
All ejection charge wires cut and twisted together	
All charge wells unscrewed	
All charge wells placed in disposable bag	
Launch Lead verifies it is safe for the recovery team to approach	
Recovery Lead photographs all components of the rocket	
Components of rocket collected	
Recovery team initiates a regular radio call indicating to the MCC that are ready to return to MCC	
Record time when Recovery Team has returned to MCC	
Communications and tracking pack returned	
Rocket presented for post-flight evaluation	

Assist Mission Manager with recording "Measured Apogee as reported by the on-board of official altitude logging system"	
Recovery team and mission manager agree all information has been recorded **Note:** MCO maintains custody of the stapled document set	
Rocket disassembled and inspected with team and packed into boxes	

Additional Recovery Comments

Purpose	The purpose of this check-sheet is to track and assure that all documentation required for launch is complete and the IREC 2026 rocket is correctly integrated on the pad and is ready for launch
Applicability	IREC 2026 Rocket "Wedgetail"
Notes	"IREC26-CI000 Pre-Assembly Checklist" and "IREC26-CI001 Assembly Checklist" must be completed prior to this checklist If a scrub occurs "IREC26-CL005 Failure to Launch Checklist" must be completed

Flight Information			
Predicted Apogee (m)	**Max Velocity (m/s)**	**Velocity Off Rail (m/s)**	**Total Weight (kg)**
Max Acceleration (m/s^2)	**CP (From Nose Cone)**	**CG (From Nose Cone)**	**Stability Calibre**

Final Sign Off			
Signing confirms the checker/s are certain all information filled out within this document is correct			
Name/Signature	**Date**	**Time**	**Ready to Launch (Yes or No)**

Procedure	**Initial and/or Value**
Rocket in flight ready condition. "IREC26-CI000 Pre-Assembly Checklist" and "IREC26-CI001 Assembly Checklist" completed **Note: "**Flight ready" does not mean "armed"	
FSR completed and rocket is marked as "nominal" or "resolved" on FSR Resolution form	
FSR Resolution form completed	
Flight card completed	
Rocket will be taken in a "safe" state to LCO inspection area with FSR Resolution form, Flight Card, motor ignitors, and all completed checklists.	
LCO inspection completed with inspector initial and "Completion" marked. LCO takes possession of flight card and staples atop FSR Resolution form **Note:** Flight cards and attached FSR from will be taken to the on duty LCO at the LCC by launch operations team members staffing the LCO	

Turn on Airbrake		
Launch Team receive permission to depart to the Launch Area		
Team Launch Lead has possession of two motor igniters		
Rocket has all shear pins after transportation to the pad. **Four aft** and **eight forward**		
Rocket loaded onto launch rail		
Launch railed erected		
Tele GPS turned on. Lock confirmed with ground station		
Confirm Airbrake GPS lock.		
Confirmation from IREC personnel that energetics can be armed		
All non-essential personnel have evacuated the pad		
Warning: The rocket is in an armed state following the Primary Raven being switched on. If a scrub occurs "IREC26-CL005 Scrub Checklist" must be completed		
Raven switched on. Audio confirmation it is on		
Telemectrum switched on. Audio confirmation it is on		
Both flight computers connected to app and showing nominal readings		
Measure and record all three battery voltages	**AV Battery 1: (V)**	
	AV Battery 2: (V)	
	AV Battery 3: (V)	
All avionics stack battery voltages **more than 4.0V** and **less than 4.3V**		
Check Telemetrum GPS lock		
Permission granted by a Range Manager to install igniters		
Igniters installed in motor		
Firing line is **not "hot"**. Leads touched together and no sparks are observed		
Igniters connected to firing line		
All team personnel have evacuated from the pad		
Tele and Telemetrum GPSs have lock		

Purpose	The purpose of this check-sheet is to track and assure that the sub-assemblies of IREC2026, that can be pre-assembled, have been correctly assembled before launch day
Applicability	IREC 2026 Rocket "Wedgetail"
Notes	All fasteners must be snug tight (slightly over hand tight)

Final Sign Off			
Signing confirms the checker/s are certain all information filled out within this document is correct			
Name	**Date**	**Time**	**Ready to Fly (Yes or No)**

Procedure	**Initial and/or Value**
Payload Pre-assembly	
Place fan into fan holder and tighten nuts with 5.5 socket **WITH LOCTITE APPLIED**	
Verify PCB securely attached to battery lid	
Attach Ballast to bottom using 2x M3x20 bolts (YELLOW). **WITH LOCTITE APPLIED**	
Verify each battery pack is at least 8V	
Verify voltage difference between packs is less than 30mV	
Verify MOLEX connectors secure to PCB	
Force blu-tack on side walls and bottom of battery container	
Force blu-tack on battery sleeves	
Using moderate force, place batteries into sleeves such that the blu-tack can hold ensuring xT-30 Connecters are exposed	
Place sleeved batteries into container and pack with blu-tack and ripped paper and pack JST Connectors between batteries	
Verify batteries don't move when shaken and verify battery sleeves can be gripped with pliers	
Erase flash – Run serial script using ST Link OR USBC Programming	

Flash payload code	
Verify blue light on board is solid ON	
Remove ST-Link OR USBC	
Mount battery container with 4x M3x20 (BLUE) **WITH LOCTITE APPLIED**	
Plug in Peltier Molex (GREEN)	
Plug in Screw Switch Molex (BLUE)	
Plug In Thermister Molex (2x6)	
Plug in battery xT-30 Connectors	
Ensure Screw Switch Bolt is backed out AWAY from contact	
Plug in short Molex (RED)	
Verify fan turns on for short period – **N/A for Serpentine**	
Verify STM Blue LED is solid ON	
Mount battery lid to structure using 4x M3x20 (BLUE) **WITH LOCTITE APPLIED**	
Place medcontainer into insulation	
Position insulation thermistors, and place thermistor holder parts in place and hand tighten bolts. • Position inside thermistor lock [ ] • Position lock spacer in place on insulation [ ] • Hand tighten nylon bolts [ ] • Use 5.5 socket to slightly over tighten the bolts [ ]	
Apply thermal paste to heatsink – **N/A For Serpentine**	
Press blank side of Peltier into heatsink – **N/A For Serpentine**	
Wipe away excess thermal paste – **N/A For Serpentine**	
Instal insulation subassembly using 3x M4x25 (PURPLE) bolts and 1x M4x20 (RED) bolt. **WITH LOCTITE APPLIED**	
Lock additional thermistors in place using brackets with a dab of superglue	

Use cable ties to neaten wires		
Ensure all fasteners work		
Avionics Pre-assembly		
Fasten flight computers to their respective PDBs nsuring F.C are connected to their appropriate power connectors.		
Verify that all steel blocks are in the right position and eye nuts are properly fastened.		
Main side bulkhead: Fasten spacer, 'M' PCB and PDB mounts with M4x30 charge wells and a M4x25.		
Drogue side bulkhead: Fasten spacer, 'D' PCB, PDB mount and battery holder with M4x35 charge wells and a M4x30		
Place charged batteries in holder.		
Measure and record all three battery voltages. Ensure voltages are measured from the Molex connector to confirm battery connection	**AV Battery 1**	
	AV Battery 2 (V):	
	AV Battery 3 (V):	
All avionics stack battery voltages are **more than 4.0V** and **less than 4.3V**		
Feed Molex plugs through the lid.		
Secure Battery lid with M3x16		
Fasten PDB with F.C to mounts with M3x10 ensuring all screw switches are facing toward main bulkhead.		
Secure Antennas through battery holder.		
Connect all coloured Molex plugs to corresponding Molex connectors.		
Verify screw switches are in the OFF position.		
Plug battery Molex's' into their assigned F.C PDB.		
Confirm ONE red LED turns on for each board.		

Disconnect all batteries.	
Fasten shroud to the steel block mounting screws.	
Verify all screw switches are accessible.	
Forward and Aft bulkhead fasteners snug tight with Loctite paint mark visible	
All avionics stack fasteners snug tight	
Molex shorting plugs placed in all ejection charge molexs	
Ignitor leads colour coded	
Airbrake Pre-Assembly	
Include Assembly Instructions	
All three M3x16 fasteners snug tight through airframe into **aft end of airbrake frame** using **Loctite** and **Primer**	
All three M3x16 fasteners snug tight through airframe into **forward end of airbrake frame** using **Loctite** and **Primer**	
All three M3x16 fasteners snug tight through airframe into **middle of airbrake frame** using **Loctite** and **Primer**	
Install **rail buttons** into airframe **Loctite** and **Primer**, ensure alignment using a section of rail	
Airbrake flaps installed and all fasteners snug tight	
Airbrake screw switches easily accessible through air frame	
Recovery Pre-Assembly	
Drogue parachute packed into deployment bag according to "NU Rocketry "Prometheus" Parachute Packing Document"	
Confirm drogue parachute has recovery hardware attached. Swivel -> 1/8" quick link and all components have Loctite applied	
Main parachute packed into deployment bag according to "NU Rocketry "Prometheus" Parachute Packing Document"	
Confirm main parachute has recovery hardware attached. M6 Jaw -> Swivel -> M6 Jaw and all components have Loctite applied	
Long end (25ft) loop of **main** shock cord attached to **lower main parachute quick link**	
Long end (25ft) loop of **drogue** shock cord attached to **lower drogue parachute quick link**	
Centre loop of **main** shock cord attached to **nose cone anchor** with an **M8 quick link**	
Short end (13.3ft) loop of **drogue** shock cord attached to **aft anchor** with an **M8 quick link**	

Note: A band of masking tape will already be on the drogue parachute and bag at this point	
Attach main shock cord to eye nut/threaded rod assembly	
Fasten eye nut/threaded rod assembly into nosecone anchor using **Loctite** and **Primer**	

Purpose	The purpose of this check-sheet is to track and assure that if IREC 2026 rocket has a CATO or other launch mishap, it is safely dealt with.
Applicability	Wedgetail
Notes	

Flight Information			
Predicted Apogee (m)	**Max Velocity (m/s)**	**Velocity Off Rail (m/s)**	**Total Weight (kg)**
Max Acceleration (m/s^2)	**CP (From Nose Cone)**	**CG (From Nose Cone)**	**Stability Calibre**

Final Sign Off		
Signing confirms the checker/s are certain all information filled out within this document is correct		
Name/Signature	**Date**	**Time**

Procedure	**Initial and/or Value**
All rocket components landed	
Launch Team given permission to go to Launch Area to inspect remanent of rocket	
Warning: Appropriate PPE used (face shield and safety glasses) **If unburnt energetics are found within the charge wells, remove unburnt charge, place in separate bag with no electronics for inspection and disassembly back at site** **Team Launch Lead to approach rocket ONLY IF COMETITION OFFICIALS ALLOW (RSO)**	
Team Launch Lead approaches rocket remnants, switch off Raven and Telemetrum. Rocket is **now Safed** **Note:** If applicable turn off Tele GPS, Airbrake and payload	
Launch Lead inspected charge wells for unburnt energetics	
Remove all ejection charge wires from terminals, cut, strip and twist each ejection charges wires together	
All charge wells unscrewed	

All charge wells placed in disposable bag	
Photographs of remnants taken	
Components of rocket collected **Note:** Discuss CATO with RSO and provide all relevant data/evidence)	

Purpose	The purpose of this check-sheet is to track and assure that conditions for IREC 2026 Rocket to launch are met
Applicability	Wedgetail
Notes	"IREC26-CL001 Assembly Checklist" and "IREC26-CL002 Preflight Checklist" must be completed prior to this checklist This checklist can be used as reference for filling out the flight card If rocket fails to launch "IREC26-CL005 Failure to Launch Checklist" must be completed

Flight Information			
Predicted Apogee (m)	**Max Velocity (m/s)**	**Velocity Off Rail (m/s)**	**Total Weight (kg)**
Max Acceleration (m/s^2)	**CP (From Nose Cone)**	**CG (From Nose Cone)**	**Stability Calibre**

Final Sign Off			
Signing confirms the checker/s are certain all information filled out within this document is correct			
Name/Signature	**Date**	**Time**	**Ready to Fly (Yes or No)**

Procedure	**Initial and/or Value**
"IREC2026-CL001 Assembly Checklist" completed	
"IREC2026-CL002 Preflight Checklist" completed	
Tele GPS has lock	
Telemetrum GPS has lock	
Wind does not exceed **32km/h** **Note:** Stability off the rail at this wind speed is 1.2 and is 1.5 after 0.1 seconds	
Air temperature does not exceed **45 °C**	
Team Launch Lead present at LCC	
Rocket Launched	

Drogue deployment successful	
Main deployment successful	
Rocket landed	
Additional Comments Regarding Launch	

Purpose	The purpose of this check-sheet is to track and assure that the IREC 2026 rocket is correctly dealt after failing to launch
Applicability	Wedgetail
Notes	A failure to launch can be from a scrub or an igniter failure. For a scrub either the Launch Director, RSO, LCO, or Flier determine the rocket should be removed from the launcher and returned to the Rocket Assembly Area(s) **(Stage 1 and 3 Only)** For an igniter failure, the igniter is replaced **(Stage 1 and 2 only)**

Flight Information			
Predicted Apogee (m)	**Max Velocity (m/s)**	**Velocity Off Rail (m/s)**	**Total Weight (kg)**
Max Acceleration (m/s^2)	**CP (From Nose Cone)**	**CG (From Nose Cone)**	**Stability Calibre**

Final Sign Off		
Signing confirms the checker/s are certain all information filled out within this document is correct		
Name/Signature	**Date**	**Time**

Procedure	**Initial and/or Value**
Stage 1 – Igniter Removal	
Launch Team receive permission to return to pad	
Launch team back on pad	
Team consulted with a Range Manager before initiating next steps	
Warning: For igniter removal procedures ensure the correct PPE is used (Face shield and safety glasses)	
Firing Line disconnected	
Igniter shunted	
Igniter removed	
Stage 2 – Igniter Installation	

Warning: For safing procedures ensure the correct PPE is used (Face shield and safety glasses) and all ladders used are stabilized by a second person and are under supervision of a pad manager **Note:** Stage 2 to only be completed if failure to launch was caused by igniters. Proceed to Stage 3 if rocket requires to be taken back to Assembly Area	
Audio Confirmation both Raven and Telemetrum are on	
Both Flight computers connected to app and showing nominal readings	
Permission granted by a Range Manager to install igniters	
Igniters installed in motor	
Firing line is **not "hot"**. Leads touched together and no sparks are observed	
All non-essential personnel have evacuated the pad **Note:** Only Launch Lead, Safety Lead and Faculty Advisor shall remain at the pad	
Igniters connected to firing line	
All team personnel have evacuated from the pad	
Tele Gps and Telemetrum have GPS lock	
Note: After replacing igniters, Launch operations must restart at "IREC26-5CL003 Launch Checklist"	
Stage 3 – Scrubbed Launch	
Warning: For safing procedures ensure the correct PPE is used (Face shield and safety glasses) and all ladders used are stabilized by a second person and are under supervision of a pad manager	
Switch off Raven and then Telemetrum. Rocket is **now Safed**	
Switch off Tele GPS	
Switch off Airbrake	
Switch off payload	
Launch rail lowered	
Rocket removed from launch rail and taken back to Assembly Area	
Is the rocket launching during the same day **(write Y or N)** **Note:** If **Yes**, ejection charges do not have to be removed and disassembled. Proceed with following steps if **No**	
All shear pins removed	
Forward bulkhead E-matches disconnected from Molex	
Aft bulkhead E-matches disconnected from Molex	
Warning: Ensure appropriate PPE is used (face shield and safety glasses) Ejection charges disassembled as per "Black Powder Preparation Standard Operating Procedure and Risk Assessment"	
Note: After a scrubbed launch, stage 5 of the "IREC26-CL001 Assembly Checklist" and checklists ahead must be recompleted	

Purpose	The purpose of this check-sheet is to track and assure that the IREC 2026 rocket is correctly assembled for flight
Applicability	IREC 2026 Rocket
Notes	This checklist can be used as a reference for assembly for FSR. This sheet includes energetics integration in the rocket which will not be completed prior to FSR on the first day of the IREC 2026 competition. All fasteners must be snug tight (slightly over hand tight)

Flight Information			
Predicted Apogee (m)	**Max Velocity (m/s)**	**Velocity Off Rail (m/s)**	**Total Weight (kg)**
Max Acceleration (m/s^2)	**CP (From Nose Cone)**	**CG (From Nose Cone)**	**Stability Calibre**

Final Sign Off			
Signing confirms the checker/s are certain all information filled out within this document is correct			
Name/Signature	**Date**	**Time**	**Ready for Preflight Checks (Yes or No)**

Procedure		**Initial and/or Value**
Stage 1 – Payload Stack Inspection		
Note: Primary structure of the payload stack is pre-assembled prior to IREC. This section is for verification it is ready to be integrated into the airframe.		
Plug in Battery Molex		
Measure and record payload battery voltages. Ensure voltages are measured from the Molex connector to confirm battery connection	**PL Battery (V):**	
Payload battery voltage is **more than 7.8V** and **less than 8.4V**		
Plug in switching Molex		
Power led's are on		
Confirm Fan switches on then off		
Blue LED is solid **on**		
Fan is turned **off**		
Screw switch in **off** position		

All payload wires retained within structure (won't clash with airframe wall)		
Stage 2 – Airbrake Stack Inspection		
Note: Primary structure of the airbrake stack is pre-assembled prior to IREC. This section is for verification it is ready to be integrated into the airframe.		
Turn on airbrake		
Measure and record AB battery voltages. Ensure voltages are measured from the Molex connector to confirm battery connection	**AB Battery (V):**	
Airbrake battery voltage is **more than 12.4V** and **less than 12.9V**		
All airbrake wires retained within structure (won't clash with airframe wall)		
Manually actuate flaps using allen key to check for inteference.		
Stage 3 – Avionics Stack Inspection		
Note: Primary structure of the avionics stack is pre-assembled prior to IREC. This section is for verification it is ready for integration into the airframe.		
Plug in all **3 molex** battery connectors		
Confirm **1 RED LED** illuminated on each PDB board		
All Avionics wires retained within structure		
Avionics shroud attached to avionics structure		
Avionics screw switches easily accessible through air frame		
Stage 4 – Aft Tube Assembly		
Mount Payload to Aft coupler using **ALL four M4x30** fasteners using **Loctite** and **Primer**		
Check the Payload screw switch hole is aligned through the coupler		
Line up coupler witness marks with aft tube witness marks		
Insert coupler into aft tube		
ALL four M5x14 fasteners snug tight through airframe into **aft coupler** using **Loctite** and **Primer**		
Payload screw switches easily accessible through air frame		
Stage 5 – Pre-BP Assembly		
Confirm Airbrake has valid GPS lock		
Decision made regarding Airbrake being turned off		
All fasteners required for final assembly are primed with Loctite primer.		
Stage 6 – Ejection Charge Preparation and Final Assembly		
Warning: Ensure appropriate PPE is used (face shield and safety glasses)		

Assemble black powder ejection charges according to the "Black Powder Preparation Standard Operating Procedure and Risk Assessment"			
Primary Drogue Charge (aft) **(2.3g nominal)** Molex Colour: **Brown, Purple**	**No. of E-matches:**		
	Grams of black powder:		
Backup Drogue Charge (aft) **(2.8g nominal)** Molex Colour: **Yellow**	**No. of E-matches:**		
	Grams of black powder:		
Primary Main Charge (forward) **(2.3g nominal)** Molex Colour: **Blue, Pink**	**No. of E-matches:**		
	Grams of black powder:		
Backup Main Charge (forward) **(2.8g nominal)** Molex Colour: **Red**	**No. of E-matches:**		
	Grams of black powder:		
Primary main and **backup main** ejection charges integrated into forward charge wells according to "Black Powder Preparation Standard Operating Procedure and Risk Assessment"			
Primary main E-matches connected to **forward molex connectors** using **Blue and Pink** Molex			
Backup main E-matches connected to **forward molex connectors** using **Red** Molex			
Excess ignitor lead cable **facing towards Molex** to avoid interference with airframe wall			
Primary drogue and **backup drogue** ejection charges integrated into aft charge wells according to "Black Powder Preparation Standard Operating Procedure and Risk Assessment"			
Primary drogue E-matches connected to **aft molex connectors** using **Brown and Purple** Molex			
Backup drogue E-matches connected to **aft molex connectors** using **Yellow** Molex			
Excess ignitor lead cable **facing towards Molex** to avoid interference with airframe wall			
Centre Loop of **drogue** shock cord attached to **lower AV-Stack anchor** with an **M8 quick link** through alignment ring with Loctite added.			
Integrate AV-Stack into airframe ensuring correct orientation with purple marks facing towards sponsors, and steel blocks locking into alignment ring			
Partially remove AV-Stack			
Short end (15ft) loop of **main** shock cord attached to **upper AV-Stack anchor** with an **M8 quick link** with Loctite applied.			

Re-Integrate AV-Stack into airframe ensuring correct orientation with purple marks facing towards sponsors, and steel blocks locking into alignment ring	
Align nose cone with witness marks on inside of forward tube.	
Integrate nose cone into forward tube	
All four M4x8 fasteners snug tight through airframe into **forward end of AV-Stack** using **Loctite** and **Primer**	
All four M4x8 fasteners snug tight through airframe into **aft end of AV-Stack** using **Loctite** and **Primer**	
Shear pin holes aligned between nose cone and forward tube.	
Eight 4-40 shear pins connecting **nose cone and forward tube.** All shear pins have small amount of superglue under head.	
Remove masking tape from drogue parachute bag.	
Integrate drogue parachute into aft end of **forward tube** ensuring excess shock cord bundled and not interfering with airframe.	
Align forward tube witness marks with aft tube.	
Integrate forward tube onto aft tube.	
Four 4-40 shear pins connecting **forward tube and aft tube**. All shear pins have small amount of superglue under head.	
Ensure masking tape band is on Recovery Leader	
CG and CP measured and labelled	
Motor has been assembled according to Aerotech Instructions	
Motor inserted into aft section	
Retainer secured snug tight	

Appendix F – Engineering Drawings

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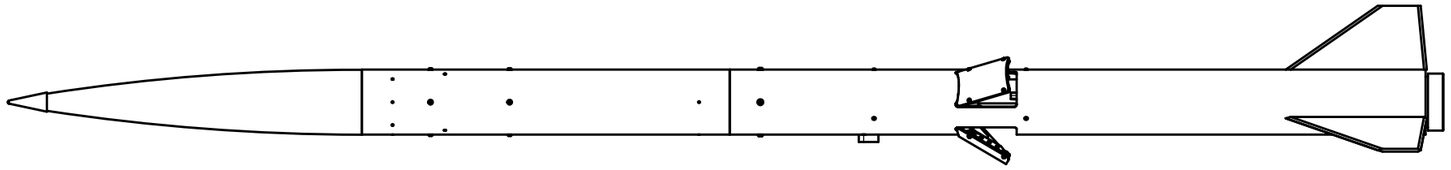
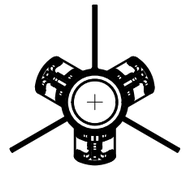
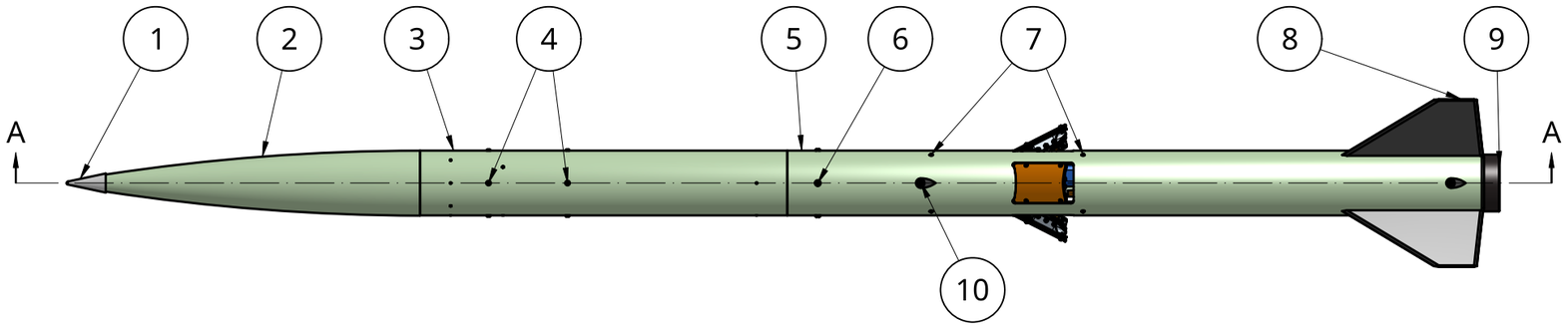
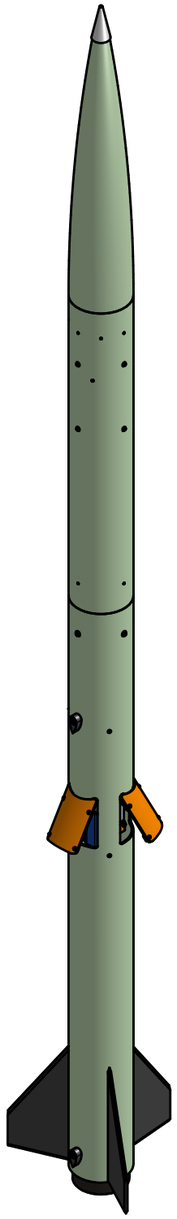
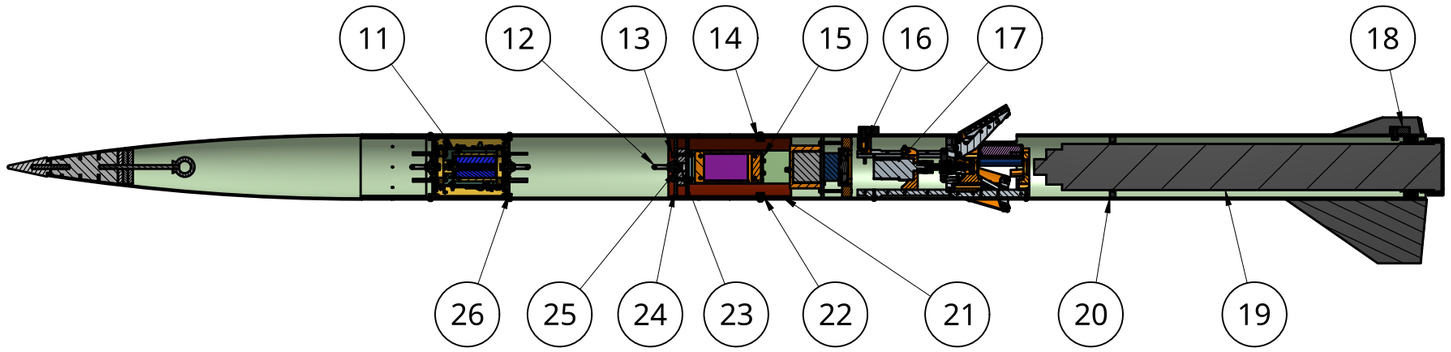
SEE PG.2, 3 FOR BOM











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REV	DESCRIPTION	BY	DATE	APPV'D

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TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: WEDGETAIL

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:10
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 1/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
1	NOSECONE ANCHOR ASSEMBLY	1	ASSEMBLY	MANUFACTURED	N/A
2	NOSECONE	1	PART	PURCHASED	FIBREGLASS
3	FOR TUBE	1	PART	MANUFACTURED	FIBREGLASS
4	SOCKET BUTTON HEAD SCREW M4x0.7 x 8	8	PART	PURCHASED	CLASS 12.9 STEEL Z/P
5	AFT TUBE	1	PART	MANUFACTURED	FIBREGLASS / CARBON FIBRE
6	SOCKET BUTTON HEAD SCREW M5X0.8 X 8	4	PART	PURCHASED	CLASS 12.9 STEEL Z/P
7	SOCKET BUTTON HEAD SCREW M3X0.8 X 8	6	PART	PURCHASED	CLASS 12.9 STEEL Z/P
8	FINS	3	PART	MANUFACTURED	CARBON FIBRE
9	MOTOR TUBE ASSEMBLY	1	ASSEMBLY	MANUFACTURED	N/A
10	HEX SOCKET COUNTERSUNK HEAD SCREW M6X1 X 20	2	PART	PURCHASED	CLASS 12.9 STEEL Z/P
11	AVIONICS STACK	1	ASSEMBLY	MANUFACTURED	N/A
12	M6 EYE NUT	1	PART	PURCHASED	CLASS 12.9 STEEL Z/P
13	HEX SOCKET HEAD CAP SCREW M4X0.70 X 30	4	PART	PURCHASED	GRADE 8 STEEL Z/P
14	HEX NUT M5X0.8	4	PART	PURCHASED	GRADE 8 STEEL Z/P
15	PAYLOAD ASSEMBLY	1	ASSEMBLY	MANUFACTURED	N/A
16	HEX NUT M6X1.00	2	PART	PURCHASED	GRADE 8 STEEL Z/P
17	AIRBRAKE ASSEMBLY	1	ASSEMBLY	MANUFACTURED	N/A
18	15X15 AERO RAIL BUTTON	2	PART	PURCHASED	ACETAL (DERLIN)

SEE PG. 3 FOR CONTINUED BOM

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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ENGINEERING CLOSE, CALLAGHAN NSW 2308
FOR ENQUIRIES PLEASE CONTACT
NU Rocketry Chief Engineer
EMAIL: tate.derrick@uon.edu.au

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DRAWN TO AS1100-1992
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DO NOT SCALE

CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: WEDGETAIL BOM 1/2

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY:		DRAWN	G. O'BRIEN		2026-04-08	SCALE: N/A
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 2/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
19	MOTOR CASE	1	PART	PURCHASED	ALUMINIUM
20	MOTOR TUBE ALIGNMENT RING	1	PART	MANUFACTUED	ABS
21	AFT COUPLER	1	PART	PURCHASED	FIBREGLASS
22	AFT COUPLER NUT HOLDER	1	PART	MANUFACTURED	ABS
23	HEX SOCKET HEAD CAP SCREW M6X1.00 X 15	1	PART	PURCHASED	CLASS 12.9 STEEL Z/P
24	AFT COUPLER BULKHEAD	1	PART	MANUFACTURED	G10 FIBREGLASS
25	M6 WASHER	1	PART	PURCHASED	STAINLESS STEEL
26	AVIONICS ALIGHNMENT PLATE	1	PART	MANUFACTURED	ABS

BOM CONTINUED FROM PG. 2

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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TOLERANCES

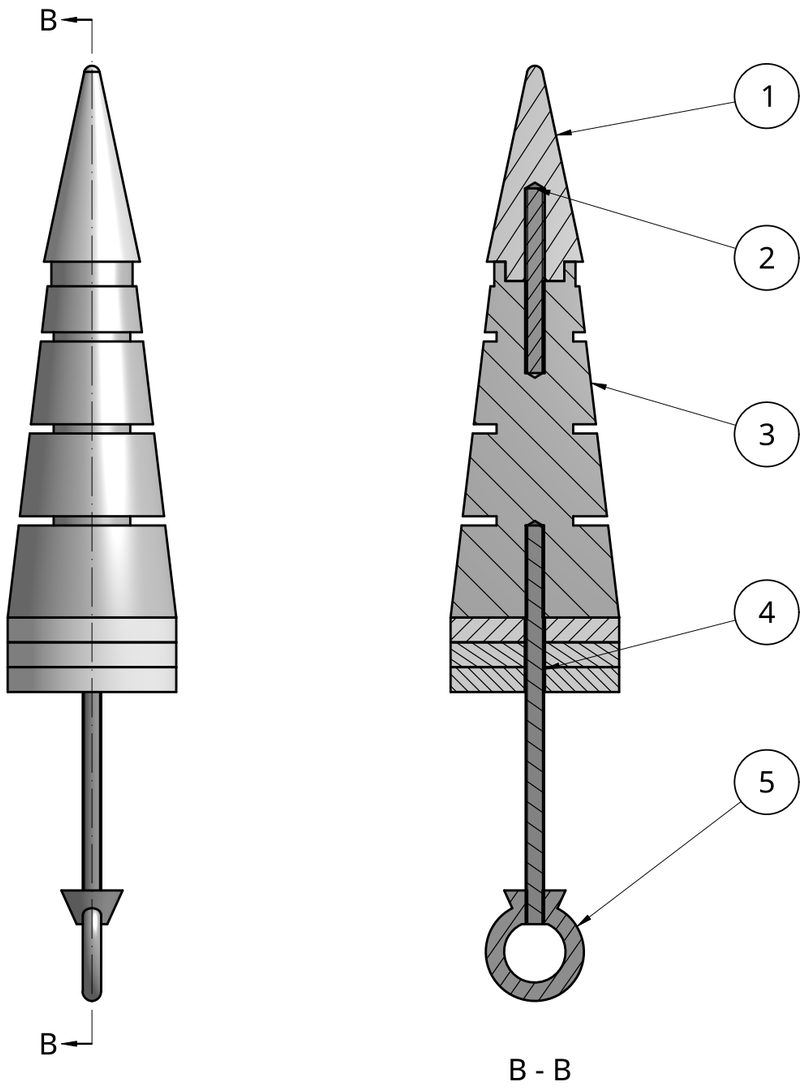
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: WEDGE TAIL BOM 2/2

PROJECT NO.: 67	APPV'D				DRAWING NO.:
FINISH: 3.2 UOS	CHK'D				REVISION:
QUANTITY:	DRAWN	G O'BRIEN		2026-04-08	SCALE: N/A
MATERIAL: N/A		NAME	SIGNATURE	DATE	SHEET: 3/66 A3

NO.	ITEM	QTY.	TYPE	SOURCE	MATERIAL
1	NOSECONE TIP	1	PART	PURCHASED	ALUMINIUM
2	UPPER THREADED ROD	1	PART	MANUFACTURED	STAINLESS STEEL
3	NOSECONE ANCHOR	1	PART	MANUFACTURED	ALUMINIUM 6061
4	LOWER THREADED ROD	1	PART	MANUFACTURED	STAINLESS STEEL
5	M6 EYE NUT	1	PART	PURCHASED	GRADE 8 STEEL Z/P



DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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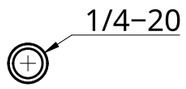
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

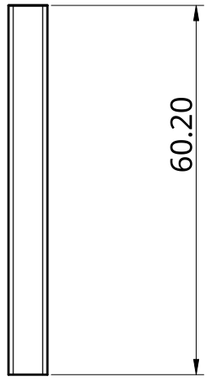
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: NOSECONE ANCHOR ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 4/66 A3



TO BE CUT FROM $\frac{1}{4}$ IN STANDARD ROD



DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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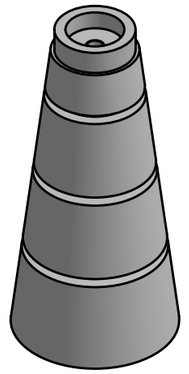
TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

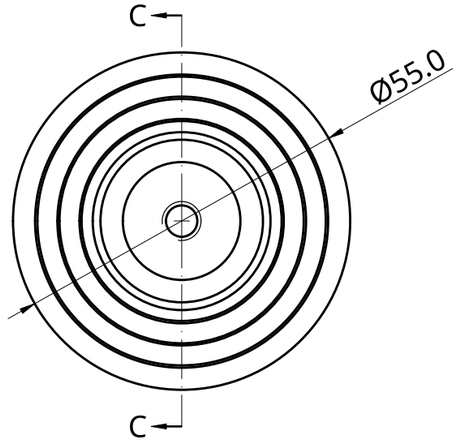
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

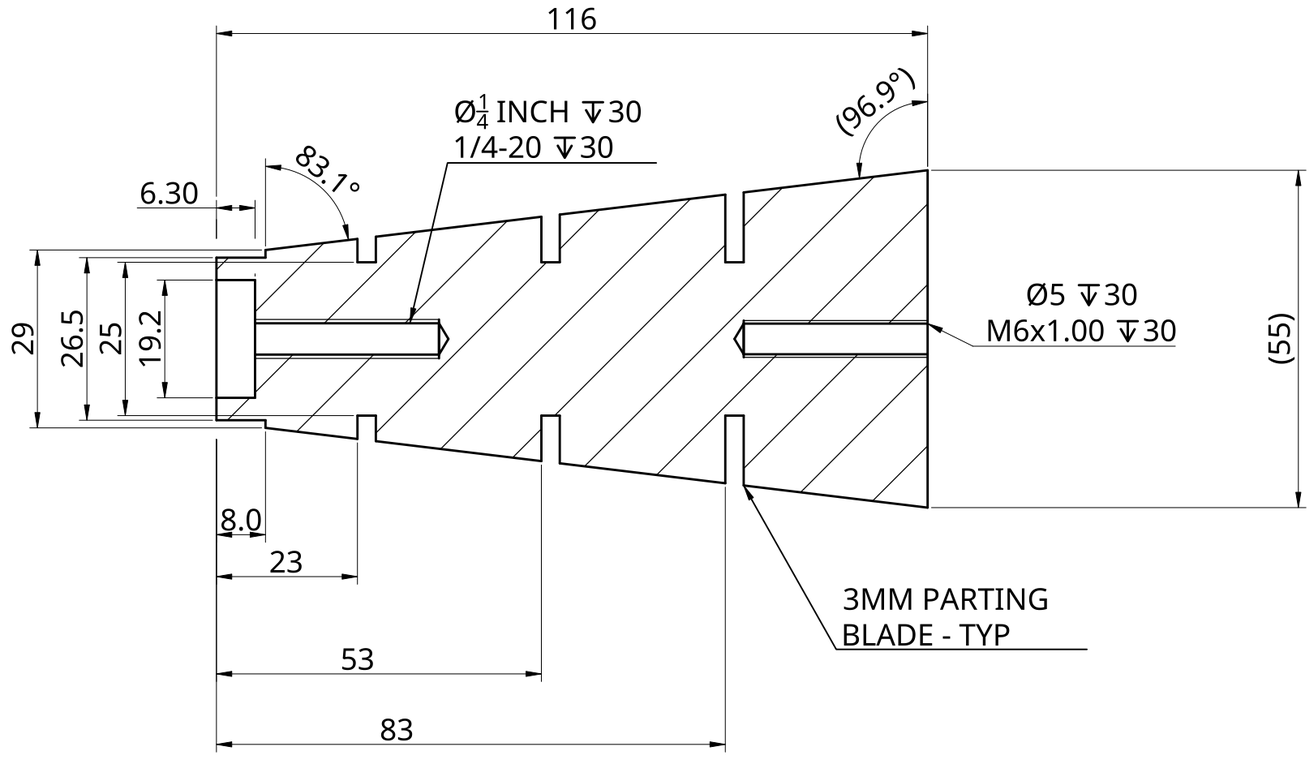
TITLE: UPPER THREADED ROD

PROJECT NO.: 67	APPV'D					DRAWING NO.:
FINISH: 3.2 UOS	CHK'D					REVISION:
QUANTITY: 1	DRAWN	G. O'BRIEN		2026-04-08		SCALE: 1:1
MATERIAL: STAINLESS STEEL		NAME	SIGNATURE	DATE	SHEET: 5/66	A3



1:2





C - C

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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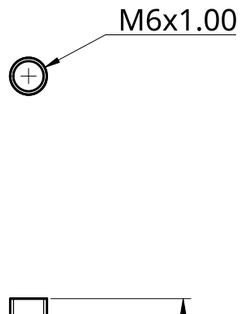
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: NOSE CONE ANCHOR

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ALUMINIUM -6061			NAME	SIGNATURE	DATE	SHEET: 6/66 A3



TO BE CUT FROM M6 STANDARD ROD

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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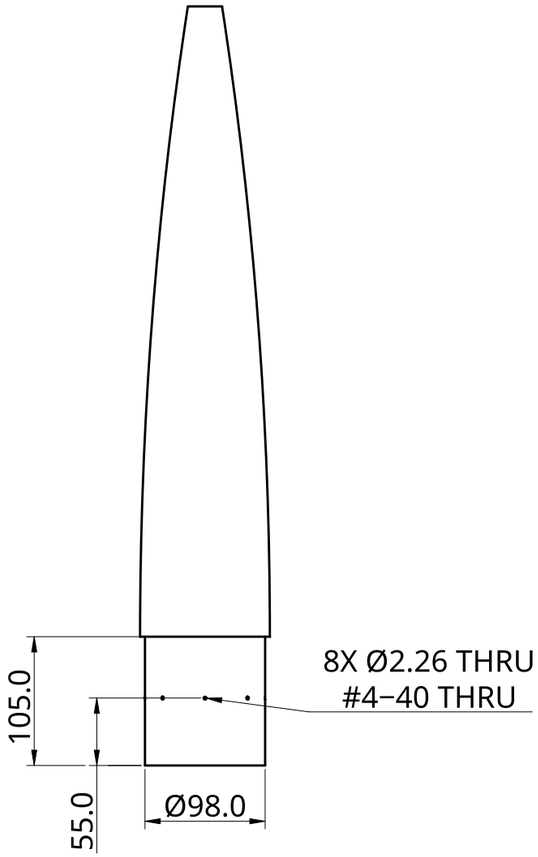
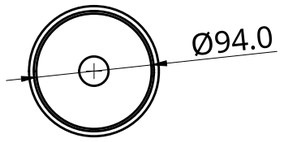
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

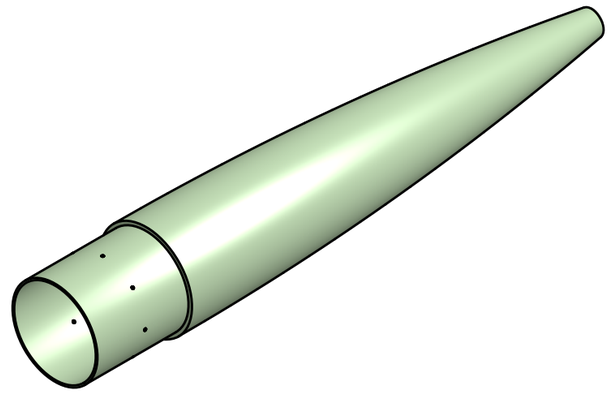
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: LOWER THREADED ROD

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: STAINLESS STEEL			NAME	SIGNATURE	DATE	SHEET: 7/66 A3







MANUFACTURING NOTE:
NOSECONE PURCHASED
COUPLER WAS COMBINED WITH EPOXY

DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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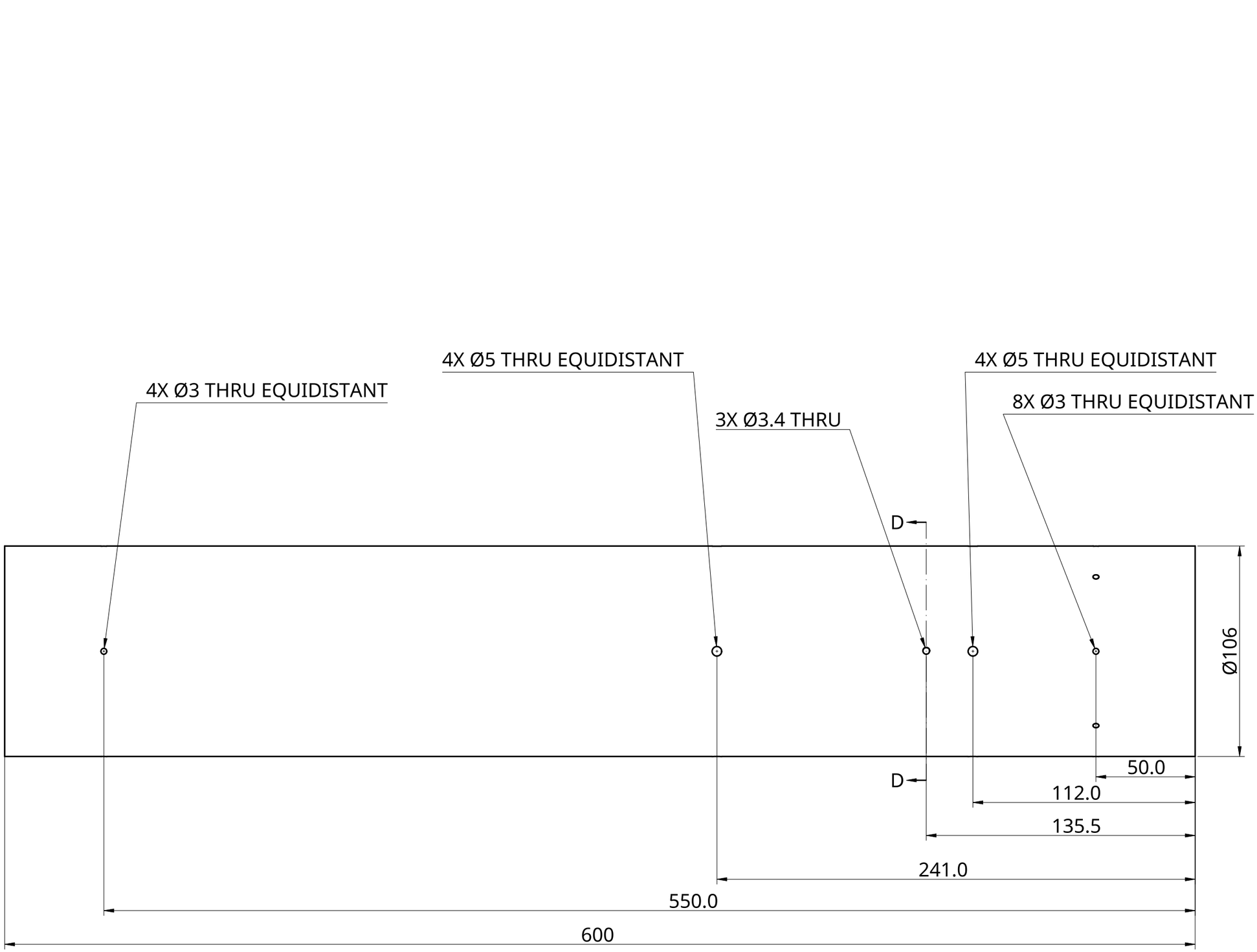
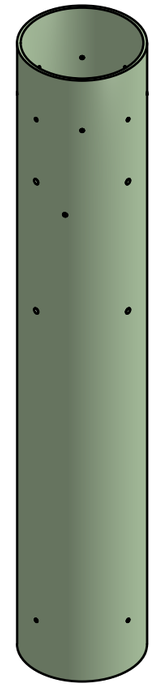
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: NOSE CONE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:5
MATERIAL: FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 8/66 A3





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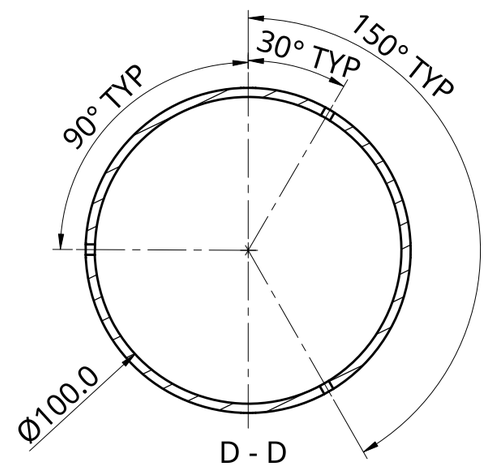

MANUFACTURING NOTE: THE SCREW SWITCH HOLES WERE DEFINED BY THE ASSEMBLED AVIONICS STACK

REV	DESCRIPTION	BY	DATE	APPV'D

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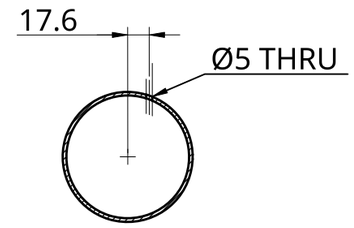
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

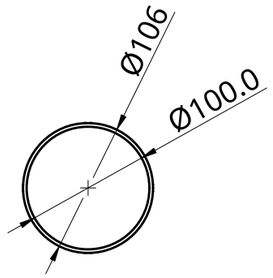
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FOR TUBE

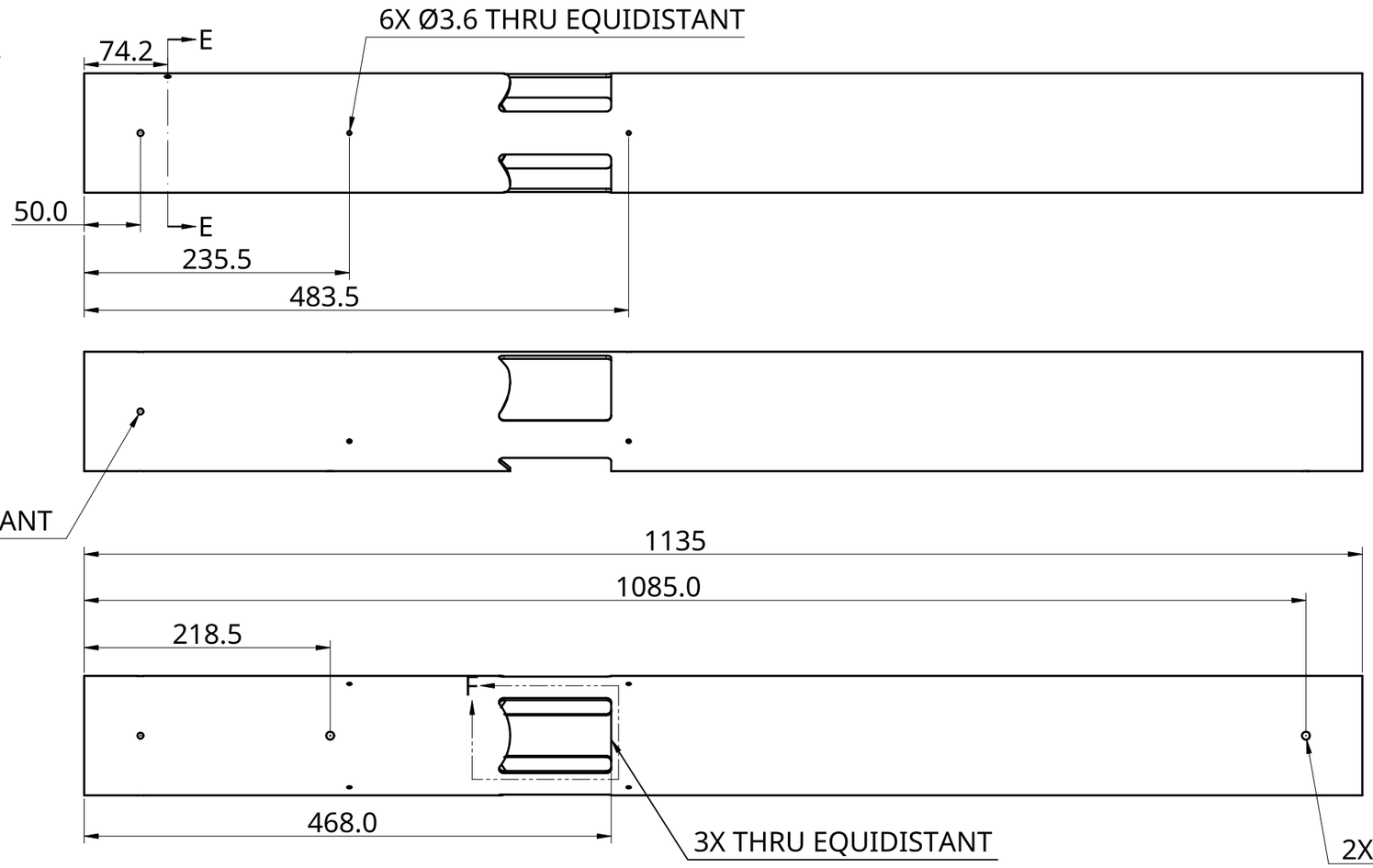
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 9/66 A3

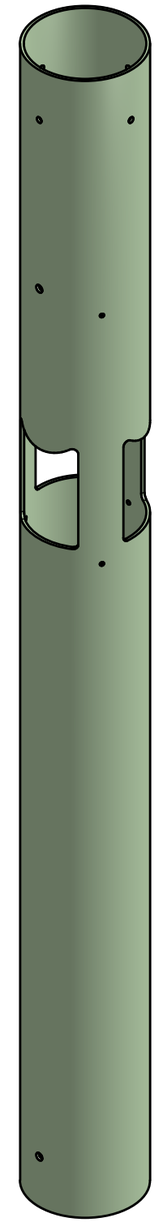


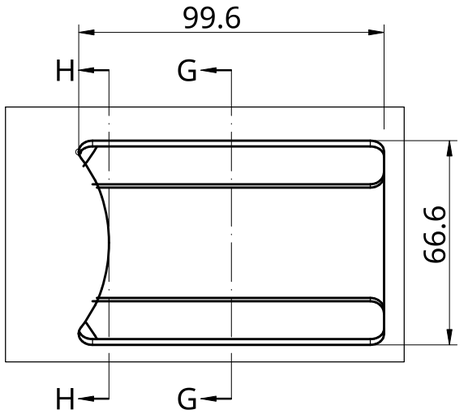
E - E



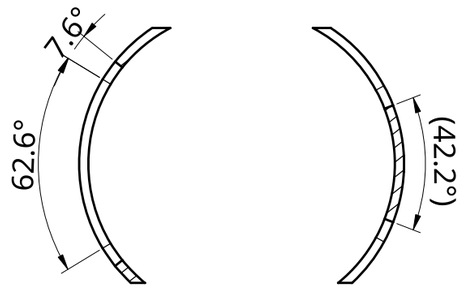
4X Ø5.3 THRU EQUIDISTANT



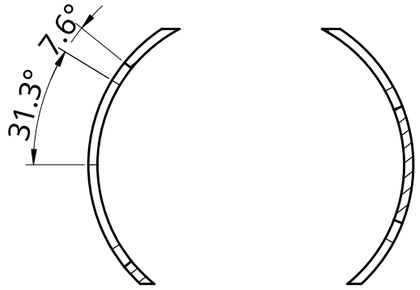




F
1:2



G - G
1:2



H - H
1:2

REV	DESCRIPTION	BY	DATE	APPV'D

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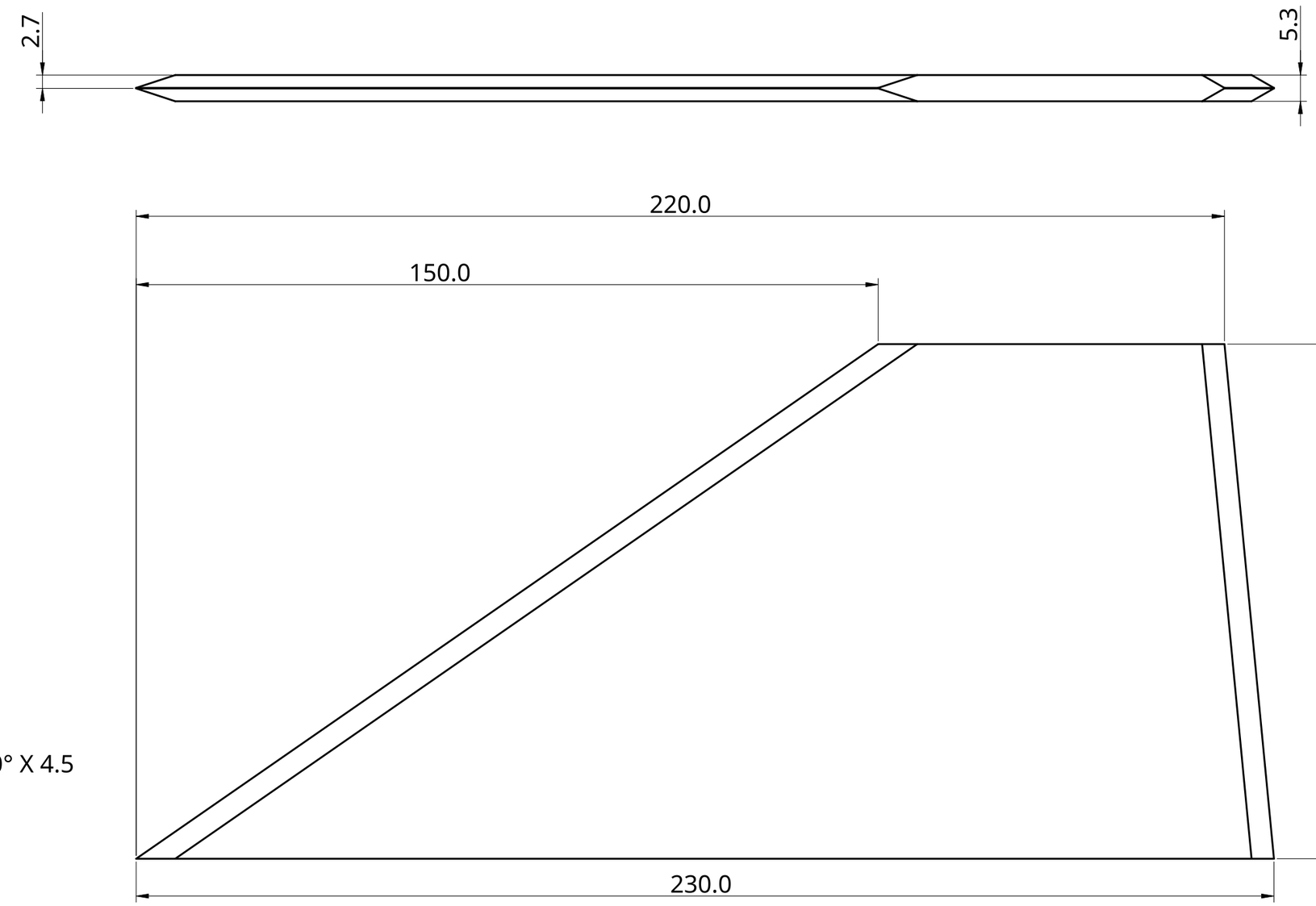
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AFT TUBE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:5
MATERIAL: FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 10/66 A3



ALL CHAMPHER 30° X 4.5

TO BE MANUFACTURED ON WATERJET CUTTER
FILE ALL ROUGH EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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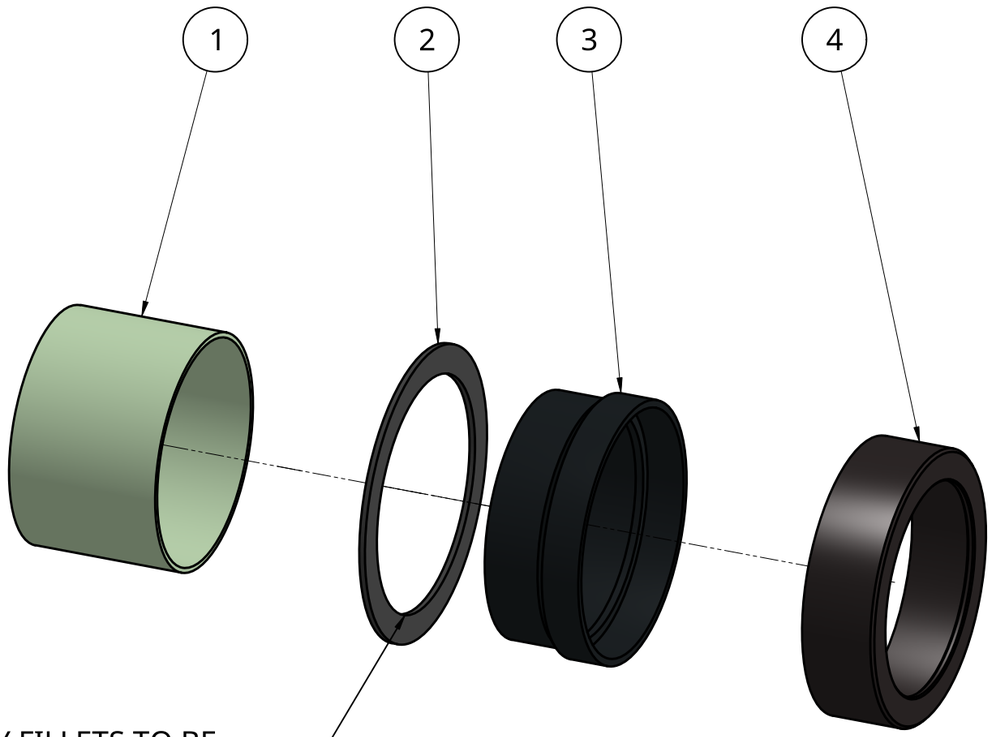
0.00±0.05

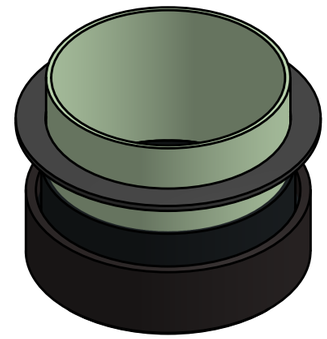
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FIN

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 3		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: CARBON FIBRE			NAME	SIGNATURE	DATE	SHEET: 11/66 A3

NO.	NAME	QTY	TYPE	SOURCE	MATERIAL
1	MOTOR TUBE	1	PART	PURCHASED	FIBREGLASS
2	MOTOR CENTERING RING	1	PART	MANUFACTURED	CARBON FIBRE
3	MOTOR RETAINER MOUNT	1	PART	PURCHASED	N/A
4	MOTOR RETAINER	1	PART	PURCHASED	N/A





2X 10MM EPOXY FILLETS TO BE USED

DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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DRAWN TO AS1100-1992
ALL DIMENSIONS IN MILLIMETRES
UNLESS OTHERWISE STATED
DO NOT SCALE

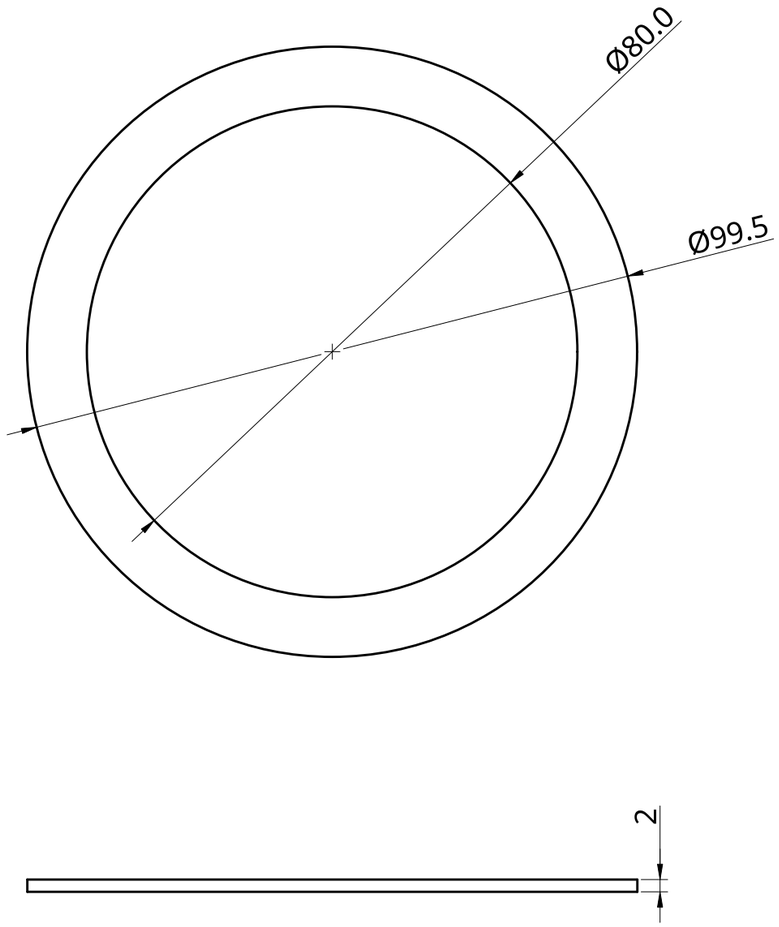
CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

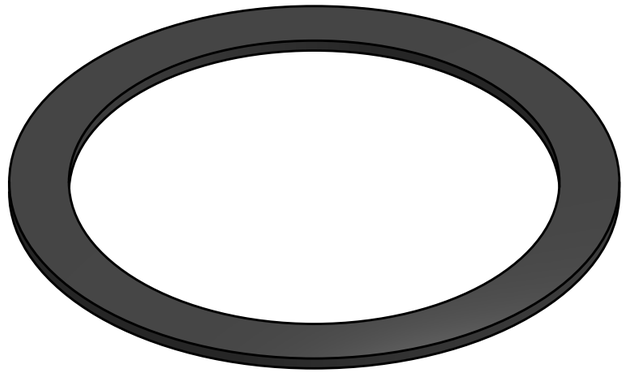
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: MOTOR TUBE ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 12/66 A3





TO BE MANUFACTURED ON WATERJET CUTTER
FILE ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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TOLERANCES

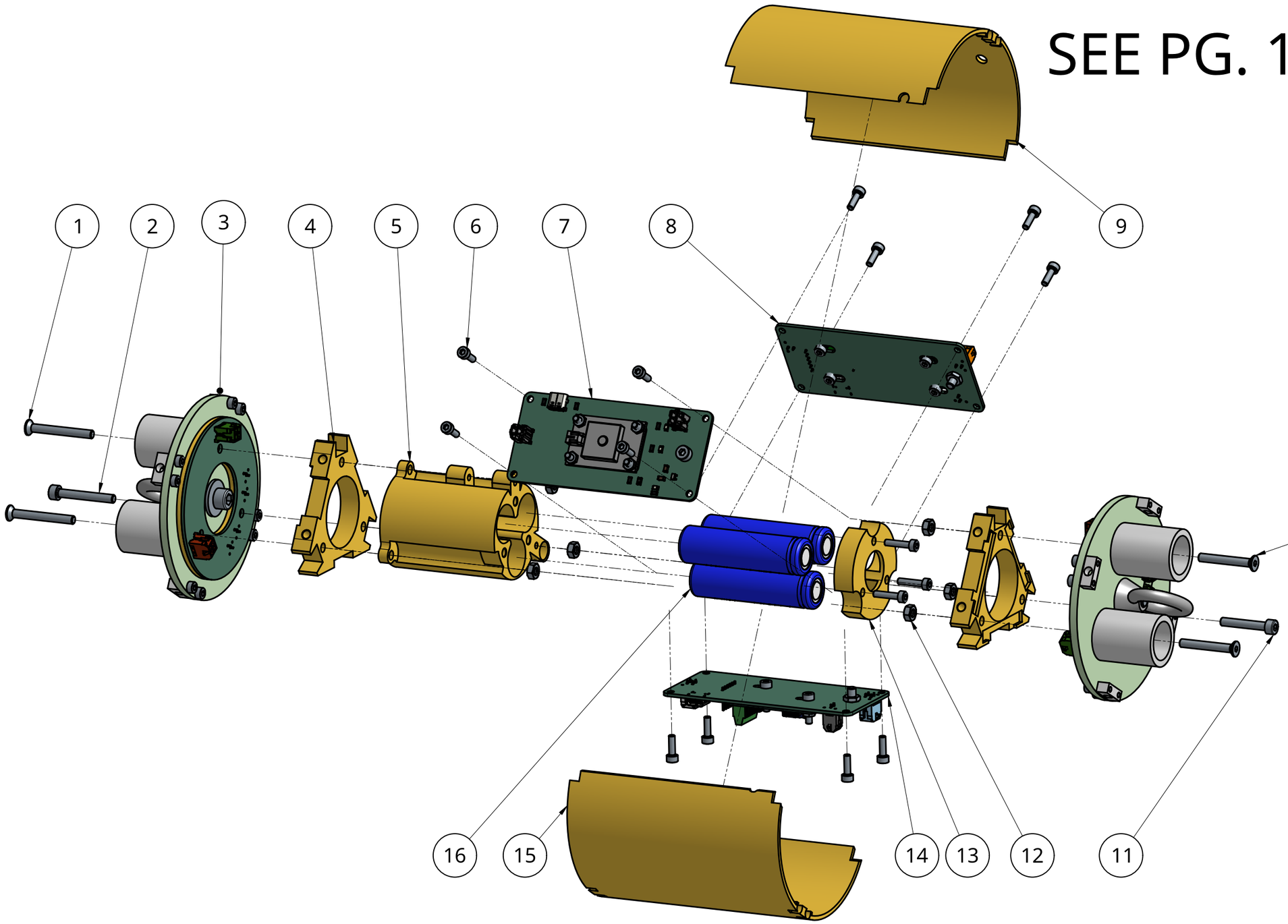
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

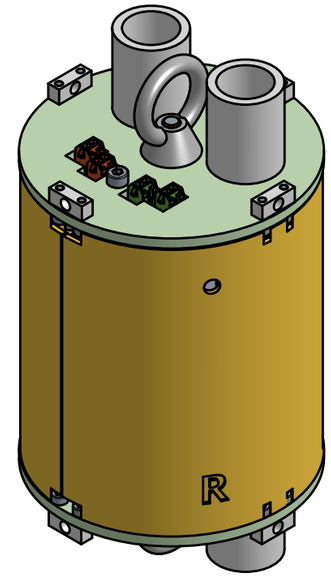
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: MOTOR CENTERING RING

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: CARBON FIBRE			NAME	SIGNATURE	DATE	SHEET: 13/66 A3

SEE PG. 12 FOR BOM





REV	DESCRIPTION	BY	DATE	APPV'D

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TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS STACK

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 14/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
1	HEX SOCKET COUNTERSUNK HEAD SCREW M4X0.70 X30	2	PART	PURCHASED	CLASS 12.9 STEEL Z/P
2	HEX SOCKET HEAD CAP SCREW M4X0.70 X 30	1	PART	PURCHASED	CLASS 12.9 STEEL Z/P
3	AVIONICS BULKHEAD ASSEMBLY	2	ASSEMBLY	MANUFACTURED	N/A
4	PDB MOUNT	2	PART	MANUFACTURED	ABS
5	AVIONICS BATTERY HOLDER	1	PART	MANUFACTURED	ABS
6	HEAD SOCKET HEAD CAP SCREW M3X0.50 X 14	12	PART	PURCHASED	CLASS 12.9 STEEL Z/P
7	TELEGPS PDB	1	ASSEMBLY	MANUFACTURED	N/A
8	TELEMENTRUM PDB	1	ASSEMBLY	MANUFACTURED	N/A
9	RHS AVIONICS SHROUD	1	PART	MANUFACTURED	ABS
10	HEX SOCKET COUNTERSUNK HEAD SCREW M4X0.70 X 25	1	PART	PURCHASED	CLASS 12.9 STEEL Z/P
11	HEX SOCKET HEAD CAP SCREW M4X0.70 X 25	1	PART	PURCHASED	CLASS 12.9 STEEL Z/P
12	HEX NUT M4X0.70	6	PART	PURCHASED	GRADE 8 STEEL
13	AVIONICS BATTERY LID	1	PART	MANUFACTURED	ABS
14	RAVEN PDB	1	ASSEMBLY	MANUFACTURED	N/A
15	LHS AVIONICS SHROUD	1	PART	MANUFACTURED	ABS
16	18650 BATTERY	3	PART	PURCHASED	N/A

REV	DESCRIPTION	BY	DATE	APPV'D

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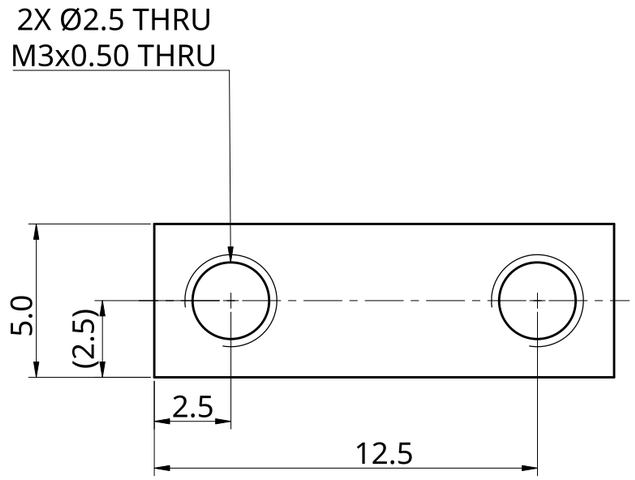
TOLERANCES

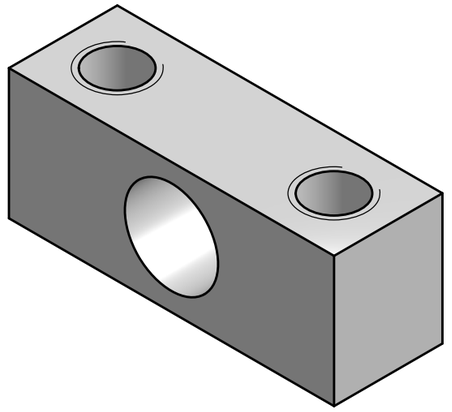
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

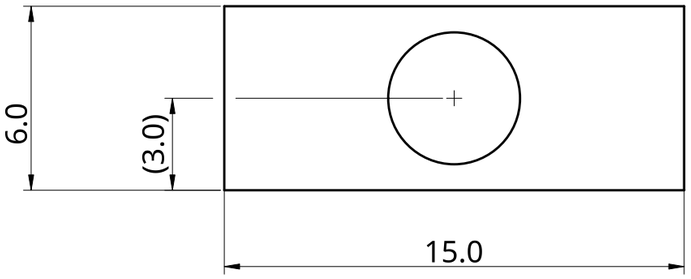
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS STACK BOM

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY:		DRAWN	G. O'BRIEN		2026-04-08	SCALE: N/A
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 15/66 A3







DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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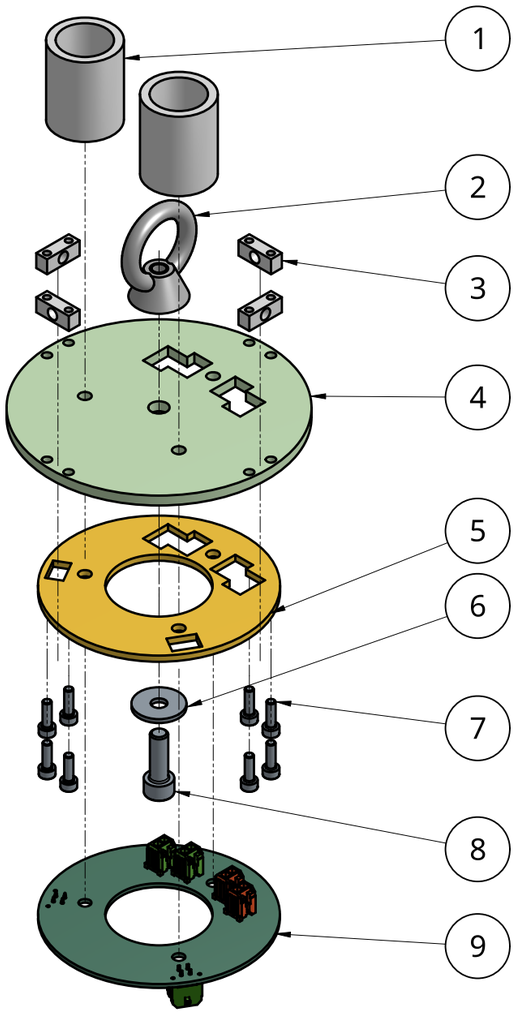
TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: BULKHEAD MOUNTING BRACKET

PROJECT NO.: 67	APPV'D					DRAWING NO.:
FINISH: 3.2 UOS	CHK'D					REVISION:
QUANTITY: 8	DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1	
MATERIAL: MILD STEEL		NAME	SIGNATURE	DATE	SHEET: 16/66	A3



NO.	NAME	QTY	TYPE	SOURCE	MATERIAL
1	CHARGE WELLS	2	PART	PURCHASED	ANODIZED ALUMINIUM
2	M6 EYE NUT	1	PART	PURCHASED	GRADE 8 STEEL
3	BULKHEAD MOUNTING BRACKET	4	PART	MANUFACTURED	316 STEEL
4	AVIONICS BULKHEAD	1	PART	MANUFACTURED	G10 FIBREGLASS
5	AVIONICS SPACER	1	PART	MANUFACTURED	ABS
6	M6 LARGE WASHER	1	PART	PURCHASED	GRADE 8 STEEL
7	HEX SOCKET HEAD CAP SCREW M3X0.70 X 10	12	PART	PURCHASED	CLASS 12.9 STEEL
8	HEX SOCKET HEAD CAP SCREW M6X1.00 X 18	1	PART	PURCHASED	CLASS 12.9 STEEL
9	AVIONICS BULKHEAD PCB	1	PART	MANUFACTURED	FR-4 FIBREGLASS

DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

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CAD GENERATED DRAWING
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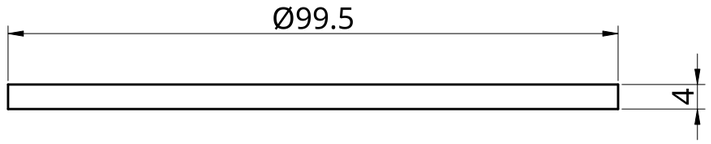
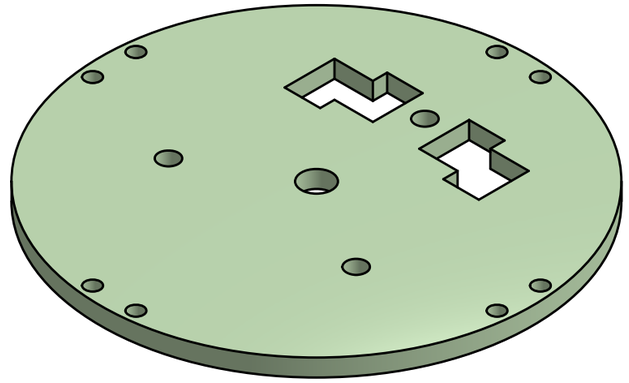
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

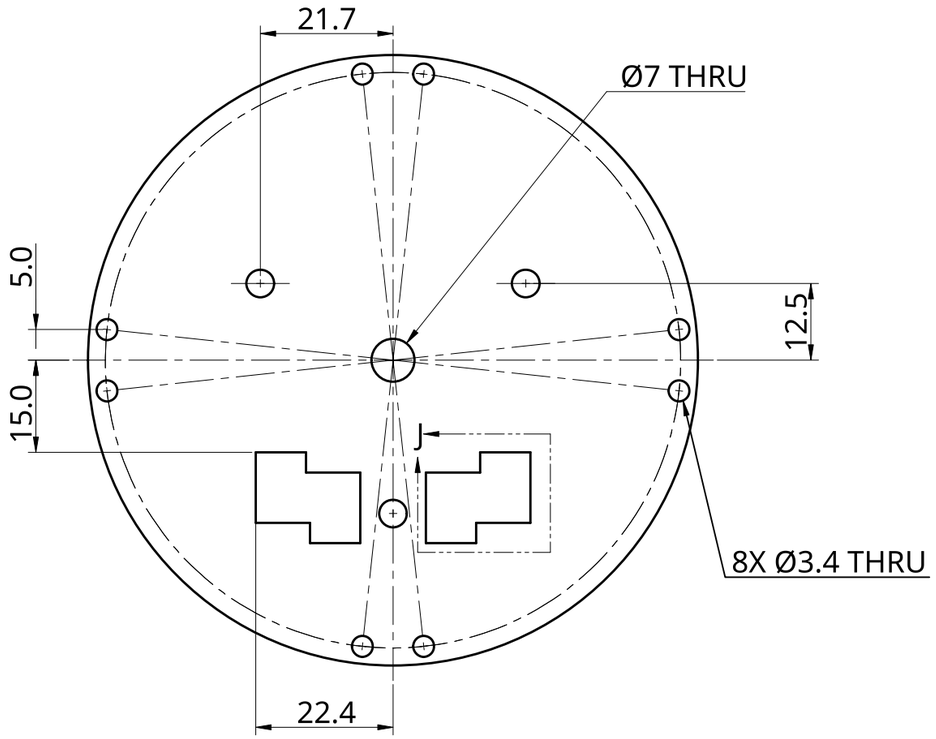
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

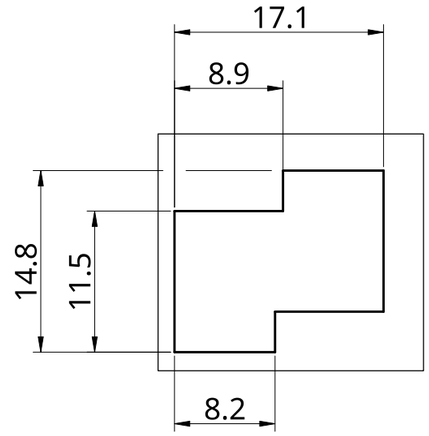
TITLE: AVIONICS BULKHEAD

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 17/66 A3









J
2:1

TO BE MANUFACTED ON WATER JET CUTTER
FILE ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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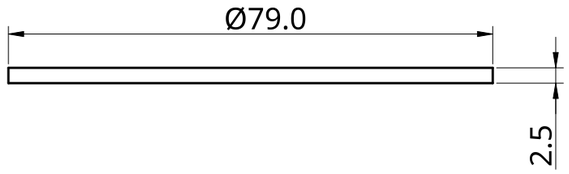
TOLERANCES

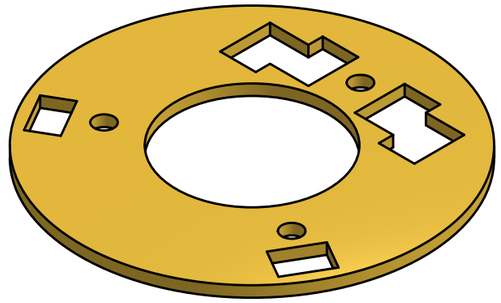
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

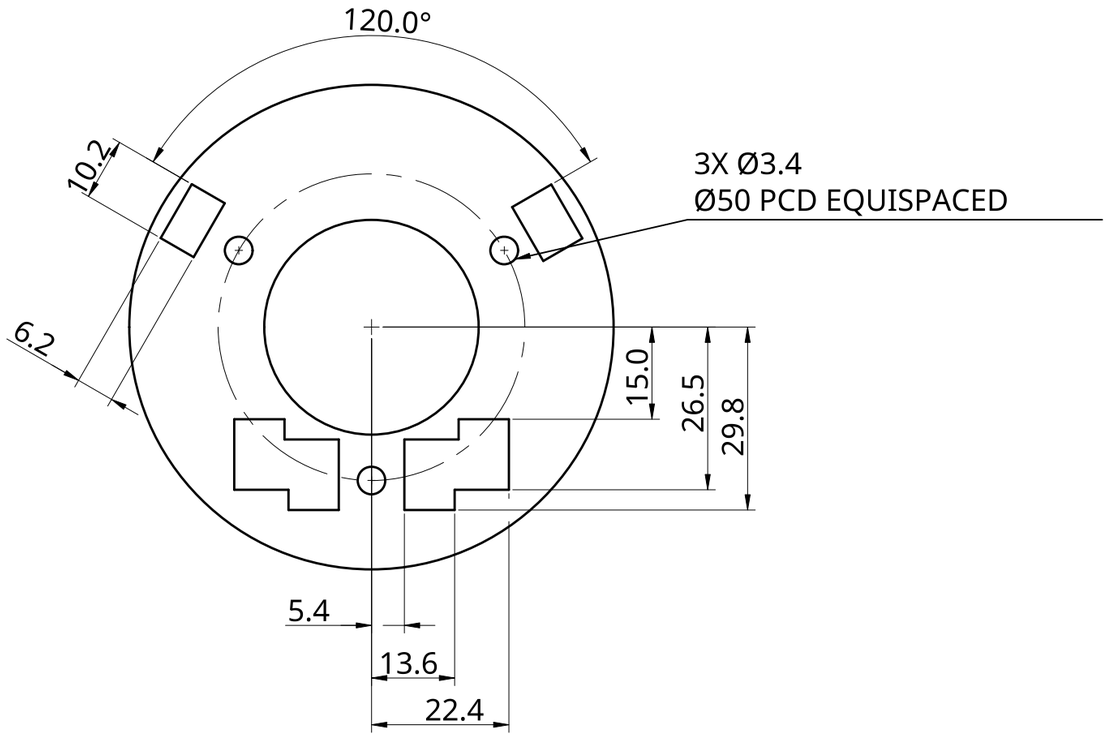
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS BULKHEAD

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: G-10 FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 18/66 A3







REV	DESCRIPTION	BY	DATE	APPV'D

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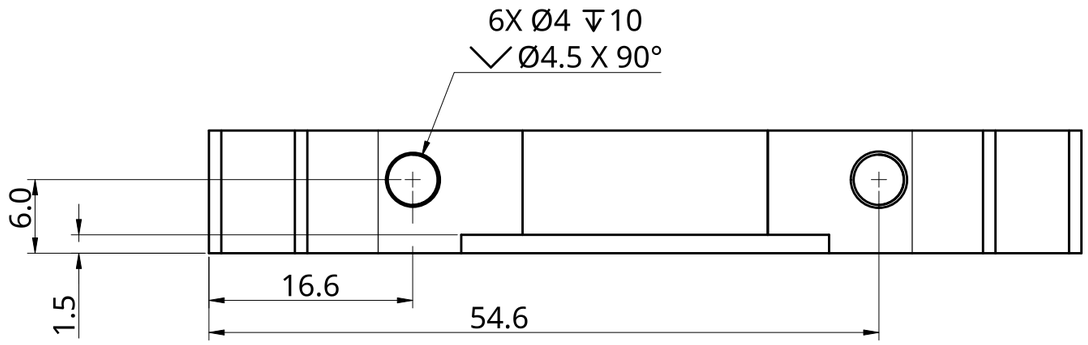
TOLERANCES

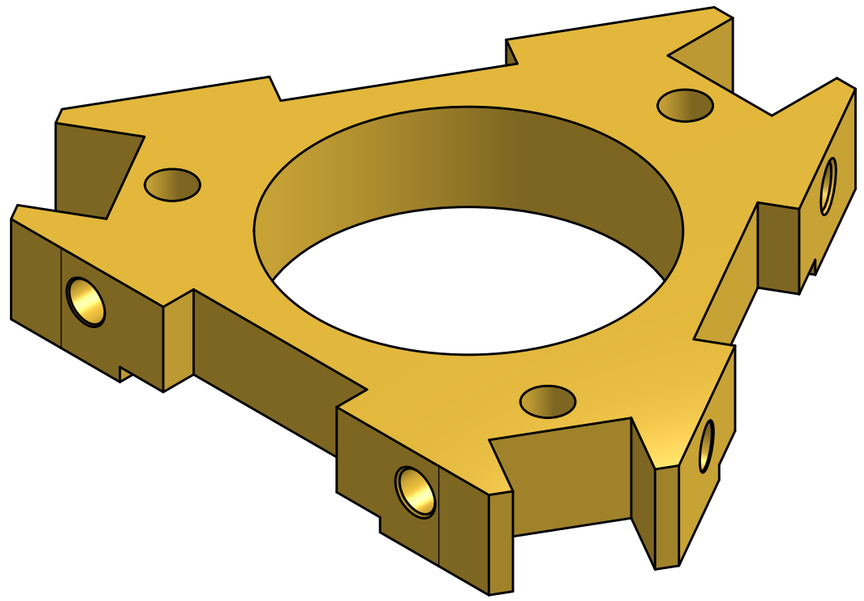
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

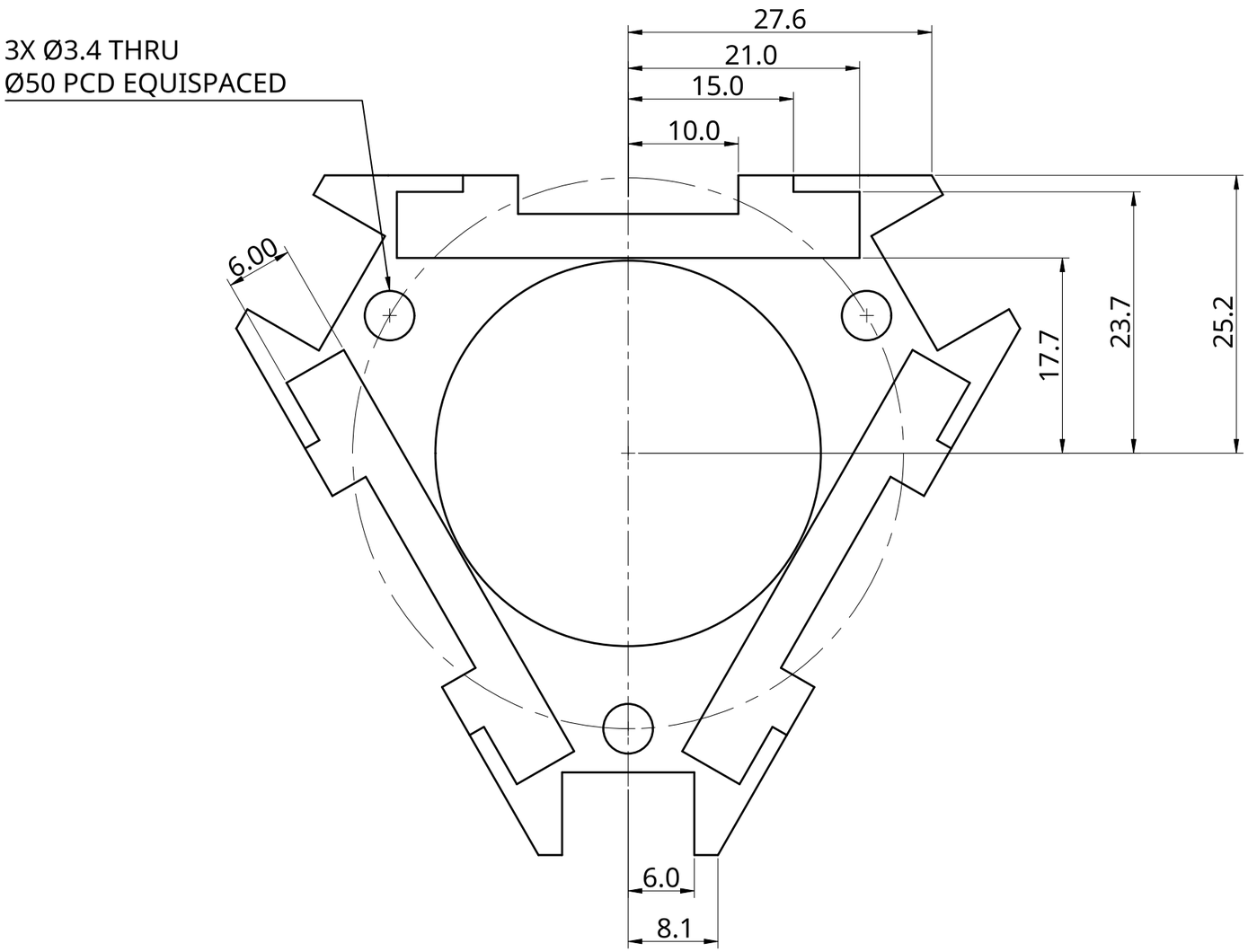
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS SPACER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 19/66 A3







DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

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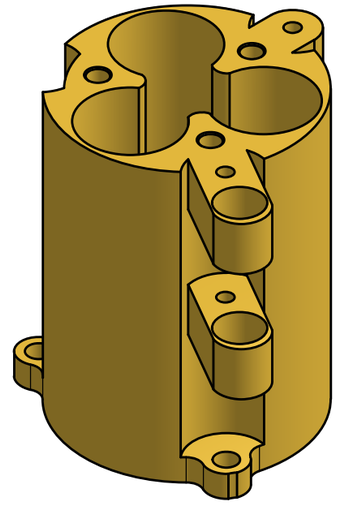
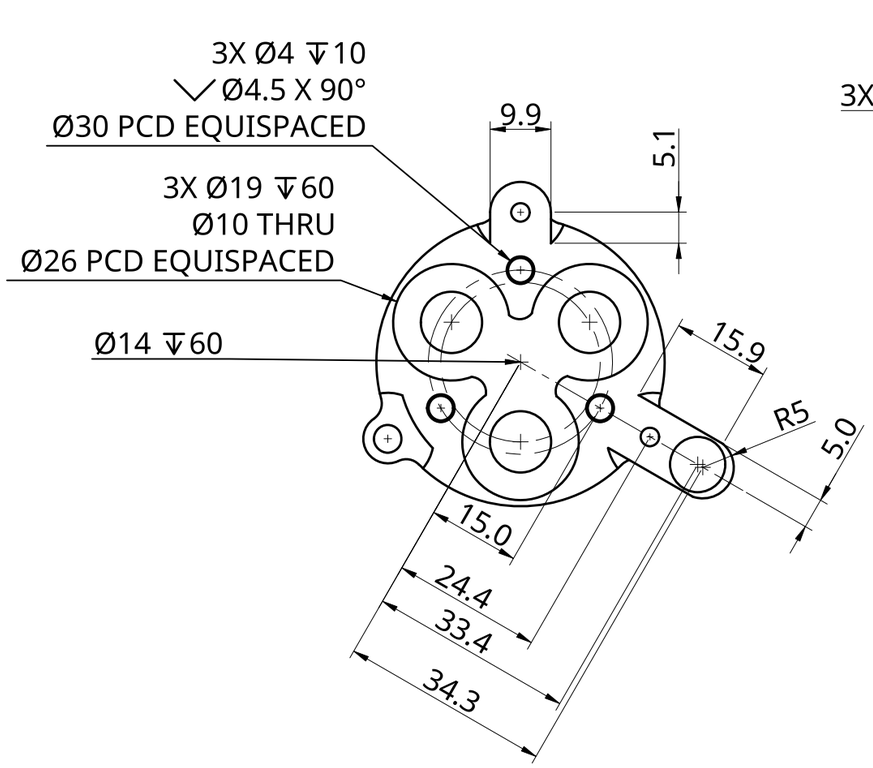
0.0±0.10
0.00±0.05

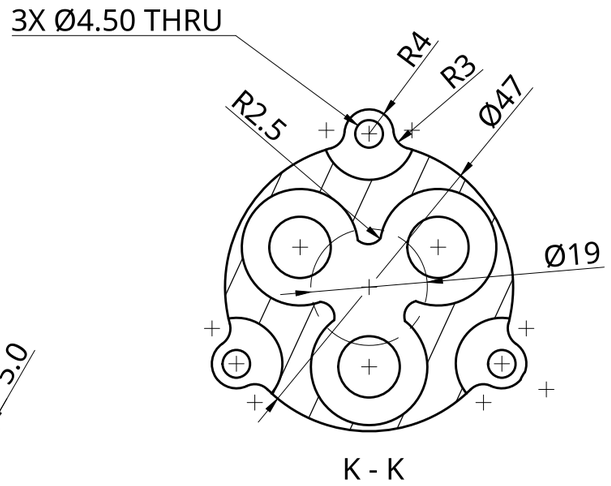
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: PDB MOUNT

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 20/66 A3







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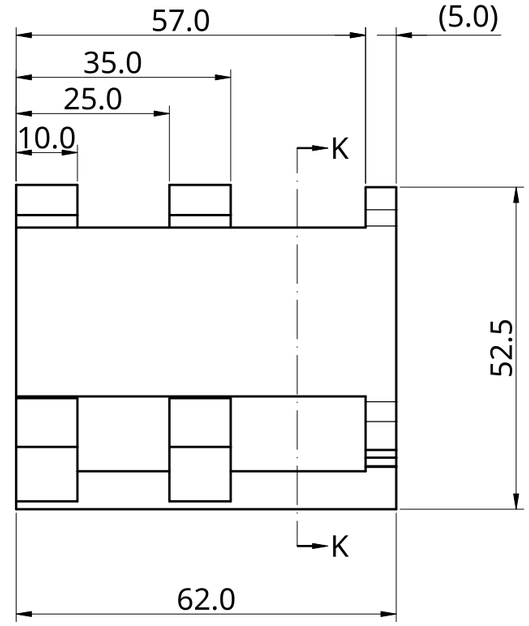

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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EMAIL: tate.derrick@uon.edu.au

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DO NOT SCALE

CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

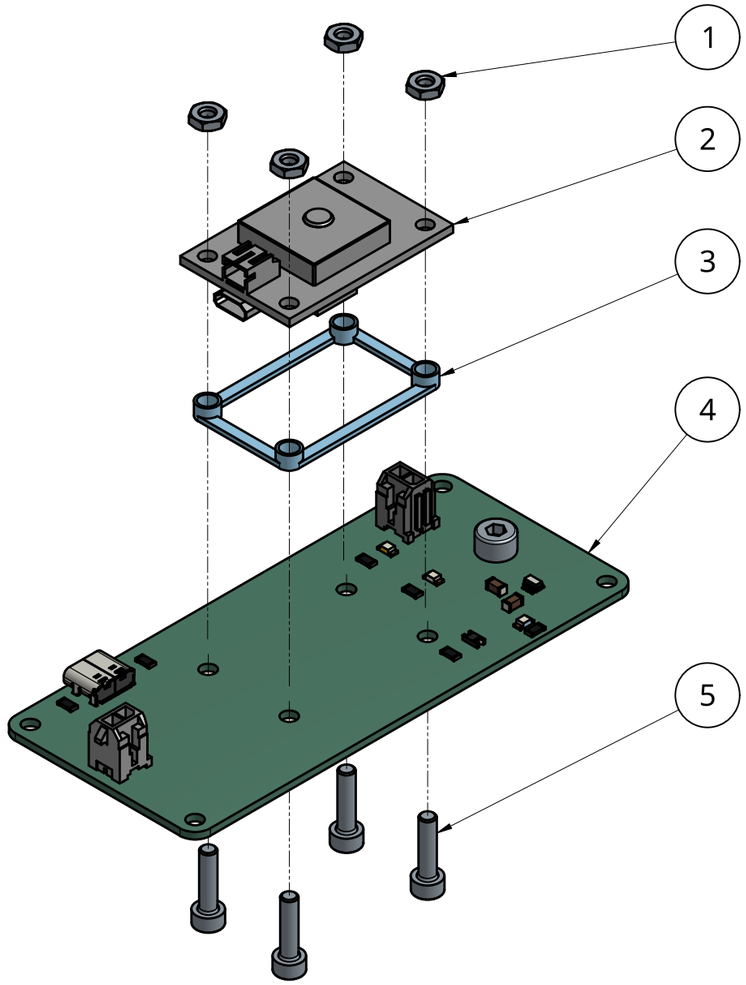
TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS BATTERY HOLDER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 21/66 A3



NO.	ITEM	QTY.	TYPE	SOURCE	MATERIAL
1	HEX M3 NUT	4	PART	PURCHASED	GRADE 8 STEEL
2	TELEGPS	1	PART	PURCHASED	N/A
3	TELEGPS SPACER	1	PART	MANUFACTURED	ABS
4	TELEGPS PDB	1	PART	PURCHASED	N/A
5	HEX SOCKET HEAD CAP SCREW M3X0.5 X 12	4	PART	PURCHASED	CLASS 12.9 STEEL

DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

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DO NOT SCALE

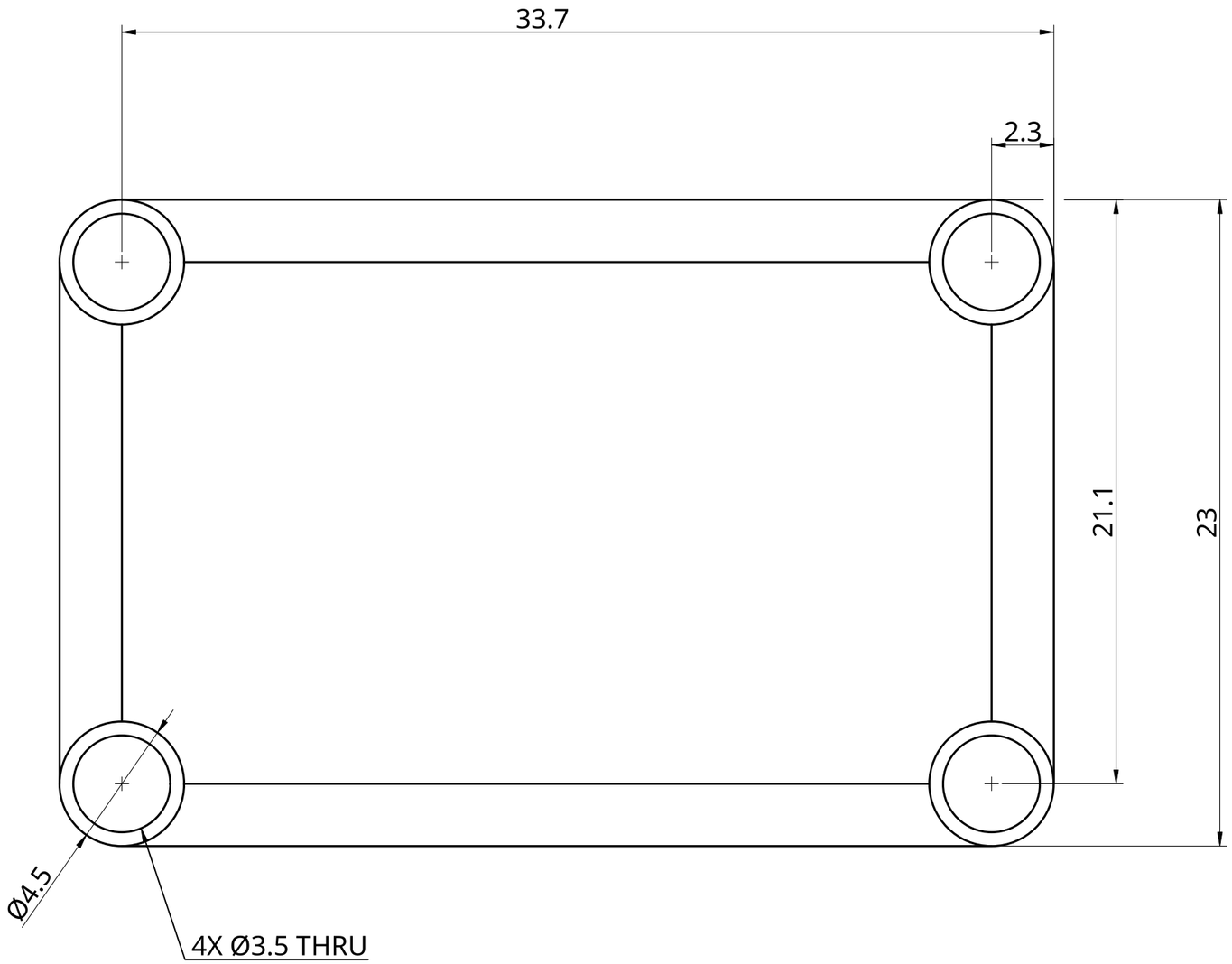
CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

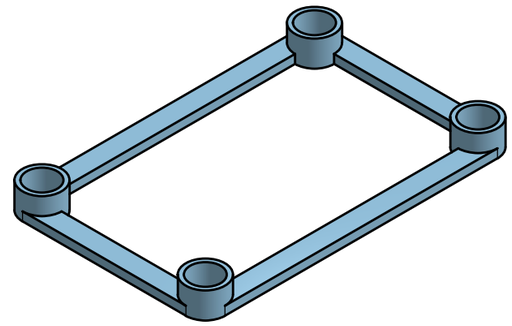
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: GPS PCB ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 22/66 A3





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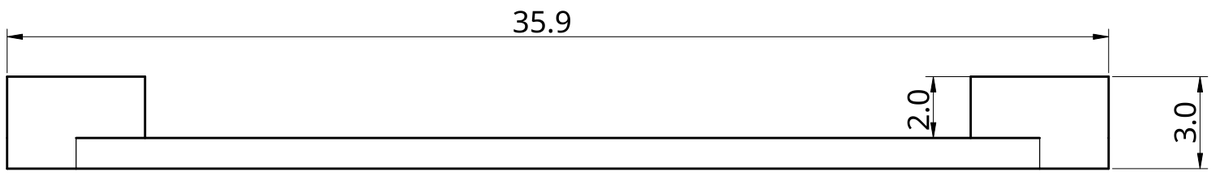

REV	DESCRIPTION	BY	DATE	APPV'D

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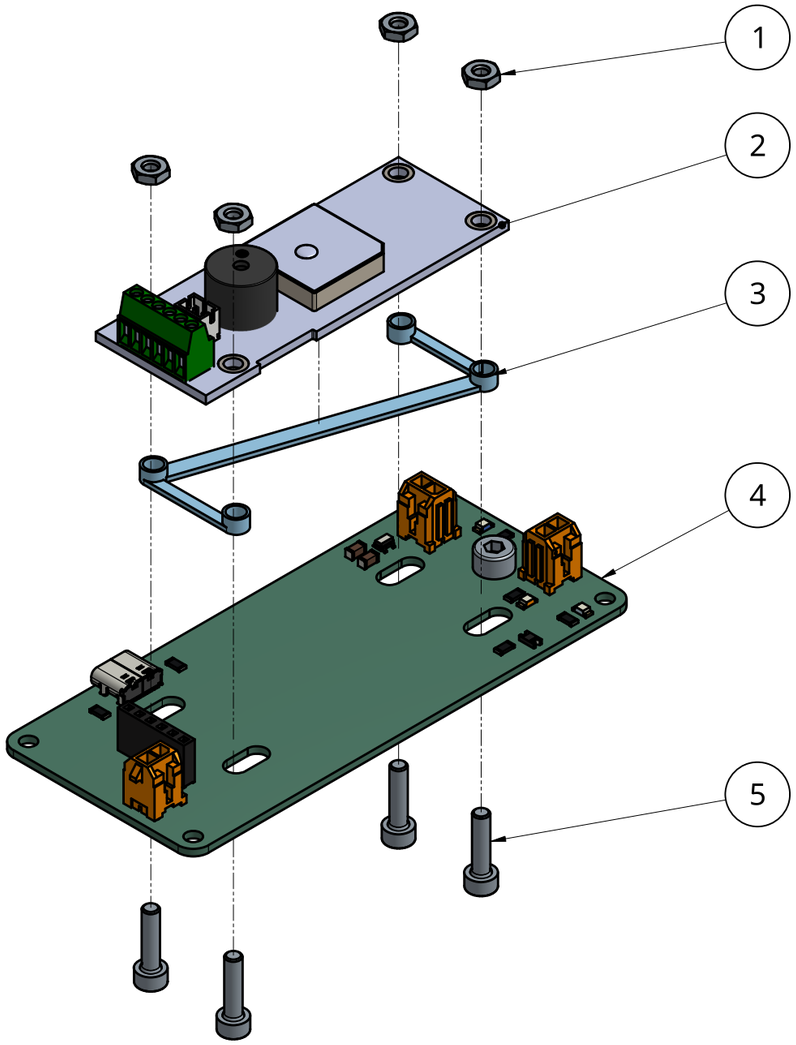
TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: TELEGPS SPACER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 23/66 A3



NO.	ITEM	QTY.	TYPE	SOURCE	MATERIAL
1	HEX M3 NUT	4	PART	PURCHASED	GRADE 8 STEEL
2	TELEMETRUM	1	PART	PURCHASED	N/A
3	TELEMETRUM SPACER	1	PART	MANUFACTURED	ABS
4	TELEMETRUM PDB	1	PART	PURCHASED	N/A
5	HEX SOCKET HEAD CAP SCREW M3X0.5 X 12	4	PART	PURCHASED	CLASS 12.9 STEEL

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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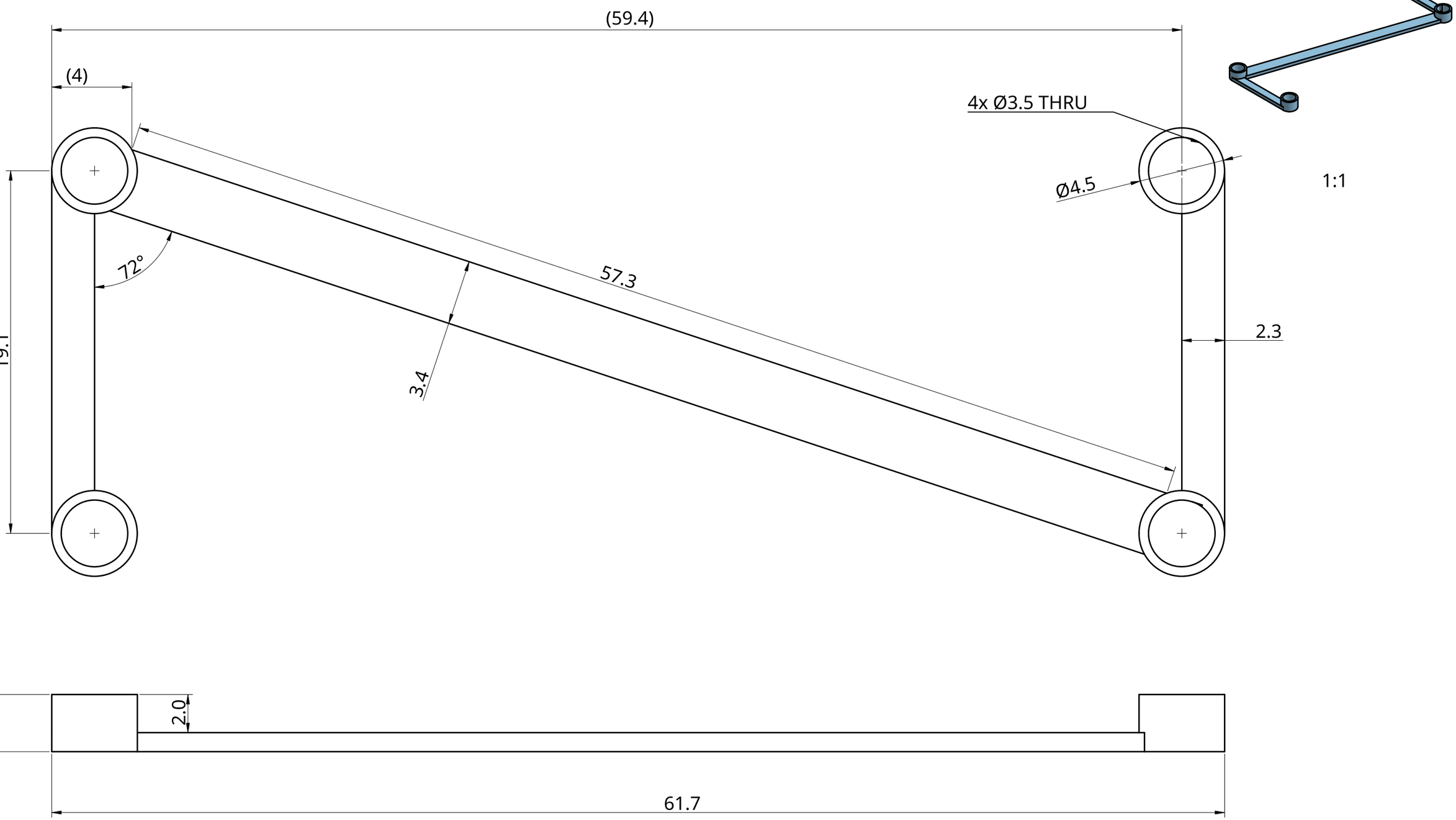
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: TELEMETRUM PDB ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 24/66 A3



REV	DESCRIPTION	BY	DATE	APPV'D

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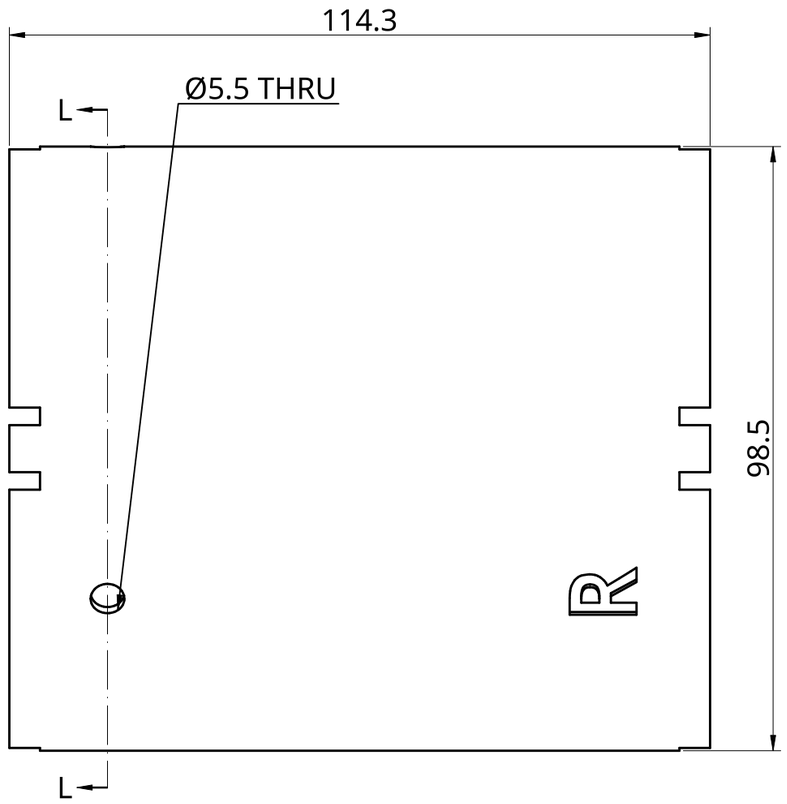
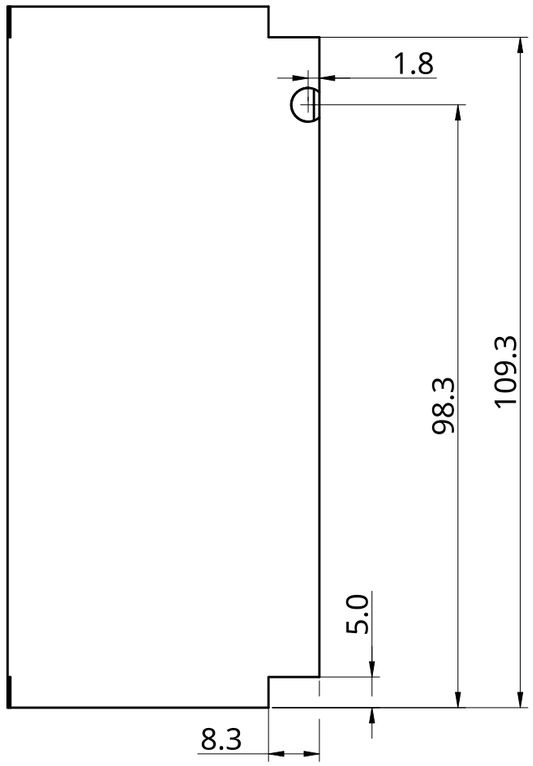
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

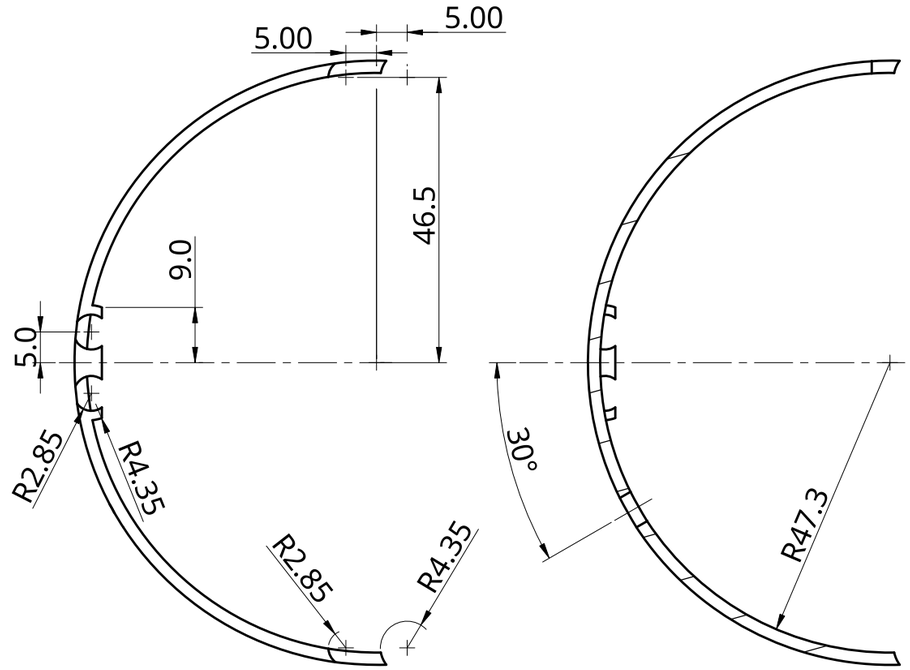
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: TELEMENTRUM GPS

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 25/66 A3







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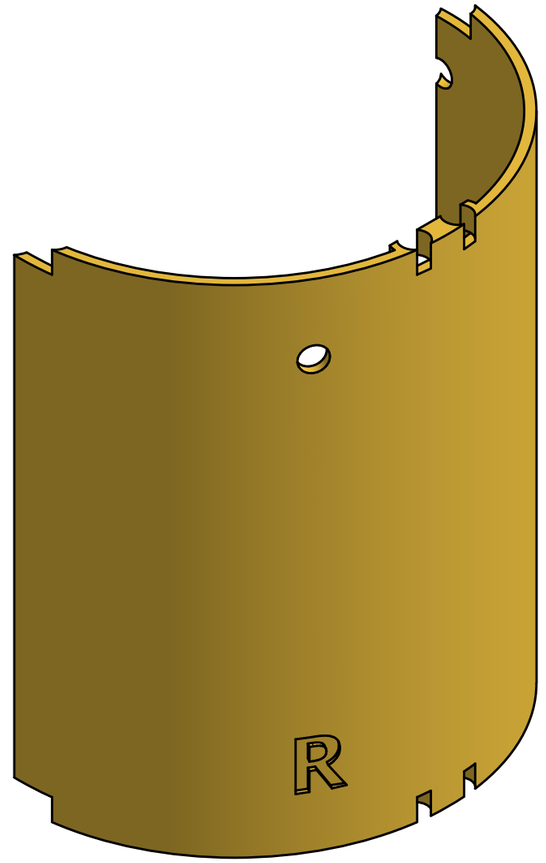

REV	DESCRIPTION	BY	DATE	APPV'D

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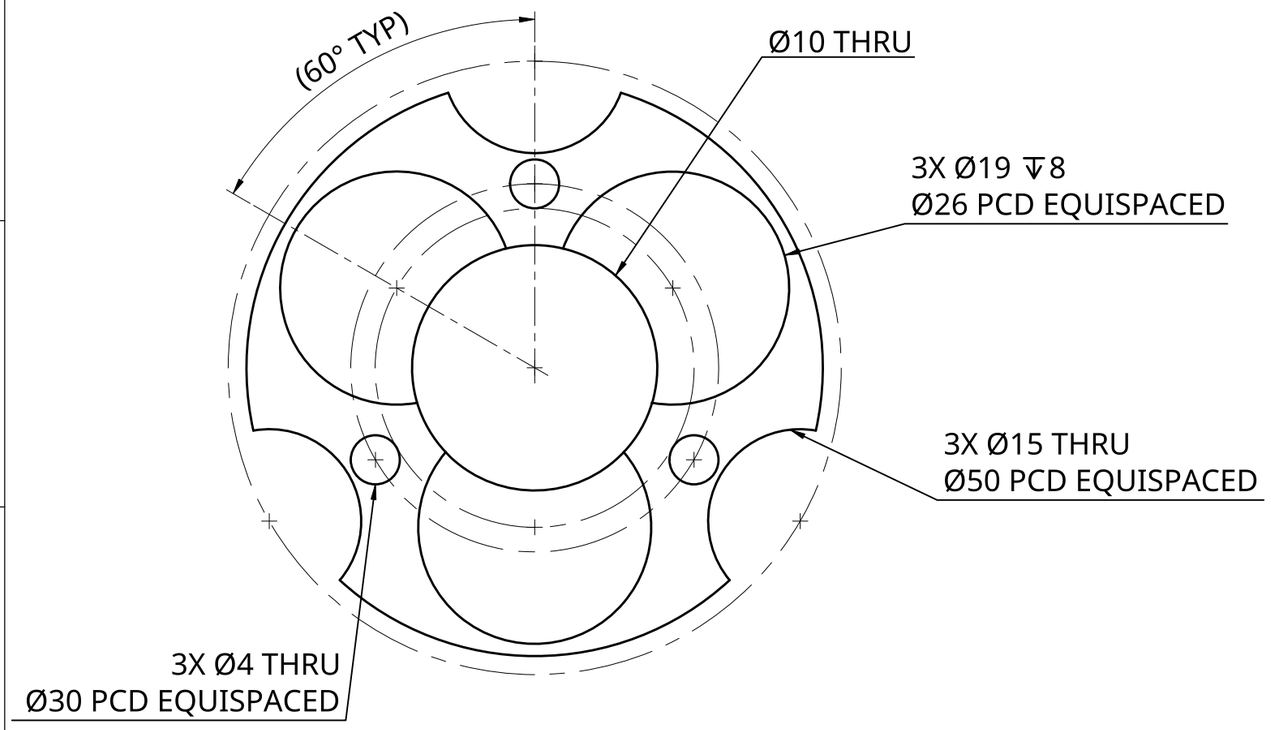
TOLERANCES

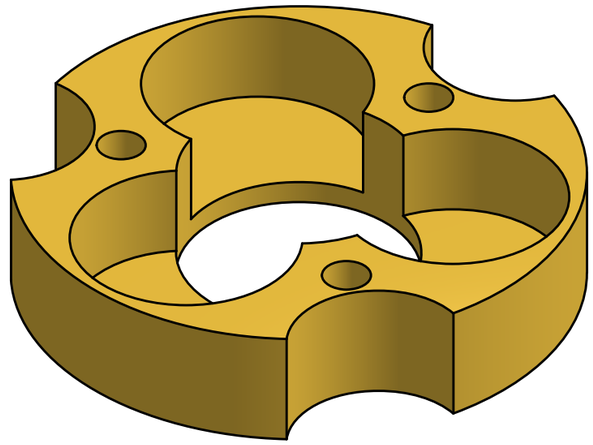
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS SHROUD RHS

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 26/66 A3





REV	DESCRIPTION	BY	DATE	APPV'D

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TOLERANCES

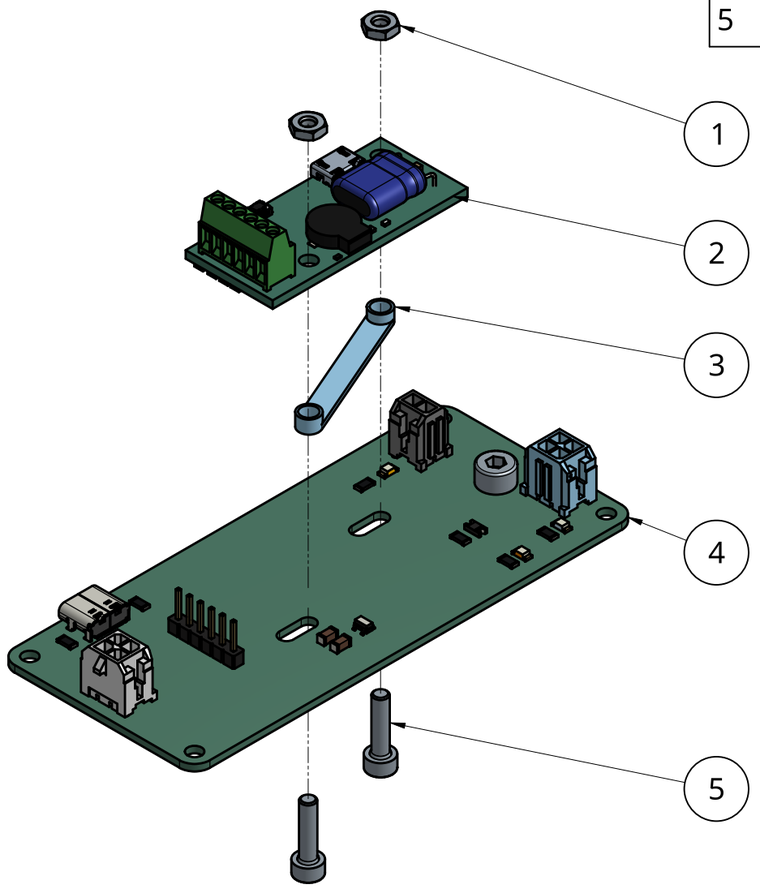
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS BATTERY LID

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 27/66 A3

NO.	ITEM	QTY.	TYPE	SOURCE	MATERIAL
1	HEX M3X0.50 NUT	2	PART	PURCHASED	GRADE 8 STEEL
2	RAVEN	1	PART	PURCHASED	N/A
3	RAVEN SPACER	1	PART	MANUFACTURED	ABS
4	RAVEN PDB	1	PART	PURCHASED	N/A
5	HEX SOCKET HEAD CAP SCREW M3X0.5 X 12	2	PART	PURCHASED	CLASS 12.9 STEEL



REV	DESCRIPTION	BY	DATE	APPV'D

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CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

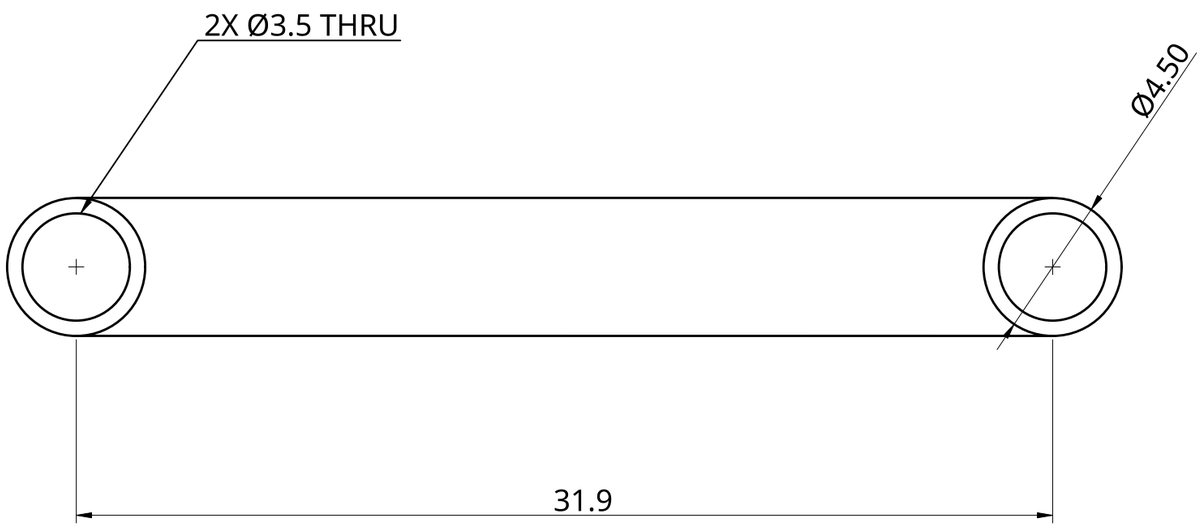
TOLERANCES

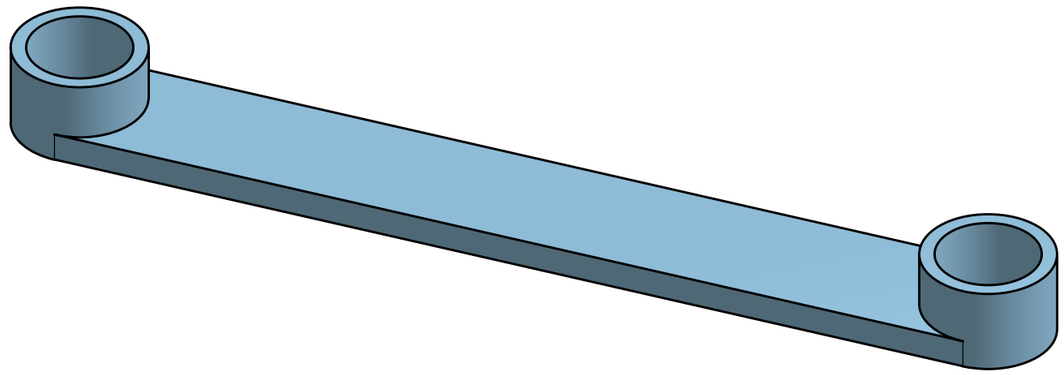
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

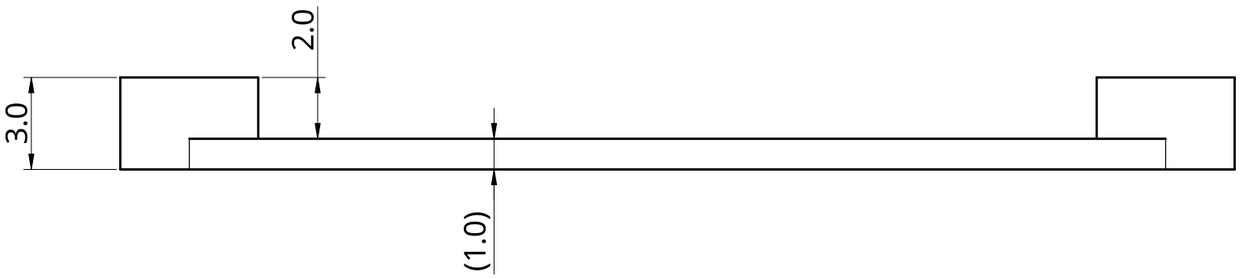
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: RAVEN PDB

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 28/66 A3







REV	DESCRIPTION	BY	DATE	APPV'D

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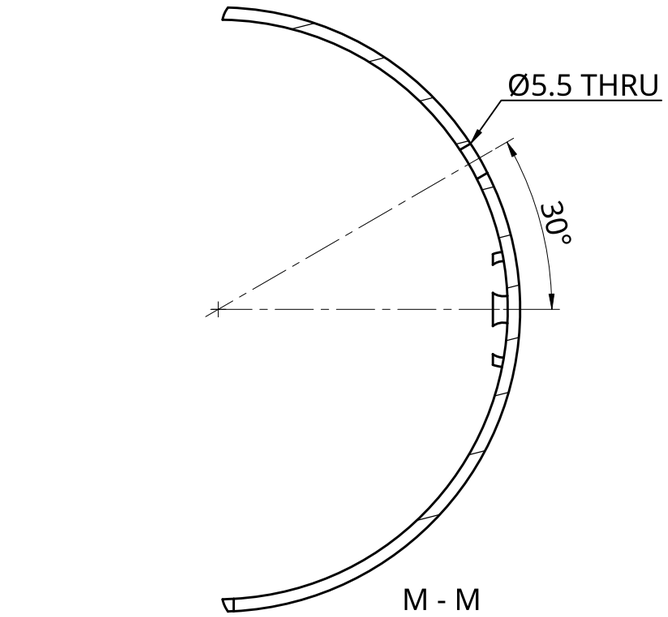
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

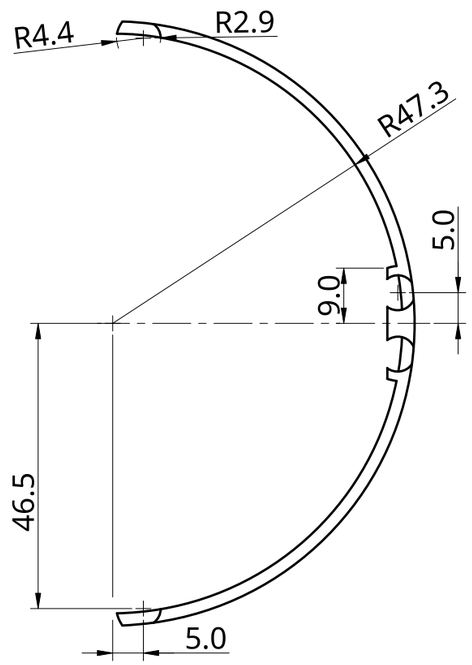
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

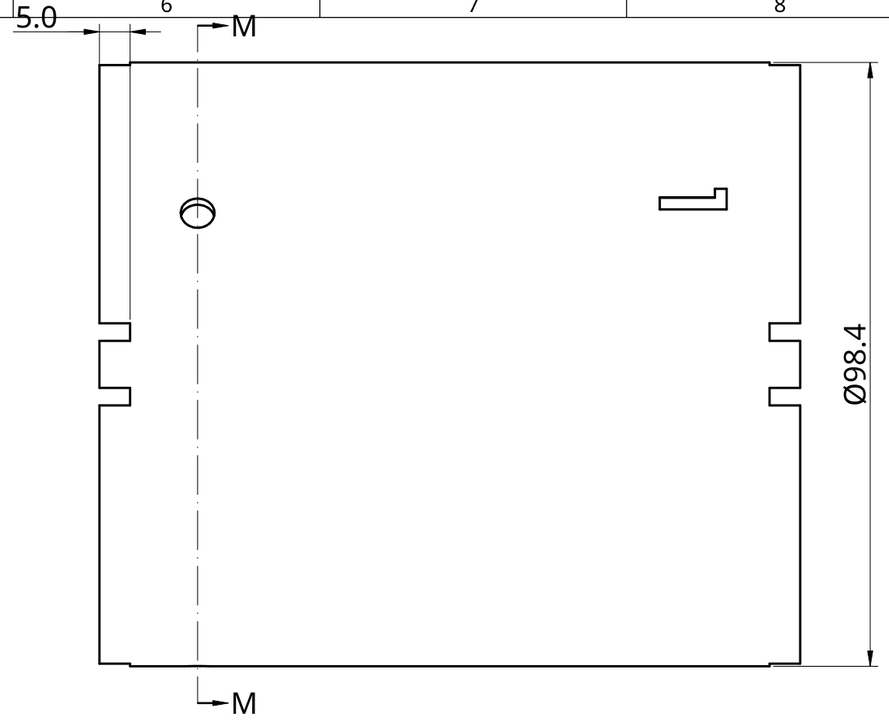
TITLE: RAVEN SPACER

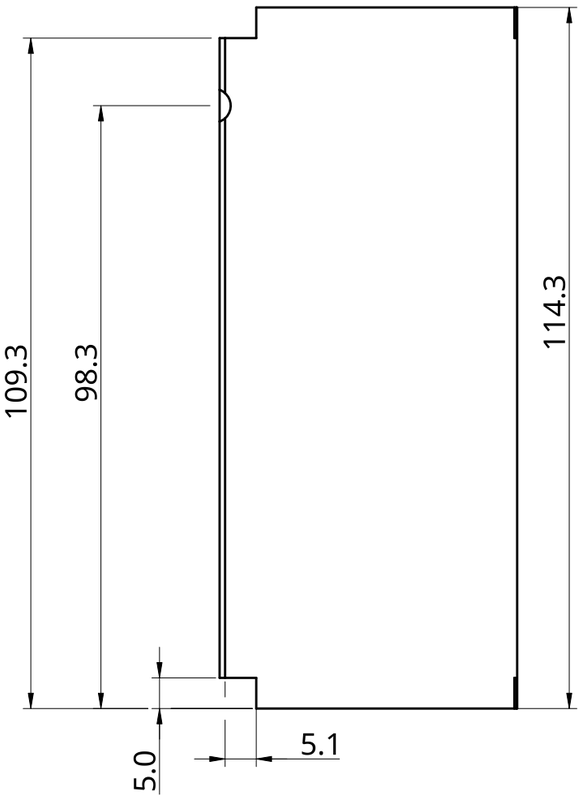
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 29/66 A3

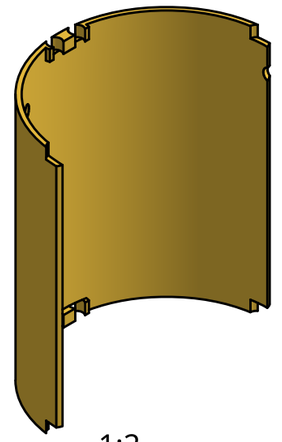


M - M









1:2

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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EMAIL: tate.derrick@uon.edu.au

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CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

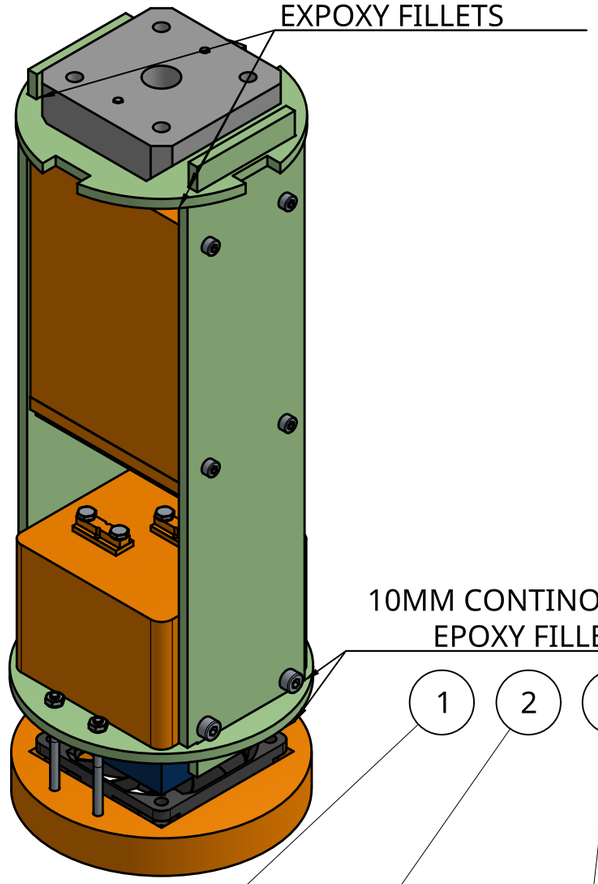
TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

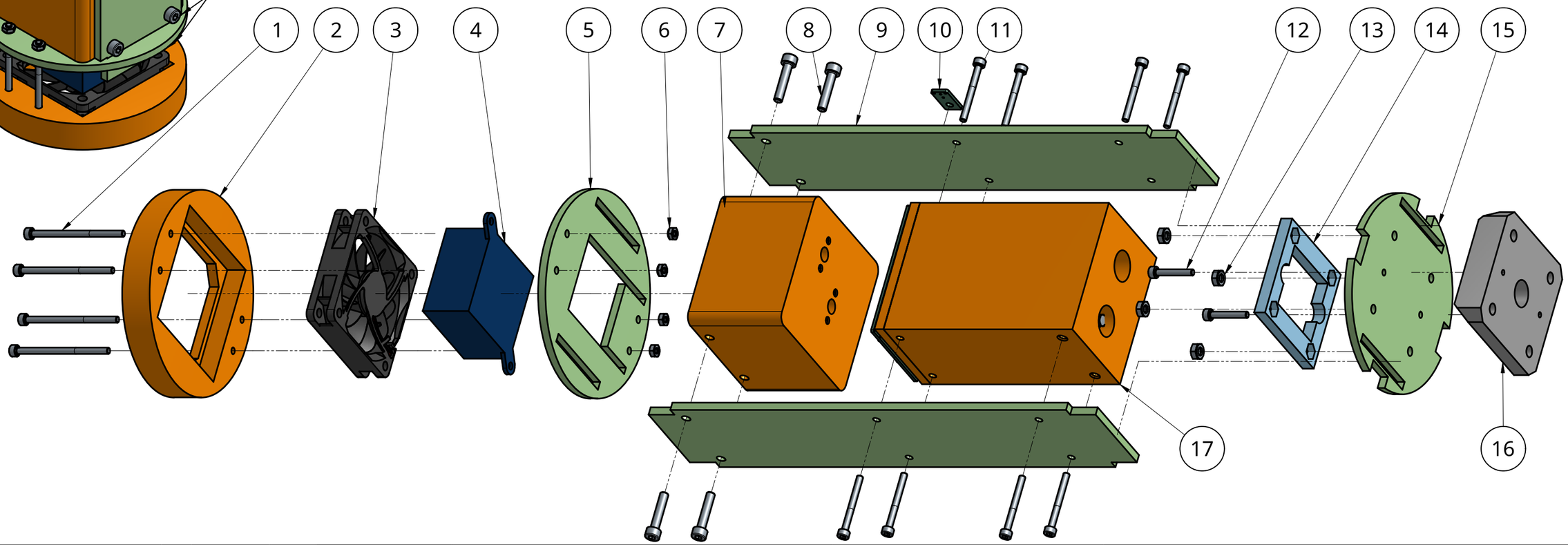
TITLE: AVIONICS SHROUD LHS

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 30/66 A3



NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
1	HEX SOCKET HEAD CAP SCREW M3X0.50 X 50	4	PART	PURCHASED	CLASS 12.9 STEEL
2	FAN HOLDER	1	PART	MANUFACTURED	ABS
3	FAN	1	PART	PURCHASED	POLYBUTYLENE TERAPTHALATE
4	HEATSINK	1	PART	PURCHASED	ANODISED ALUMINIUM
5	PAYLOAD LOWER PLATE	1	PART	MANUFACTURED	G10 FIBREGLASS
6	HEX NUT M3X0.50	4	PART	PURCHASED	GRADE 8 STEEL
7	THERMAL CHAMBER	1	ASSEMBLY	MANUFACTURED	N/A
8	HEX SOCKET HEAD CAP SCREW M4X0.7 X 20	4	PART	PURCHASED	CLASS 12.9 STEEL

SEE PG. 29 FOR CONTINUED BOM



REV	DESCRIPTION	BY	DATE	APPV'D

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TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: PAYLOAD ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 31/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
9	PAYLOAD SIDE PLATE	2	PART	MANUFACTURED	G10 FIBREGLASS
10	SCREW SWITCH PCB	1	PART	MANUFACTURED	FR-4 FIBREGLASS
11	HEX SOCKET HEAD CAP SCREW M3X0.50 X 30	8	PART	PURCHASED	CLASS 12.9 STEEL
12	HEX SOCKET HEAD CAP SCREW M3X0.50 X 20	2	PART	PURCHASED	CLASS 12.9 STEEL
13	HEX NUT M3X0.50	4	PART	PURCHASED	GRADE 8 STEEL
14	PAYLOAD NUT RETAINER	1	PART	MANUFACTURED	ABS
15	PAYLOAD TOP PLATE	1	PART	MANUFACTURED	G10 FIBREGLASS
16	PAYLOAD SPACER	1	PART	MANUFACTRUED	MILD STEEL
17	BATTERY HOLDER ASSEMBLY	1	ASSEMBLY	MANUFACTURED	N/A

REV	DESCRIPTION	BY	DATE	APPV'D

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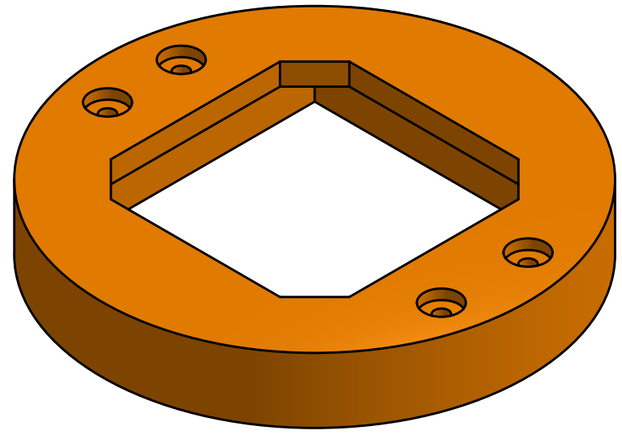
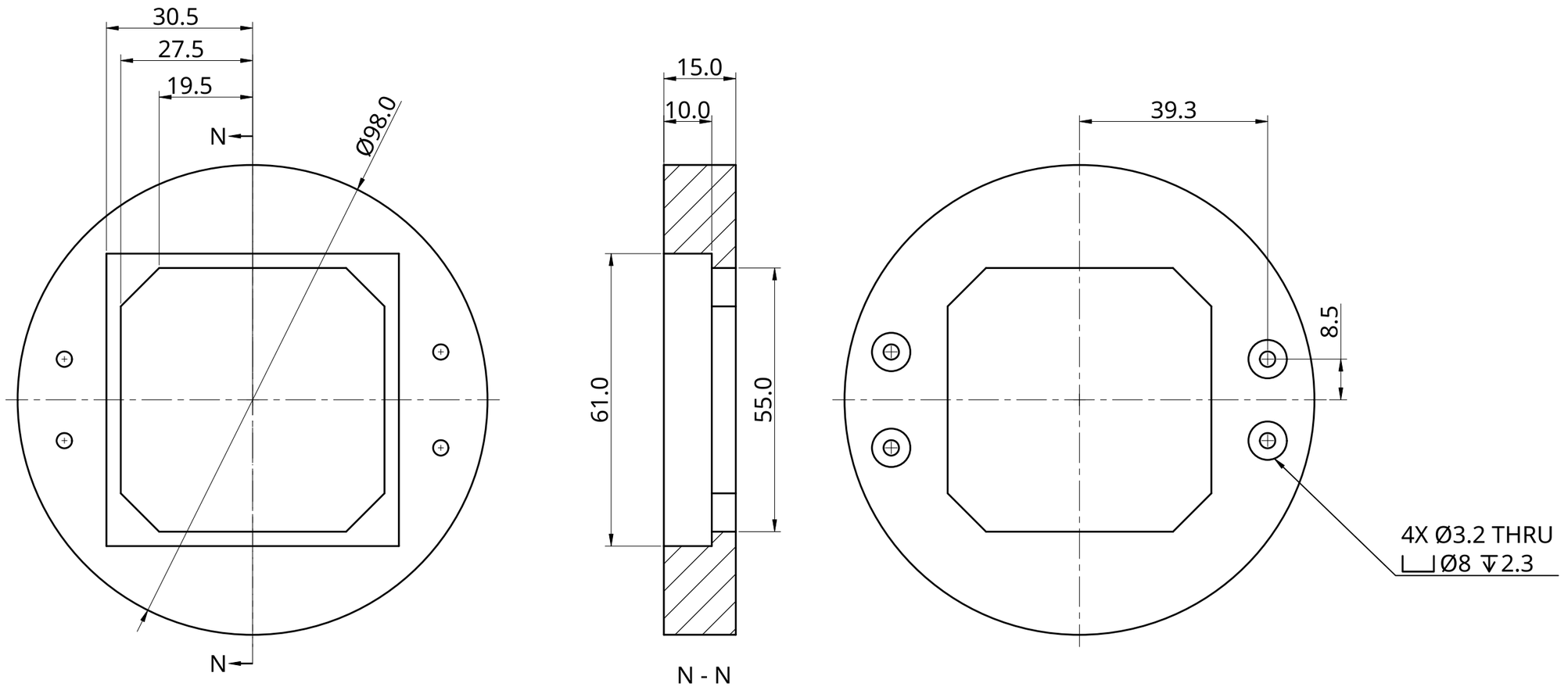
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: PAYLOAD BOM CONTINUED

PROJECT NO.: 67	APPV'D				DRAWING NO.:	
FINISH: 3.2 UOS	CHK'D				REVISION:	
QUANTITY:	DRAWN	G. O'BRIEN		2026-04-08	SCALE: N/A	
MATERIAL: N/A		NAME	SIGNATURE	DATE	SHEET: 32/66	A3





REV	DESCRIPTION	BY	DATE	APPV'D

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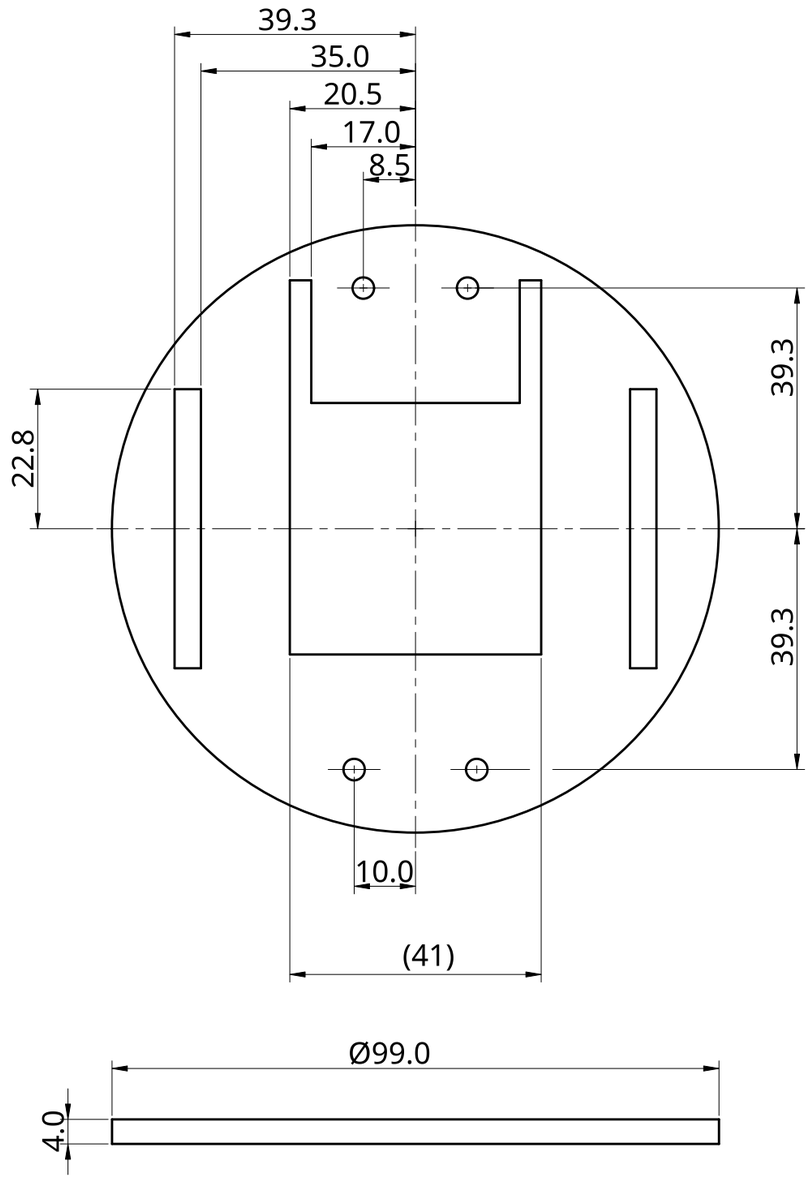
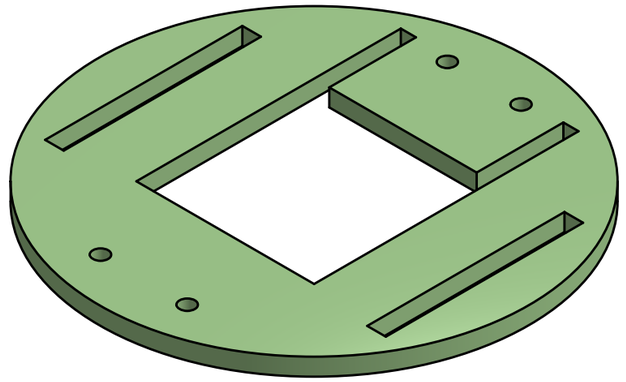
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FAN HOLDER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 33/66 A3





REV	DESCRIPTION	BY	DATE	APPV'D

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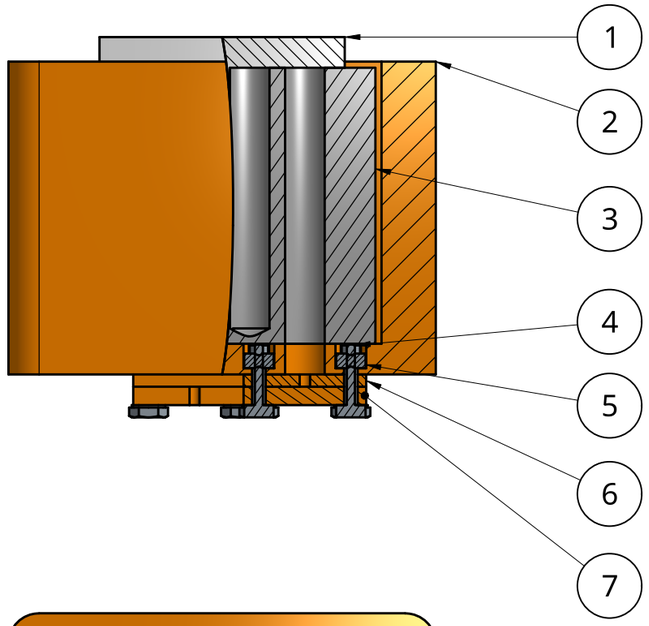
0.±0.50
0.0±0.10
0.00±0.05

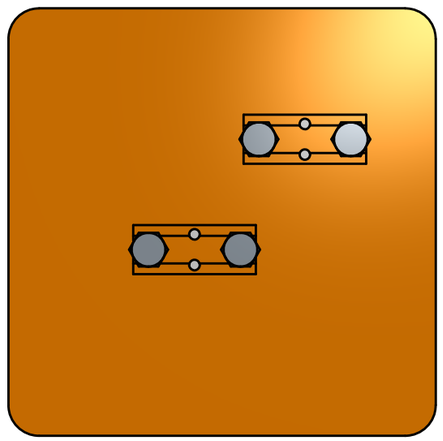
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

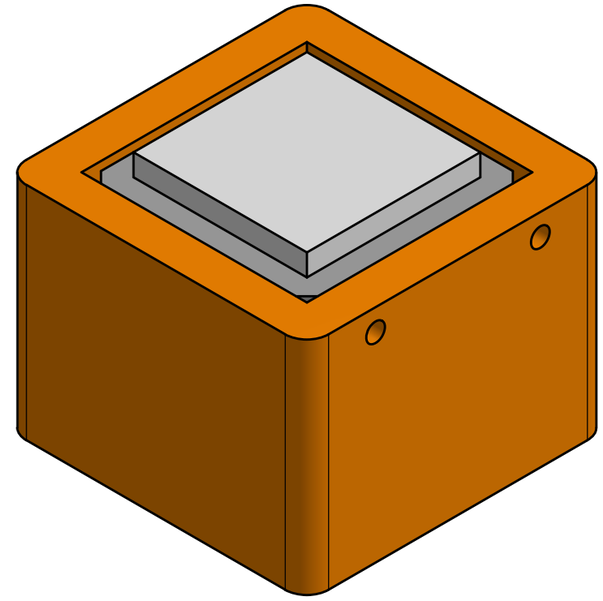
TITLE: PAYLOAD LOWER PLATE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: G10 FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 34/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
1	PELTIER	1	PART	PURCHASED	N/A
2	PAYLOAD INSULATION	1	PART	MANUFACTURED	ABS
3	MEDICAL CONTAINER	1	PART	MANUFACTED	316 STEEL
4	HEX HEAD BOLT M3X0.50 X 10	4	PART	PURCHASED	NYLON
5	HEX NUT M3X0.50	4	PART	PURCHASED	NYLON
6	THERMISTER SPACER	2	PART	MANUFACTURED	ABS
7	THERMISTER HOLDER	2	PART	MANUFACTURED	ABS







REV	DESCRIPTION	BY	DATE	APPV'D

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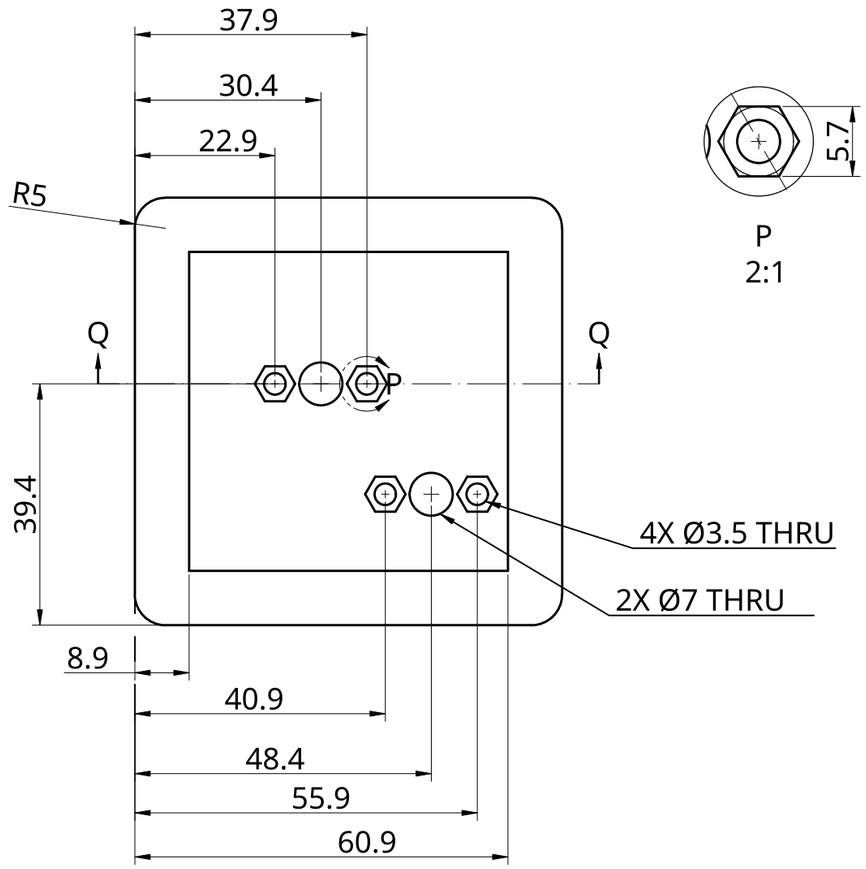
TOLERANCES

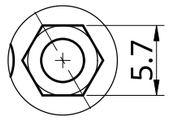
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: THERMAL CHAMBER ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 35/66 A3





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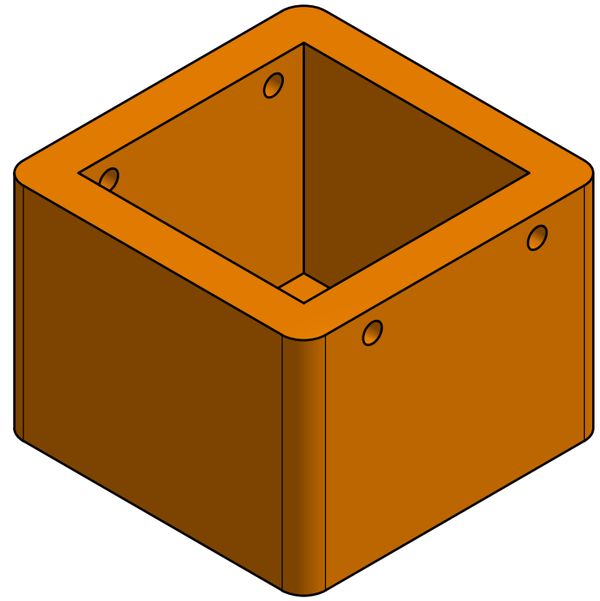

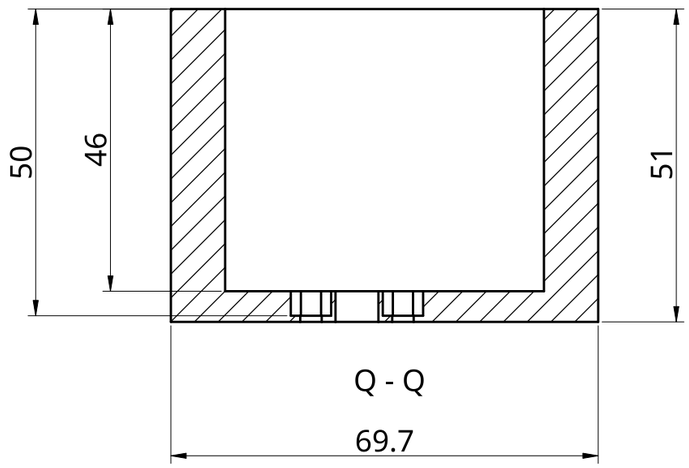

REV	DESCRIPTION	BY	DATE	APPV'D

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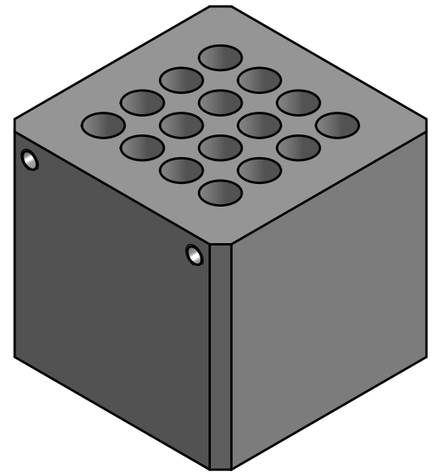
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: PAYLOAD INSULATION

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 36/66 A3



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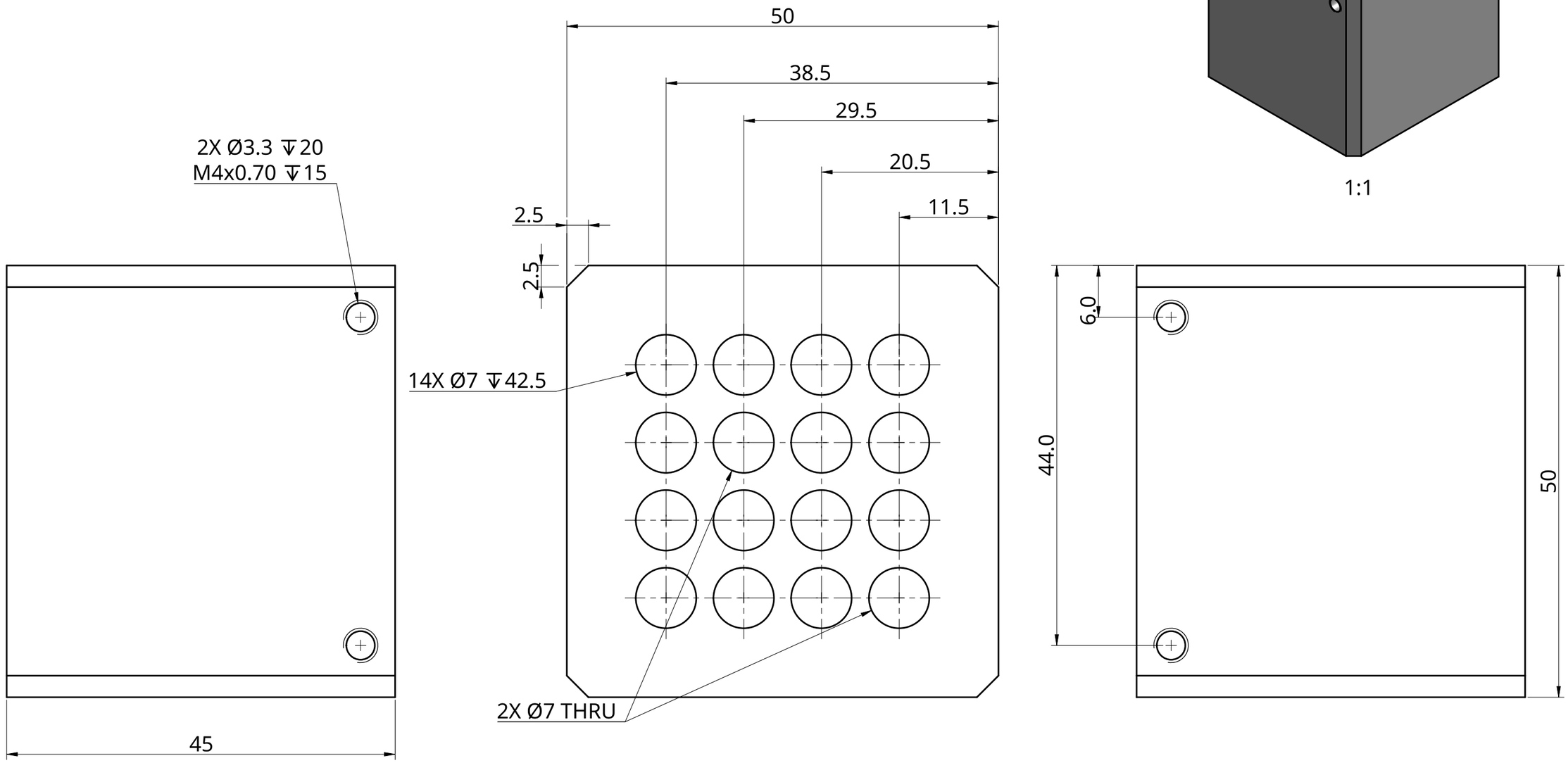

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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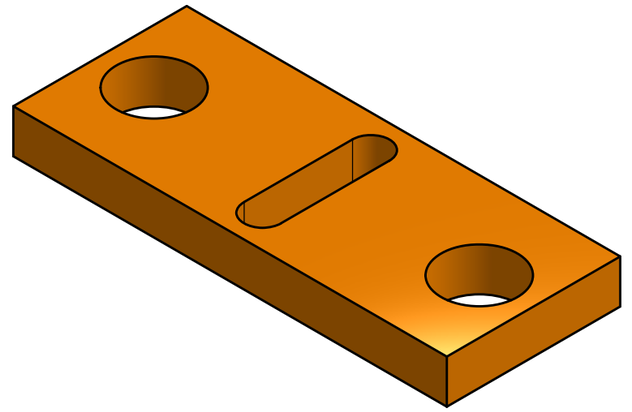
TOLERANCES

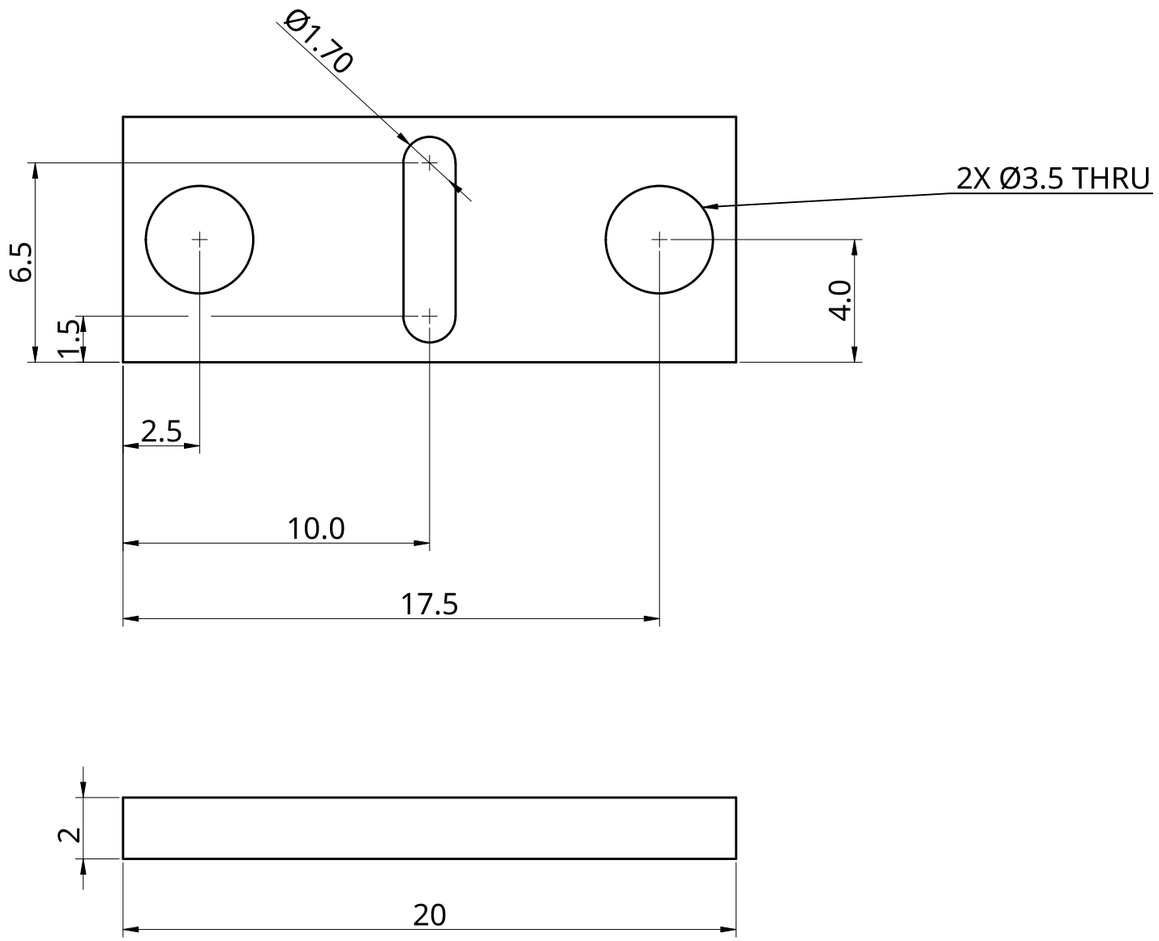
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: MEDCONTAINER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: MILD STEEL			NAME	SIGNATURE	DATE	SHEET: 37/66 A3





CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

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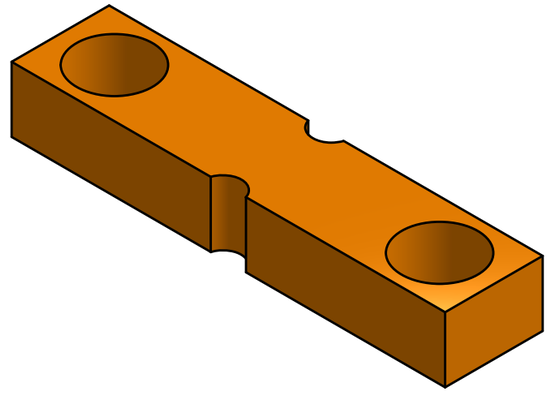
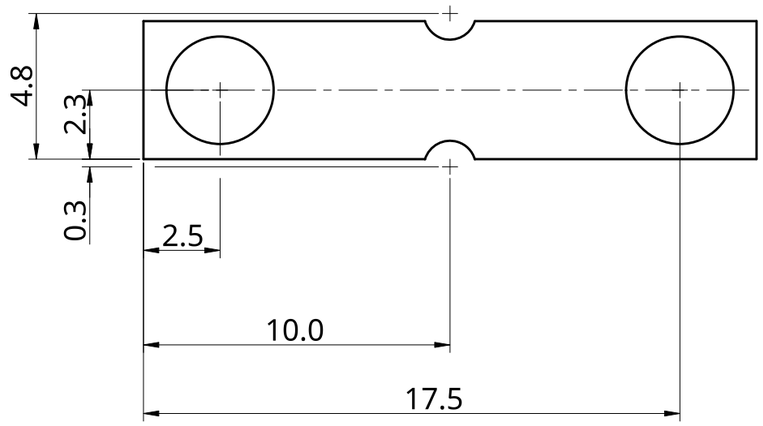
0.0±0.10
0.00±0.05

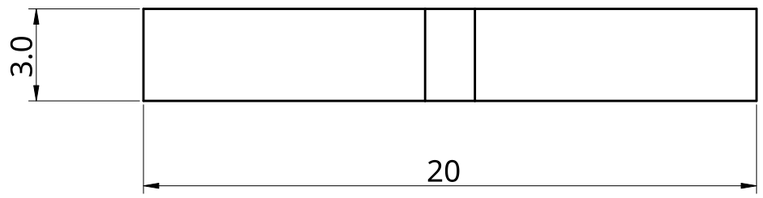
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: THERMISTER SPACER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 38/66 A3







REV	DESCRIPTION	BY	DATE	APPV'D

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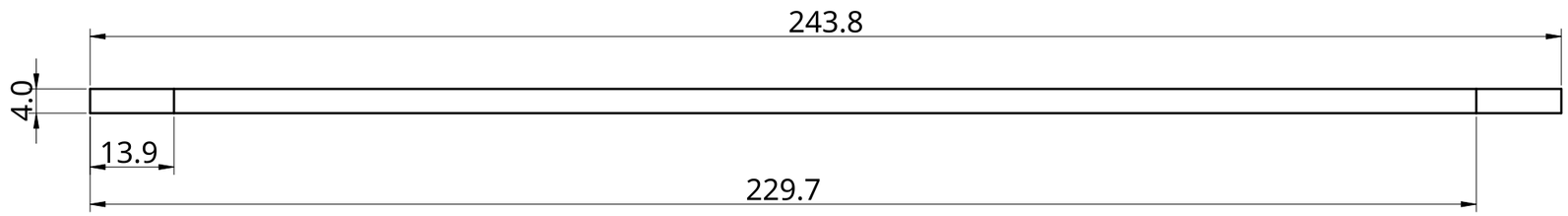
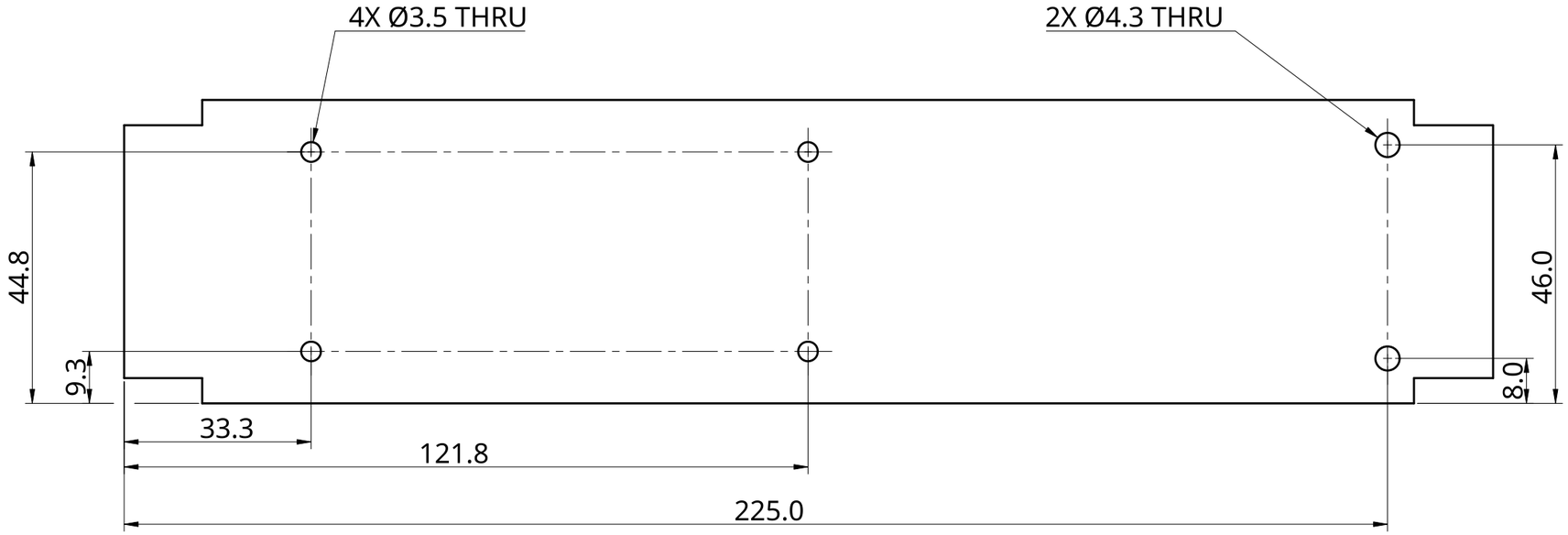
0.00±0.05

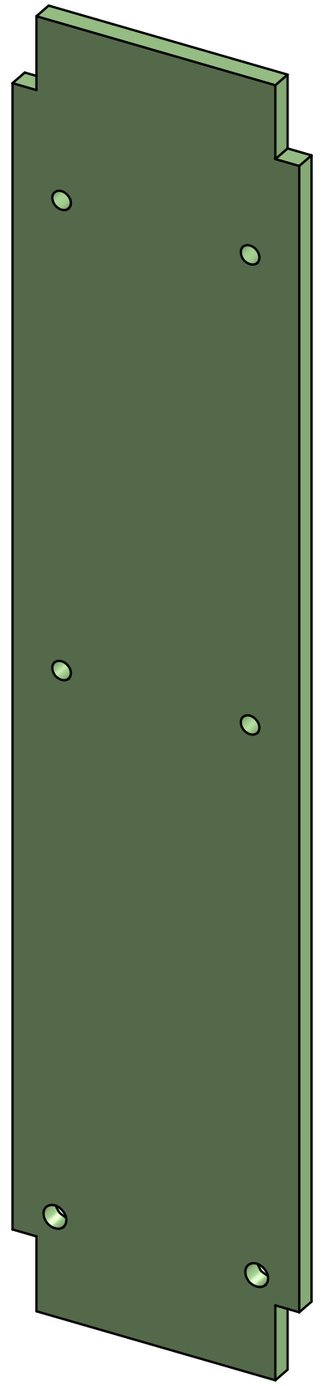
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: THERMISTER HOLDER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 39/66 A3







REV	DESCRIPTION	BY	DATE	APPV'D

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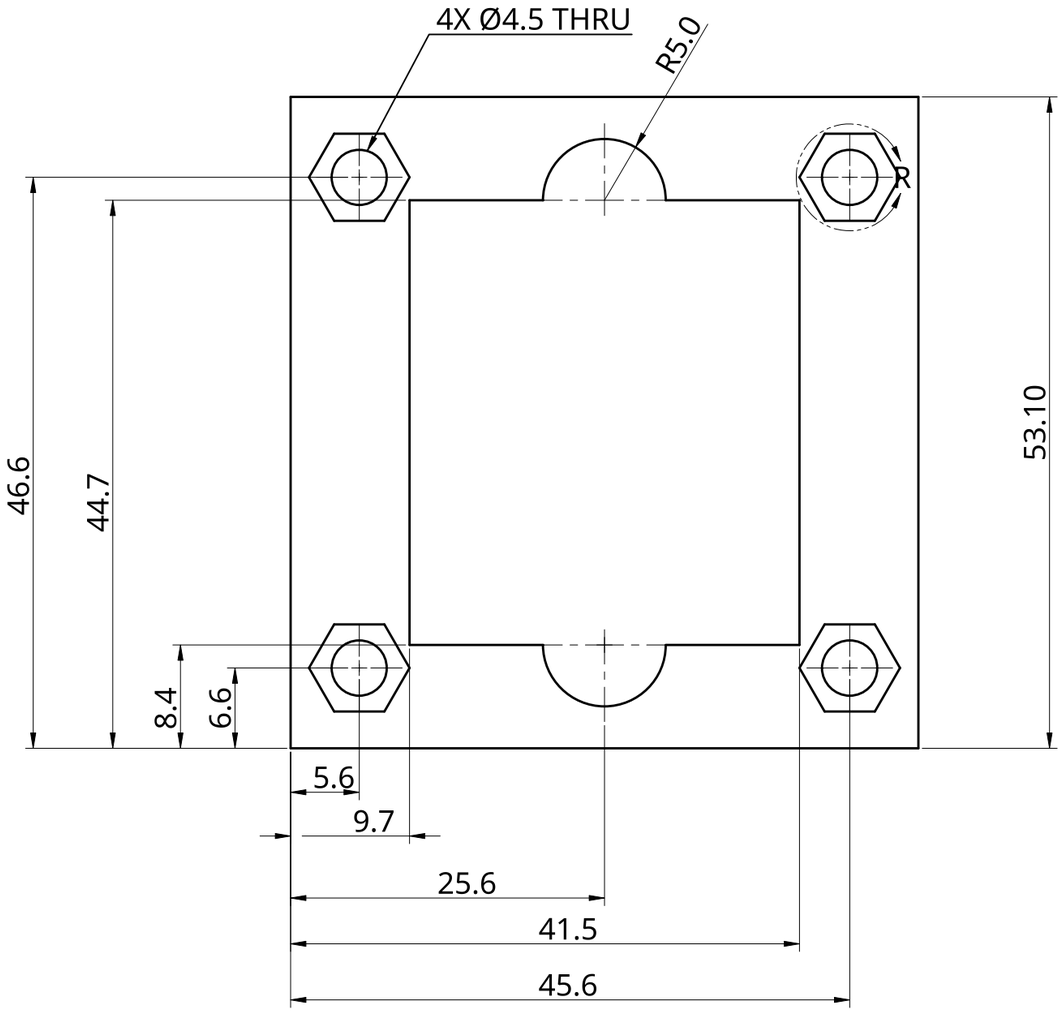
TOLERANCES

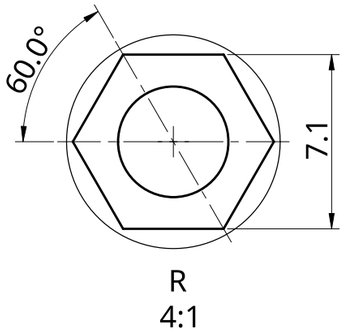
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

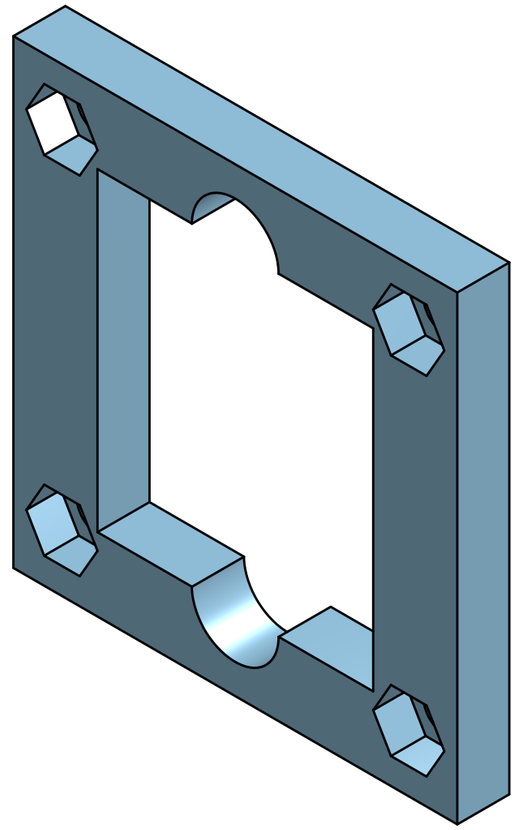
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

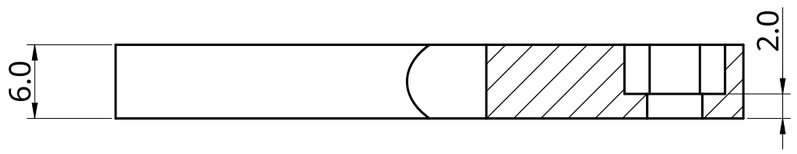
TITLE: PAYLOAD SIDE PLATE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: G10 FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 40/66 A3









REV	DESCRIPTION	BY	DATE	APPV'D

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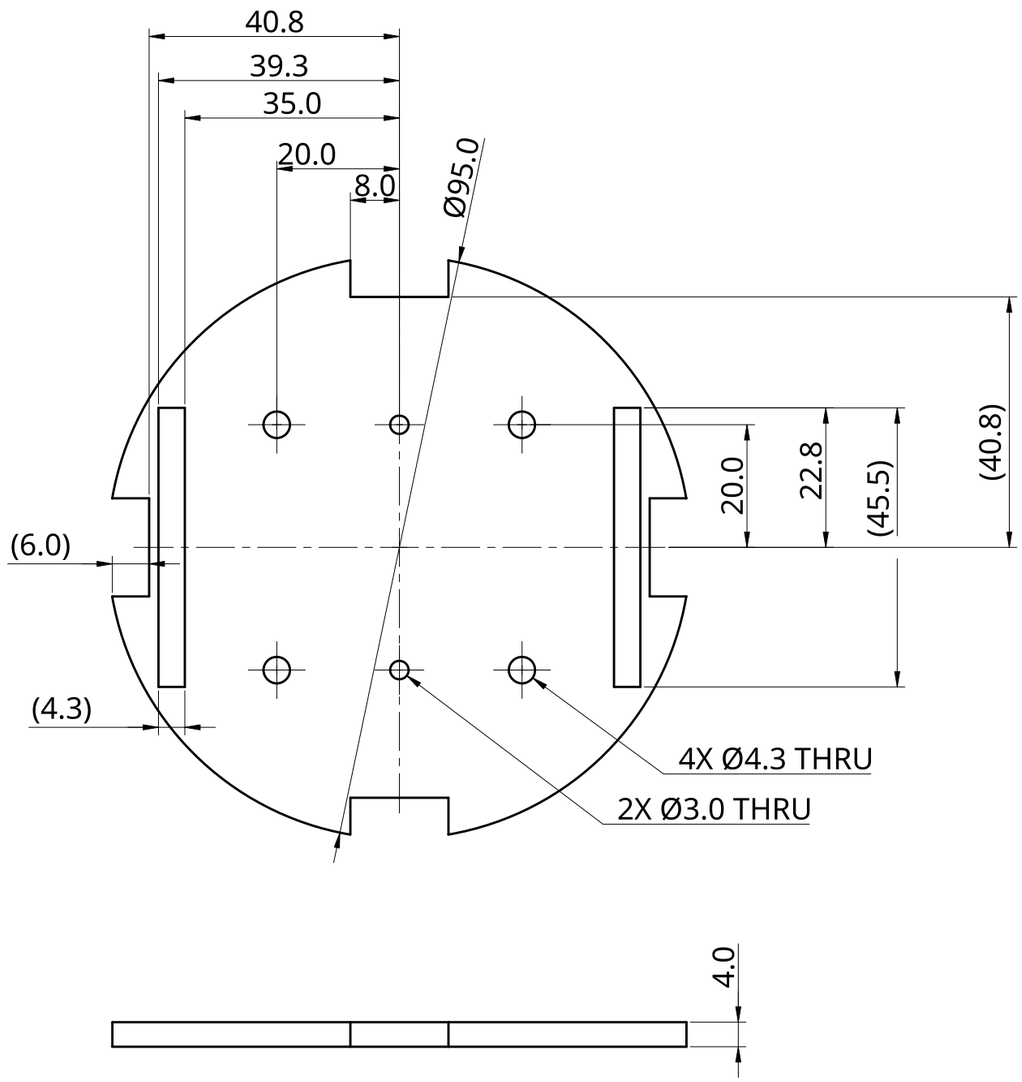
TOLERANCES

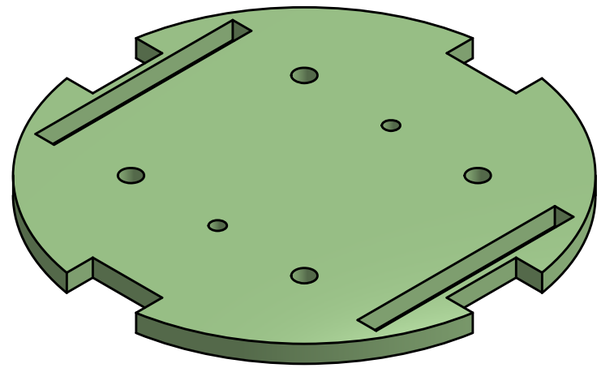
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: PAYLOAD NUT RETAINER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 41/66 A3





REV	DESCRIPTION	BY	DATE	APPV'D

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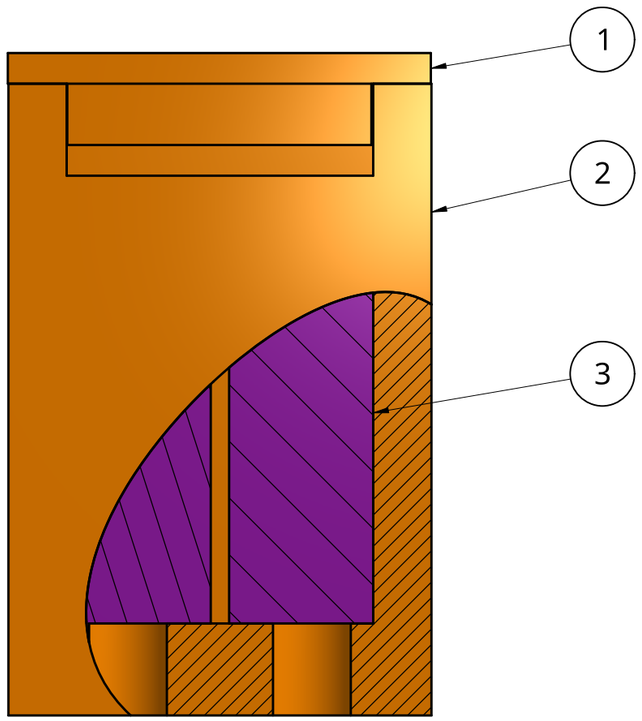
0.0±0.10
0.00±0.05

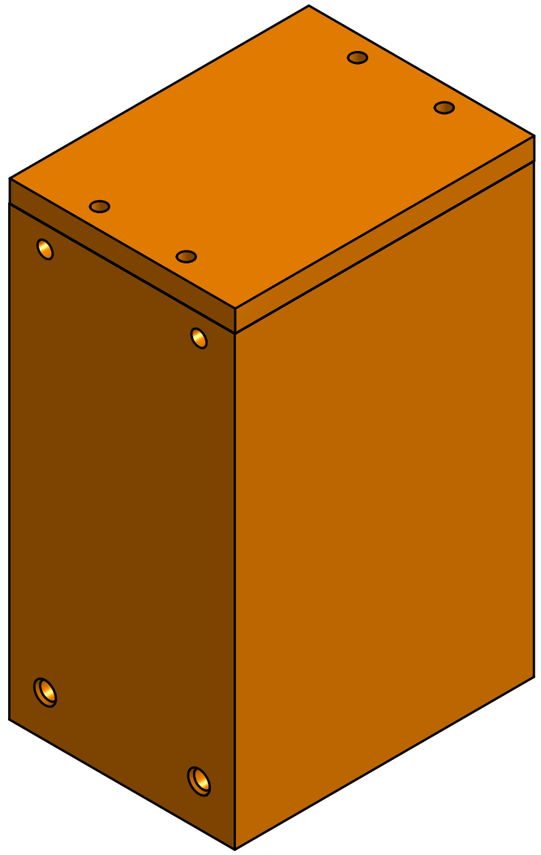
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: PAYLOAD BASEPLATE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: G10 FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 42/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
1	BATTERY LID	1	PART	MANUFACTURED	ABS
2	PAYLOAD BATTERY CONTAINER	1	PART	MANUFACTURED	ABS
3	BATTERY	2	PART	PLACEHOLDER	N/A





REV	DESCRIPTION	BY	DATE	APPV'D

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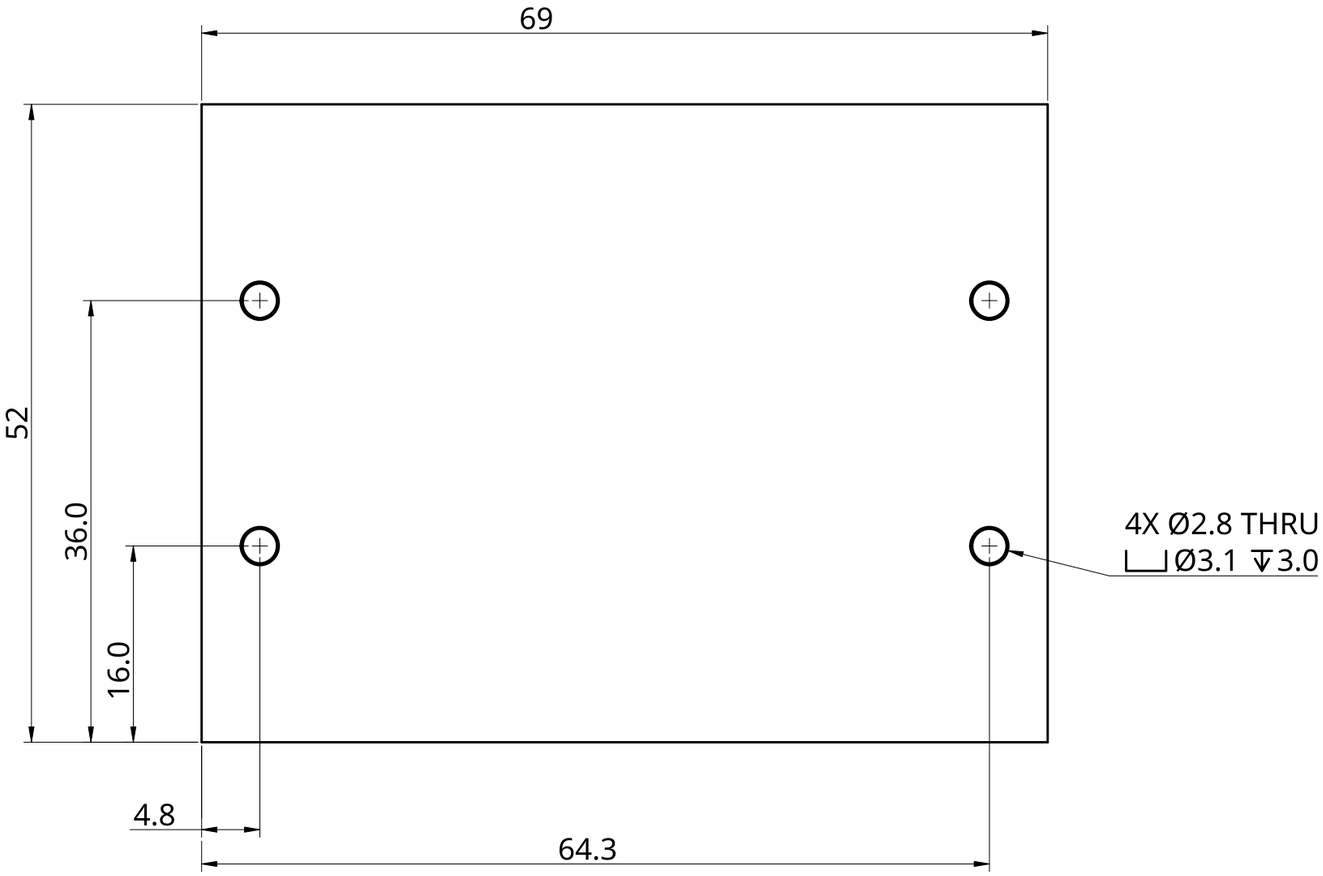
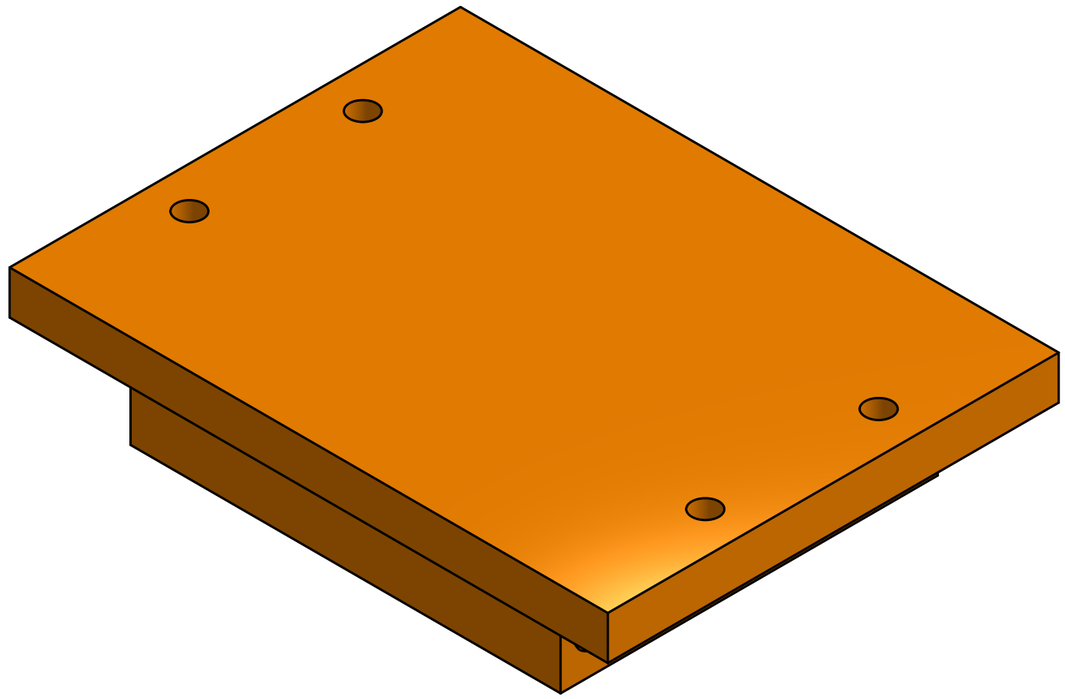
0.0±0.10
0.00±0.05

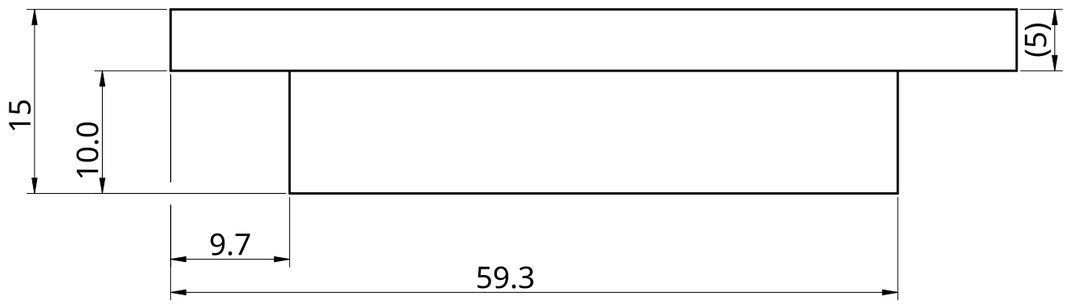
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

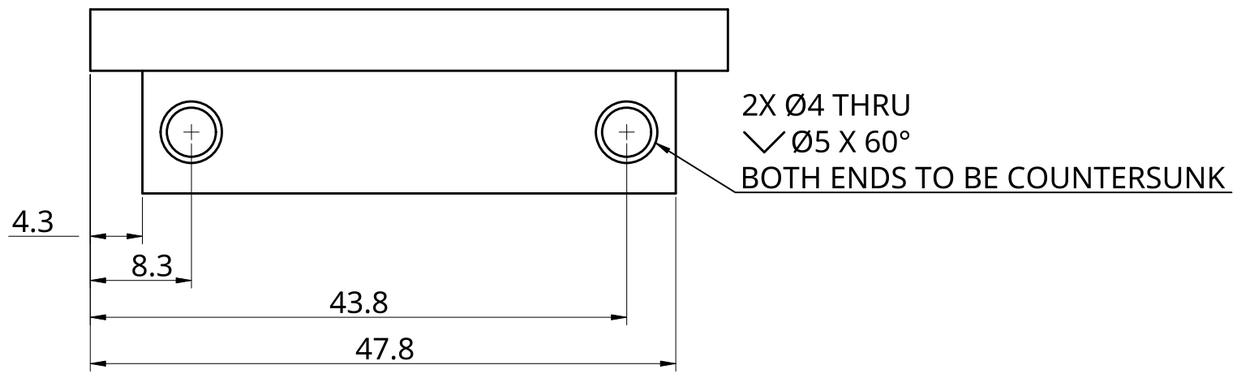
TITLE: PAYLOAD BATTERY ASSEMBLY

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 43/66 A3









REV	DESCRIPTION	BY	DATE	APPV'D

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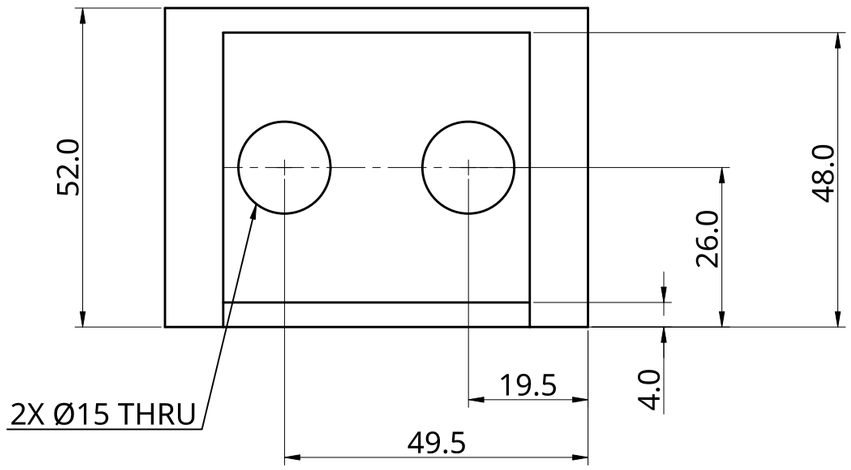
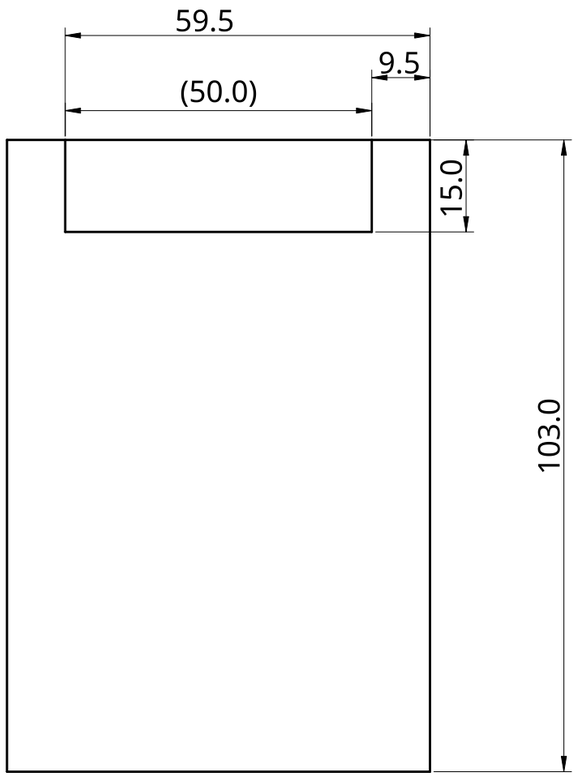
0.±0.50
0.0±0.10
0.00±0.05

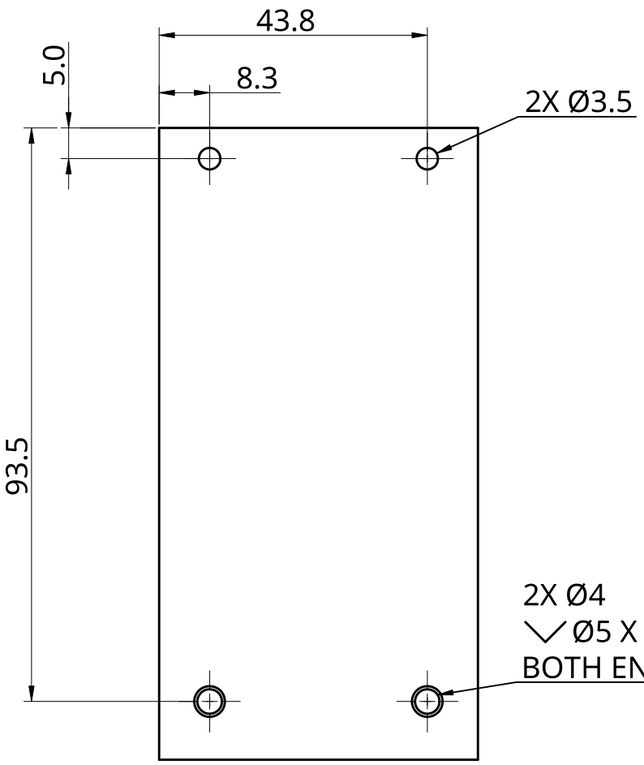
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

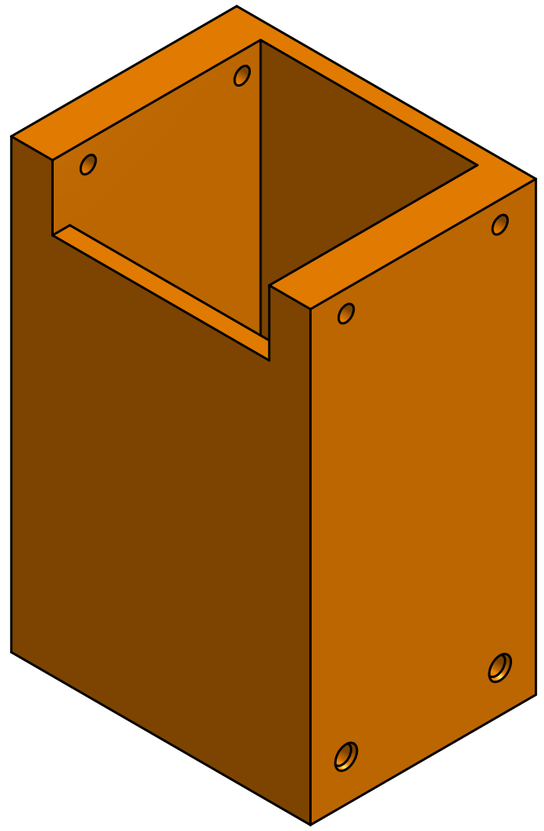
TITLE: PAYLOAD BATTERY LID

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 44/66 A3









REV	DESCRIPTION	BY	DATE	APPV'D

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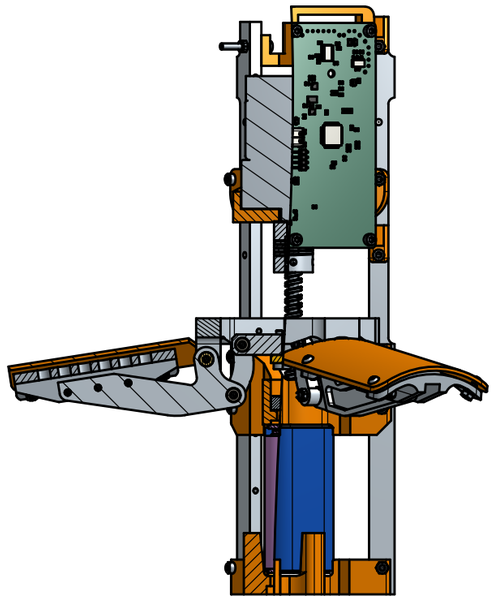
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

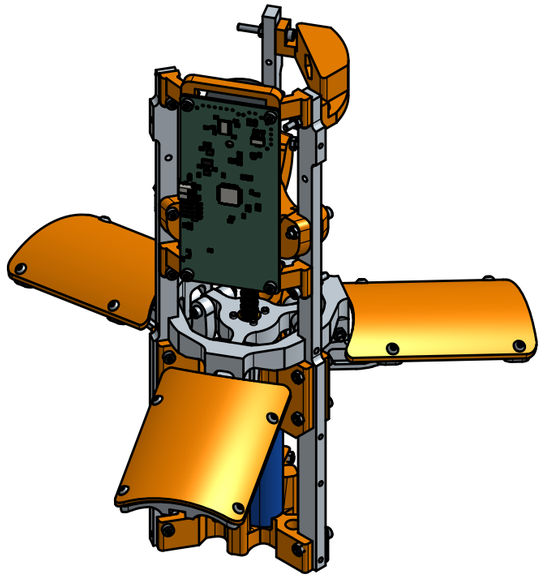
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

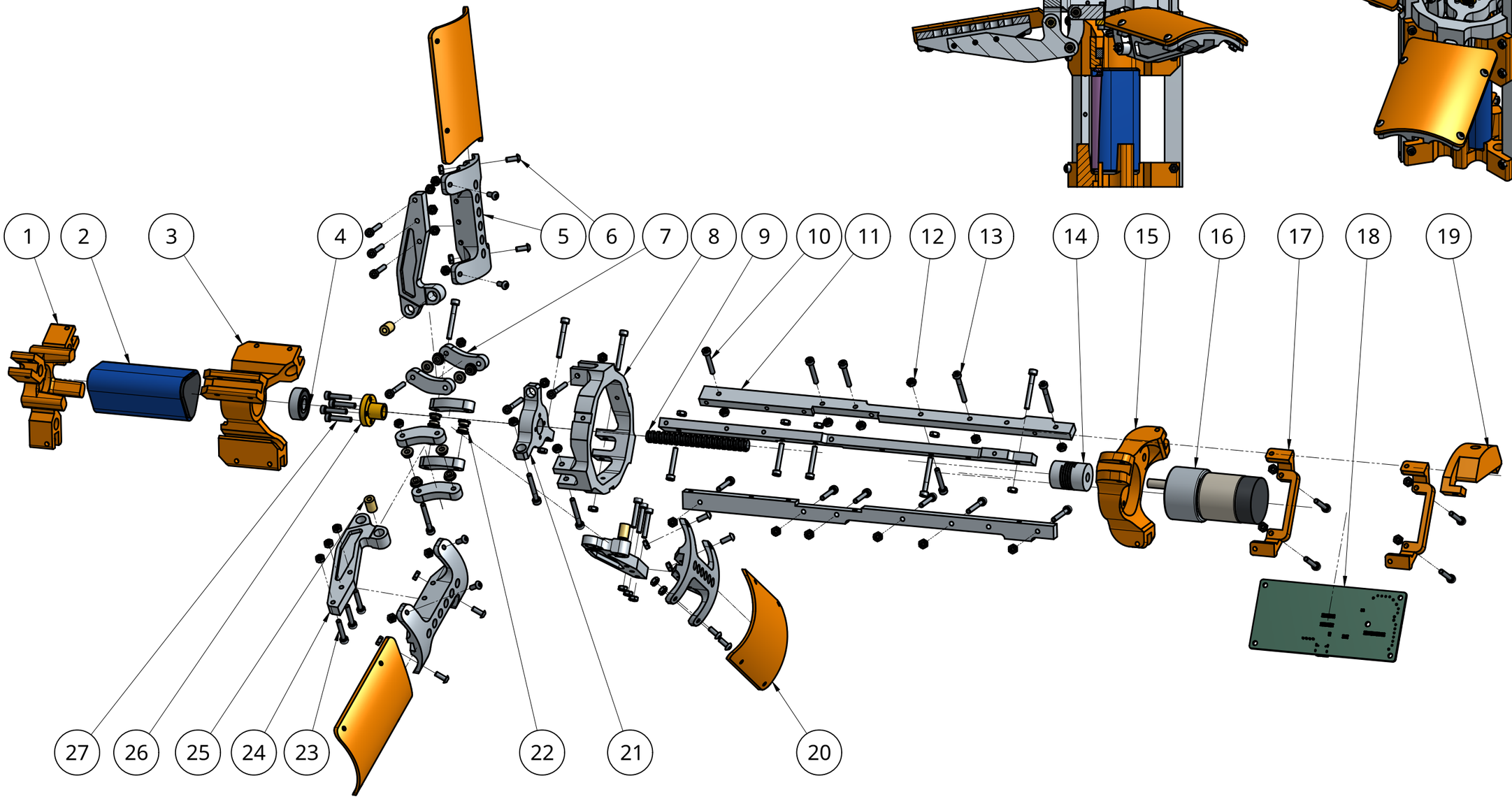
TITLE: PAYLOAD BATTERY CONTAINER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 45/66 A3

SEE PG. 44, 45 FOR BOM







REV	DESCRIPTION	BY	DATE	APPV'D

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0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AIRBRAKE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:3
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 46/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
1	LOWER BATTERY HOLDER	1	PART	MANUFACTURED	ABS
2	3S 18650 PACK	1	PART	PURCHASED	N/A
3	UPPER BATTERY HOLDER	1	PART	MANUFACTURED	ABS
4	MAIN BEARING	1	PART	PURCHASED	N/A
5	FLAP REINFORCEMENT	3	PART	MANUFACTURED	ALUMINIUM - 6061
6	HEX SOCKET HEAD BUTTON SCREW M3X0.50 X 8	12	PART	PURCHASED	CLASS 12.9 STEEL
7	ACTUATOR ARM	6	PART	MANUFACTURED	ALUMINIUM
8	ARM FLAP MOUNT	1	PART	MANUFACTURED	ALUMINIUM
9	LEAD SCREW	1	PART	PURCHASED	N/A
10	HEX SOCKET HEAD CAP SCREW M3X0.50 X 20	15	PART	PURCHASED	CLASS 12.9 STEEL
11	SPINE	3	PART	MANUFACTURED	ALUMINIUM - 6061
12	HEX NUT M3X0.50	51	PART	PURCHASED	GRADE 8 STEEL
13	HEX SOCKET HEAD CAP SCREW M3X0.50 X 25	15	PART	PURCHASED	CLASS 12.9 STEEL
14	FLEXABILE COUPLER	1	PART	PURCHASED	ALUMINIUM - 6061
15	MOTOR MOUNT	1	PART	MANUFACTURED	ABS
16	MOTOR	1	PART	PURCHASED	N/A
17	PCB MOUNTING BRACKET	2	PART	MANUFACTURED	ABS
18	AIRBRAKE PCB	1	PART	MANUFACTURED	FR-4 FIBREGLASS

SEE PG. 45 FOR CONTINUED BOM

REV	DESCRIPTION	BY	DATE	APPV'D

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0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AIRBRAKE BOM 1/2

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: N/A		DRAWN	G. O'BRIEN		2026-04-08	SCALE: N/A
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 47/66 A3

NO.	NAME	QTY.	TYPE	SOURCE	MATERIAL
19	FORWARD RAIL BUTTON MOUNT	1	PART	MANUFACTURED	ABS
20	FLAP	3	PART	MANUFACTURED	CARBON FIBRE WRAPPED ABS
21	ACTUATOR	1	PART	MANUFACTURED	ALUMINIUM - 6061
22	ARM BEARING	12	PART	MANUFACTURED	BRASS
23	HEX SOCKET HEAD CAP SCREW M3X0.50 X 18	9	PART	PURCHASED	CLASS 12.9 STEEL
24	FLAP ARM	3	PART	MANUFACTURED	ALUMIINUM - 6061
25	ARM BUSHINGS	3	PART	MANUFACTURED	BRASS
26	LEADSCREW NUT	1	PART	PURCHASED	BRASS
27	HEX SOCKET HEAD CAP SCREW M3X0.50 X 16	4	PART	PURCHASED	CLASS 12.9 STEEL

REV	DESCRIPTION	BY	DATE	APPV'D

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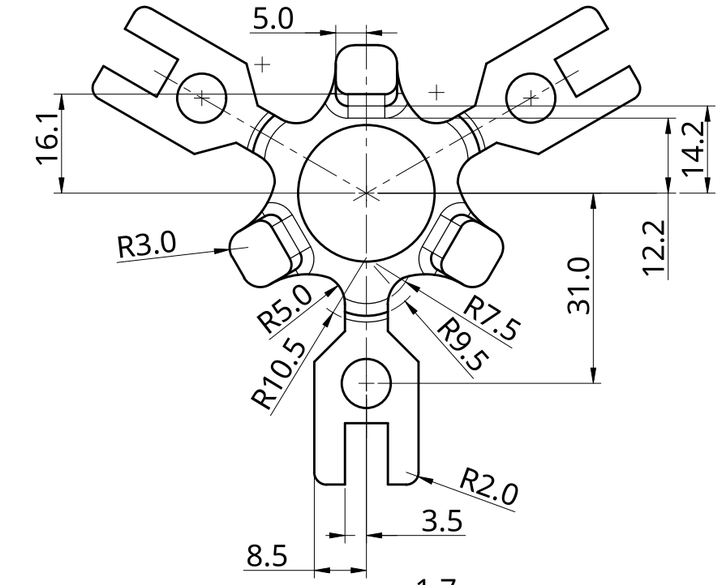
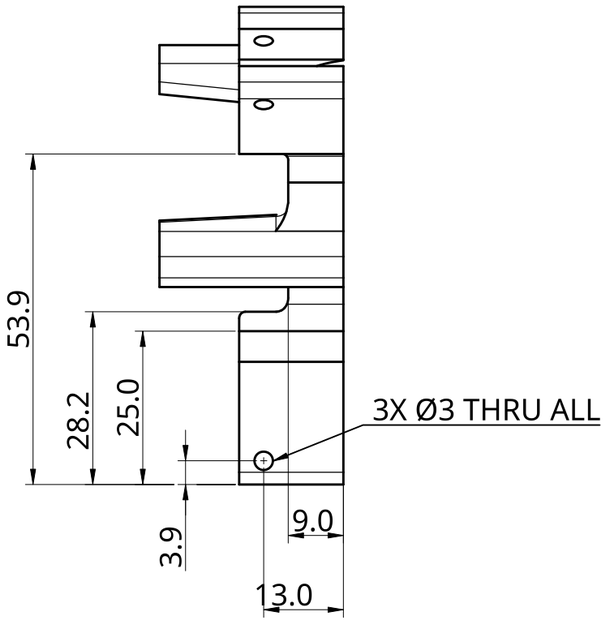
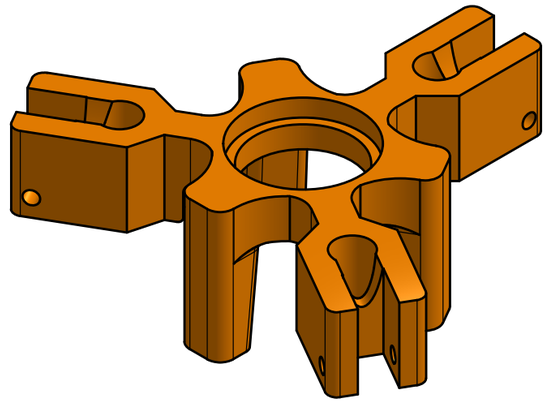
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

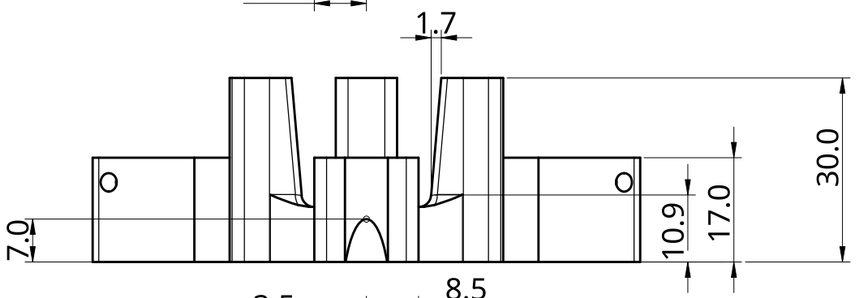
TITLE: AIRBRAKE BOM 2/2

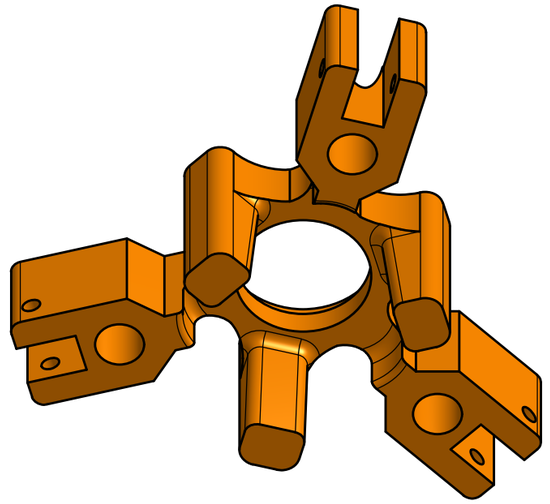
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: N/A		DRAWN	G. O'BRIEN		2026-04-08	SCALE: N/A
MATERIAL: N/A			NAME	SIGNATURE	DATE	SHEET: 48/66 A3

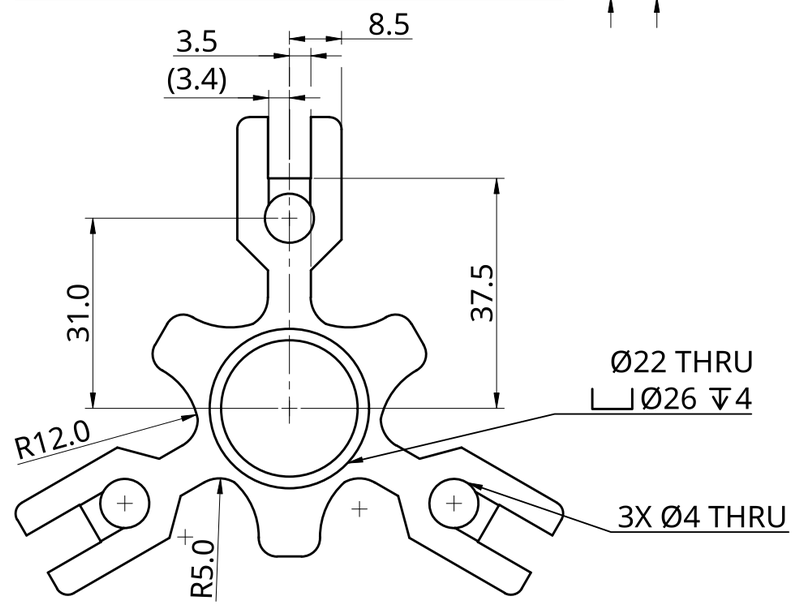












REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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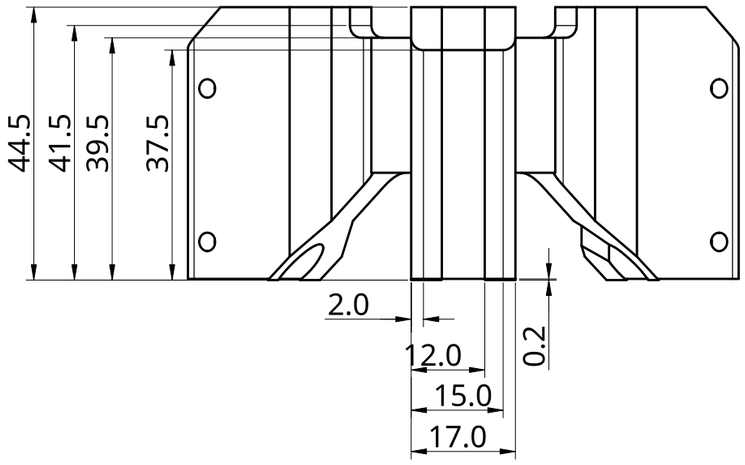
TOLERANCES

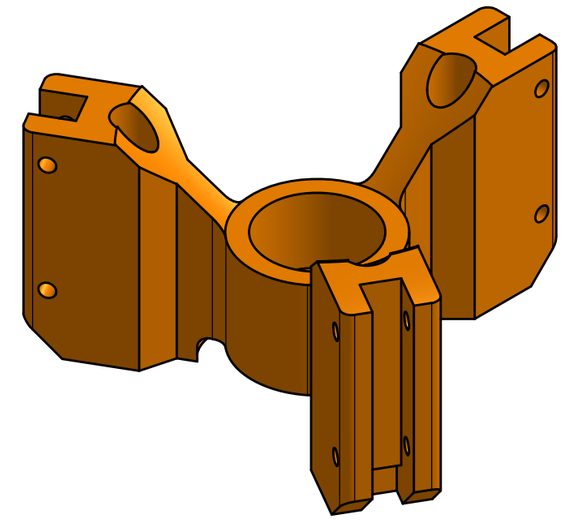
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

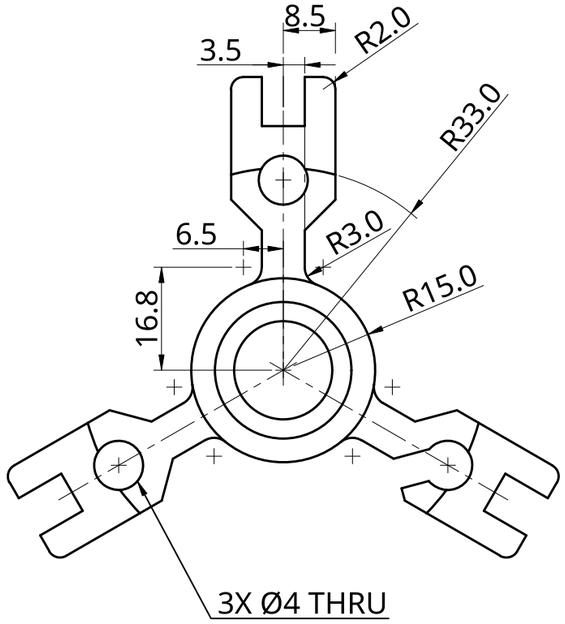
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

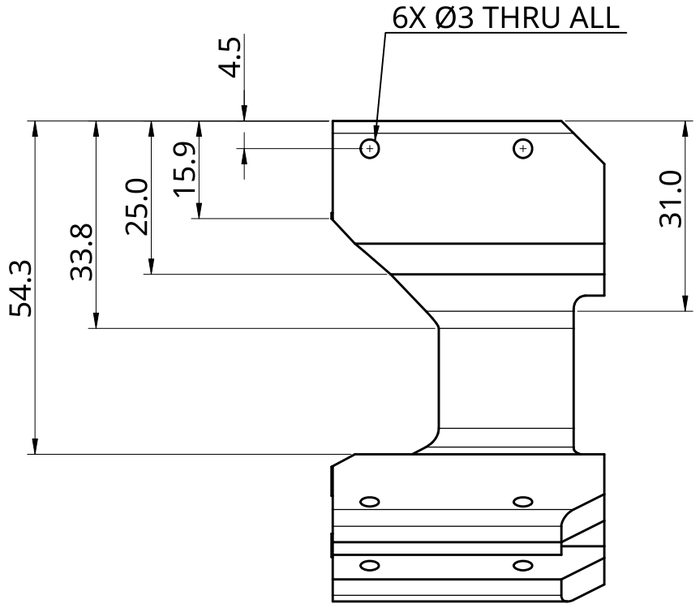
TITLE: BOTTOM BATTERY MOUNT

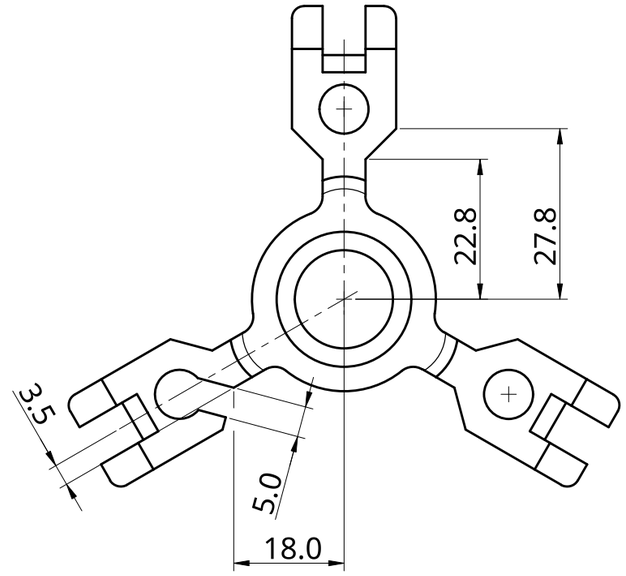
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 49/66 A3











REV	DESCRIPTION	BY	DATE	APPV'D

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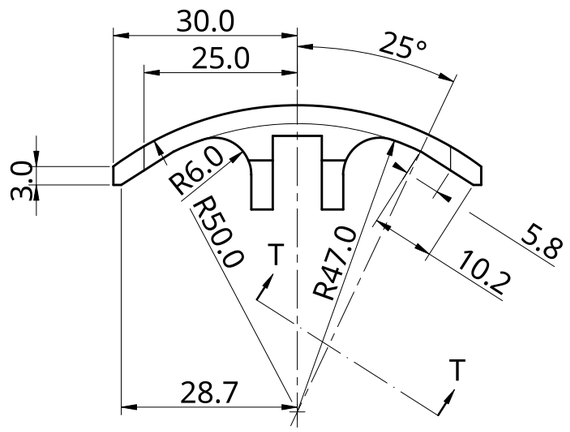
TOLERANCES

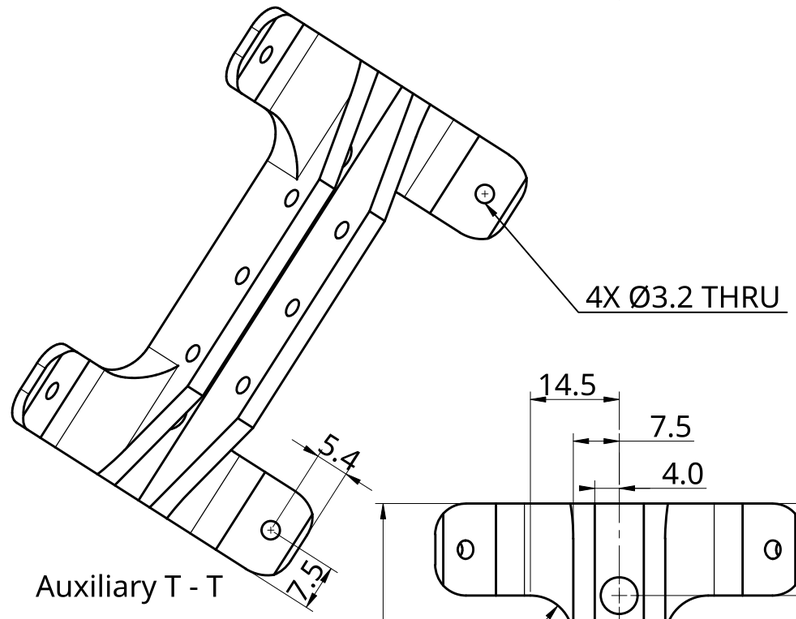
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

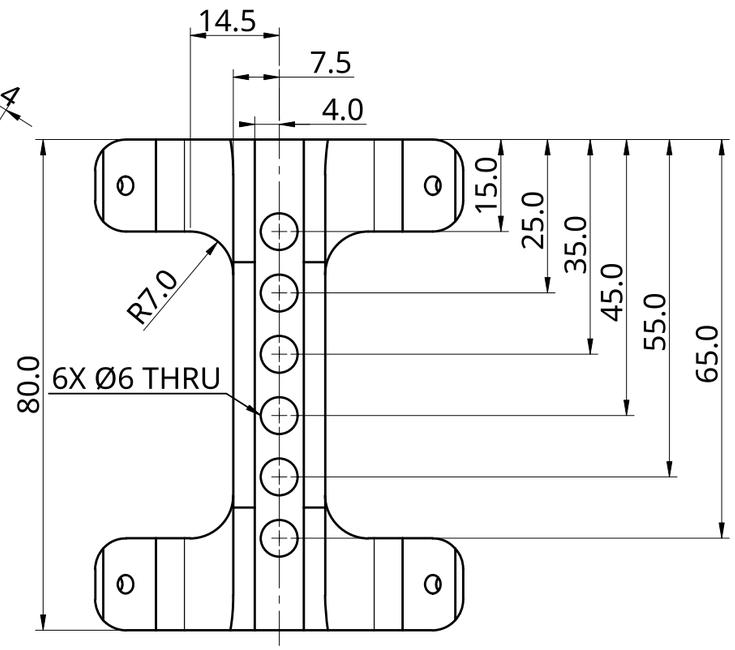
TITLE: BEARING MOUNT

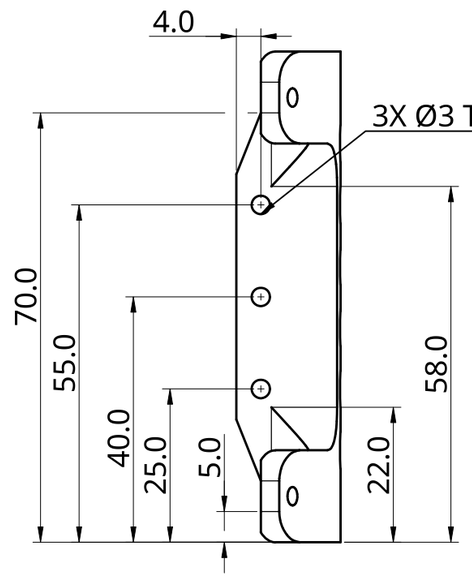
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 50/66 A3

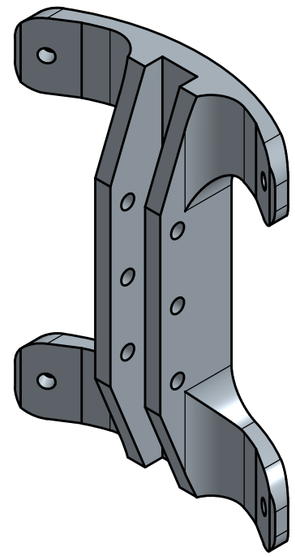


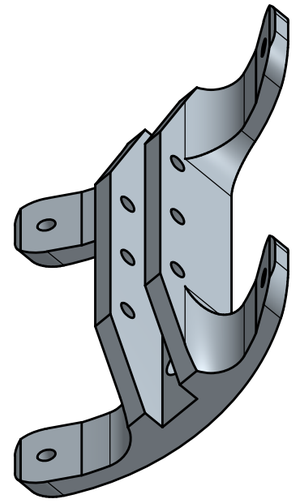


Auxiliary T - T









DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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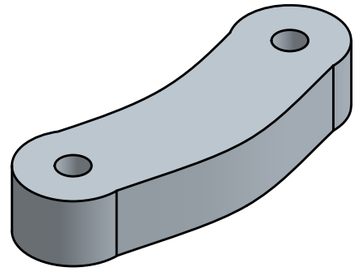
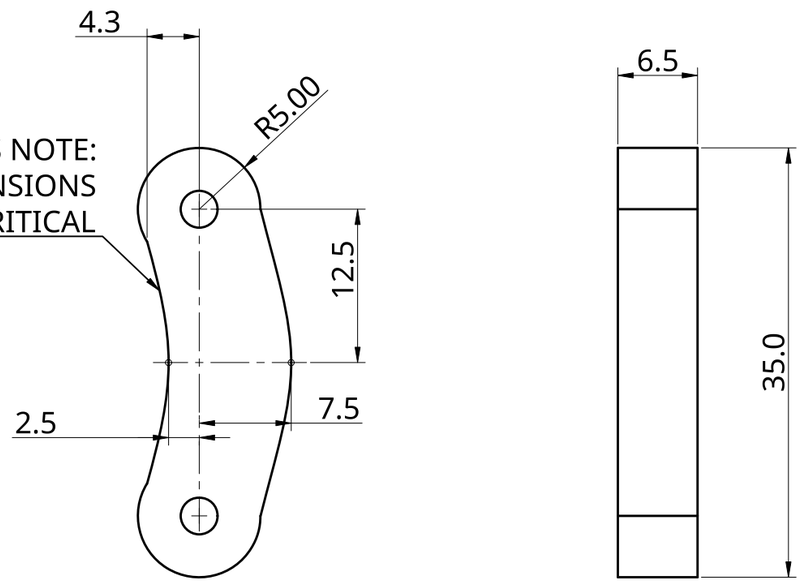
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FLAP REINFORCEMENT

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 3		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ALUMINUM - 6061			NAME	SIGNATURE	DATE	SHEET: 51/66 A3





TO BE MANUFACTURED ON WATERJET CUTTER
FILE ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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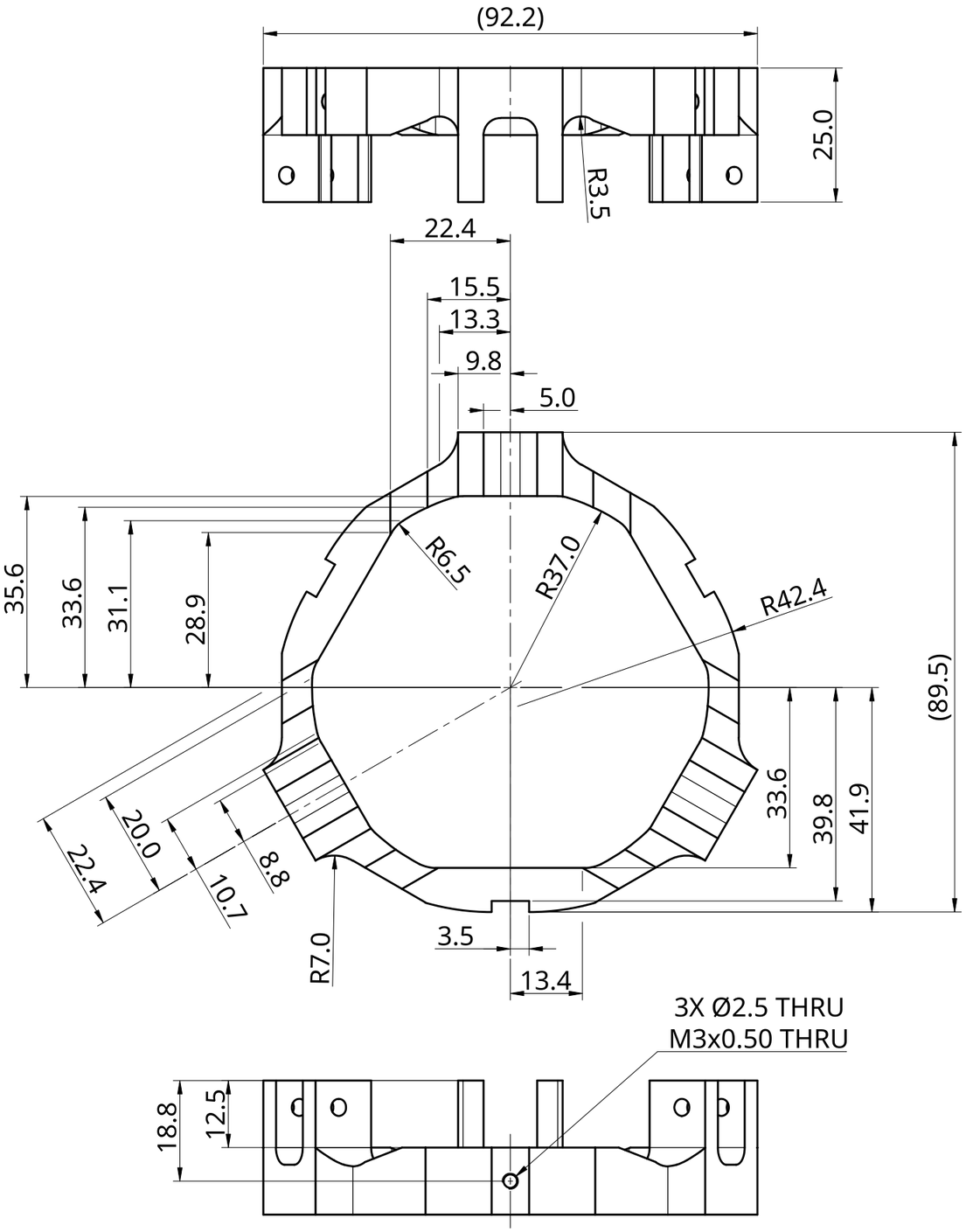
TOLERANCES

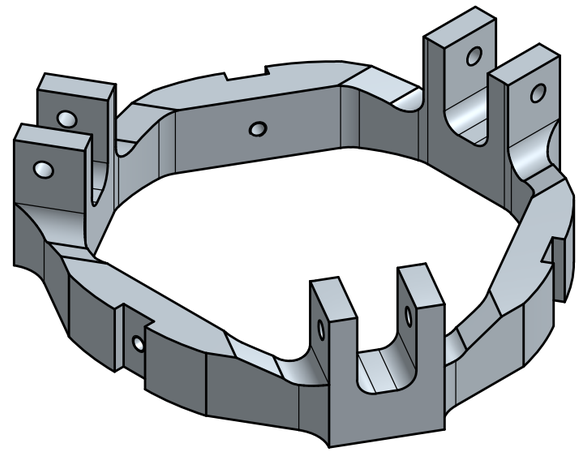
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: ACTUATOR ARM

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 6		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: ALUMINUM - 6061			NAME	SIGNATURE	DATE	SHEET: 52/66 A3





DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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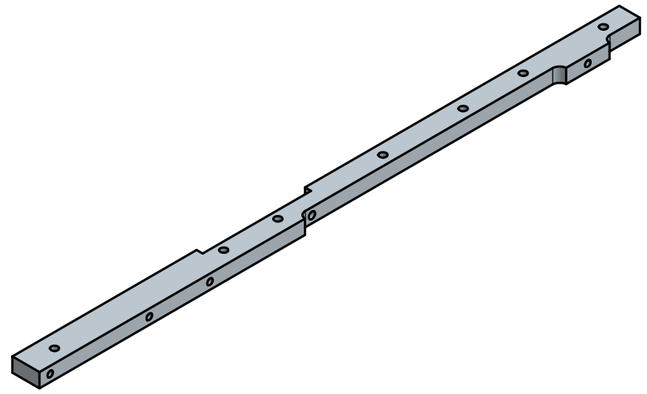
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: ARM FLAP MOUNT

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BIREN		2026-04-08	SCALE: 1:1
MATERIAL: ALUMINUM - 6061			NAME	SIGNATURE	DATE	SHEET: 53/66 A3



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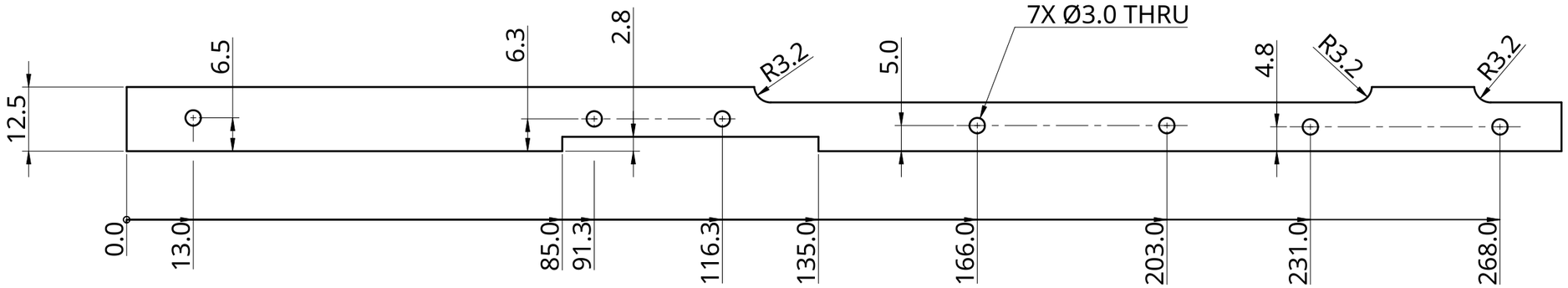

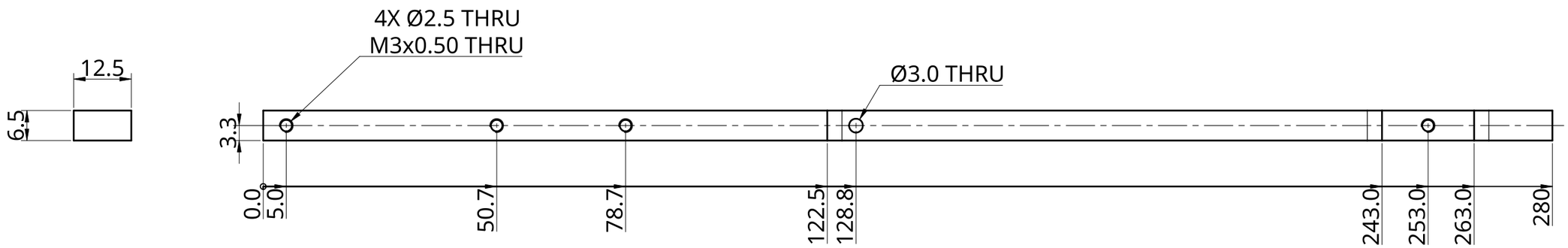

DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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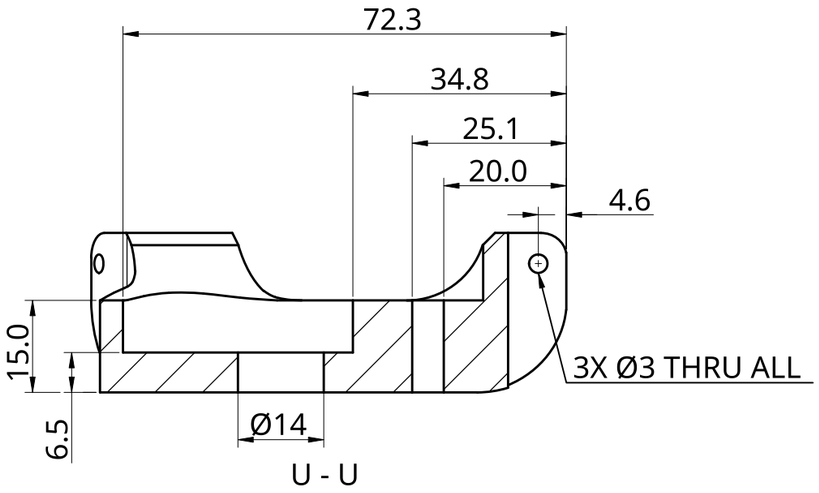
TOLERANCES

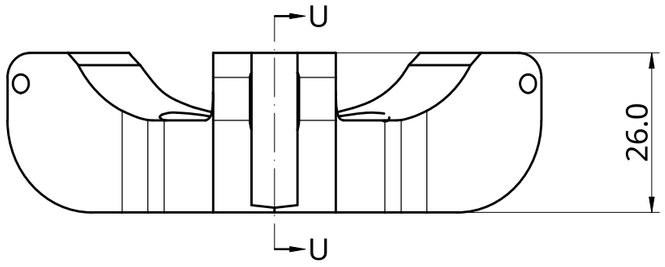
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

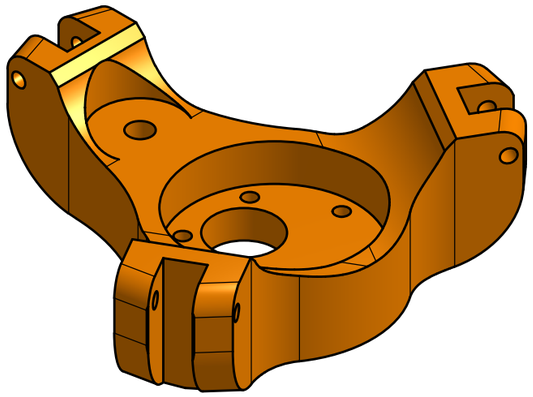
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

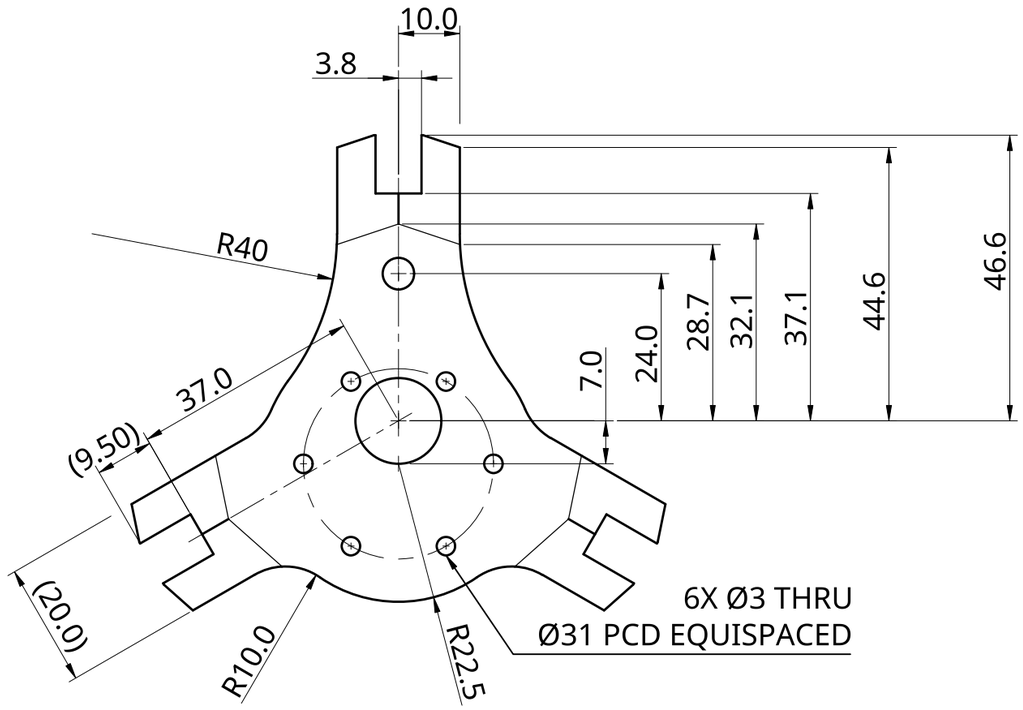
TITLE: STRUCTURE SPINE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 3		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ALUMINIUM - 6061			NAME	SIGNATURE	DATE	SHEET: 54/66 A3









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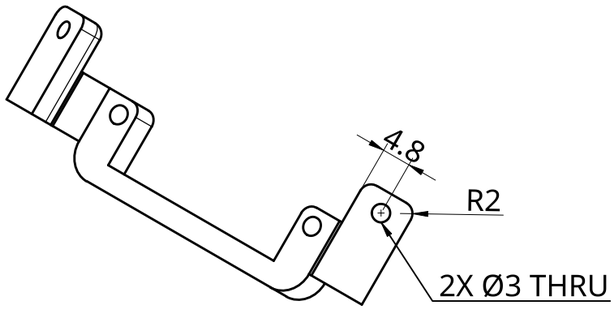
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

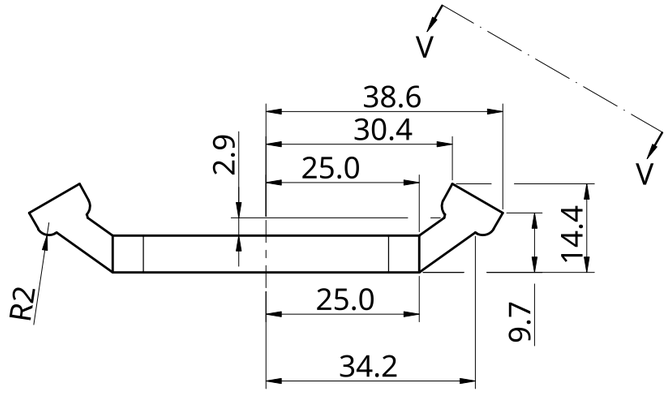
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

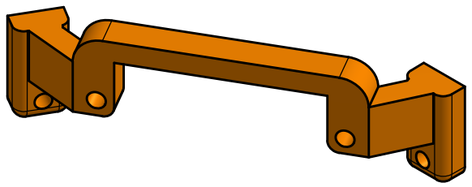
TITLE: MOTOR MOUNT

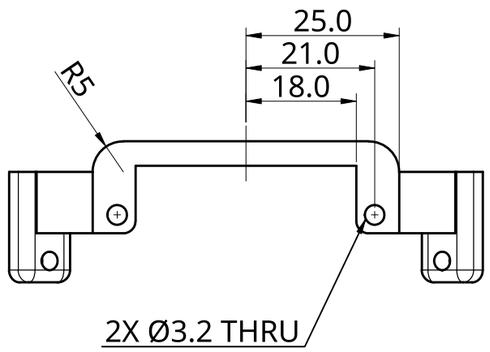
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 55/66 A3



Auxiliary V - V







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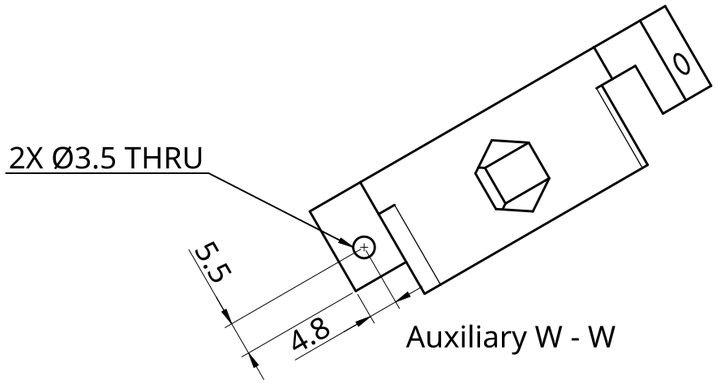
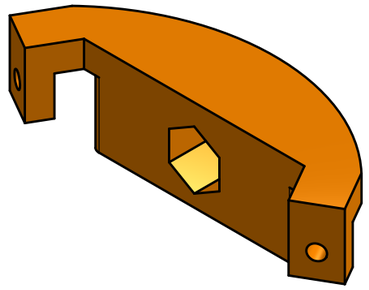
0.0±0.10
0.00±0.05

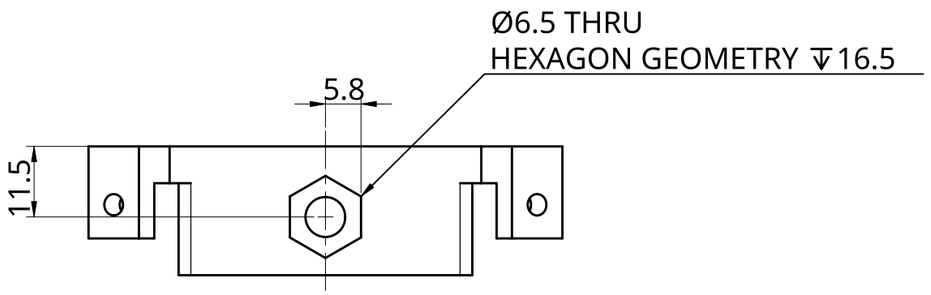
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

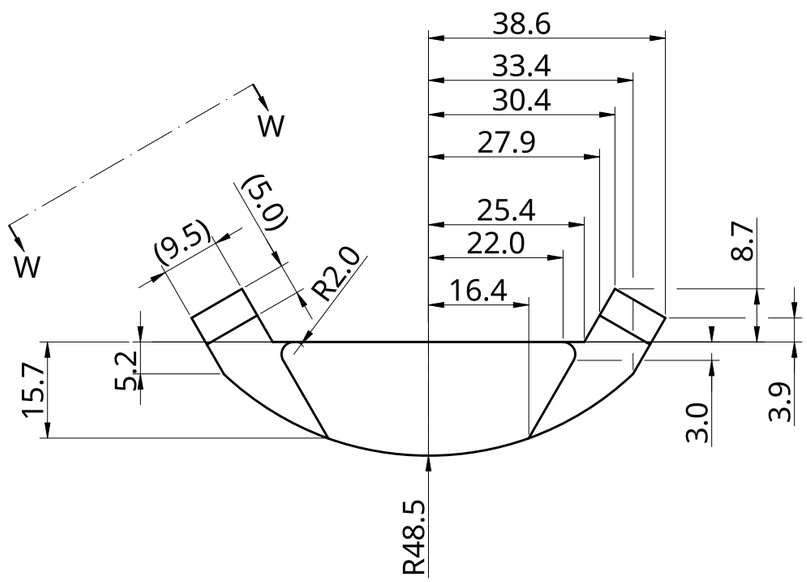
TITLE: PCB MOUNT

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 2		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 56/66 A3









DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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NU Rocketry Chief Engineer
EMAIL: tate.derrick@uon.edu.au

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DRAWN TO AS1100-1992
ALL DIMENSIONS IN MILLIMETRES
UNLESS OTHERWISE STATED
DO NOT SCALE

CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

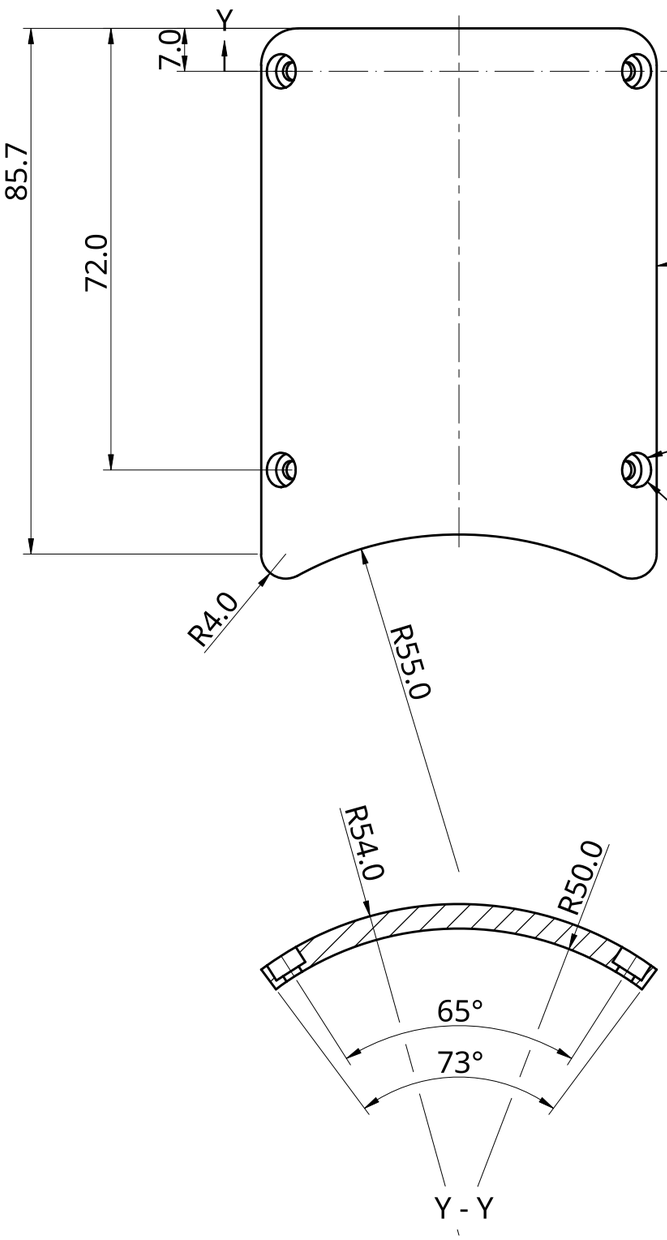
TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FORWARD RAIL BUTTON MOUNT

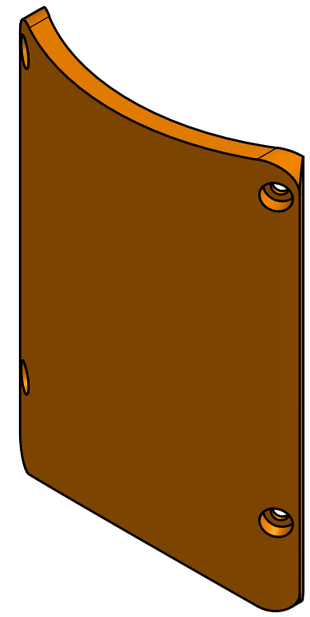
PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 57/66 A3



MANUFACTURING NOTE:
ON SURFACE WRAP WITH SINGLE LAYER CARBON FIBRE

ALL HOLES NORMAL TO SURFACE

4X Ø3 THRU
⌴ Ø5.6 ↧2.9



REV	DESCRIPTION	BY	DATE	APPV'D

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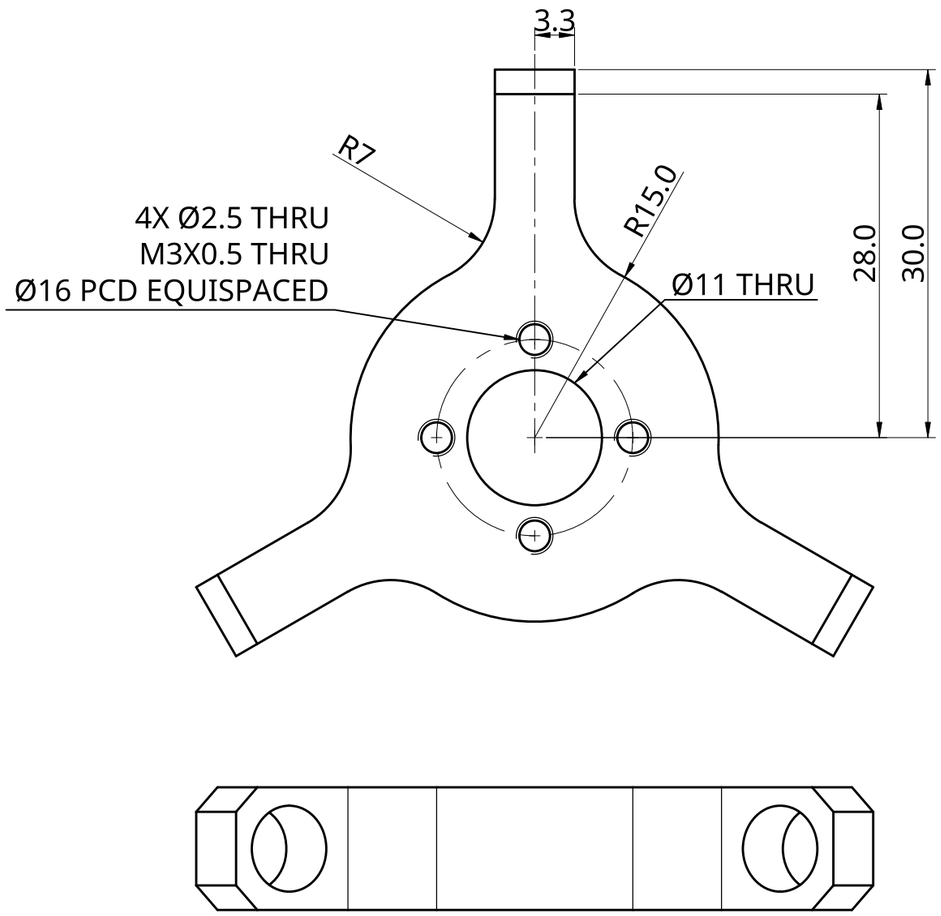
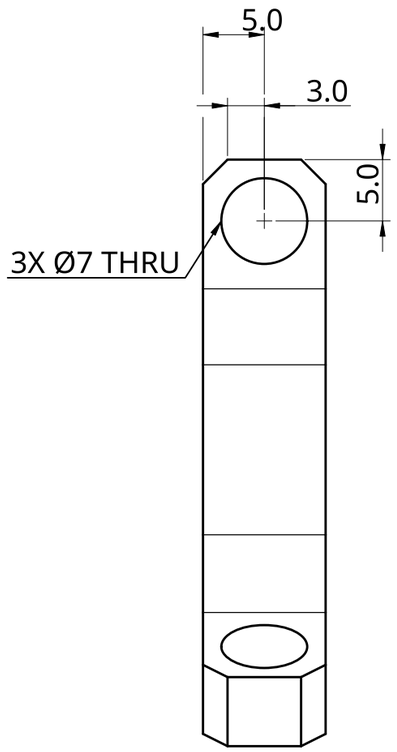
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

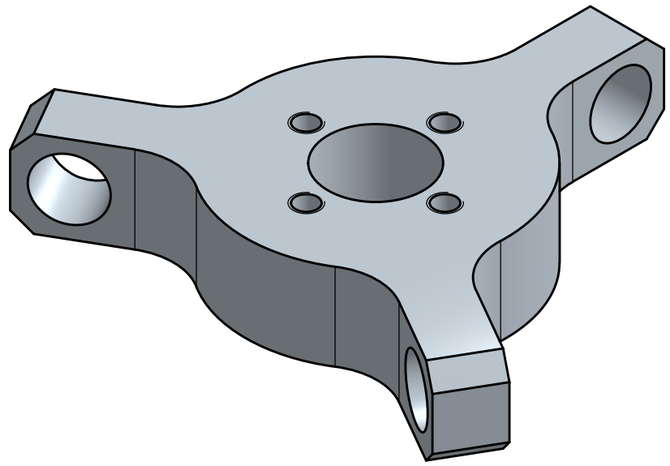
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FLAP

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 3		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 58/66 A3







INITIAL GEOMETRY TO BE WATERJET CUT
FILE/DEBUR ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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CAD GENERATED DRAWING
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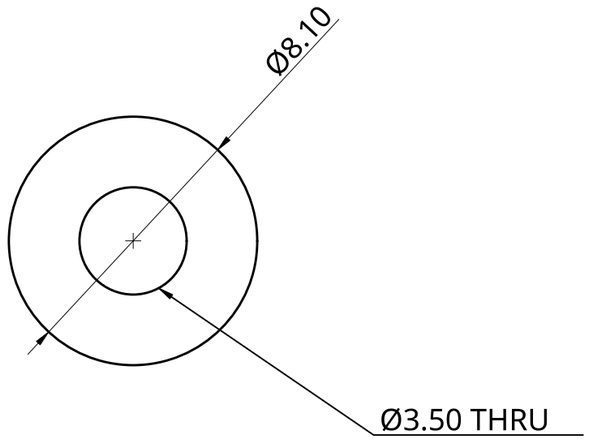
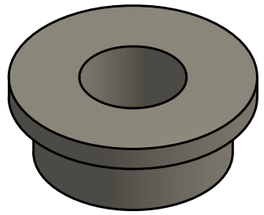
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

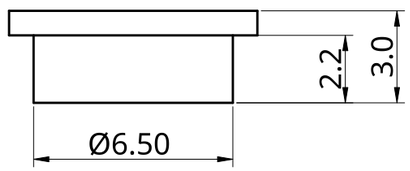
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: ACTUATOR

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 2:1
MATERIAL: ALUMINUM - 6061			NAME	SIGNATURE	DATE	SHEET: 59/66 A3







DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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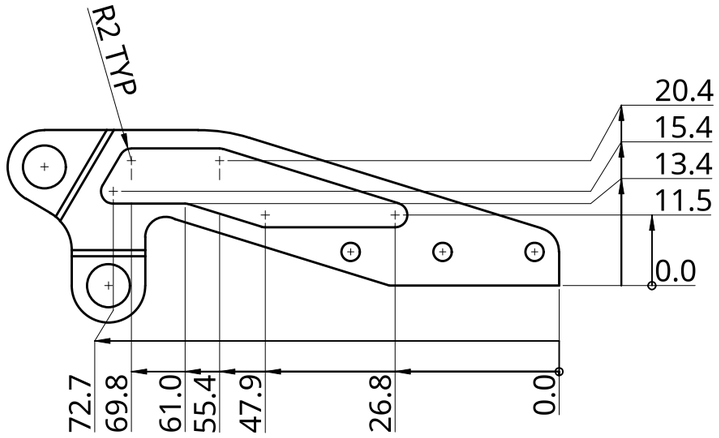
TOLERANCES

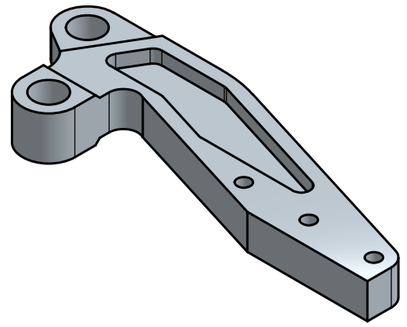
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

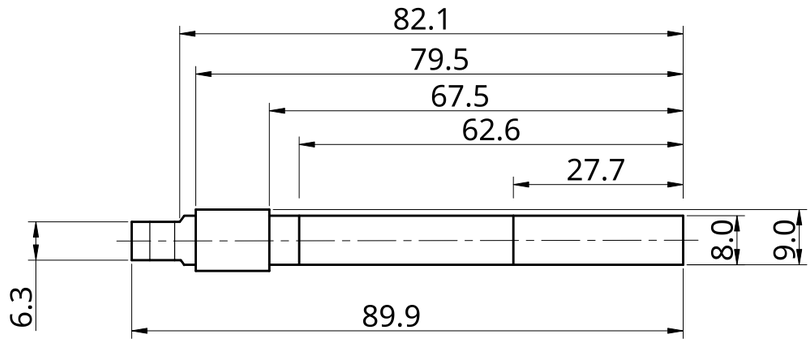
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

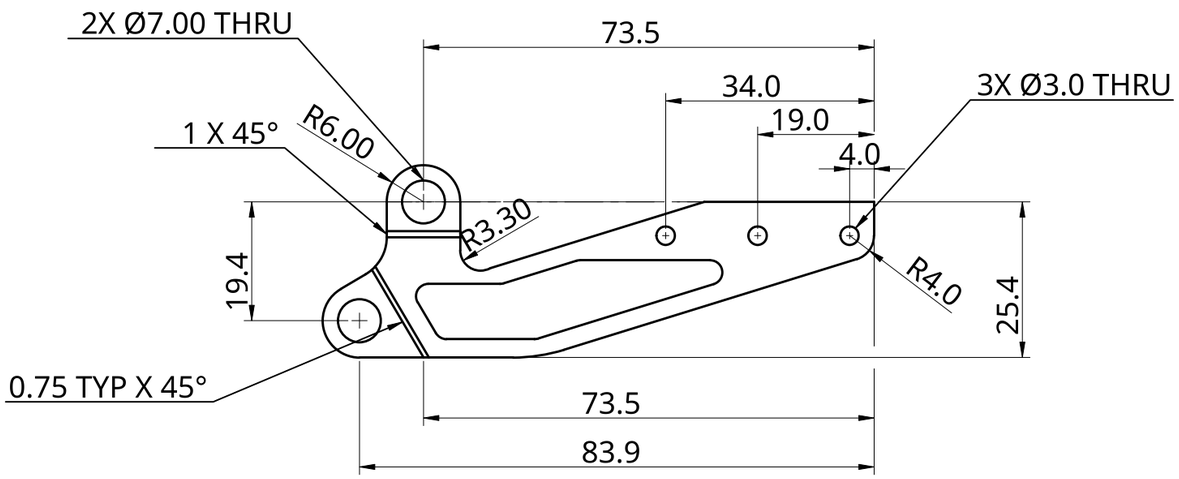
TITLE: ARM BEARING

PROJECT NO.: 67	APPV'D					DRAWING NO.:
FINISH: 3.2 UOS	CHK'D					REVISION:
QUANTITY: 12	DRAWN	G. O'BRIEN		2026-04-08		SCALE: 5:1
MATERIAL: BRASS		NAME	SIGNATURE	DATE		SHEET: 60/66 A3









DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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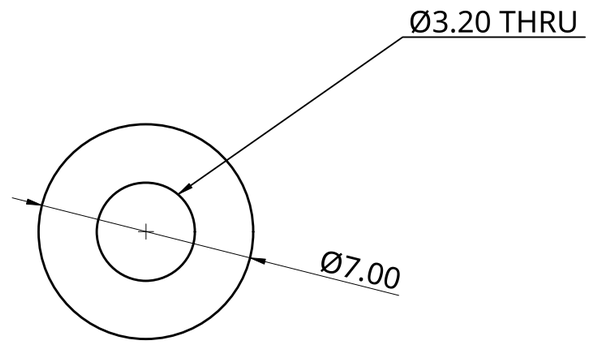
CAD GENERATED DRAWING
DO NOT MANUALLY UPDATE

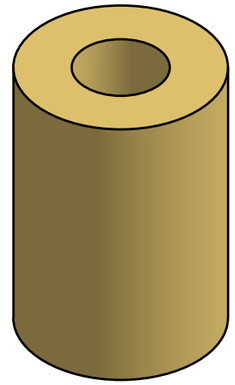
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

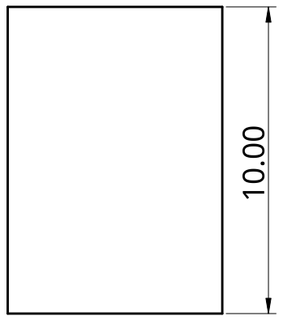
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: FLAP ARM

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 3		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ALUMINUM - 6061			NAME	SIGNATURE	DATE	SHEET: 61/66 A3







DEBUR ALL SHARP EDGES
CONTINUOUS 6mm FILLET WELD UNLESS OTHERWISE STATED

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

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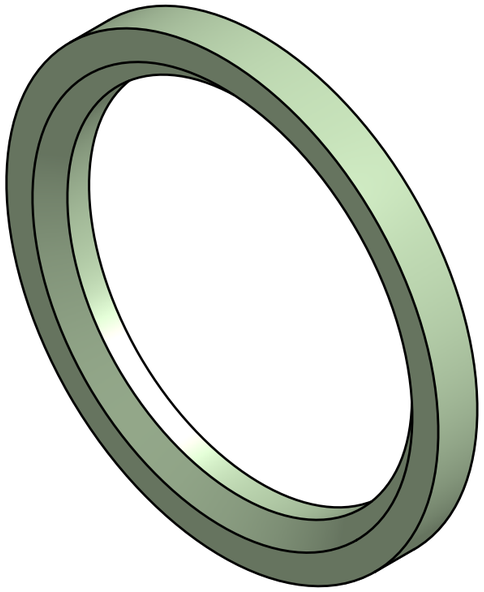
TOLERANCES

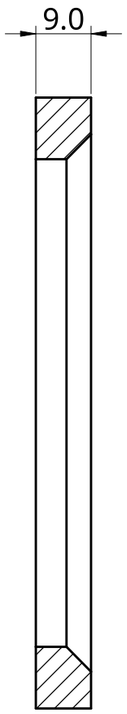
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

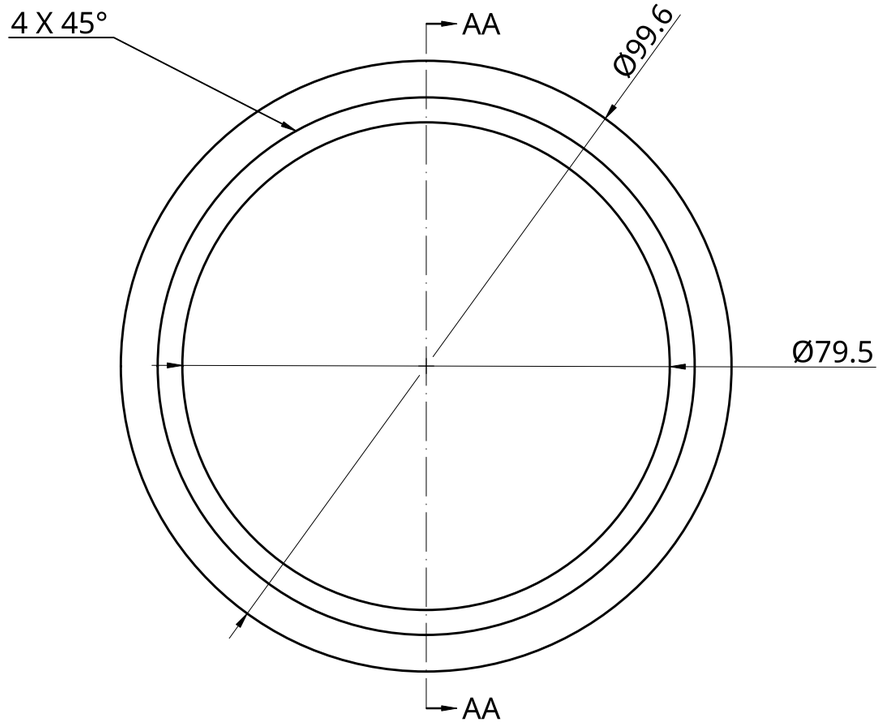
TITLE: BUSHING

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 3		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 5:1
MATERIAL: BRASS			NAME	SIGNATURE	DATE	SHEET: 62/66 A3





AA - AA



DEBUR ALL SHARP EDGES
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REV	DESCRIPTION	BY	DATE	APPV'D

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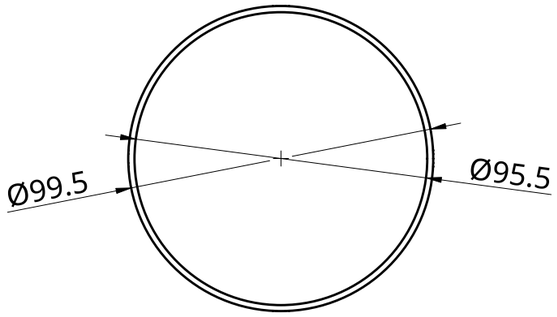
TOLERANCES

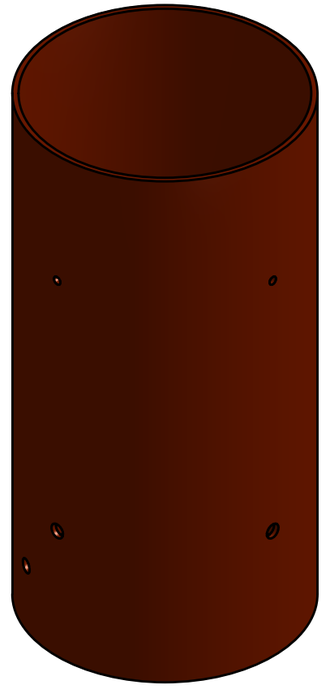
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

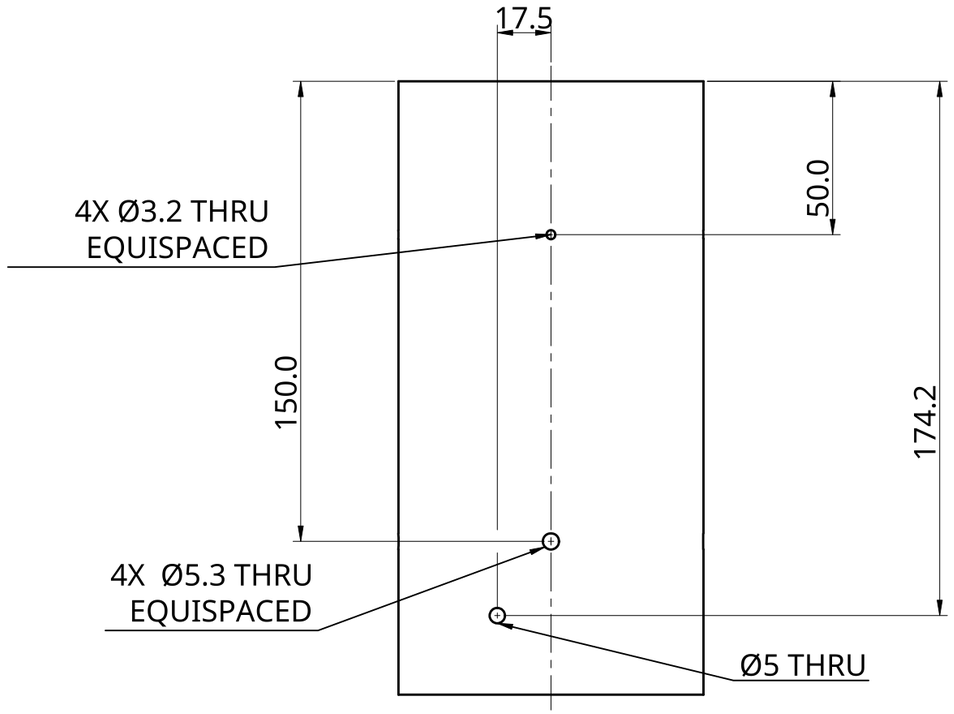
ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: MOTOR TUBE ALIGNMENT RING

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 63/66 A3







DEBUR ALL SHARP EDGES
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REV	DESCRIPTION	BY	DATE	APPV'D

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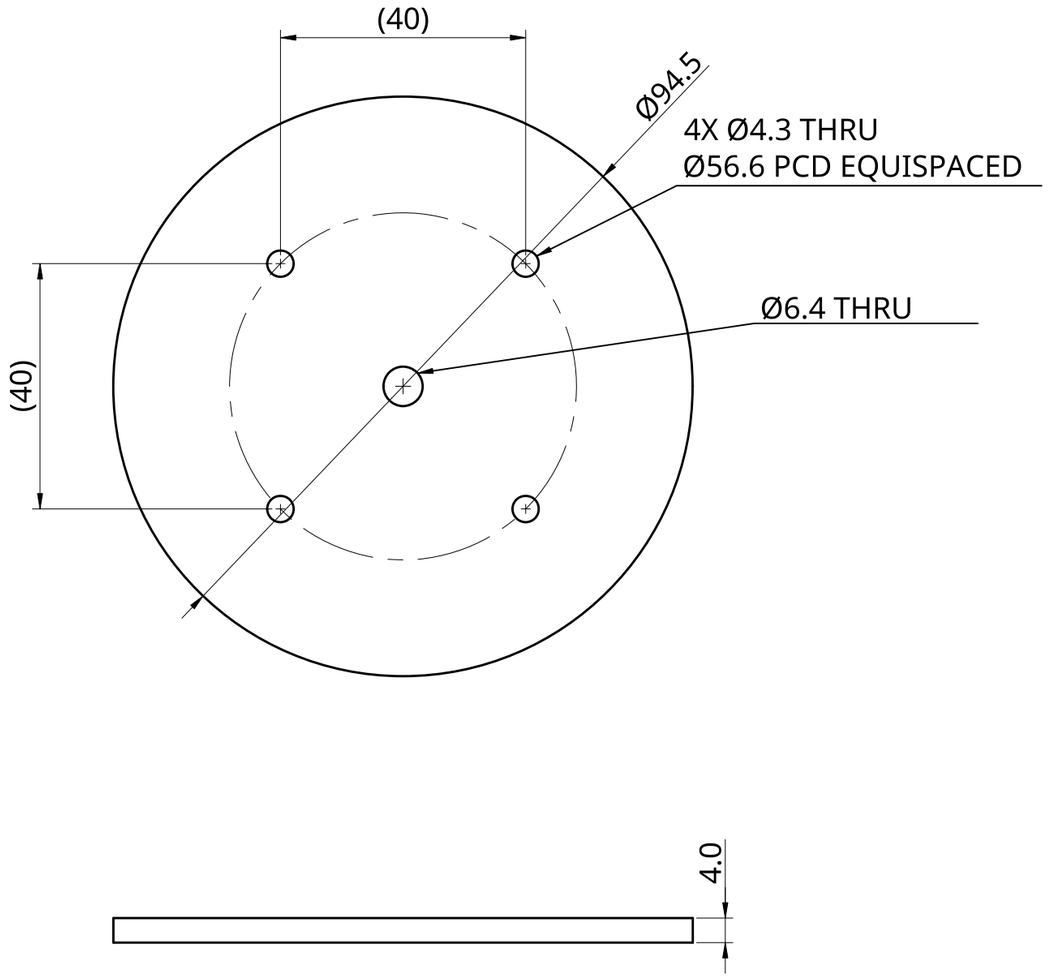
TOLERANCES

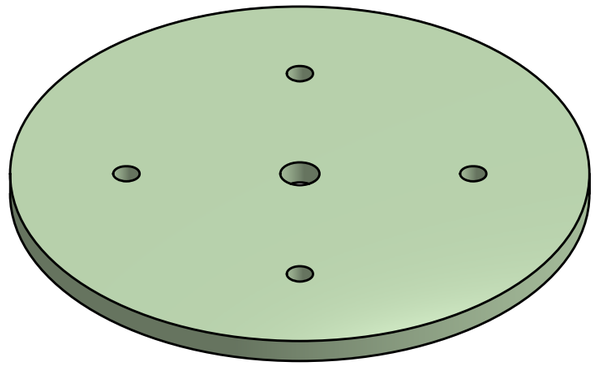
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: COUPLER

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:2
MATERIAL: FIBREGLASS			NAME	SIGNATURE	DATE	SHEET: 64/66 A3





TO BE MANUFACTURED ON WATERJET CUTTER
FILE ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

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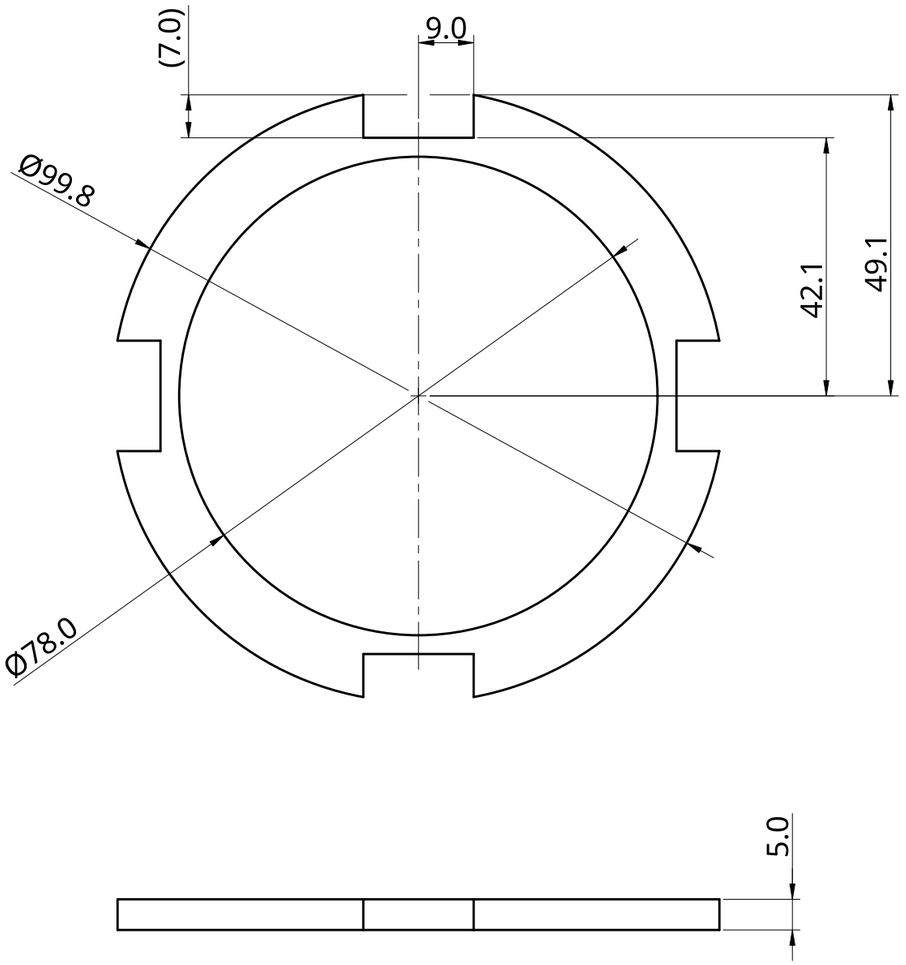
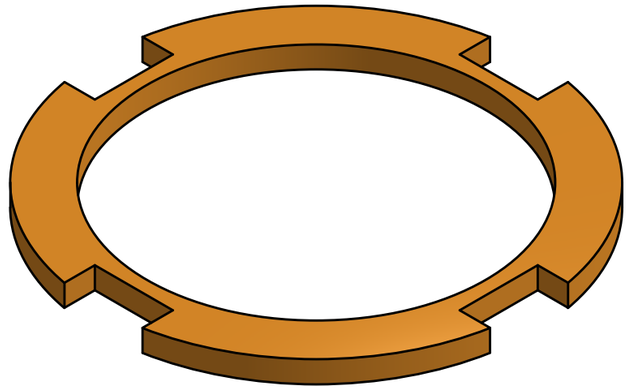
TOLERANCES
LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AFT COUPLER BULKHEAD

PROJECT NO.: 67	APPV'D					DRAWING NO.:
FINISH: 3.2 UOS	CHK'D					REVISION:
QUANTITY: 1	DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1	
MATERIAL: G10 FIBREGLASS		NAME	SIGNATURE	DATE	SHEET: 65/66	A3





TO BE MANUFACTURED ON WATERJET CUTTER
FILE ALL SHARP EDGES

REV	DESCRIPTION	BY	DATE	APPV'D

REVISIONS

NOTICE

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TOLERANCES

LINEAR
0.±0.50
0.0±0.10
0.00±0.05

ANGULAR
0.±0.20DEG
0.0±0.15DEG
0.00±0.10DEG

TITLE: AVIONICS STACK ALIGNMENT PLATE

PROJECT NO.: 67		APPV'D				DRAWING NO.:
FINISH: 3.2 UOS		CHK'D				REVISION:
QUANTITY: 1		DRAWN	G. O'BRIEN		2026-04-08	SCALE: 1:1
MATERIAL: ABS			NAME	SIGNATURE	DATE	SHEET: 66/66 A3

Appendix G - Calculations

Pressure Port

$$D_{Vent} = 0.004396 \cdot L \cdot \sqrt{\frac{D_{Tube}}{N}} \quad \text{Eq 8}$$

$$3.12 = 0.004396 \cdot 150 \cdot \sqrt{\frac{100}{3}} \quad \text{Eq 9}$$

Force on bulkhead

$$\Delta P = P_{sl} - P_{apogee} = 101325 - 57477 = 43848 \quad \text{Eq 10}$$

$$A = \pi r^2 = \pi \times 0.05^2 = 0.00785 \quad \text{Eq 11}$$

$$F = P \times A = 43848 \times 0.00785 = 344.2 \quad \text{Eq 12}$$

$$FoS = \frac{Failure\ stress}{working\ stress} = \frac{676}{344.2} = 1.96 \quad \text{Eq 13}$$

Aft Coupler Fastener Calculations

$$F_{Aft} = P_{shear} \times A_{minor} = 280 \times 12.7 = 3556 \quad \text{Eq 14}$$

$$FOS_{main} = \frac{N \times F_{AftCup}}{F_{main}} = \frac{4 \times 3556}{5200} = 2.7 \quad \text{Eq 15}$$

$$FoS_{Drouge} = \frac{N \times F_{AftCup}}{F_{Drouge}} = \frac{4 \times 3556}{1500} = 9.5 \quad \text{Eq 16}$$

Fore-Tube Bulkhead Fastener Calculation

$$F_{Aft} = P_{shear} \times A_{minor} = 420 \times 7.75 = 3255 \quad \text{Eq 17}$$

$$FOS_{main} = \frac{N \times F_{AftCup}}{F_{main}} = \frac{4 \times 3255}{5200} = 2.5 \quad \text{Eq 18}$$

$$FoS_{Drouge} = \frac{N \times F_{AftCup}}{F_{Drouge}} = \frac{4 \times 3255}{1500} = 8.7 \quad \text{Eq 19}$$

Fore-Tube Bulkhead Fastener Calculation – Nuts

$$F_{Aft} = P_{shear} \times A_{minor} = 420 \times 8.78 = 3688 \quad \text{Eq 20}$$

$$FOS_{main} = \frac{N \times F_{AftCup}}{F_{main}} = \frac{4 \times 3688}{5200} = 2.8 \quad \text{Eq 21}$$

$$FoS_{Drouge} = \frac{N \times F_{AftCup}}{F_{Drouge}} = \frac{4 \times 3688}{1500} = 9.8 \quad \text{Eq 22}$$

Aft Bulkhead Loading – landing deceleration

$$F = M_{payload} \times a_{landing} = 4000 = 2 \times a_{landing} \quad \text{Eq 23}$$

$$a_{landing} = \frac{F}{m_{payload}} = \frac{4800}{2} = 2400$$ Eq 24

Aft Bulkhead Loading – landing deceleration

$$F_{Aft} = P_{shear} \times A_{minor} = 420 \times 8.78 = 3688$$ Eq 25

$$FOS_{main} = \frac{N \times F_{AftCup}}{F_{main}} = \frac{4 \times 3688}{5200} = 2.8$$ Eq 26

$$FoS_{Drouge} = \frac{N \times F_{AftCup}}{F_{Drouge}} = \frac{4 \times 3688}{1500} = 9.8$$ Eq 27

Appendix H – Standard Layup Procedure

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NUROCKETRY MYLAR TUBE LAYUP INSTRUCTIONS V2.3

What you'll need before laying up:

1. Mandrel (where OD is equivalent to the desired tube ID)
2. Fiber cloth of choice
3. 2x Big Brushes and a Cylindrical Roller Figure 1 A)
4. West Systems 105 Epoxy & 206 Figure 1 D)
5. Paddle Pop Stick Figure 1 F)
6. Minimum 8 standard paper Cups or Bowls Figure 1 F)
7. Roll of Masking Tape Figure 1 B)
8. Thermo-Shrinking Tape Figure 1 B)
9. Acetone + 3 empty 2L Ice cream tubs Figure 1 I)
10. Permanent Marker
11. At least 5 meter Tape Measure
12. Composite Scissors Figure 1 I)
13. Pen and Paper (For recording batches)
14. 1-2 Friendly Unpaid Helpers
15. 2kg capacity Scales Figure 1 I)
16. 3x 6 foot white camping tables Figure 1 H)
17. Large Drop sheet Figure 1 H)
18. Clean Chux Wipes
19. Mould Release Wax Figure 1 E)
20. Rotisserie Equipment Figure 1 C)
21. PPE (Gloves, Safety Glasses, Masks(optional)
22. TA Heat Gun Figure 1 G)

Mylar-Related specifics:

23. Mylar film (.04mm thick) 10m x 1m Roll Figure 2 A)
24. 1m long sacrificial Board
25. 1m Metal Ruler/ Straight Edge
26. Scalpel or Stanley Knife
27. Flash/ blue release tape Figure 1 B)

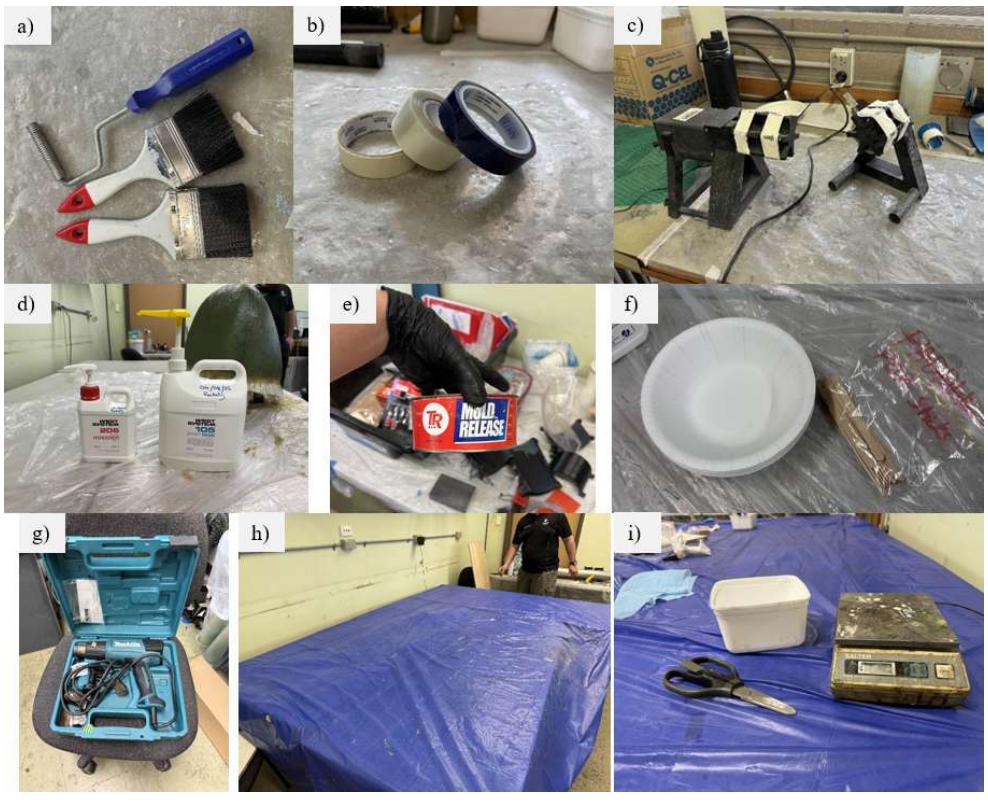

Figure 1 - A) Brushes and Rollers, B) From L-R Masking Tape, Thermo-Shrinking Tape, Blue/Flash Release Tape, C) Rotisserie, D) West Systems Resin & Hardener, E) TR Mold Release Wax, F) Paper Bowl & Paddle Pop Sticks, G) TA Heat Gun, H) Drop Sheet & Table set up, I) Composite Scissors, Acetone Tub & Scales.

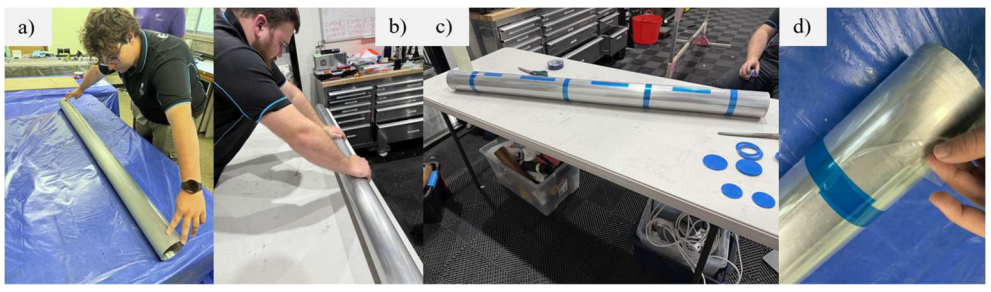

Figure 2 - A) Aligning Mylar, B) Pushing down & Rolling out bubbles, C) Example Flash Taping, D Tape toward end warning

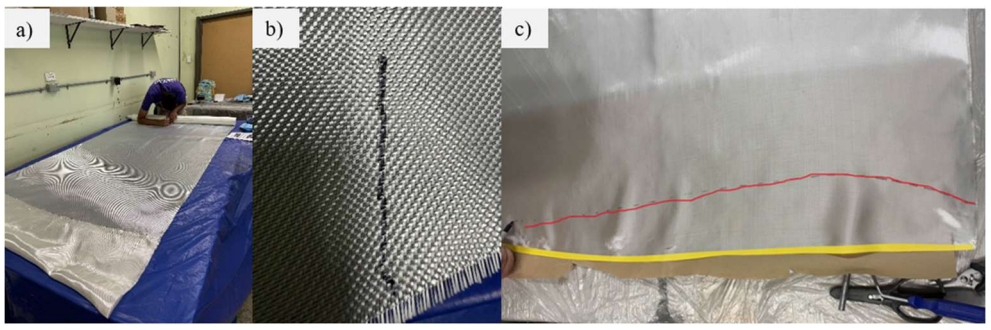

Figure 3 - A) Rolled out Fibreglass, B) Zoomed in Twill, C) Zoomed out Twill Line

Figure 4 - A) Weighing Glass, B) Rotisserie & Glass Positioning, C) Streaky Mixing Bowl

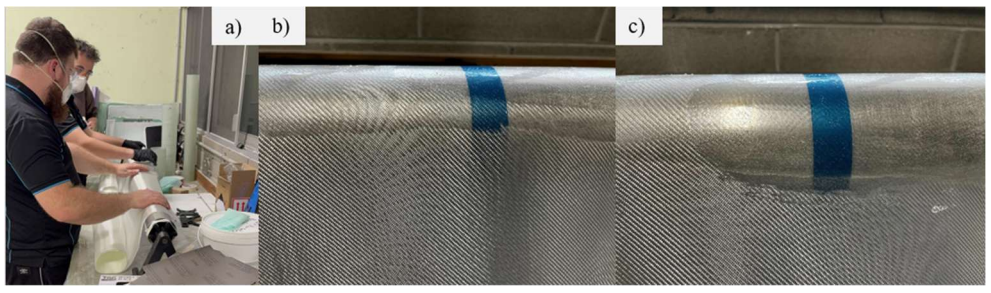

Figure 5 - A) Start of Layup, B) Dry Fiberglass, C) Wetted Out Fibreglass

Figure 6 -A) Last Layer, B) Taping shrinking tape, C) Masking tape end secural.

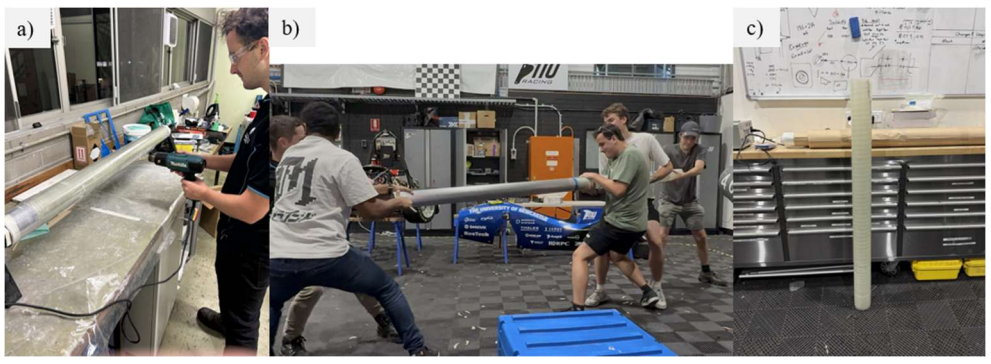

Figure 7 - A) Heat gun use, B) Tug of War, C) Produced Tube

Use of Instructions.

- Print all pages separately and on A3 Paper, keeping apart
- Have read a copy of Mitch Jeffery's 2025 3200 Tube manufacture report. If unsure about a section, ask for help. Recommended to keep a copy available on laptop back in TA
- Check local weather temps & Organise helpers
- Read through this in full before starting

Cutting Mylar

1. Using three tables in the comp lab, lay down a clean and tight drop sheet as shown in Figure 1 H), and place sacrificial board on top.
2. Calculate Mylar needed, **length of desired tube + 90mm**.
3. Being super careful, roll out Mylar Sheet over the top of board.
4. Mark out the length, perpendicular to the roll of Mylar, on both sides.
5. Going parallel to the roll, align a straight edge on the marks and cut away from yourself, using either a scalpel or razor blade.
6. Once cut return unused Mylar back into storage tube.

Wrapping Mylar

7. Make sure Mandrel is clean and in good condition before next steps. Ensuring there are no scratches or contamination on there. If Mandrel has either of these things, deal with as per '5.2 Mandrel Restoration' section of Mitch J's report.
8. Apply 2 coats of wax to Mandrel if not done recently. Wax on, wait for the first side to set (doesn't streak when brushed) and wax off with clean CHUX. Reapply before waxing off second side. Repeat.
9. Place Mandrel on the long end of the cutout, in line with the end. As is shown in Figure 2 A).
10. With the help of two assistants, roll the Mylar up tight around the Mandrel so there are no wrinkles, making sure the wrap is rolled on equally between sides brushing contaminants out along the way (dirt and dust) with your finger.
11. If the Mylar creases (likely) roll the mandrel back and try again.
12. Roll the Mylar up all the way and then roll down and push hard on the mandrel, think thick rolling pin. Figure 2 B)
13. Continue till rolled tightly and air bubbles have been pushed out.
14. Holding the mandrel, apply 4 Equi-spaced strips of flash tape around the axis of the mandrel, securing with small overlap. It is best practice to add two of these at the very end of the Mylar, unlike Figure 2 D). The spacing does not need to be precise just roughly close to Figure 2 C).
15. Along the seam, along the length of the pipe, apply strips in between the previous strips. Here it is better to do sections than in long strips.
16. Secure Mandrel to Rotisserie, out of the way on bench. Place one end onto spinner and the other onto spinning stand. It is recommended to roll on some masking tape onto the end plugs. Also placing something heavy at the opposite end to the motor as an end weight is advised. See Figure 4 B)

Layup Prep

17. Check rotisserie is plugged in at motor and wall.
18. Use $\boldsymbol{L = 2\pi r \times N + 55mm}$ to determine the length of fibre needed. **N** is the number of wraps desired, usually 8 for 2x2 twill. **R** is the radius of the mandrel. (A standard 8 Wrap 4in tube is 2,570mm.)
19. Roll the cloth out on the tables just like the Mylar earlier. Figure 3 A)
20. With the help of assistants, measure this length by putting the tape measure on a corner of the fibre cloth.
21. Mark the calculated length on a 'thread' of Twill with a permanent marker, follow the thread parallel to the roll drawing over it with a marker, then cut along the line with composite scissors. Figure 3 B&C) Show what is meant by following the twill.
22. Fold up the glass, weigh the glass, using a wide flat piece of carboard or metal to sit it on. Remember to Tare the scales after placing the cardboard. Figure 4 A)
23. Unfold and roll into a cylinder, along short edge, keeping sides square. Place in front of the rotisserie parallel to the rotisserie. It should be set up so the roll will unspool like Figure 4 B), Figure 5 A)
24. Add masking tape to both ends of the Mandrel. Just a 3 loops of tape along the edges.
25. Apply gloves, safety glasses and masks before next steps. As well as topping up any water bottles, doing any bathroom trips and place valuables away from the layup.
26. One last thing to do is to check the temperature. If the temperature is above 27*C, highly consider waiting for a cool temperature change later in the day or postpone completely. Epoxy may flash cure.

Mixing Resin

27. Fill 2 2L ice cream tubs with Acetone, using a third flipped upside as a lid for the final clear one. Fill them up about a quarter of the way full. One of these, 'the dirty bucket' is to wash the brushes mid layup. They will likely go hard around the mid-point. Here dunk it in getting acetone all through the brush. Dry it off thoroughly by brushing it off on a table leg. Use your arm as a guide if it feels dry. If any acetone is left on the brush, it may destroy the epoxy between layers.
28. Work out a 1:1.2 ratio of Glass:Epoxy. An example calc for a standard 4in tube (~520g) is below:

$$W_{glass} \times 1.2 \rightarrow 516 \times 1.2 = 619.2\ 'Goal\ Budget'$$

$$\frac{619.2}{6} = 103.2g\ hardener \rightarrow 103.2 \times 5 = 516g\ Epoxy\ Resin$$

Basically, this serves as 'Budget' to know immediately after the layup roughly what the Glass:Epoxy ratio is.

The next equation is an example of a 'during layup' quantity. Sticking to West Systems 5:1 Ratio for their products. Using this is as your guide if you over/under shoot the resin by 5g add/subtract a gram of hardener.

$$E.g\ Bowl: Resin\ 150g \div 5 = 28g\ Hardener$$

This is your absolute last chance to have everything prepped before laying up for the next 1.5-2hours. Check again now.

29. Place your cup or bowl (bowl is preferred) on the scale and tare, before pumping your resin and then the hardener. Record weights of each in a table for record keeping.
30. Thoroughly mix with paddle pop stick, Resin should be clear, no 'streaky shadows'. Sometimes Hardener does gets stuck on the sides so make sure to occasionally brush the sides while stirring. Figure 4C)
31. *WHEN EPOXY EVENTUALLY GOES HARD DO NOT DISPOSE OF, KEEP ASIDE IT'S A 🔥 HAZARD. IF SMOKING PUT IN SINK AND WATER.*
32. Apply a thick base coat to the entire mandrel. Using the rotisserie to slowly spin the mandrel while the coat goes on. Application should be done via brush and use about ~100g of Epoxy as a guide.

Lay Up & Onwards

33. If you have three people, two should be continually laying up while another mixes up new resin a bit before the current batch goes off or runs out.
34. Lay the first 50mm of fibre on to the mandrel, keeping the leading-edge square with the help of friends. Figure 5 A)
35. Fully saturate fibre (fibreglass = colour change, carbon = glossy), using the brushes to spread the resin through, before moving onto the next section. Keeping sides straight each roll. Only after the first 2 layers should you use the roller. A photo is given to show how fibreglass should look when wetted out. Figure 5 B&C)
36. Keep saturating, mixing up new batches of around ~150g and rolling more cloth on until done.
37. When the final layer comes about reduce how much epoxy is needed. However more epoxy is always better than none. It might have some loose strands hanging from it, just trim them with composite scissors to clean it up. Figure 6 A)
38. Immediately after, apply thermosetting tape, letting the mandrel slowly spool the tape on. By attaching the tape to one end and creeping along after turning the rotisserie on. Aim for half overlap between layers of tape. As the taper gets more confident speed can be increased gradually at their request as this is monotonous. Figure 6 B) Shows this in progress.
39. Once the taper is 3/4 of the way along, get someone else to begin using the heat gun on the first end, Figure 7 A). Again, slowly working their way across (at 330*C). A guide is to try and follow the spiral of tape moving along with it. The heater should notice resin coming through under the tape.
40. Once the taper reaches the end of the mandrel, stop the spinning rotisserie. Cut the shrinking tape and attach a small bit of masking tape to the end, before sticking the masking tape down onto the mandrel. Figure 6 C)

Cleanup area and equipment.

41. First put the heat gun off to the side in a safe spot and let cool down.
42. Start by washing out brushes and rollers in dirty acetone tub that likely has been used once already, before placing in ‘clean’ acetone tub. Don’t remove until Acetone has evaporated in either tub
43. Put resin and hardener back in flammable cupboard and dispose of PPE and other rubbish to make the workspace usable again by others.

Tube Removal

44. Do not remove shrink tape until after 24hours.
45. When removing the tape. Lift up masking tape and pull the tape off as the mandrel spins on the stands.
46. Wait at least 48hours before removing from Mandrel.
47. With Mylar assuming it all went well should only require 2-5 people. With one holding the mandrel with a bolt through it and the other tugging against them holding onto the tube. It should slide off quickly. Figure 7 B) Depicts one that needed 5 sets of hands.
48. If this doesn’t work grab more people. If it’s still stuck try freezing the mandrel or using Nicks removal method in Mitch J’s report, would be the next recommended steps of action.

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